Routing among Strategic LLM Providers: A Contextual Mechanism without Monetary Transfers

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Abstract

Routing queries among competing language-model providers is increasingly common, yet existing approaches typically assume fixed provider quality and overlook strategic behavior. Providers may benefit from deploying cheaper models or reducing effort, while their black-box nature and stochastic outputs make such deviations difficult to detect. Two additional challenges arise: provider capabilities vary across tasks, and without monetary payments, the platform can incentivize effort only through future query allocations. We propose a mechanism that elicits provider capabilities, sets a price based on the second-highest reported value, and monitors the selected provider's cumulative performance against this price using an evidence-dependent tolerance. If the test fails, the mechanism restarts adversarial learning over all providers. To accommodate task heterogeneity, it operates independently across regions of task space, with parameters determined by the global horizon. Each region's price is adjusted by its diameter to ensure that the selected provider can meet it on every task, rather than merely on average. We establish an additive approximate equilibrium against arbitrary nonanticipating behavior, revenue guarantees at the second-highest capability, and performance guarantees relative to the best honest provider under arbitrary joint play. The bounds scale as $n^{(d+2)/(d+3)}$, where d is the task-space dimension, with an exponent independent of the number of providers. The latter two guarantees hold for arbitrary task arrival sequences. Experiments show that a deviation undercutting the pointwise runner-up yields a per-round gain of 0.0648 against a quality-estimation router that selects the estimated best provider, compared with just 0.0001 against our mechanism.

CCS Concepts

• **Theory of computation** → **Algorithmic mechanism design**;
• **Computing methodologies** → *Multi-agent systems.*

Keywords

strategic agents; multi-armed bandits; mechanism design; moral hazard; contextual bandits

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1 Introduction

Routing platforms select among language-model providers to balance response quality and cost [5, 7, 28]. Existing approaches typically treat provider quality as a property to estimate [17, 26]. Yet this quality depends on the backend model, reasoning budget, and sampling effort used to produce each answer, which the providers control. These choices affect both inference cost and response quality, but the improvement in quality from additional computation varies across tasks [33, 34]. A router therefore does more than predict quality: by allocating future business, it influences the quality providers choose to deliver.

Even when a model and reasoning-effort level are specified, inference systems may retain flexibility in backend execution, such as adapting computation budgets or early-exit decisions across requests [11, 32, 35]. These decisions remain under a provider's private control, and reducing these computation can lower costs, potentially at the expense of response quality. With fixed compensation per query, a provider benefits from reducing effort unless the resulting loss of future traffic outweighs its cost savings. This problem has motivated performance-based contracts [30, 34], but accurate quality estimation alone does not resolve it. For example, a provider capable of 90% success may continue receiving queries over a 70%-capable competitor even after reducing its delivered success rate to 71%. Thus, even perfect estimates of delivered quality need not induce providers to operate at their full capabilities.

Recent audits provide concrete evidence that advertised services and delivered behavior can diverge. Zhang et al. [38] examine three third-party LLM API services and find that 11 of 24 tested model endpoints produce responses inconsistent with their advertised model identities, suggesting possible model substitution. A separate audit of ten commercial API gateways documents discrepancies between advertised and observed services, including substitution-like behavior [24]. Earlier, Gao et al. [12] found that 11 of 31 commercial endpoints served distributions differing from the released reference weights, without notifying users. These findings do not establish that providers systematically best-respond to routing policies. They establish a practical reason to account for hidden serving choices rather than assume fixed quality.

Allocation is already a practical quality-control instrument. OpenRouter's Auto Exacto periodically evaluates provider performance and adjusts routing priority accordingly [27]. However, providers are typically compensated at predetermined usage-based rates rather than with delivered quality [29]. While a router can redirect queries immediately, introducing quality-dependent payments requires a separate compensation arrangement. We therefore ask whether future query allocations alone can sustain performance.

A second difficulty is that no single performance standard fits every task. Models' relative capabilities vary across queries and task families [17, 19, 26]. In our measurements across eleven models, pairwise rankings reverse between task families in 30.1% of comparisons (Section B.2). Thus, neither a provider's capability nor an appropriate performance standard is generally a single number. A threshold calibrated to average performance may be unattainable

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on difficult tasks and unnecessarily weak on easier ones. The central question therefore is: *What performance can a routing platform guarantee when providers privately choose costly inference effort, capability varies across tasks, and compliance is observed only through noisy outcomes?*

Model and objective. Over n rounds, the platform observes a task context x and selects one of K providers. Provider i has a private intrinsic mean $\mu_i(x)$, its zero-effort baseline, and a private capability $M_i(x)$, its maximum attainable mean. When selected, it chooses a target $m \in [0, M_i(x)]$ and pays an incremental cost $\varphi_i(m) - \varphi_i(\mu_i(x))$. We assume private, normalized, nondecreasing, convex Lipschitz cost functions. We assume a convex cost function, reflecting that improving response quality requires increasingly greater computational effort [34]. Costs are measured relative to the provider's baseline performance, so delivering lower quality represents a cost saving. The platform observes only a Bernoulli success signal with mean m, not the target or inference effort, and seeks to maximize cumulative expected reward.

We assume a nonempty set of *honest* providers that never deliver below their intrinsic means, potentially including a trusted in-house endpoint. The platform need not know their identities. Our goal is to design a routing mechanism that admits an approximate equilibrium in which no provider can gain substantially by deviating alone, while the platform achieves cumulative expected reward close to the sum of the second-highest provider capabilities across tasks. Independently of equilibrium, we require protection relative to the best honest intrinsic performance under arbitrary joint behavior. Achieving this goal raises three major challenges.

(a) A single failed response does not imply reduced effort. Providers control their probability of success, not the outcome of each query. Therefore, even a provider delivering the required quality may produce an unsuccessful response. Unlike settings where providers directly manipulate observed rewards [3, 8], permanently excluding providers based on individual outcomes is unreliable. We prove that such exclusion rules, when they maintain a uniformly positive chance of removal, can yield only constant expected cumulative reward even when all providers are honest (Theorem 3.1). Instead, we monitor cumulative performance to distinguish sustained underdelivery from random failures.

(b) A provider's achievable quality depends on the task. Even at full effort, a provider might answer 90% of coding questions correctly but only 60% of mathematics questions. A router expecting the same success rate on both could mistake failures on mathematics questions for reduced effort, while overlooking reduced effort on coding questions. In our simulations, ignoring task differences causes the router to incorrectly flag a provider delivering its maximum achievable quality in 27% of runs (Section B.6). Since new queries may not fit predefined categories, the router must learn from similar tasks while accounting for differences in what providers can achieve.

(c) Providers can weaken the incentives faced by their competitors. Since payments do not depend on response quality, the router must encourage effort through future query allocations. A natural approach is to select the best-performing provider and require it to deliver quality comparable to its strongest competitor. However, a weaker provider with little chance of being selected may save computation by delivering lower-quality responses when its performance is evaluated. This makes the competition appear weaker, allowing the selected provider to reduce effort without losing future queries. Consequently, even when no provider has much incentive to change its behavior, the router may receive substantially lower-quality responses than intended.

1.1 Contributions

1. Mechanism and guarantees. We develop MESP, which estimates providers' performance through repeated queries, selects the highest-performing provider, and sets its required quality using the second-highest estimate, adjusted for statistical uncertainty. A cumulative confidence-based test distinguishes sustained underperformance from random failures, triggering a restart of EXP3 over all providers while preserving access to honest ones. To handle task-dependent capabilities, we run separate instances within small regions of the task space, combining observations from similar tasks and adjusting quality requirements for variation within each region. We establish three guarantees: unilateral deviations yield at most sublinear utility gains; at the proposed approximate equilibrium, platform reward approaches the taskwise second-highest capability; and under arbitrary joint behavior, reward remains close to the best honest provider's taskwise baseline. All three errors are $O(Qn^{2/3}(\log n)^{1/3})$ for Q discrete task types under fixed-instance conditions, and $O(n^{(d+2)/(d+3)} \log(eKn))$ for a d-dimensional continuous task space under the stated smoothness assumptions (Theorem 4.2 and the full version). Incentives require exogenous task arrivals, while performance guarantees allow adaptive arrivals with fresh reward draws. The continuum rate's exponent depends on task dimension, not the number of providers.

2. Tightness and equilibrium characterization. We construct explicit provider deviations that match the leading elicitation losses and establish the tight growth threshold for the number of task types supported by MESP under the stated conditions. We also characterize how competition determines platform reward across a family of equilibria (full version). When top providers have equal capabilities, competition preserves the second-highest-quality benchmark; when one provider is uniquely strongest, a continuum of equilibria can yield reward as low as the best honest-provider baseline. These results extend task by task to finite contexts. We further analyze coordinated provider behavior and extend our guarantees to noisy task labels (Theorem A.1).

3. Evaluation. We evaluate MESP through exact best-response computations, algorithmic ablations, and simulations using measured LLM capabilities. Across 142 finite-instance configurations, optimal deviations computed by dynamic programming yield up to 38.65 times the gain of hand-designed attacks, while remaining within our theoretical bounds. We vary the horizon, number of task types, and task-space dimension to examine predicted performance thresholds, and isolate the effects of monitoring, fallback, and context-dependent pricing. Using RouterBench outcomes for eleven models across seven task families [17], we simulate strategic effort and compare MESP with quality-based routers and adaptations of existing monitoring methods. Under a tested strategy of delivering just above the strongest competitor's capability, a provider gains up to 0.0648 utility per round against ε-greedy routing, versus 0.0001 against MESP.

1.2 Related Work

LLM routing and hidden-effort incentives. Existing LLM routing methods seek to balance response quality and inference cost by selecting models according to the input query. Chen et al. [5] develop cascades that invoke additional models when needed, while Ding et al. [7] route between smaller and larger models based on predicted query difficulty. Other approaches learn routing decisions from human preferences [28] or reward-model evaluations [26], while Jiang et al. [19] rank and combine outputs from multiple models. These methods exploit differences in model capabilities but do not account for providers strategically adjusting their inference effort in response to query allocations. A related line of work addresses hidden inference decisions through monetary incentives. Saig et al. [30] design statistical contracts that tie payments to evaluated output quality, while Sreenivas and Larson [34] study contracts that incentivize both model selection and computational effort. We bring these two directions together by studying routing among competing providers whose privately chosen effort determines delivered quality.

Strategic bandits and effort incentives. Strategic bandit models study how self-interested agents influence learning and allocation decisions. Prior work considers agents that inflate observed rewards under limited manipulation budgets [9], misreport contextual information [14, 20, 36], or choose actions in response to monetary incentives [25, 31]. Closer to our setting, Braverman et al. [3] study arms that retain part of their realized rewards and show that standard no-regret algorithms can perform poorly when arms behave strategically. They develop mechanisms that use future allocations to incentivize reward sharing. Esmaeili et al. [8] extend this setting to agents that can withhold or increase realized rewards at a cost, combining second-price allocation with performance guarantees that remain meaningful when some agents behave arbitrarily. Another line of work models strategic choices of actual effort and quality. Ghosh and Hummel [13] study how bandit-based allocation of user attention incentivizes content creators to produce higher-quality contributions, while Hu et al. [18] show that standard online learning algorithms can discourage effort in content recommendation and develop alternatives that incentivize higher quality. In crowdsourcing, Ho et al. [16] use bandit learning to design monetary contracts that incentivize unobservable worker effort. Our work builds on these directions by considering providers that privately choose their effort-dependent success probabilities before outcomes are realized. We study how future query allocations alone can sustain high performance across heterogeneous tasks, while providing guarantees both at an approximate equilibrium and under arbitrary joint strategic behavior.

Closest related work. Cao et al. [4] study repeated LLM routing when providers may secretly switch to cheaper models or overstate token usage. Their mechanism estimates providers' service quality and cost, selects the best provider, and sets its required performance based on its strongest competitor. By monitoring delivery and adjusting future allocations, they limit incentives for strategic behavior and achieve performance close to the second-best provider's net value, with $O(T^{2/3}\log T)$ error. We consider providers that privately choose their inference effort, where the platform observes only noisy successes and failures and achievable quality varies across tasks. Our mechanism, MESP, learns from similar tasks, sets achievable quality requirements, and monitors cumulative performance from the first regular selection, returning to learning over all providers when sustained underperformance is detected. We establish approximate equilibrium incentives, near-second-highest quality at the proposed equilibrium, and performance guarantees relative to the best honest provider under arbitrary joint behavior. All three errors are $O(Qn^{2/3}(\log n)^{1/3})$ for Q discrete task types and $O(n^{(d+2)/(d+3)}\log(eKn))$ for a d-dimensional continuous task space (Theorem 4.2).

2 Model

2.1 Providers, effort, and noisy delivery

Interaction. Over a known horizon n, a platform routes tasks among K providers, indexed by $\mathcal{K} = [K]$. At each round t, all providers face the same task, represented by a context $x_t \in \mathcal{X}$. The platform observes x_t and selects a provider I_t. Each provider i has private intrinsic-mean and capability functions $\mu_i, M_i : \mathcal{X} \to [0,1]$, with $\mu_i(x) \le M_i(x)$, representing its zero-effort performance and maximum attainable mean, respectively. The selected provider also observes x_t before choosing a target $m_{I_t,t} \in [0, M_{I_t}(x_t)]$. The reward is then drawn according to $Y_t \mid \mathcal{F}_t^- \sim \text{Bernoulli}(m_{I_t,t})$, where $\mathcal{F}_t^-$ contains everything determined before the reward draw, including the current context, selection, and target. The platform only observes contexts and rewards. The single-context model corresponds to $\mathcal{X} = \{x_0\}$, in which case we omit the context argument.

Effort costs. Each provider has a private, nondecreasing, convex, Λ-Lipschitz cost function $\varphi_i : [0,1] \to \mathbb{R}$, normalized by $\varphi_i(1) - \varphi_i(0) \le 1$. Delivering target $m_{i,t}$ incurs incremental cost $c_{i,t} := \varphi_i(m_{i,t}) - \varphi_i(\mu_i(x_t))$. Costs are measured relative to intrinsic delivery and may be negative when a provider lowers its mean below that baseline. The linear model $\varphi_i(m) = m$ is a special case. Convexity models diminishing returns to computation and is used only in the incentive analysis.

Utilities. Each selection is worth one unit to its provider, net of effort cost. For a strategy profile $S = (S_1, \dots, S_K)$, let $\mathbb{E}_S$ denote expectation over all resulting randomness. Define provider i's selection count as $T_i(n) := \sum_{t=1}^n \mathbf{1}\{I_t = i\}$ and its cumulative cost by $C_i(n) := \sum_{t \le n : I_t = i} c_{i,t}$. Provider utility is then defined as $u_i(n; S) := \mathbb{E}_S[T_i(n) - C_i(n)]$, and platform revenue as $P(n; S) := \mathbb{E}_S\left[\sum_{t=1}^n Y_t\right] = \mathbb{E}_S\left[\sum_{t=1}^n m_{I_t,t}\right]$.

By monotonicity and cost normalization, every selection yields provider utility $1 - c_{i,t} \in [0,2]$. Thus, allocation is the platform's incentive instrument, and its revenue measures successful delivery rather than monetary receipts.

2.2 Strategies and performance criteria

Information and behavioral strategies. A provider observes its own selections, the corresponding task contexts, its chosen targets, and the resulting rewards, but not other providers' histories or the current global round. Its strategy may depend on the known horizon n, its private information, past observations and decisions, any information communicated by the platform, and independent private randomness. We denote by $\mathcal{B}_{i,n}^{\text{ctx}}$ the class of all feasible strategies using this information. For stronger incentive guarantees, we also consider an enlarged class $\mathcal{B}_{i,n}^{\text{ctx}+}$, in which a provider

additionally knows the entire sequence of task contexts, their visit counts, and the current global round.

Honesty. A fixed, nonempty set $\mathcal{K}_H \subseteq \mathcal{K}$ consists of *honest* providers, whose identities are unknown to the platform. Every strategy of an honest provider h must satisfy $m_{h,t} \geq \mu_h(x_t)$ whenever selected, after every possible history.

Behavioral equilibrium. A sequence of feasible behavioral profiles $S^\star = (S_n^\star)_{n\geq 1}$ is an *additive behavioral asymptotic equilibrium* if, for every provider i, $\sup_{S'_{i,n} \in \mathcal{B}_{i,n}^{\mathrm{ctx}}} \left\{u_i(n; S'_{i,n}, S^\star_{-i,n}) - u_i(n; S_n^\star)\right\} = o(n)$. Thus, no adaptive unilateral deviation yields a linear utility gain. When a result covers $\mathcal{B}_{i,n}^{\mathrm{ctx}+}$, the supremum is taken over that larger class.

Performance benchmarks. For $K \geq 2$, let $M_{(2)}(x)$ be the second-highest entry of $(M_1(x), \ldots, M_K(x))$, counting ties, and define $b(x) := \max_{h\in\mathcal{K}_H} \mu_h(x)$. At the proposed profile, we compare platform revenue with the taskwise second-capability benchmark $\mathbb{E}_{S^\star}\left[\sum_{t=1}^n M_{(2)}(x_t)\right]$. Under arbitrary feasible joint behavior S, we compare it with the honest benchmark $\mathbb{E}_S\left[\sum_{t=1}^n b(x_t)\right]$. The latter allows the best honest provider to change with the task. Both benchmarks are evaluated along the arrival path generated under the profile being assessed. For a single context, the honest benchmark is $\mu_H^\star n$, where $\mu_H^\star := b(x_0)$.

2.3 Task spaces and arrival assumptions

Task spaces. For finite contexts, $\mathcal{X}_F$ is a fixed set of size Q, with both the set and Q unknown to the platform and no assumed relationship between its elements. For continuous contexts, $\mathcal{X} = [0,1)^d$, and all μ_i and M_i are L-Lipschitz in the Euclidean norm, with L unknown to the platform.

Context arrival assumptions. We consider three assumptions on how tasks arrive and rewards are generated, strongest to weakest.

(A1) Stochastic arrivals. Task contexts are independently drawn from a fixed distribution ν, independently of providers' behavior, reward randomness, and the mechanism's randomization.

(A2) Exogenous arrivals. Providers cannot influence the sequence of task contexts. Moreover, conditional on the entire context sequence and all information available before each draw, the mechanism's randomization and reward randomness retain their prescribed distributions. Contexts need not be independent or identically distributed.

(A3) Fresh draws. At each round, the mechanism's random choices and the reward are generated according to their prescribed conditional distributions, given all information available before each draw, including the current context and the selected provider's chosen target.

These assumptions satisfy (A1) $\Rightarrow$ (A2) $\Rightarrow$ (A3). Our incentive guarantees require (A2), while the platform's performance guarantees require only (A3). The stronger assumption (A1) is used only for supplementary results with publicly known provider capabilities.

3 Incentivizing effort with a single context

We first consider only one task type, so providers' means and capabilities are fixed. This isolates the central incentive problem: rewarding sustained effort through future selections when only noisy outcomes are observed and payments cannot depend on quality. We establish why permanent exclusion based on individual outcomes fails, then introduce a mechanism that sets a competitor-based quality requirement and monitors cumulative quality estimates. Section 4 extends it to a contextual setting.

3.1 Failure of permanent exclusion

A failed response may reflect random variation rather than reduced effort. Consequently, punishing providers after individual outcomes can eliminate those that consistently maintain high quality. We formalize this failure for a broad class of adaptive exclusion rules. An *exclusion rule* maintains a set of providers eligible to receive queries. For each query, it selects a provider and, based on past observations, specifies the success rate that the provider is expected to maintain. After observing whether the response succeeds or fails, the rule may permanently exclude the provider based on that outcome, the specified success rate, and an independent random choice. We call an exclusion rule *ρ-responsive* if, whenever it assigns a query to a provider, a particular outcome leads to that the permanent exclusion of the provider with probability at least $\rho > 0$, regardless of its past performance. Thus, even a provider with a long history of successful responses continues to face a risk of permanent exclusion based on the outcome of a single query.

Theorem 3.1 (A ρ-responsive rule loses honest benchmarks)**.** *Fix $\kappa > 0$ and an all-honest instance satisfying $\min\{\mu_i, 1-M_i\} \geq \kappa$ for all i. Every ρ-responsive exclusion rule satisfies $P(n;S) \leq (1-\kappa)K/(\kappa\rho)$ for every horizon n, uniformly over strategy profiles S. For fixed K and κ, achieving $\mu_H^\star n - o(n)$ therefore requires $\rho = O(1/n)$.*

Proof sketch. Honesty and capability bounds imply that every selected mean lies in $[\kappa, 1-\kappa]$, so either reward outcome occurs with conditional probability at least κ. Each round therefore removes its provider with probability at least $\kappa\rho$. The probability of surviving r rounds is at most $(1-\kappa\rho)^r$, giving at most $1/(\kappa\rho)$ expected rounds per provider. Summing over providers and bounding each reward mean by $1-\kappa$ proves the claim. □

Therefore, for constant ρ, expected platform reward remains constant independently of n, without any strategic manipulation.

3.2 Monitored empirical second price (MESP)

Algorithm 1 presents the *monitored empirical second-price* mechanism (MESP). It uses competition to set a required success rate, monitors the providers' cumulative performance, and switches to adversarial learning if irrational behavior is detected. The mechanism has three stages: bidding, monitored service, and fallback.

Stage 1: Bidding. The mechanism assigns L_n queries to each provider and uses its observed success rate as an empirical *bid* (Lines 1–4). The highest bidder wins the next stage (Line 6). Its required success rate τ is the second-highest bid $\beta_{(2)}$, counting ties, reduced by the statistical allowance s_n to account for random errors in the estimates (Lines 7–8), where $(z)_+ := \max\{z, 0\}$. For fixed bidding targets, all bids lie within s_n of their means with probability at least $1 - n^{-4}$. Bids estimate delivered quality, coinciding with capabilities when providers bid at capacity. The choice of L_n balances the KL_n bidding rounds against the later estimation loss ns_n, leaving $N_n := n - KL_n$ rounds.

Stage 2: Monitored service. The winner continues receiving queries while the mechanism tracks its cumulative successes Z_r after r queries (Line 11). The monitoring allowance $b_n(r)$ accounts

Algorithm 1 Monitored empirical second price (MESP)

Require: Horizon n, number of providers $K \geq 2$

1: Set $L_n := \min\{\lfloor n/(2K) \rfloor, \lceil n^{2/3}(\log(en))^{1/3} \rceil\}$
2: **for** $i = 1, \ldots, K$ **do** ▷ Stage 1: Bidding
3: Select i for L_n rounds; observe $Y_{i,1}, \ldots, Y_{i,L_n}$.
4: Set $\beta_i := L_n^{-1} \sum_{r=1}^{L_n} Y_{i,r}$.
5: **end for**
6: Choose $W \in \arg\max_i \beta_i$, using independent uniform tie break.
7: Set $s_n := \sqrt{\log(2Kn^4)/(2L_n)}$
8: Announce $\tau := (\beta_{(2)} - s_n)_+$ to all providers
9: $b_n(r) := \sqrt{r \log(2n^4)/2}$, $r \leftarrow 0$, $Z_0 \leftarrow 0$ ▷ Stage 2: Monitoring
10: **while** rounds remain **do**
11: Select W, observe Y, set $r \leftarrow r + 1$ and $Z_r \leftarrow Z_{r-1} + Y$
12: **if** $Z_r < r\tau - b_n(r)$ **then** ▷ Stage 3: Fallback
13: Let H be the remaining horizon, if $H > 0$, run fresh explicit-exploration EXP3 over all K providers for H rounds.
14: **end if**
15: **end while**

for random fluctuations in these outcomes (Line 9). If Z_r falls more than $b_n(r)$ below the $r\tau$ successes expected at the required quality, fallback is triggered (Line 12). The confidence bound holds simultaneously for all primary counts, allowing monitoring from the first query without another evaluation period, although detecting underperformance still requires sufficient evidence.

Stage 3: Fallback. When monitoring detects underperformance, the mechanism starts EXP3 afresh over all providers for the remaining H rounds (Line 13). It discards previous statistics, uses equal initial weights and fresh randomness, and learns from subsequent rewards even when delivered means change, without permanently excluding potentially honest providers [2]. The EXP3 parameters are specified in the full version.

Proposed strategy. Under $S^\star$, provider i delivers M_i during bidding, fallback, and cyclic allocation when $n < 2K$. A stage-2 winner outside $\mathcal{K}_H$ delivers $\min\{\tau, M_i\}$; an honest winner delivers $\max\{\mu_i, \min\{\tau, M_i\}\}$. These rules respect capabilities and honesty after every history, including when estimation error makes the required success rate unattainable. Although the profile depends on the stage and horizon, the incentive guarantee allows deviations adapting to the provider's entire available history.

3.3 Incentives and performance guarantees

Providers may save effort during bidding or after winning, or deliberately trigger fallback to regain allocations. The central challenge is to bound all three opportunities while protecting platform reward even if several providers deviate simultaneously.

Theorem 3.2 (Equilibrium, revenue, and robustness)**.** *Fix $K \geq 2$ and a single-context instance following Assumption* (A3)*.* MESP *satisfies the following expected guarantees.*

(i) *For every provider i, the maximum gain from any unilateral deviation in $\mathcal{B}_{i,n}$, against $S^\star_{-i,n}$, is $O_{K,\Lambda}(n^{2/3}(\log n)^{1/3}) = o(n)$.*

(ii) *At $S^\star$, $P(n; S^\star) \geq M_{(2)} n - O_K(n^{2/3}(\log n)^{1/3})$.*

(iii) *Every feasible profile S satisfies $P(n; S) \geq \mu_H^\star n - O_K(n^{2/3}(\log n)^{1/3})$.*

Thus, $S^\star$ is an additive approximate equilibrium approaching second-capability performance, while the best honest intrinsic mean remains protected against arbitrary joint behavior, without identifying honest providers.

Proof sketch: unilateral incentives. Fix provider i deviating while others follow $S^\star$. We bound its gains from reducing bidding effort, underdelivering during service, and deliberately triggering fallback.

Stage 1 (bidding). The deviator may adapt its targets, so its bid need not concentrate around a fixed mean. Instead, coupling its rewards with hypothetical capacity-delivery rewards using the same uniform draws gives $\beta_i' \leq \beta_i^{\max}$ on every execution. Thus, with identical tie-breaking, any deviating win is also a capacity-bidding win at the same required success rate $\tau_i := (\max_{j \neq i} \beta_j - s_n)_+$, determined entirely by competitors' bids. Concentration of their bids and the hypothetical capacity bid ensures $\tau_i \leq M_i$ except with probability n^{-4}. This requires neither fixed deviating targets nor capability gaps, and normalized costs bound bidding savings by L_n.

Stage 2 (monitored service). For adaptive targets m_r, the differences $Y_r - m_r$ have conditional mean zero. With probability at least $1 - n^{-3}$, simultaneous concentration gives $\left|Z_r - \sum_{\ell \leq r} m_\ell\right| \leq b_n(r)$ for every monitored query count. Thus, if service lasts q queries, including a possible triggering query, then $\sum_{r \leq q} m_r \geq q\tau_i - 1 - 2b_n(q)$. The simultaneous bound handles the random switch time; the extra unit accounts for its final query. Convexity and Lipschitz continuity imply $\sum_{r \leq q} \varphi_i(m_r) \geq q\varphi_i(\tau_i) - \Lambda(1 + 2b_n(q))$, so prolonged service requires nearly the prescribed average effort cost.

Stage 3 (fallback). Because switching and effort are adaptive, we prove an EXP3 guarantee conditional on the fallback-start history. Let $V_i := \max_{j \neq i} M_j$ and $R_K(H) := 3\sqrt{KH \log K} + 2K \log K$ for $H > 0$, with $R_K(0) := 0$. A nondeviating competitor delivers V_i, so expected fallback reward is at least $V_i H - R_K(H)$. Since other providers deliver at most V_i, receiving many fallback queries requires comparable delivery from the deviator. Convexity bounds its conditional utility by $(1 + \varphi_i(\mu_i) - \varphi_i(V_i))H + \Lambda R_K(H)$. As $\tau_i \leq V_i$ when competitors' bids are accurate, triggering fallback cannot improve the utility rate of continued service beyond this sublinear error. An honest winner with $\tau_i \leq \mu_i$ already earns its maximum feasible utility of one per query.

Combining the stages. A losing deviator receives further queries only after a rare erroneous switch by the compliant winner. For a winner, combining the bounds limits total unilateral gain to $L_n + \Lambda(1 + 2b_n(n) + R_K(n)) + O(1)$. No ns_n term appears because both executions face the same requirement τ_i. For fixed K, the bidding term dominates, proving part (i). Full proof in the full version.[1]

Proof sketch: platform performance. At $S^\star$, capacity bidding gives $(M_{(2)} - 2s_n)_+ \leq \tau \leq M_{(2)}$ and $\tau \leq M_W$ with high probability. The winner can therefore meet a requirement near the second-highest capability, while EXP3 protects performance after a switch. Charging the KL_n bidding rounds, statistical adjustment, and fallback regret proves part (ii).

For part (iii), choose an honest provider h with $\mu_h = \mu_H^\star$. Even under arbitrary joint behavior, honesty ensures $\beta_h \geq \mu_H^\star - s_n$ with high probability. If h wins, it supplies the benchmark directly; otherwise, its bid forces $\tau \geq (\mu_H^\star - 2s_n)_+$. Monitoring limits the winner's

[1] Full version, code, and results: [link to be added].



(a) Largest tested unilateral gain per round. **(b) Largest shortfall below the honest benchmark per round.**

Figure 1: Changing one component of MESP at a time.

shortfall, while fallback retains h as a reliable comparator, establishing robustness without equilibrium play. Full proof in the full version.[1]

Figure 1 illustrates these roles: removing monitoring increases the largest tested deviation gain, while replacing EXP3 with round-robin allocation or an unmonitored second bidder increases the shortfall below the honest benchmark. Removing the noise allowance instead flags providers following $S^\star$ in 99.5% of runs.

Matching incentive rates. For a unique strongest provider i with $M_i > V_i$, bidding at $m' := (M_i + V_i)/2$, then following $S^\star$, preserves the win and requirement with high probability once $4s_n < M_i - V_i$. For an honest provider, use $m' := \max\{(M_i + V_i)/2, \mu_i\}$, assuming $\mu_i < M_i$. The deviation saves $L_n[\varphi_i(M_i) - \varphi_i(m')]$ up to negligible error, yielding $\Theta(L_n) = \Theta(n^{2/3}(\log n)^{1/3})$ when the cost difference is positive. Thus, the leading incentive loss is inherent to MESP's bidding stage, not a lower bound for all allocation-only mechanisms.

4 Task-dependent performance guarantees

We extend MESP to task-dependent capabilities, learning from similar queries while keeping the required quality achievable on each task. The main result combines near-second-highest reward at an approximate equilibrium with protection relative to the best honest provider. Noisy labels are treated in Appendix A.

4.1 Routing over a continuous task space

Continuous contexts need not repeat, so the platform must combine observations from nearby tasks. However, even an accurately estimated average may exceed a provider's capability on harder tasks in the same region. We therefore adjust each regional requirement to be achievable at every task where it is enforced, even when the winner's bidding effort is strategic.

Separate instances for nearby tasks. Partition $[0,1)^d$ into $J_n := g_n^d$ half-open cubic *cells*, with $g_n := \lceil n^{1/(d+3)} \rceil$, side length $h_n := g_n^{-1}$, and diameter $\sqrt{d}\, h_n$. Fix the grid before play and initialize a separate MESP instance at each cell's first query. Each instance stores its own bids, winner, required success rate, stats, and EXP3 weights, updated only by queries in that cell. The platform need not know future visit counts or the arrival distribution. Within each cell, the three stages of Algorithm 1 operate as follows.

Contextual mechanism. Run Algorithm 1 separately in each cell of the grid above, initializing each instance on its first query and updating it only on queries in that cell. All instances use the global horizon n and stop at round n, with three changes. **Stage 1 (bidding).** Cycle through providers for $L_n := \lceil n^{2/(d+3)} \rceil$ queries each; cells receiving fewer than KL_n queries remain in bidding. Use $s_n := \sqrt{\log(2Kn^5)/(2L_n)}$ for bidding uncertainty and set the winner's required success rate to $\tau_C := (\beta_{(2),C} - 3s_n - \rho_n)_+$, where $\beta_{(2),C}$ is the cell's second-highest bid counting ties. The additional allowance $\rho_n := \sqrt{d}\, h_n \log(en)$ accounts for capability variation within a cell of side length h_n. **Stage 2 (monitoring).** Apply the same cumulative test using $b_n(r) := \sqrt{r \log(2Kn^5)/2}$ to allow for reward noise, with r counting monitored queries in that cell. **Stage 3 (fallback).** Run EXP3 over all providers in blocks of 1, 2, 4, ... subsequent queries in that cell. Restart with fresh weights at each block and use its length in place of H when setting EXP3's parameters, since the remaining number of cell visits is unknown. Assigned providers receive their cell's stage and required success rate.

Proposed strategy. Under $S^\star$, provider i delivers $M_i(x_t)$ during bidding and fallback. During monitored service, it delivers $\min\{\tau_C, M_i(x_t)\}$, or $\max\{\mu_i(x_t), \min\{\tau_C, M_i(x_t)\}\}$ if honest.

Lemma 4.1 (Quality requirements are feasible throughout each cell). *Assume* (A3) *and that every capability function M_i is L-Lipschitz, and $n \geq e^{3L}$. Under any feasible behavioral profile, let $\mathcal{E}$ be the event that each provider's bid in every cell differs by at most s_n from that its average bidding target. Then $\Pr(\mathcal{E}) \geq 1 - J_n n^{-5}$, and on E, every such cell satisfies $\tau_C \leq \inf_{x \in C} M_{W_C}(x)$. If $J_n \leq n$, then $\Pr(E) \geq 1 - n^{-4}$. At $S^\star$, the same event gives $\tau_C \geq M_{(2)}(x) - 4s_n - \frac{4}{3}\rho_n$ for every $x \in C$.*

Proof sketch. Even when a provider adjusts its effort using earlier outcomes, its observed success rate differs from the average success probability it chose by more than s_n with probability at most $1/(Kn^5)$. Adding these error probabilities across all KJ_n provider-cell pairs gives $\Pr(E) \geq 1 - J_n n^{-5}$. For any $x \in C$, feasibility and within-cell variation imply $\beta_{(2),C} \leq \beta_{W_C,C} \leq M_{W_C}(x) + s_n + \rho_n/3$ on E. Subtracting $3s_n + \rho_n$ and taking positive parts proves the upper bound. At $S^\star$, capacity bidding also gives $\beta_{i,C} \geq M_i(x) - s_n - \rho_n/3$ for every provider; applying this to the second-highest entries proves the lower bound. Full proof in the full version.[1] □

The upper bound holds even for a strategically bidding winner, without estimating its capability function. The lower bound shows that, at $S^\star$, making the requirement achievable everywhere sacrifices only the stated statistical and task-variation allowances.

Theorem 4.2 (Contextual incentives and honest-provider protection). *Assume $K \geq 2$, L-Lipschitz intrinsic means and capabilities, a nonempty honest set, and $n \geq \max\{2K, e^{3L}\}$, with L unknown to the platform. Set $R_{d,n} := n^{(d+2)/(d+3)} \log(eKn)$. The continuum mechanism has the following expected guarantees.*
(i) *Under* (A2) *and the standing cost assumptions, every unilateral deviation from $S^\star$ in $\mathcal{B}_{i,n}^{\mathrm{ctx}+}$ gains at most $O_{d,K,\Lambda}(R_{d,n}) = o(n)$ for fixed d, K, Λ.* (ii) *Under* (A3), $P(n; S^\star) \geq \mathbb{E}_{S^\star} \sum_{t \leq n} M_{(2)}(x_t) - O_{d,K}(R_{d,n})$. (iii) *Under* (A3), *every feasible joint behavioral profile S satisfies* $P(n; S) \geq \mathbb{E}_S \sum_{t \leq n} b(x_t) - O_{d,K}(R_{d,n})$.

The honest benchmark may use a different provider for each task. Both performance guarantees allow arrivals to depend on past interactions under (A3), with benchmarks evaluated on that same realized task sequence. Incentives require (A2) because a deviation is compared with $S^\star$ on the same tasks; a linear-gain deviation exists when providers can influence future arrivals.

Proof sketch: unilateral incentives. Separate mechanism states do not restrict provider strategies, which may combine observations

across cells. Under (A2), the fixed grid and reward-independent bidding schedule give the same bidding queries in the proposed and deviating executions. Sharing reward randomness as in Section 3 ensures that a deviating win is also a capacity-bidding win at the same required success rate. The feasibility lemma makes that requirement attainable throughout the cell, avoiding any need to separate winners by capability gaps.

The fallback argument also needs a fixed competitor, although the best provider may change within a cell. When provider i deviates and wins, concentration and within-cell variation give $\tau_C \le \max_{j\neq i} \inf_{x\in C} M_j(x)$. A provider attaining this maximum can meet the requirement on every fallback query. Monitoring and conditional EXP3 bounds therefore limit the queries obtainable without comparable delivery, as in the scalar proof. Convexity converts delivery into effort costs, while Lipschitz continuity bounds the effect of changing intrinsic means and capabilities by $O(\Lambda\rho_n)$ per round. The bounds sum across cells without assuming independence. Full proof in the full version.[1]

Proof sketch: performance under adaptive arrivals. Under (A3), conditioning on future contexts may reveal earlier rewards and invalidate concentration. Instead, analyze rewards in chronological order, counting each toward its cell using information available before the reward draw. This preserves zero-mean reward differences even when arrivals depend on past outcomes. At a fallback start, the remaining visits stay inside conditional expectations; the doubling blocks bound regret using visits that actually occur.

At $S^\star$, the feasibility and lower bounds on τ_C give near-second-capability delivery on each monitored query, with erroneous switches confined to concentration failures. For arbitrary joint behavior, let $v_C := \max_{h\in\mathcal{K}_H} \inf_{x\in C} \mu_h(x)$. An honest provider attaining v_C always delivers at least that mean in C, while smoothness gives $b(x) \le v_C + \rho_n/3$. Its bid either wins directly or sets a lower bound on the winner's requirement; monitoring controls subsequent delivery, and EXP3 preserves access to that honest provider after a switch. This recovers the taskwise honest benchmark without identifying honest providers or requiring minimum cell visit counts. Full proof in the full version.[1]

The rate. Smaller cells reduce task variation but require more bidding rounds; longer bidding improves accuracy but leaves fewer queries for monitored service. The leading losses are KJ_nL_n for bidding, ns_n for statistical adjustment, and $n\rho_n$ for within-cell variation. Ignoring logarithms and fixed K, these scale as $h_n^{-d}L_n$, $nL_n^{-1/2}$, and nh_n; balancing gives $L_n \asymp h_n^{-2}$, $h_n \asymp n^{-1/(d+3)}$, and exponent $(d+2)/(d+3)$. Monitoring and fallback have smaller powers of n, since cell visit counts sum to n. The exponent depends on task dimension, not provider count. Figure 2 illustrates this dimension dependence: quality requirements become positive near the predicted horizons, which increase sharply with task dimension. A provider's bidding savings are at most J_nL_n, without the factor K; the leading incentive coefficient becomes independent of K once $n \gtrsim (K\log K)^{d+3}$. Per query, locating the cell takes $O(d)$ time, bidding or monitoring $O(1)$, and fallback $O(K)$.

5 Experiments

We test incentive control, delivered quality, and task-specific feasibility using measured LLM capabilities, recorded query arrivals, and a self-hosted model. Appendix B adds scaling, monitoring, and



(a) Cells with a positive requirement. **(b) Per round revenue.** **(c) Detection at $d = 1$.**

Figure 2: Continuum MESP across n and d.

sensitivity experiments that explain how allocation losses offset effort savings and when model substitution weaken performance.

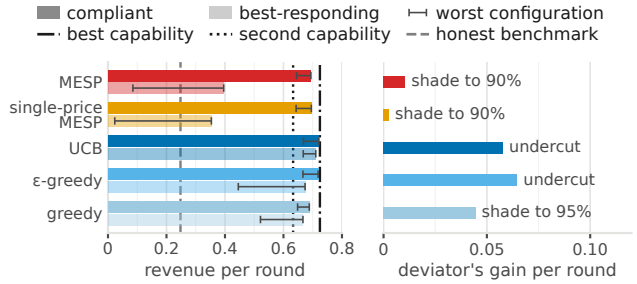
Data and protocol. RouterBench provides over 405,000 outcomes for eleven models across seven task families [17]; an independent replication uses twenty models on six Open LLM Leaderboard benchmarks [10]. We also replay windows from 5,167,182 WildChat-4.8M turns classified into five task types [39]. Benchmark outcomes and replayed arrival order are observed; provider effort is simulated. Unless stated otherwise, targets lie in $[\mu_i(x), M_i(x)]$, with $\mu_i(x) = \alpha M_i(x)$, $\alpha \in \{0, 0.25, 0.5, 0.75\}$, and $n \in \{3\times10^5, 3\times10^6\}$. We use ten to twenty paired seeds and shared random numbers across rules. **Baselines and metrics.** We compare MESP, run separately by task type, with its pooled, single-price variant; task-specific UCB1, ε-greedy and greedy routers; a static router given true capabilities but never updated; and a reimplementation of Cao et al. [4]. The latter pools tasks: unmonitored delivery is at the intrinsic baseline under its lenient profile and zero under its equilibrium profile. Appendix B.1 specifies implementations and additional exclusion and monitoring baselines. These are controlled comparisons under our feedback model, not evaluations of deployed routing services. Reward measures delivered quality per query; unilateral gain measures utility improvement per round over the reference profile. A false switch flags a provider following its prescribed strategy.

5.1 Incentives and different routing scenarios

Profitable effort reduction. Delivering just above the strongest competitor gains up to 0.0648 utility per round against ε-greedy and 0.0578 against UCB1, versus at most 0.0001 against MESP. The largest gain found against MESP in the declared strategy set is below 0.011 (Figure 3); these searches do not certify unrestricted equilibria. Quality estimation can preserve traffic after effort falls, whereas MESP substantially reduces this opportunity in the tested comparisons.

Reward without infeasible requirements. Across 80 leaderboard configurations, mean reward is 0.418 for MESP, 0.368 for the best quality router, and 0.247/0.169 for Cao et al.'s lenient/equilibrium profiles, both leaving 21% unserved (Figure 4). Single-price MESP earns 0.426 but has a 48% mean false-switch rate, versus no observed false switch for task-local MESP. Pooling can demand unattainable quality on weaker tasks; task-specific requirements avoid this failure without a large reward loss in this comparison.

Changing workloads and arrivals. MESP maintains similar reward across independent, drifting, bursty, and grouped arrivals (Figure 5a). At $n = 3\times10^6$, the pooled Cao et al. baseline leaves 50%/58%



(a) Platform reward. (b) Provider's gain.

Figure 3: Reward and largest tested deviation gains on 15 three-provider RouterBench configurations at $n = 3 \times 10^5$.

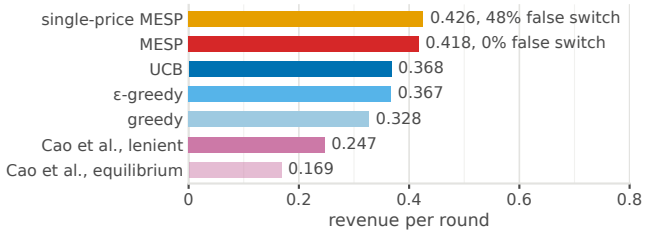
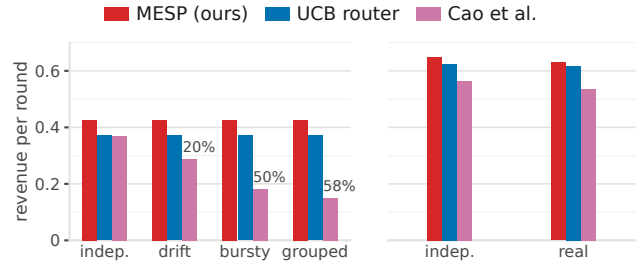


Figure 4: Mean reward over 80 leaderboard configurations.



(a) Synthetic orders, leaderboard data. (b) Indep. vs. WildChat.

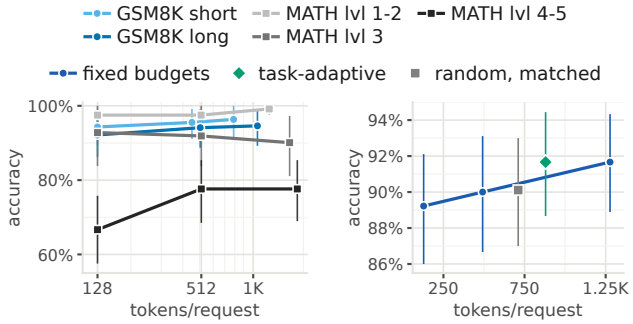
Figure 5: Reward across arrival orders.

unserved under bursty/grouped arrivals. On WildChat replay, MESP earns 0.625/0.635 at the two horizons against second-capability benchmark 0.687, with no false switch or unserved query (Figure 5b). Local states and fallback maintain service despite changes in task order. Open chat supplies 48% of arrivals and uses the least certain capability estimates (Appendix B.3).

Changing delivery and feedback. At $n = 3 \times 10^5$, slow/fast effort reductions on real arrivals leave MESP reward at 0.49/0.39, versus 0.211/0.091 for static routing; UCB1 retains about 0.51. Reallocation helps, but incentive control does not imply maximal reward under every delivery schedule. Separate tests with graded scores and up to 63 task types find no false switches; 10% label error causes false switches in 18% of runs under the unchanged strategy (Appendices B.2 and B.5).

5.2 Private inference effort in an LLM service

Qwen3-4B [37], served through vLLM [21], answers 150 GSM8K and 150 MATH-500 questions with three seeds per budget [6, 15, 23]. Only the private reasoning budget changes among 128, 512, and



(a) Accuracy by task subgroup. (b) Five gateway policies.

Figure 6: Accuracy vs. private reasoning budgets.

2,048 tokens; the model, service tier, prompts, sampling settings, and output limit remain fixed. Increasing the budget improves accuracy by 2.2 percentage points on GSM8K and 5.1 on MATH-500, with paired 95% intervals [0.4, 4.7] and [2.4, 8.2], respectively (Figure 6). Giving competition mathematics the high budget and arithmetic the low budget matches the all-high policy's observed 91.7% accuracy with 31% fewer reasoning tokens. This does not demonstrate a commercial strategic behavior, as reasoning-token savings need not reduce total tokens (Appendix B.7).

5.3 Checks of the theoretical guarantees

Certified incentives. Under linear costs, dynamic programming optimizes bidding and monitored service over permitted adaptive deviations; bounds on fallback utility give a certified interval for total gain. Across 142 configurations, every certified upper bound is below the theoretical bound; the ratio is 0.105–0.540 for the 56 configurations completing bidding in every context (Appendix B.6). This checks incentives beyond selected attacks.

Monitoring and feasibility. At matched false-switch rates, cumulative monitoring detects a 0.05 reduction in a median of 798 monitored queries, versus 1,000 for the adapted two-sample test [12]. Removing only the task-variation allowance ρ_n makes requirements unattainable on some tasks in 367 of 4,000 cells; retaining it yields no violation, isolating its role in contextual feasibility.

Use of AI tools. An AI assistant (Claude, Anthropic) conducted all experiments, made every figure, and reviewed and fully edited the appendix and the proofs of the full version.[1]

6 Conclusion

We show that future query allocations alone can incentivize private effort despite noisy outcomes and task-dependent capabilities. MESP combines achievable quality requirements, cumulative monitoring, and fallback learning. It supports an approximate equilibrium with reward approaching the taskwise second-highest capability, while protecting the honest-provider benchmark under arbitrary joint behavior. The incentive guarantee requires exogenous arrivals, and the continuum calibration can require large horizons. Reducing these horizon requirements, accommodating provider-influenced arrivals, and characterizing continuum equilibria are directions for future work.

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Supplementary Material

A Extension: noisy task labels

The platform observes a classifier label a_t, while the selected provider sees the true task type x_t before choosing effort. A label can group tasks with different capabilities: applying the scalar delivery rule to each type may fall short on difficult tasks without compensating on easier ones.

Mechanism and strategy. Run one MESP instance per label, initialized on its first arrival and updated only by that label's queries. Use Section 3.2's scalar parameters with the global horizon n: cycle through providers for L_n bidding queries each, and leave labels with

fewer than KL_n visits in bidding. After a switch, run local EXP3 tuned once to n, without restarts.

During monitored service, minimize expected cost subject to meeting the label's average requirement and each type's capability and honesty constraints. For convex costs, *water filling* achieves this by setting a common target, clipped at each type's capability and any intrinsic honesty floor. Deliver at capacity if the requirement exceeds average capability, and during bidding and fallback. The provider uses the label's true-type distribution; the platform needs neither that distribution nor the classifier's error rates. A variant estimates the distribution from bidding observations and adds an estimation allowance.

For each label, average each provider's capability and intrinsic mean, then take the second-highest average capability and highest honest average intrinsic mean. Summing over arriving labels defines the cumulative benchmarks.

Theorem A.1 (Performance with noisy labels)**.** *Assume $K \geq 2$, $n \geq 2K$, the standing model conditions, finitely many true types, $Q_{\mathcal{L}}$ labels, and a nonempty honest set. The label sequence is exogenous; conditional on it and preceding information, each true type follows a fixed label-specific distribution, and all other draws retain their prescribed conditional laws. Deviations may use the entire label sequence and observations across labels, but not future true types. For fixed $K, Q_{\mathcal{L}}$ and provider characteristics, three errors are $O(Q_{\mathcal{L}} n^{2/3}(\log n)^{1/3}) = o(n)$: unilateral gains against the water-filling profile, second-capability shortfall at that profile, and honest-benchmark shortfall under arbitrary joint behavior, using the label-level benchmarks. Only incentives require the cost assumptions. For each label, a provider bidding at capacity and meeting its requirement wins and is later switched out with probability $O(n^{-3})$. When labels name true types with average error probability ε, the cumulative second-capability benchmark loses at most εn relative to the true-type benchmark, and coefficient one is attained. With one honest provider, its benchmark is unchanged.*

Proof sketch. Average the targets the strategy would choose across the label's possible true types before the current type is revealed. This gives the platform's conditional success probability. The minimum expected cost of attaining this mean is nondecreasing, convex, and Λ-Lipschitz; water filling attains it, so the bidding, monitoring, and fallback arguments apply. For the benchmark comparison, take the two strongest providers at the type named by the label. The label benchmark is at least the average of their smaller taskwise capability, which equals the true second capability on correctly labeled tasks and loses at most one otherwise. Averaging gives εn; with one honest provider, averaging preserves its cumulative intrinsic mean.

A fixed mixture matters. Even independent classification errors allow task bursts to change the true-type distribution within a label. A quality drop can then reflect inability rather than reduced effort.

Proposition A.2 (Inability and reduced effort can be indistinguishable)**.** *For every fixed $\varepsilon \in (0, 1]$, an instance with three providers, three types, two labels, and per-type classification error at most ε admits two exogenous type processes. The scalar profile of Section 3.2, applied to each true type on the first process, and a feasible effort-reducing strategy on the second induce identical laws of labels, allocations, and rewards. Under label-local* MESP*, applying the scalar profile on both processes gives the affected provider at least $\varepsilon n/4 - O(n^{2/3}(\log n)^{1/3})$ fewer expected queries on the first, where the changing mixture triggers a switch despite adherence to the profile.*

Proof sketch. Both processes initially present tasks on which high quality is feasible. Later, one introduces zero-capability tasks that sometimes retain the old label; the other keeps feasible tasks but reduces effort to match their outcomes. Shared randomness makes the observable histories and allocations identical. Under linear costs, the deviation saves roughly $\varepsilon n/4$ on the stable-mixture process; its sublinear net gain therefore requires offsetting allocation losses. Identical histories transfer these losses to the scalar profile on the changing-mixture process. Full details appear in the full version; experiments are in Section 5.

B Additional experiments and protocol details

Full configurations, seeds, and further plots appear in the full version.

B.1 Data, baselines, and reporting

Data. RouterBench sweeps cover all 165 three-model subsets and pools up to eleven models [17]. The leaderboard replication uses IFEval, BBH, MATH level 5, GPQA, MuSR, and MMLU-Pro [10]. WildChat spans April 2023–July 2025; each seed replays a fixed-offset window without repetition [39]. We redistribute only aggregate statistics.

Baselines. Quality-learning routers update task-specific empirical means and represent estimated-quality routing [5, 7, 17, 26, 28]. Static routing always chooses the highest-capability provider for each type. We adapt one-query exclusion [8], two-sample testing [12], and fixed-tolerance monitoring after an unmonitored period [3] to our feedback model. Our pooled Cao et al. implementation [4] uses the smallest capability gap as its price and a requirement on average quality across tasks. Unmonitored delivery is zero under its equilibrium profile and the intrinsic mean under its lenient profile, shown unless stated otherwise. The trajectory experiment uses a different reference profile; Appendix B.5 evaluates a task-specific implementation.

Strategies and configurations. The figures' *compliant* providers deliver at capacity; the proposed profile follows the mechanism's stage-dependent rule. Deviations deliver below the requirement (*shading*), just above the strongest competitor (the *undercut*), or at capacity during bidding before reducing delivery (*bait-and-switch*). Best-response iteration searches a declared strategy set; joint-response tests allow all strategic providers to change behavior. Unserved queries are assigned to nobody.

Figure 3 uses three model triples and five intrinsic levels; bars show means and whiskers the worst configuration. The expanded search uses four triples, five levels, and six seeds. The undercut is the best tested response in 49/60 router configurations and 0/40 mechanism configurations, reducing reward by up to 0.098 against ε-greedy and 0.072 against UCB1 per round. The 80 leaderboard configurations combine ten subsets, four orders, and two horizons. For single-price MESP, 84% contain a false switch among twenty seeds, whereas 48% is the mean within-configuration rate (Figure 4).

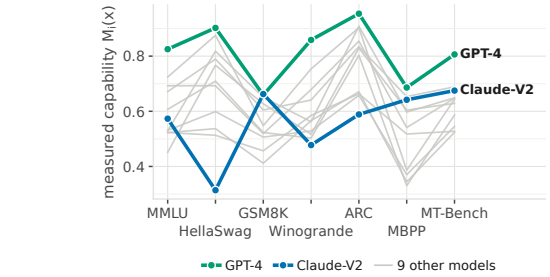
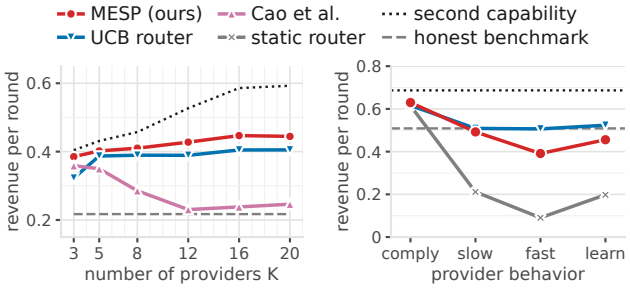


Figure 7: Measured model capabilities across seven RouterBench task families.



(a) Provider-pool size. **(b) Provider behavior.**

Figure 8: Platform reward under changing competition and delivery.

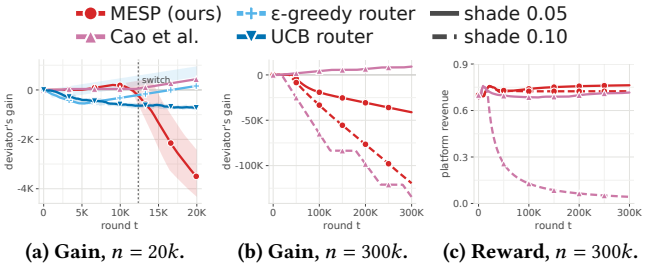
B.2 Competition and strategic providers

Pairwise rankings reverse across task families in 30.1% of comparisons, although GPT-4 leads on six of seven families and only two models are ever best (Figure 7). Thus, competition varies even when the winner does not. As the pool grows, MESP reward rises and exceeds UCB1, while the evaluated Cao et al. baseline declines (Figure 8a).

Effort reductions. Static-router rewards (independent/real arrivals) fall from 0.621/0.615 at capacity to 0.224/0.211 under slow reduction, 0.111/0.091 under fast reduction, and 0.210/0.198 under learned responses, below the honest benchmark; UCB1 retains about 0.51 under both reductions (Figure 8b). On real arrivals, MESP earns 0.49 under slow reduction, switching to fallback in 99.8% of runs. Under fast reduction, it earns 0.39: strategic providers reach zero delivery before bidding ends, rarely win, and trigger switches in only 2.5% of runs. Across ten subsets, rewards rise to 0.51/0.42 at $n = 3 \times 10^6$; independent-arrival rewards are 0.50/0.41. Under learned responses, UCB1 earns 0.524 versus 0.456 for MESP. These tests do not evaluate deployed commercial services.

B.3 Additional arrival-order results

Figure 5a uses $n = 3 \times 10^6$; at 3×10^5, pooled Cao et al. leaves 18%/22% unserved under bursty/grouped arrivals. WildChat replay lowers MESP reward by 0.018 relative to independent arrivals, yet it exceeds both Cao et al. profiles and the exploited router in all twenty subset–horizon configurations, and full-capacity learning routers in thirteen to fourteen. At the larger horizon, Cao et al. leaves 8.1% unserved; one-query exclusion leaves 85% unserved and earns 0.023. Open chat accounts for 48% of arrivals and uses



(a) Gain, $n = 20k$. **(b) Gain, $n = 300k$.** **(c) Reward, $n = 300k$.**

Figure 9: Reduced delivery after capacity bidding.

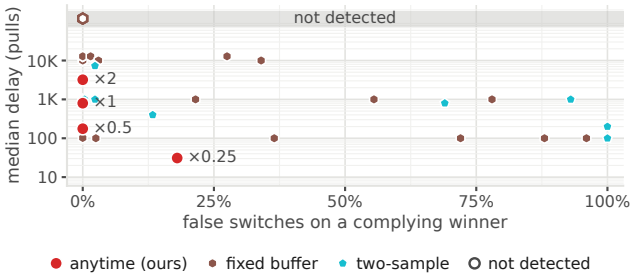


Figure 10: False-switch rate vs median delay for a 0.05 reduction.

benchmark means we could not reproduce, limiting confidence in replayed quality.

B.4 Deviation dynamics and monitoring

Allocation losses offset savings. Figure 9 follows capacity bidding and delivery 0.05 or 0.10 below each rule's reference target. Under MESP, allocation losses outweigh effort savings by the horizon while fallback maintains service. In this single-type test, Cao et al.'s reference delivers capacity outside the priced phase, with price equal to the smallest capability gap. Monitoring starts only beyond approximately $75k$ rounds, so is absent at $n = 20k$. At $n = 300k$, tolerance 0.08 misses the 0.05 reduction, yielding 9,186 additional utility; detecting the 0.10 reduction instead leaves 94% unserved. Detection therefore needs a recovery rule that continues serving queries.

Detection and false switches. Across 300 replicates per setting at $n = 3 \times 10^5$, cumulative monitoring detects the 0.05 reduction in a median of 798 monitored queries and the tested bait-and-switch in 199, versus $1k$ for the adapted two-sample test at matched false-switch rates (Figure 10). Fixed tolerance 0.2 misses the reduction; 0.005 falsely switches compliant winners in 96% of runs.

One-query exclusion removes 2.2–3.0 of three full-capacity providers on average and earns 0.000–0.012, versus 0.600 for MESP. Its requirement is 1 in 73% of seeds and 0 otherwise, illustrating Theorem 3.1's noisy-feedback obstruction.

Detection scale. Equating expected shortfall $r(\tau - m)$ for constant delivery $m < \tau$ with tolerance $b_n(r)$ gives the first-order delay estimate $\log(2n^4)/(2(\tau - m)^2)$. Figure 11 switches at query 7,552. Median delays are 0.93–1.02 times the estimate across 42 zero-delivery configurations and 0.67–1.03 across 23 partial-reduction configurations.

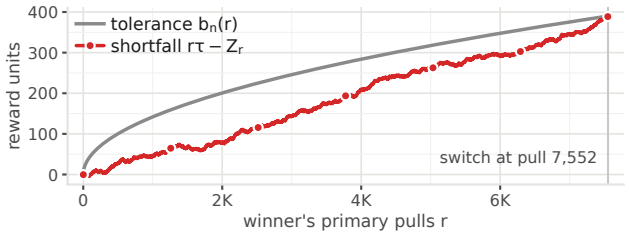


Figure 11: A 0.05 delivery reduction triggers fallback when cumulative shortfall crosses $b_n(r)$.

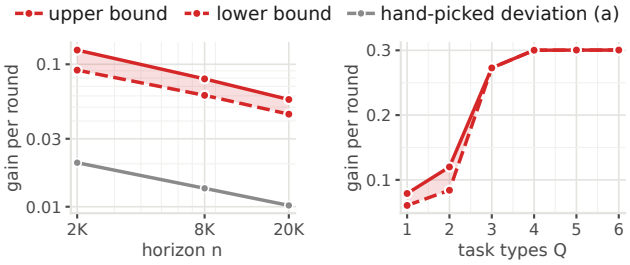
B.5 Feedback, labels, and model substitution

Graded feedback. With empirical-variance and ten-level scores in $[0, 1]$, MESP earns 0.659 without observed false switches. Single-price MESP earns 0.687 but falsely switches in 76% of runs; UCB1 earns 0.627 at capacity and 0.507 under reduction; Cao et al. earns 0.424 with 27% unserved. High reward and reliable monitoring remain distinct. **Label noise.** The unchanged strategy has no observed false switches at 5% label error, but rates reach 18% and 58% at 10% and 20%, despite reward 0.656 at 10%. Single-price MESP falsely switches in about 80% of runs throughout. **More types.** From seven to 63 types, MESP retains reward 0.685–0.694 without false switches. Single-price MESP falsely switches in 33%–65% of runs; UCB1 and Cao et al. fall to 0.646 and 0.525. Finer task separation preserves reward at these horizons while avoiding infeasible pooled requirements. **Substitution.** Learned substitution gives MESP reward 0.348, versus 0.397 for UCB1 and benchmark 0.583. Always choosing the cheapest model gives 0.327 and 0.343, respectively; Cao et al. earns 0.164 with 53% unserved. Reduced bidding quality weakens the requirement itself, so detecting later underdelivery need not maximize reward. **Task-specific Cao et al.** Its two profiles earn 0.557 and 0.368, versus 0.586 pooled and 0.659 for MESP, on ten subsets at $n = 3 \times 10^6$. Repeated exploration can outweigh task-specific requirements' benefit; further plots appear in the full version.

B.6 Certified computations and bidding

Certification. Under linear costs, each dynamic-programming objective is affine in the target, so an optimal action is an endpoint of its feasible interval. Backward induction optimizes bidding and monitored service exactly; bounds from zero continuation utility and the EXP3 guarantee bracket fallback utility. Summing per-context bounds covers the full cross-context behavioral class by additivity. Figure 12 therefore shows a certified interval, not sampling uncertainty. Across 142 configurations, covering eight instance families, two or three providers, one to eight contexts, and horizons up to $50k$, all upper bounds are below the theoretical bound. For the 56 configurations completing bidding everywhere, the ratio is 0.105–0.540.

Context-count and detection thresholds. Panel (b), with three providers and $n = 8k$, tests the construction-specific threshold $Q = o(n^{1/3}(\log n)^{-1/3})$ under the conditions stated in the full version. For $\tau > 0$, even zero delivery cannot trigger monitoring before $\log(2n^4)/(2\tau^2)$ queries have been exceeded: completing bidding alone does not ensure useful detection.



(a) Horizon; one context. **(b) Number of contexts.**

Figure 12: Certified bounds on the largest unilateral gain per.

Table 1: Predicted and evaluated thresholds.

	$d = 1$	$d = 2$	$d = 3$
predicted activation	$10^{6.53}$ to $10^{6.58}$	$10^{9.10}$ to $10^{9.16}$	$10^{11.78}$ to $10^{11.85}$
first positive requirement	$10^{6.50}$	$10^{9.10}$	$10^{11.80}$
all cells positive	$10^{6.70}$	$10^{9.20}$	$10^{11.90}$
At $d = 1$: horizon at which each error bound falls below its trivial bound			
incentive, predicted / evaluated	$10^{5.25}$ / $10^{5.25}$		
revenue, predicted / evaluated	$10^{6.76}$ / between $10^{6.70}$ and $10^{6.80}$		
robustness, predicted / evaluated	$10^{6.93}$ / between $10^{6.90}$ and $10^{7.00}$		

Pooling observations. One pooled bidding block reduces tested gains from 0.025 to 0.012 with two contexts and from 0.225 to 0.037 with six. But its requirement 0.513, versus local requirements 0.737 and 0.287, causes false switches in 27% of replicates versus none locally.

B.7 Additional private-effort measurements

Every low-budget completion uses all 128 reasoning tokens. The MATH-500 gain is concentrated in the hardest third and already achieved at 512. With the same budget mix, random allocation reaches 90.1% accuracy versus 91.7% adaptively; the paired 1.6-point difference has interval $[-0.1, 3.2]$, leaving the advantage unresolved. Interrupted reasoning continues in the visible answer: the lowest MATH-500 budget uses more total tokens than the middle budget. So fewer reasoning tokens need not mean less total computation.

B.8 Horizon thresholds and competition

Continuum thresholds. CG-MESP is the continuum mechanism of Section 4.1. At short horizons, statistical and task-variation allowances can reduce requirements to zero. Without parameter tuning, positive requirements emerge near their predicted horizons, 2.6–2.7 orders of magnitude later per added dimension (Table 1). The lower block separately reports when theoretical errors fall below trivial bounds. The table quantifies the large horizons in Figure 2.

Near-tie competition. Across 2,036 RouterBench subsets, exact top ties have zero arrival mass, but mass with a top-two gap below 0.005 rises from 2% for two providers to 21% for eleven. At $n = 3{\times}10^5$ on a constructed instance, the lowest-reward profile found earns 0.725 through gap 0.0060, dropping to 0.440 at 0.0065 (figure in the full version). The predicted crossover 0.0068 and empirical scale $0.65/\sqrt{L_n}$ reflect instance-specific bid uncertainty. The stationary-strategy search passes fixed-point checks at 17 of 19 gaps, not 0.05 or 0.1.

C Public capabilities

Single-context notation. We use Section 2's notation and write $\mathcal{B}_{i,n}$ for $\mathcal{B}_{i,n}^{\text{ctx}}$ when $\mathcal{X} = \{x_0\}$. Let $M_{\text{top}} := \max_i M_i$, $\mathcal{K}_{\text{top}} := \{i : M_i = M_{\text{top}}\}$, $K_{\text{top}} := |\mathcal{K}_{\text{top}}|$, and $V_i := \max_{j \neq i} M_j$. Thus, M_{top} is the highest capability, K_{top} counts providers attaining it, and V_i is i's strongest competitor's capability. The benchmarks $M_{(2)}$ and $\mu_H^\star$ are as in Section 2. The cost normalization is

$$\varphi_i(1) - \varphi_i(0) \le 1. \tag{1}$$

Together with monotonicity, it bounds per-query utility $1 + \varphi_i(\mu_i) - \varphi_i(m)$ in $[0, 2]$.

The *deterministic horizon-independent class* $\mathcal{S}_{\text{HI}}$ fixes one target $m_i \in [0, M_i]$ per provider, used whenever selected regardless of history or horizon; honest providers satisfy $m_h \ge \mu_h$. Call profiles in $\mathcal{S}_{\text{HI}}^K$ *fixed*. For a declared class $\mathcal{S}$, a profile $S \in \mathcal{S}^K$ is an *$\mathcal{S}$-asymptotic equilibrium* if $u_i(n; S_i', S_{-i}) \le u_i(n; S) + o(n)$ for every provider i and deviation $S_i' \in \mathcal{S}$.

We extend the SAMF class of MAB algorithms [8] through stochastic versions of sharp adaptivity, monotonicity, and fairness.

Stochastic sharp adaptivity. Define the cumulative *mean lift* $\Delta_i(n) := \sum_{t \le n : I_t = i} (m_{i,t} - \mu_i)$, measuring delivery above provider i's intrinsic baseline. An algorithm is *stochastically sharply adaptive* on $\mathcal{S}$ if a universal constant C gives, for every $S \in \mathcal{S}^K$, provider i, and horizon n,

$$\mathbb{E}_S[\Delta_i(n)] \ge (\mu_H^\star - \mu_i)\mathbb{E}_S[T_i(n)] - C\sqrt{n \log(en)}.$$

Thus, receiving many queries requires enough delivery to compensate for a low intrinsic mean. When $\varphi_i(m) = m$, mean lift equals cumulative cost $C_i(n)$.

Theorem C.1 (Sharp adaptivity implies robustness)**.** *Every sharply adaptive algorithm satisfies, for every profile in its declared class,*

$$P(n; S) \ge \mu_H^\star n - O(K\sqrt{n \log(en)}).$$

Proof idea. For any profile and cost functions, $P(n; S) = \sum_i (\mathbb{E}_S[\Delta_i(n)] + \mu_i \mathbb{E}_S[T_i(n)])$. Insert the mean-lift bounds and use $\sum_i \mathbb{E}_S[T_i(n)] = n$. Details appear in Appendix D.

Stochastic monotonicity. For fixed $S \in \mathcal{S}_{\text{HI}}^K$, let $S^{i\uparrow}$ raise only provider i's target to M_i. An algorithm is *stochastically monotone* if, for every fixed S, i, and $j \neq i$,

$$\mathbb{E}_{S^{i\uparrow}}[T_j(n)] \le \mathbb{E}_S[T_j(n)] + o(n).$$

Improving one provider cannot increase another's asymptotic allocation share; the remainder is pointwise in the compared profiles.

Fairness among equal top agents. An algorithm satisfies *fairness among equal top agents* (FATA) if providers i, j with the same top reward law under S satisfy

$$\left|\mathbb{E}_S[T_i(n)] - \mathbb{E}_S[T_j(n)]\right| = o(n).$$

Sharp adaptivity, monotonicity, and FATA together define stochastic SAMF.

A symmetric UCB1 *rule.* Initialize providers in uniform random order, then maximize $\widehat{m}_i + \sqrt{4 \log n / T_i}$, breaking ties independently and uniformly. Random initialization removes label asymmetry; using the known horizon n, rather than the current round, makes each index depend only on that provider's reward sequence and count. This enables monotonicity. For the anytime width $\sqrt{2 \log t / T_i}$, Appendix D gives a three-provider reward sequence violating the corresponding pathwise property.

Theorem C.2 (Scalar UCB1 is stochastic SAMF)**.** *On $\mathcal{S}_{\text{HI}}$, symmetric* UCB1 *is stochastic* SAMF*, with sharp-adaptivity error $4\sqrt{n \log n} + 3$, zero monotonicity remainder, and exactly equal expected counts for equal-law top providers. Raising one provider's fixed mean weakly reduces every other's expected count at every horizon. If initialization and ties instead use one uniform priority drawn before play, this comparison holds on every sample path under the reward coupling.*

Proof idea. An honest provider delivers at least $\mu_H^\star$; splitting the UCB1 fixed-gap bound at $4\sqrt{\log n / n}$ controls each provider's delivery deficit [1, 22]. For monotonicity, shared reward draws make a higher mean raise only that provider's index sequence. The resulting index comparison cannot increase another provider's count. Symmetry gives equal expected counts for equal-law top providers. Full proofs appear in Appendix D.

Suppose M_{top} is public and the *two-top condition*, $K_{\text{top}} \ge 2$, holds. A uniquely strongest provider might save effort while remaining best; with another provider at M_{top}, any fixed reduction instead makes it suboptimal.

Theorem C.3 (Public restricted equilibrium)**.** *Under $K_{\text{top}} \ge 2$ and symmetric* UCB1*, the capacity-delivery profile $S^{\max}$ is an $\mathcal{S}_{\text{HI}}$-asymptotic equilibrium: every fixed unilateral deviation gains at most $o(n)$. The bound is deviation-dependent: a top provider lowering its mean to $m' < M_{\text{top}}$ receives $O(\log n / (M_{\text{top}} - m')^2)$ queries, so it is not uniform as the gap vanishes. Moreover, $P(n; S^{\max}) \ge M_{\text{top}} n - O(\log n)$, with a constant depending on fixed positive capability gaps.*

Proof idea. At $S^{\max}$, non-top providers receive $O(\log n)$ queries and top providers have equal expected counts. A top provider either keeps its target and utility unchanged or falls below another top provider, receiving only $O(\log n)$ queries; a non-top provider remains separated by its capability gap. Each query yields at most two utility units, while baseline utility $1 + \varphi_i(\mu_i) - \varphi_i(M_{\text{top}})$ is nonnegative. Hence no fixed deviation gains linearly, and all but $O(\log n)$ expected queries achieve M_{top}. Only monotonicity and cost normalization are needed: fixed targets require no convexity argument. See Appendix E.

Private capabilities require a different comparison because effort changes empirical bids and noisy outcomes cannot justify permanent exclusion. Section 3.2's MESP addresses both issues.

Proof skeleton for Theorem 3.2. Fix a unilateral deviation by provider i, with all others following $S^\star$.

Bid domination. Generate each bidding reward as $Y = \mathbf{1}\{U \le m\}$, sharing the uniform draw U with hypothetical capacity delivery. Since $m \le M_i$, the deviating bid cannot exceed the capacity bid. With common tie-breaking, every deviating win is also a capacity-bidding win at the same competitor-determined requirement $\tau_i := (\max_{j \neq i} \beta_j - s_n)_+$. Normalization bounds bidding savings by L_n.

Monitoring. Let $A_r := \sum_{\ell \le r} m_\ell$ be the cumulative target over r monitored queries. Since each target precedes its outcome, $Z_r - A_r$ is a martingale with increments in an interval of length one. Each tail at $b_n(r)$ is at most $\exp(-2b_n(r)^2/r) = (2n^4)^{-1}$; summing over both tails and $r \le n$ gives failure probability n^{-3}. Otherwise, service lasting q queries satisfies $A_q \ge q\tau_i - 1 - 2b_n(q)$, including

the triggering query. Thus, sustained allocations require average delivery close to the requirement, even under adaptive effort.

Fallback. Restart EXP3 over all providers for the remaining H rounds. A nondeviating competitor supplies V_i, giving conditional expected reward at least $V_iH - R_K(H)$, with R_K as in Section 3.3. Convexity converts this delivery bound into a cost bound (Lemma F.1); normalization and $\tau_i \le V_i$ on accurate-bid histories then bound the deviator's expected fallback utility by $(1 + \varphi_i(\mu_i) - \varphi_i(\tau_i))H + \Lambda R_K(H)$. An honest provider with $\tau_i \le \mu_i$ already earns the maximum honest utility of one per monitored query.

Accounting. For a winning strategic provider, or an honest one with $\tau_i > \mu_i$, monitoring and fallback bound post-bidding utility by $(1 + \varphi_i(\mu_i) - \varphi_i(\tau_i))N_n + \Lambda(1 + 2b_n(n) + R_K(n))$. The reference profile earns the leading term unless a false switch occurs, whose conditional probability is at most $(2n^3)^{-1}$. The honest case $\tau_i \le \mu_i$ follows from the one-unit utility bound. A losing deviator receives post-bidding queries only after a rare false switch. Including bidding savings and exceptional events gives total gain at most $L_n + \Lambda(1 + 2b_n(n) + R_K(n)) + O(1)$. This covers non-top winners: concentration of the hypothetical capacity bid ensures $\tau_i \le M_i$ whenever the deviation wins on the accurate-bid event, without a capability-gap assumption. Monitoring and fallback likewise yield the reward and honest-benchmark bounds. Full proofs and equilibrium characterization appear in Appendix F and Appendix I, respectively.

C.1 The bidding bound is attained

On a unique-top instance, bid at the midpoint m' of the two largest capabilities, raised to the intrinsic mean if the provider is honest, then follow $S^\star$. Once s_n is below one quarter of the capability gap, this preserves the win and requirement with high probability and saves $L_n[\varphi_i(M_i) - \varphi_i(m')]$, up to negligible error (Proposition F.4 in Appendix F). When the cost difference is positive, this attains $\Theta(L_n) = \Theta(n^{2/3}(\log n)^{1/3})$: the leading loss belongs to the mechanism, not loose analysis. Whether the monitoring term is attained remains open; following just inside the monitoring boundary may yield $\Omega(\sqrt{n \log n})$, but no such lower bound is proved.

D Scalar UCB1 proofs

We prove Theorems C.1 and C.2. Throughout, fix a deterministic profile in $\mathcal{S}_{\mathrm{HI}}$, so that agent k delivers the constant mean $m_k \in [0, M_k]$ on every pull and its rewards are i.i.d. Bernoulli(m_k), and write $m^\star := \max_k m_k$.

Lemma D.1 (UCB1 deficit bound)**.** *For every agent i and every comparator mean $a \le m^\star$,*

$$(a - m_i)_+ \mathbb{E}T_i(n) \le 4\sqrt{n \log n} + 3.$$

Proof. Put $\Delta := m^\star - m_i \ge 0$. The width $\sqrt{4 \log n / T_i} = \sqrt{2 \log(1/\delta)/T_i}$ with $\delta = n^{-2}$ is the UCB1(δ) index of Lattimore and Szepesvári [22, Chapter 7]; Bernoulli rewards are 1/2-subgaussian, so their Theorem 7.1, whose proof does not depend on the tie rule or on the order of initialization, gives $\mathbb{E}T_i(n) \le 3 + 16 \log n / \Delta^2$ whenever $\Delta > 0$. Since $a \le m^\star$, we have $(a - m_i)_+ \le \Delta$. If $\Delta \le 4\sqrt{\log n / n}$, the trivial bound $T_i(n) \le n$ gives $(a - m_i)_+ \mathbb{E}T_i(n) \le \Delta n \le 4\sqrt{n \log n}$. If $\Delta > 4\sqrt{\log n / n}$, then $\Delta \mathbb{E}T_i(n) \le 3\Delta + 16 \log n / \Delta \le 3 + 4\sqrt{n \log n}$. For $\Delta = 0$ the left side is 0. □

Sharp adaptivity in Theorem C.2. Choose an honest agent h with $\mu_h = \mu_H^\star$. Honesty gives $m_h \ge \mu_H^\star$, hence $m^\star \ge \mu_H^\star$, so Lemma D.1 applies with $a = \mu_H^\star$. Since agent i delivers the constant m_i, its mean lift is $\Delta_i(n) = (m_i - \mu_i)T_i(n)$, and

$$\begin{aligned}\Delta_i(n) &= (\mu_H^\star - \mu_i)T_i(n) - (\mu_H^\star - m_i)T_i(n) \\ &\ge (\mu_H^\star - \mu_i)T_i(n) - (\mu_H^\star - m_i)_+ T_i(n).\end{aligned}$$

Taking expectations and applying the lemma gives $\mathbb{E}_S[\Delta_i(n)] \ge (\mu_H^\star - \mu_i)\mathbb{E}_S[T_i(n)] - 4\sqrt{n \log n} - 3$, and $4\sqrt{n \log n} + 3 \le 7\sqrt{n \log(en)}$ for all $n \ge 1$, so the definition holds with the universal constant $C = 7$. The honest agent need not deliver exactly its intrinsic mean. □

Lemma D.2 (Index races are monotone)**.** *Give each agent a an arbitrary sequence $f_a(0), f_a(1), \dots$ of index values in $(-\infty, +\infty]$, fix one strict priority order on the agents, and at each step pull an agent maximizing $f_a(T_a)$, where T_a is the agent's current count, breaking ties by the priority. Let $T_a(t)$ be the count after t steps. If $f_1' \ge f_1$ pointwise while the other sequences are unchanged, then $T_j'(t) \le T_j(t)$ for every $j \ne 1$ and every t, and $T_1'(t) \ge T_1(t)$.*

Proof. Write $(v, a) > (v', a')$ if $v > v'$, or $v = v'$ and a has priority over a'. Let $e_a(k) := \min_{k' \le k} f_a(k')$ be the running-minimum envelope, which is nonincreasing in k.

Step 1: the race on (f_a) and the race on (e_a) make the same choices. By induction on the step, suppose the two races have made the same pulls so far, so the counts agree, and let $a^\star$ be the agent the f-race pulls next. Either $f_{a^\star}(T_{a^\star}) = e_{a^\star}(T_{a^\star})$, and then for every rival a, $(e_{a^\star}(T_{a^\star}), a^\star) = (f_{a^\star}(T_{a^\star}), a^\star) > (f_a(T_a), a) \ge (e_a(T_a), a)$; or the minimum was attained at an earlier count $k' < T_{a^\star}$, at the step when $a^\star$ had count k' and was pulled, so that $(f_{a^\star}(k'), a^\star) > (f_a(T_a^{\mathrm{then}}), a) \ge (e_a(T_a^{\mathrm{then}}), a) \ge (e_a(T_a), a)$ for every rival a, using $T_a^{\mathrm{then}} \le T_a$ and the monotonicity of e_a. In both cases $a^\star$ is also the e-race's choice.

Step 2: the e-race is a merge. Since each e_a is nonincreasing, the e-race consumes the items $(e_a(k), a)$ in decreasing order of (v, a): at each step the pulled value is the current maximum, and that agent's next value is no larger. Hence the step at which agent j receives its kth pull equals $k + \sum_{a \ne j} \#\{k' : (e_a(k'), a) > (e_j(k-1), j)\}$, and if this count is infinite, j never receives a kth pull.

Step 3: raising agent 1 delays every other pull. Raising f_1 to $f_1' \ge f_1$ raises e_1 to $e_1' \ge e_1$ pointwise, which can only increase the counting term in Step 2 for every $j \ne 1$ and every k. So each pull of j occurs no earlier and $T_j'(t) \le T_j(t)$ for all t. Since the counts sum to t, $T_1'(t) \ge T_1(t)$. □

This is the known-horizon race argument of Feng et al. [9] and Esmaeili et al. [8]; what is new here is only that the index sequences need not be monotone in the count, and the coupling below, which raises a target rather than a realized reward.

Monotonicity in Theorem C.2. Realize rewards as $Y_{k,r} = \mathbf{1}\{U_{k,r} \le m_k\}$ with independent uniforms indexed by the agent k and its own pull count r. Agent k's index after r pulls, $f_k(r) := r^{-1} \sum_{r' \le r} Y_{k,r'} + \sqrt{4 \log n / r}$ for $r \ge 1$ and $f_k(0) := +\infty$, is a

function of r and of its own rewards alone; the random initialization is the tie rule applied at the value $+\infty$. Raising m_i to M_i raises agent i's rewards pointwise on the same uniforms, hence raises f_i pointwise, while every other agent's index sequence is unchanged.

One priority drawn before play. If the initialization order and all ties use one uniformly random strict priority drawn before play, Lemma D.2 applies on every sample point and gives $T_j(n)$ weakly smaller for every $j \neq i$ and every horizon, so the monotonicity remainder is zero and the sample-point statement of the theorem holds.

Independent per-round ties. Realize the uniform tie rule of the algorithm as follows. Let $(A_k)_{k\geq 1}$ be independent uniform choices, with A_k used at the kth pull of an agent other than i to pick uniformly among the agents other than i in the argmax set, and let $(B_t)_{t\geq 1}$ be independent uniform $[0,1]$ variables, with i pulled at round t if and only if i is in the argmax set $\mathcal{A}_t$ and $B_t \leq 1/|\mathcal{A}_t|$; otherwise the candidate chosen by the next A_k is pulled. At every round this selects a uniformly random element of $\mathcal{A}_t$, so the execution has the law of the algorithm. Under this representation the sequence of agents receiving the pulls not given to i is determined by the index sequences of the agents other than i and by (A_k) alone: the kth such pull goes to the agent maximizing its index at the counts produced by the previous $k-1$ such pulls, with ties broken by A_k, and does not depend on when the pulls of i are interleaved. Call this fixed sequence $(a_k)_{k\geq 1}$. Now compare the execution with m_i and the one with M_i on the same uniforms, the same (A_k) and the same (B_t). Suppose $T_i'(t) \geq T_i(t)$ fails for some t, and let t_0 be the first such round; then $T_i'(t_0-1) = T_i(t_0-1) =: c$, and at round t_0 agent i is pulled in the original execution but not in the raised one. Both executions have the same counts before round t_0, hence the same next candidate a_{t_0-c} at the same count. In the original execution $i \in \mathcal{A}_{t_0}$ and $B_{t_0} \leq 1/|\mathcal{A}_{t_0}|$. In the raised execution $f_i'(c) \geq f_i(c)$: if $f_i'(c) > f_i(c)$ then i is the strict maximizer and is pulled; if $f_i'(c) = f_i(c)$ the argmax set is the same and the same B_{t_0} selects i. Either way i is pulled at round t_0 in the raised execution, a contradiction. Hence $T_i'(t) \geq T_i(t)$ for all t, and since $T_j(t) = \#\{k \leq t - T_i(t) : a_k = j\}$ for $j \neq i$ with the same sequence (a_k) in both executions, $T_j'(t) \leq T_j(t)$ for every $j \neq i$ and every t. Taking expectations gives the monotonicity statement with zero remainder for the algorithm as specified. □

Remark D.3 (The anytime width is not monotone). *With the width $\sqrt{2\log t/T_i}$, where t is the index of the round being decided, the conclusion fails on a sample point. Take $K=3$, $n=8$, initialization in label order, ties to the lowest label, and reward tapes, indexed by each agent's own pull count, $(0,0,0,0,1,0,0,0)$, $(0,0,1,0,0,1,1,1)$, $(1,1,1,1,1,0,0,1)$. Raising the first agent's tape to $(1,0,0,1,1,1,0,1)$, which dominates it pointwise, changes the counts after 8 rounds from $(2,1,5)$ to $(2,2,4)$: the second agent gains a selection although only the first improved. The reason is that the anytime index is not a function of the agent's own count and own rewards alone, which is what Lemma D.2 requires. Whether the anytime width is monotone in expectation is open.*

Fata in Theorem C.2. If agents i and j have the same Bernoulli law, then relabelling the two agents, that is, swapping their reward tapes, their positions in the uniform initialization order, and their roles in every later tie break, leaves the joint law of the execution invariant, because each index depends only on the agent's own count and own rewards and the tie rules are uniform. Thus $(T_i(n), T_j(n))$ is exchangeable and $\mathbb{E}T_i(n) = \mathbb{E}T_j(n)$ exactly. □

Proof of Theorem C.1. Suppose an algorithm satisfies the stochastic sharp-adaptivity inequality with constant C on a class $\mathcal{S}$, and fix $S \in \mathcal{S}^K$. By the definition $\Delta_i(n) = \sum_{t\leq n: I_t=i}(m_{i,t} - \mu_i)$ and the revenue identity $P(n;S) = \mathbb{E}_S \sum_{t\leq n} m_{I_t,t}$,

$$P(n;S) = \sum_i \big(\mathbb{E}_S[\Delta_i(n)] + \mu_i \mathbb{E}_S[T_i(n)]\big),$$

for every profile, whether or not the mean lifts are negative, and for every cost function, since both sides are statements about delivered means. Adding $\mu_i \mathbb{E}_S[T_i(n)]$ to the sharp-adaptivity inequality of agent i and summing over the K agents, with $\sum_i \mathbb{E}_S[T_i(n)] = n$,

$$P(n;S) \geq \mu_H^\star n - KC\sqrt{n\log(en)}.$$

□

We will also use the standard gap-free consequence of the same UCB1 analysis. In an s-round fixed-mean instance with best mean $m^\star$, run with the width $\sqrt{4\log s/T_i}$,

$$\mathbb{E}\sum_{r=1}^{s}(m^\star - m_{I_r}) \leq C\sqrt{Ks\log(es)} + CK. \tag{2}$$

Indeed, for $\varepsilon > 0$, agents with gap below ε contribute at most $s\varepsilon$, while by the fixed-gap bound each agent with gap $\Delta_k > \varepsilon$ contributes $\Delta_k \mathbb{E}T_k(s) \leq 3 + 16\log s/\varepsilon$; choosing $\varepsilon = \sqrt{16K\log s/s}$ gives $8\sqrt{Ks\log s} + 3K$, so $C = 8$ works.

E Public-capacity proof

Proof of Theorem C.3. Throughout, UCB1 denotes the symmetric rule of Section C with width $\sqrt{4\log n/T_i}$, and we use the fixed-gap bound of Lemma D.1's proof: in a fixed-mean instance, an agent whose mean is $\Delta > 0$ below the best mean receives $\mathbb{E}T_k(n) \leq 3 + 16\log n/\Delta^2$ pulls.

Allocation at $S^{\max}$. Under $S^{\max}$ every agent delivers its capacity, so each $k \notin \mathcal{K}_{\text{top}}$ has fixed gap $M_{\text{top}} - M_k > 0$ and $\mathbb{E}T_k(n) \leq 3 + 16\log n/(M_{\text{top}} - M_k)^2 = O(\log n)$. The K_{top} top agents have identical Bernoulli(M_{top}) reward laws, and the rule is symmetric in the agents' labels, so their counts are exchangeable and each satisfies

$$\mathbb{E}T_i(n) = \frac{n - \sum_{k\notin\mathcal{K}_{\text{top}}}\mathbb{E}T_k(n)}{K_{\text{top}}} = \frac{n}{K_{\text{top}}} - O(\log n), \qquad i \in \mathcal{K}_{\text{top}}.$$

A top agent's utility is $(1 + \varphi_i(\mu_i) - \varphi_i(M_{\text{top}}))\mathbb{E}T_i(n)$, and the coefficient lies in $[0,1]$: $\varphi_i(M_{\text{top}}) - \varphi_i(\mu_i) \in [0,1]$ by monotonicity, $\mu_i \leq M_i \leq M_{\text{top}}$ and (1).

Deviations. Fix an agent i and a deviation in $\mathcal{S}_{\text{HI}}$, that is, a fixed mean $m' \in [0, M_i]$ delivered on every pull (with $m' \geq \mu_i$ if i is honest). If i is a top agent and $m' = M_{\text{top}}$, the execution law is unchanged and so is its utility. If i is a top agent and $m' < M_{\text{top}}$, another top agent still delivers M_{top} because $K_{\text{top}} \geq 2$, so i has the fixed gap $M_{\text{top}} - m' > 0$ and receives $\mathbb{E}T_i(n) \leq 3 + 16\log n/(M_{\text{top}} - m')^2$ pulls, each worth at most 2; its deviating utility is therefore $O(\log n/(M_{\text{top}} - m')^2)$, while its baseline utility is nonnegative, so the gain is $O(\log n)$ for each fixed m', with a constant that grows as

$m' \uparrow M_{\text{top}}$, as the theorem states. If i is not a top agent, every feasible $m' \le M_i < M_{\text{top}}$ leaves a gap of at least $M_{\text{top}} - M_i > 0$, so both the baseline and the deviating utility are at most $2(3 + 16 \log n/(M_{\text{top}} - M_i)^2) = O(\log n)$. In all cases the gain is $o(n)$, which is the $\mathcal{S}_{\text{HI}}$-equilibrium inequality. The argument uses nothing about φ_i beyond monotonicity and (1): a strategy in $\mathcal{S}_{\text{HI}}$ commits to one mean, so there is no splitting of effort across pulls for convexity to control.

Revenue. Every pull of a top agent yields expected reward M_{top}, and every other pull yields at least $0 \ge M_{\text{top}} - 1$. Hence $P(n; S^{\max}) \ge M_{\text{top}} n - \sum_{k \notin \mathcal{K}_{\text{top}}} \mathbb{E}T_k(n) = M_{\text{top}} n - O(\log n)$, where the constant depends on the fixed positive gaps $M_{\text{top}} - M_k$. □

Without further restrictions, the two-top assumption cannot be dropped. If a unique top agent has enough feasible downward slack to remain above the runner-up, it can lower its mean while remaining uniquely best and save a linear amount of effort. Knife-edge cases without such slack can still be equilibria.

F Private-capacity proofs

This appendix proves Theorem 3.2. We write $\mathcal{B}_{i,n}$ for the behavioural class $\mathcal{B}^{\text{ctx}}_{i,n}$ of Section 2 in the single-context case $\mathcal{X} = \{x_0\}$; since there is no context sequence to influence, Assumptions (A2) and (A3) coincide here, which is why the theorem assumes only (A3).

Standing conventions. For $n < 2K$ the mechanism has $L_n = 0$; it then cycles through the agents and $S^\star$ delivers capacity (Section 3.2). Any agent's utility at such a horizon is at most $2n < 4K$, so these finitely many horizons are absorbed into the constants, and we fix $n \ge 2K$ from now on. Stage 1 then uses exactly $KL_n \le n/2$ rounds, so $N_n := n - KL_n \ge n/2$. Throughout, each randomisation of the mechanism (the tie-breaking priority and the EXP3 selections) has its specified law conditionally on everything realised before it, and each round's reward is Bernoulli with parameter equal to the selected agent's target conditionally on everything determined before the draw, the target included; this is Assumption (A3). The uniform tie break of Algorithm 1 is realised as follows: a uniformly random strict priority order over the agents is drawn before play, and the winner is the highest-priority agent among those attaining the maximal bid. This has the same law as a uniform choice among the tied agents, and it allows two executions to share one priority. Put

$$R_K(H) := 3\sqrt{KH \log K} + 2K \log K \quad (H \ge 1), \qquad R_K(0) := 0,$$

a nondecreasing function of H. For a phase of the mechanism, U_i^{phase} denotes agent i's realised utility in that phase, the sum of $1 + \varphi_i(\mu_i) - \varphi_i(m_{i,t})$ over its pulls in the phase. By monotonicity and the normalisation (1), every summand lies in $[0, 2]$.

F.1 From delivered means to cost

Every accounting step below first bounds a sum, or a conditional expectation, of *delivered means*, through the monitor or through the fallback, and then converts that bound into a bound on the corresponding sum of *costs*. The conversion is always the same two lines, recorded once.

Lemma F.1 (cost of a monitored tenure). *Let φ be nondecreasing, convex, and Λ-Lipschitz on $[0, 1]$, and let $\theta \in [0, 1]$ and $\Delta \ge 0$.*

(a) *If $m_1, \dots, m_q \in [0, 1]$ satisfy $\sum_{r \le q} m_r \ge q\theta - \Delta$, then $\sum_{r \le q} \varphi(m_r) \ge q\varphi(\theta) - \Lambda\Delta$.*
(b) *If π is a measure on $[0, 1]$ of total mass $\bar{Q} > 0$ with $\int m \, d\pi \ge \bar{Q}\theta - \Delta$, then $\int \varphi \, d\pi \ge \bar{Q}\varphi(\theta) - \Lambda\Delta$.*

Proof. We prove (b). The function φ is continuous, hence bounded on $[0, 1]$ and integrable against π. Put $\bar{m} := \bar{Q}^{-1} \int m \, d\pi \in [0, 1]$. Jensen's inequality for the probability measure $\pi/\bar{Q}$ and the convex function φ gives $\int \varphi \, d\pi \ge \bar{Q}\varphi(\bar{m})$. If $\bar{m} \ge \theta$, then $\varphi(\bar{m}) \ge \varphi(\theta)$ because φ is nondecreasing. If $\bar{m} < \theta$, the hypothesis gives $0 < \theta - \bar{m} \le \Delta/\bar{Q}$, and the Lipschitz property gives $\varphi(\bar{m}) \ge \varphi(\theta) - \Lambda\Delta/\bar{Q}$. In both cases $\bar{Q}\varphi(\bar{m}) \ge \bar{Q}\varphi(\theta) - \Lambda\Delta$, which is (b). Part (a) is (b) applied to $\pi := \sum_{r \le q} \delta_{m_r}$, of mass q, when $q \ge 1$; for $q = 0$ both sides of (a) reduce to $0 \ge -\Lambda\Delta$. □

Convexity enters the proof of Theorem 3.2 only through this lemma, and in the favourable direction: the cost of an average target is at most the average of the costs. As a remark without proof, a concave cost reverses the inequality, and the resulting loss is of the order of the tenure times the gap between φ and its convex envelope, which is linear in n.

F.2 Monitoring predictable targets

The monitor of Stage 2 compares the winner's cumulative reward Z_r with $r\tau - b_n(r)$, where $b_n(r) = \sqrt{r \log(2n^4)/2}$. The next lemma is the only property of the monitor that the proofs use. It is stated conditionally on a σ-field $\mathcal{G}_0$ that is determined before the first primary reward draw and with respect to which the price τ is measurable; in the proofs $\mathcal{G}_0$ is the Stage-1 history, possibly enlarged by Stage-1 reward seeds.

Lemma F.2 (Anytime primary monitor). *Condition on $\mathcal{G}_0$. For $r \ge 1$ let $\mathcal{G}_r$ be generated by $\mathcal{G}_0$, the primary targets $m_1, \dots, m_{r+1}$, the private randomness used to choose them, and the primary rewards $Y_1, \dots, Y_r$; the winner's rth primary target $m_r \in [0, M_i]$ is thus $\mathcal{G}_{r-1}$-measurable, and it may depend on the winner's whole available history. Put $A_r := \sum_{q \le r} m_q$, $Z_r := \sum_{q \le r} Y_q$, $A_0 = Z_0 := 0$ and $b_n(0) := 0$. Then:*

(1) $\Pr\{\exists r \le n : |Z_r - A_r| > b_n(r) \mid \mathcal{G}_0\} \le n^{-3}$.
(2) *If the primary phase lasts q pulls, the pull that triggers the switch included when there is one, then on the complement of the event in (1),*

$$A_q \ge q\tau - 1 - 2b_n(q) \quad \text{and} \quad U_i^{\text{pri}} \le q\big(1 + \varphi_i(\mu_i) - \varphi_i(\tau)\big) + \Lambda\big(1 + 2b_n(q)\big).$$

(3) *If every primary target satisfies $m_r \ge \tau$, the probability of a switch is at most $(2n^3)^{-1}$.*

Proof. By Assumption (A3), $\mathbb{E}[Y_r \mid \mathcal{G}_{r-1}] = m_r$, so the differences $D_r := Y_r - m_r$ are martingale differences with respect to $(\mathcal{G}_r)$, and each D_r lies in the interval $[-m_r, 1 - m_r]$ of length one whose endpoints are $\mathcal{G}_{r-1}$-measurable. The Azuma-Hoeffding inequality for such differences gives, for every fixed $r \ge 1$ and $b > 0$, $\Pr\{Z_r - A_r > b \mid \mathcal{G}_0\} \le \exp(-2b^2/r)$ and the same bound for the lower tail. At $b = b_n(r)$ each tail is at most $\exp(-\log(2n^4)) = (2n^4)^{-1}$. A union bound over the two tails and over $r = 1, \dots, n$ gives $2n \cdot (2n^4)^{-1} = n^{-3}$, which is (1); the case $r = 0$ is trivial. The primary phase has at most $N_n \le n$ pulls, so (1) covers every primary count.

For (2), work on the complement of the event in (1), so that $|Z_r - A_r| \le b_n(r)$ for all $r \le n$. If no switch occurs during the q primary pulls, the test did not trigger at $r = q$, so $Z_q \ge q\tau - b_n(q)$, and $Z_q \le A_q + b_n(q)$; together $A_q \ge q\tau - 2b_n(q)$, which implies the first display. If the switch occurs at pull q, the test did not trigger at $q-1$, so the same two inequalities at $q-1$ give $A_{q-1} \ge (q-1)\tau - 2b_n(q-1)$; since $m_q \ge 0$, $\tau \le 1$ and b_n is nondecreasing, $A_q = A_{q-1} + m_q \ge q\tau - 1 - 2b_n(q)$. For the cost bound, $U_i^{\mathrm{pri}} = q(1+\varphi_i(\mu_i)) - \sum_{r\le q}\varphi_i(m_r)$, and Lemma F.1(a) with $\theta = \tau$ and $\Delta = 1 + 2b_n(q)$ gives $\sum_{r\le q}\varphi_i(m_r) \ge q\varphi_i(\tau) - \Lambda(1+2b_n(q))$.

For (3), if $m_r \ge \tau$ for every r then $A_r \ge r\tau$, so a switch at pull r means $Z_r < r\tau - b_n(r) \le A_r - b_n(r)$, a lower-tail deviation at r. The n lower tails have total probability at most $n\cdot(2n^4)^{-1} = (2n^3)^{-1}$. □

F.3 A selected-reactive EXP3 lemma

At a fallback start let $\mathcal{F}_{\mathrm{fb}}$ be the σ-field generated by the full preceding history, Stage-1 seeds included; the remaining horizon H is $\mathcal{F}_{\mathrm{fb}}$-measurable. Conditional on $\mathcal{F}_{\mathrm{fb}}$, the fallback runs EXP3 with fresh weights $w_{i,1} := 1$ for all i. For $H \ge 1$, at local time $t = 1, \dots, H$, define

$$\gamma := \min\left\{\frac{1}{2}, \sqrt{\frac{K\log K}{H}}\right\}, \qquad \eta := \frac{\gamma}{K},$$
$$q_{i,t} := \frac{w_{i,t}}{\sum_k w_{k,t}}, \qquad p_{i,t} := (1-\gamma)q_{i,t} + \frac{\gamma}{K}.$$

The mechanism draws $I_t \sim p_t$; the selected agent then chooses its target and Y_t is observed; finally

$$\widehat{Y}_{i,t} := \frac{Y_t \mathbf{1}\{I_t = i\}}{p_{i,t}}, \qquad w_{i,t+1} := w_{i,t}\exp(\eta\widehat{Y}_{i,t}).$$

Let $\mathcal{F}_{t-1}$ be the σ-field generated by $\mathcal{F}_{\mathrm{fb}}$, the fallback selections and rewards before local time t, and the private randomness the agents have used so far; the weights $w_{\cdot,t}$ and the law p_t are $\mathcal{F}_{t-1}$-measurable. For each agent i write $r_{i,t} := \mathbb{E}[Y_t \mid \mathcal{F}_{t-1}, I_t = i] \in [0,1]$ for the conditional mean of the target that i would deliver if selected at time t.

Lemma F.3 (Selected-reactive EXP3)**.** *Suppose an $\mathcal{F}_{\mathrm{fb}}$-measurable comparator j satisfies $r_{j,t} \ge v$ for every $t \le H$. The other agents may choose arbitrary feasible targets after being selected, using any information available to them. Then*

$$\mathbb{E}\left[\sum_{t=1}^{H} Y_t \,\middle|\, \mathcal{F}_{\mathrm{fb}}\right] \ge vH - R_K(H).$$

Proof. If $H = 0$ both sides are 0. Let $H \ge 1$ and condition on $\mathcal{F}_{\mathrm{fb}}$ throughout. Only $r_{j,t} \ge v$ is used about the comparator; no reward of an unselected agent is ever needed.

The potential. Put $W_t := \sum_i w_{i,t}$, so $W_1 = K$. Since $p_{i,t} \ge \gamma/K = \eta$ and $Y_t \le 1$, we have $0 \le \eta\widehat{Y}_{i,t} \le 1$, and $e^x \le 1 + x + x^2$ on $[0,1]$ gives

$$\frac{W_{t+1}}{W_t} = \sum_i q_{i,t} e^{\eta\widehat{Y}_{i,t}} \le 1 + \eta\sum_i q_{i,t}\widehat{Y}_{i,t} + \eta^2\sum_i q_{i,t}\widehat{Y}_{i,t}^2.$$

Taking logarithms, using $\log(1+u) \le u$, and summing over $t \le H$,

$$\log\frac{W_{H+1}}{W_1} \le \eta\sum_{t=1}^{H}\sum_i q_{i,t}\widehat{Y}_{i,t} + \eta^2\sum_{t=1}^{H}\sum_i q_{i,t}\widehat{Y}_{i,t}^2.$$

On the other hand $W_{H+1} \ge w_{j,H+1} = \exp(\eta\sum_{t\le H}\widehat{Y}_{j,t})$, so $\log(W_{H+1}/W_1) \ge \eta\sum_{t\le H}\widehat{Y}_{j,t} - \log K$. Combining the two bounds and dividing by η,

$$\sum_{t=1}^{H}\widehat{Y}_{j,t} - \frac{\log K}{\eta} \le \sum_{t=1}^{H}\sum_i q_{i,t}\widehat{Y}_{i,t} + \eta\sum_{t=1}^{H}\sum_i q_{i,t}\widehat{Y}_{i,t}^2. \tag{3}$$

This holds on every sample path.

Conditional expectations. By Assumption (A3), I_t has law p_t conditionally on $\mathcal{F}_{t-1}$, and $Y_t^2 = Y_t$. Hence, conditionally on $\mathcal{F}_{t-1}$,

$$\mathbb{E}\widehat{Y}_{j,t} = p_{j,t}\cdot\frac{r_{j,t}}{p_{j,t}} = r_{j,t} \ge v,$$
$$\mathbb{E}\sum_i q_{i,t}\widehat{Y}_{i,t} = \sum_i q_{i,t}r_{i,t} =: Q_t,$$
$$\mathbb{E}\sum_i q_{i,t}\widehat{Y}_{i,t}^2 = \sum_i \frac{q_{i,t}r_{i,t}}{p_{i,t}} \le \frac{K}{1-\gamma},$$

the last step because $p_{i,t} \ge (1-\gamma)q_{i,t}$ and $r_{i,t} \le 1$. The conditional mean of the realised reward is $G_t := \mathbb{E}[Y_t \mid \mathcal{F}_{t-1}] = \sum_i p_{i,t}r_{i,t} = (1-\gamma)Q_t + (\gamma/K)\sum_i r_{i,t} \ge (1-\gamma)Q_t$.

Assembling. Take conditional expectations given $\mathcal{F}_{\mathrm{fb}}$ in (3), using the tower property for each t:

$$vH - \frac{K\log K}{\gamma} \le \mathbb{E}\sum_{t=1}^{H} Q_t + \frac{\gamma H}{1-\gamma}.$$

Multiply by $1-\gamma$ and use $(1-\gamma)Q_t \le G_t$ and $\mathbb{E}\sum_t G_t = \mathbb{E}\sum_t Y_t$:

$$\mathbb{E}\sum_{t=1}^{H} Y_t \ge (1-\gamma)vH - (1-\gamma)\frac{K\log K}{\gamma} - \gamma H \ge vH - 2\gamma H - \frac{K\log K}{\gamma},$$

where $v \le 1$ was used. If $H \ge 4K\log K$, then $\gamma = \sqrt{K\log K/H} \le 1/2$ and the loss $2\gamma H + K\log K/\gamma$ equals $3\sqrt{KH\log K}$. If $H < 4K\log K$, then $\gamma = 1/2$ and the loss is $H + 2K\log K \le 3\sqrt{KH\log K} + 2K\log K$, because $H < 4K\log K \le 9K\log K$ gives $H \le 3\sqrt{KH\log K}$. In both cases the loss is at most $R_K(H)$.

The bound holds for every value of $\mathcal{F}_{\mathrm{fb}}$, and H is $\mathcal{F}_{\mathrm{fb}}$-measurable, so integrating over the fallback-start history gives the unconditional statement for the random start without any optional-stopping argument. □

F.4 Bidding coupling and fallback utility

Coupling the bids. Fix a unilateral deviation $S_i' \in \mathcal{B}_{i,n}$ by agent i and condition on its private coins, so that its targets are functions of its history. Realise the Stage-1 rewards of every agent through uniform seeds $U_{k,r}$, indexed by the agent k and its rth Stage-1 pull, as $Y_{k,r} = \mathbf{1}\{U_{k,r} \le m_{k,r}\}$. Write $m'_{i,r}$ for the deviator's target on its rth Stage-1 pull and $\beta_i' := L_n^{-1}\sum_{r\le L_n} Y'_{i,r}$ for its bid. Couple each deviating reward with the reward of the capacity target on the same seed:

$$Y'_{i,r} = \mathbf{1}\{U_{i,r} \le m'_{i,r}\} \le \mathbf{1}\{U_{i,r} \le M_i\} =: Y_{i,r}^{\max},$$

which holds pathwise because feasibility gives $m'_{i,r} \le M_i$, whatever history $m'_{i,r}$ depends on. Define $\beta_i^{\max} := L_n^{-1} \sum_{r \le L_n} Y_{i,r}^{\max}$; then $\beta'_i \le \beta_i^{\max}$ on every sample path. Under $S^\star$ every agent $j \ne i$ delivers the constant M_j on each Stage-1 pull, and Stage 1 runs on a fixed schedule, so the bids β_j, $j \ne i$, are the same random variables in the deviating execution and in the execution in which i bids at capacity. With the common priority, every deviating win is also a capacity-bidding win: if β'_i is the strict maximum then so is $\beta_i^{\max} \ge \beta'_i$; if β'_i ties for the maximum and the priority selects i, then either $\beta_i^{\max}$ is the strict maximum or $\beta_i^{\max} = \beta'_i$ and the tie set is the same, so the same priority selects i. On $\{i \text{ wins}\}$ the second order statistic counting ties is $\max_{j \ne i} \beta_j$, so the announced price is

$$\tau_i := \Big(\max_{j \ne i} \beta_j - s_n\Big)_+,$$

a function of the other agents' Stage-1 rewards alone, identical in both executions.

Good bidding events. Define

$$\mathcal{G}_{-i} := \{|\beta_j - M_j| \le s_n \text{ for all } j \ne i\}, \qquad \mathcal{G}_i := \mathcal{G}_{-i} \cap \{\beta_i^{\max} \le M_i + s_n\}.$$

Both are functions of the Stage-1 seeds alone and do not depend on the deviation. Each β_j, $j \ne i$, is an average of L_n independent Bernoulli(M_j) variables, and $\beta_i^{\max}$ of L_n independent Bernoulli(M_i) variables, so Hoeffding's inequality bounds each of the $2(K-1)$ one-sided tails of $\mathcal{G}_{-i}$ and the single upper tail of $\beta_i^{\max}$ by $\exp(-2L_n s_n^2) = (2Kn^4)^{-1}$; hence $\Pr(\mathcal{G}_i^c) \le (2K-1)(2Kn^4)^{-1} \le n^{-4}$. On $\mathcal{G}_{-i}$, every β_j with $j \ne i$ lies in $[M_j - s_n, M_j + s_n]$, so $\max_{j \ne i} \beta_j \in [V_i - s_n, V_i + s_n]$ and

$$(V_i - 2s_n)_+ \le \tau_i \le V_i. \tag{4}$$

On $\mathcal{G}_i \cap \{i \text{ wins}\}$, moreover, $\max_{j \ne i} \beta_j \le \beta'_i \le \beta_i^{\max} \le M_i + s_n$, so $\tau_i \le M_i$ and

$$\tau_i \le \min\{V_i, M_i\}. \tag{5}$$

Only the upper tail of the capacity bid $\beta_i^{\max}$ is used. The realised bid β'_i is not concentrated around anything, since the deviator chooses its targets; it enters only through the pathwise domination $\beta'_i \le \beta_i^{\max}$.

Fallback utility of the deviator. Suppose fallback begins with $H \ge 1$ rounds remaining, and condition on $\mathcal{F}_{\text{fb}}$. Let $Q_i := \sum_{t=1}^{H} \mathbf{1}\{I_t = i\}$ be the deviator's number of fallback pulls and put

$$\bar{Q}_i := \mathbb{E}[Q_i \mid \mathcal{F}_{\text{fb}}],$$
$$\bar{A}_i := \mathbb{E}\Big[\sum_{t \le H : I_t = i} m_{i,t} \,\Big|\, \mathcal{F}_{\text{fb}}\Big],$$
$$\pi(B) := \mathbb{E}\Big[\sum_{t \le H : I_t = i} \mathbf{1}\{m_{i,t} \in B\} \,\Big|\, \mathcal{F}_{\text{fb}}\Big]$$

for Borel $B \subseteq [0,1]$; π is the expected occupation measure of the deviator's fallback targets, of total mass $\bar{Q}_i$ and first moment $\int m \, \mathrm{d}\pi = \bar{A}_i$. Every other agent follows $S^\star$ and delivers its capacity in fallback. An agent $j^\star \ne i$ with $M_{j^\star} = V_i$ is therefore a comparator with $r_{j^\star,t} = V_i$ for all t, and every other agent's conditional mean is at most V_i. Lemma F.3 with $v = V_i$, together with the upper bound $\mathbb{E}[\sum_{t \le H} Y_t \mid \mathcal{F}_{\text{fb}}] \le \bar{A}_i + V_i(H - \bar{Q}_i)$, gives

$$V_i H - R_K(H) \le \bar{A}_i + V_i(H - \bar{Q}_i), \qquad \text{that is,} \qquad \bar{A}_i \ge V_i \bar{Q}_i - R_K(H). \tag{6}$$

If $\bar{Q}_i = 0$ the deviator's fallback utility is 0 almost surely and the bound below is trivial. Otherwise, since $\mathbb{E}[U_i^{\text{fb}} \mid \mathcal{F}_{\text{fb}}] = (1 + \varphi_i(\mu_i))\bar{Q}_i - \int \varphi_i \, \mathrm{d}\pi$, Lemma F.1(b) with $\theta = V_i$ and $\Delta = R_K(H)$ gives $\mathbb{E}[U_i^{\text{fb}} \mid \mathcal{F}_{\text{fb}}] \le (1 + \varphi_i(\mu_i) - \varphi_i(V_i))\bar{Q}_i + \Lambda R_K(H)$. The coefficient of $\bar{Q}_i$ is nonnegative, because $\varphi_i(V_i) \le \varphi_i(1) \le \varphi_i(0) + 1 \le \varphi_i(\mu_i) + 1$ by monotonicity and (1), so $\bar{Q}_i \le H$ may be replaced by H:

$$\mathbb{E}[U_i^{\text{fb}} \mid \mathcal{F}_{\text{fb}}] \le (1 + \varphi_i(\mu_i) - \varphi_i(V_i))H + \Lambda R_K(H). \tag{7}$$

For $H = 0$ both sides are 0.

F.5 Behavioral equilibrium

Equilibrium part of Theorem 3.2. Fix agent i and a deviation $S'_i \in \mathcal{B}_{i,n}$, and condition on i's private coins. Compare the deviating execution with the baseline execution, in which i follows $S_i^\star$, on the coupled space of the previous subsection: the two executions share the Stage-1 seeds and the priority, and their post-bid randomness is fresh. Let $\mathcal{F}_{\text{bid}}$ be the σ-field generated by the Stage-1 seeds of both executions and the priority. The event {the deviation wins} is $\mathcal{F}_{\text{bid}}$-measurable, and $\mathcal{G}_i \in \mathcal{F}_{\text{bid}}$. We bound the gain separately on the two pieces {the deviation loses} and {the deviation wins}. On the first piece the baseline may win or lose, and only the sign of its post-bid utility is used; on the second piece the baseline wins as well, at the same price τ_i. All conditional probabilities below are conditional on $\mathcal{F}_{\text{bid}}$; Lemma F.2 applies with $\mathcal{G}_0 = \mathcal{F}_{\text{bid}}$, and (7) is unaffected because the fallback coins and all post-bid seeds are fresh.

The deviation loses. Let $W \ne i$ be the winner. On $\mathcal{G}_{-i}$, $\tau = (\beta_{(2)} - s_n)_+ \le (\beta_W - s_n)_+ \le M_W$, so W's primary target $\min\{\tau, M_W\}$, or $\max\{\mu_W, \min\{\tau, M_W\}\}$ if W is honest, is at least τ, and Lemma F.2(3) bounds the probability of a switch by $(2n^3)^{-1}$. The deviator is selected after Stage 1 only in fallback. Its post-bid utility is therefore 0 unless $\mathcal{G}_{-i}^c$ occurs or a switch occurs, and it is at most $2n$ in any case, so its expected post-bid utility is at most $2n\big(n^{-4} + (2n^3)^{-1}\big)$. Its Stage-1 utility is at most $L_n(1 + \varphi_i(\mu_i) - \varphi_i(0))$, while the baseline's Stage-1 utility is $L_n(1 + \varphi_i(\mu_i) - \varphi_i(M_i))$ and its post-bid utility is nonnegative. On this piece the gain is at most $L_n(\varphi_i(M_i) - \varphi_i(0)) + 2n(n^{-4} + (2n^3)^{-1}) \le L_n + 2n^{-2}$, using (1) and $n \ge 4$.

The deviation wins. Let q be the number of primary pulls and $H := N_n - q$ the fallback length. On $\mathcal{G}_i$, the price satisfies $\tau_i \le \min\{V_i, M_i\}$ by (5), so $\varphi_i(\tau_i) \le \varphi_i(V_i)$ and (7) gives $\mathbb{E}[U_i^{\text{fb}} \mid \mathcal{F}_{\text{fb}}] \le (1 + \varphi_i(\mu_i) - \varphi_i(\tau_i))H + \Lambda R_K(H)$. Let $\mathcal{E}_{\text{mon}}$ be the complement of the event in Lemma F.2(1) at $\tau = \tau_i$; it has probability at least $1 - n^{-3}$, and its restriction to the primary pulls actually made is $\mathcal{F}_{\text{fb}}$-measurable, so the pathwise primary bound and the conditional fallback bound can be added. On $\mathcal{G}_i \cap \mathcal{E}_{\text{mon}}$, Lemma F.2(2) with $\tau = \tau_i$ and the fallback bound give an expected post-bid utility of at most

$$(1 + \varphi_i(\mu_i) - \varphi_i(\tau_i))(q + H) + \Lambda\big(1 + 2b_n(q) + R_K(H)\big)$$
$$\le (1 + \varphi_i(\mu_i) - \varphi_i(\tau_i))N_n + \Lambda\big(1 + 2b_n(n) + R_K(n)\big),$$

since $q + H = N_n$ and b_n, R_K are nondecreasing. Off $\mathcal{G}_i \cap \mathcal{E}_{\text{mon}}$ the post-bid utility is at most $2n$.

In the baseline on this piece, i wins at the same price τ_i. If i is strategic, its primary target is $\min\{\tau_i, M_i\} = \tau_i$ on $\mathcal{G}_i$; by Lemma F.2(3) a switch has probability at most $(2n^3)^{-1}$, and otherwise the primary phase lasts all N_n pulls and yields exactly

$N_n(1+\varphi_i(\mu_i)-\varphi_i(\tau_i))$. Since post-bid utility is nonnegative, the baseline post-bid utility is at least $N_n(1+\varphi_i(\mu_i)-\varphi_i(\tau_i))-2N_n(2n^3)^{-1}$. If i is honest and $\tau_i > \mu_i$, its target is again τ_i and the same bound holds. If i is honest and $\tau_i \le \mu_i$, its target is $\mu_i \ge \tau_i$, so it earns 1 per primary pull and a switch again has probability at most $(2n^3)^{-1}$; every honest deviation delivers $m \ge \mu_i$ and so earns $1+\varphi_i(\mu_i)-\varphi_i(m) \le 1$ per selected pull, with at most N_n post-bid selections, so its post-bid utility is at most $N_n \le$ baseline $+ N_n(2n^3)^{-1}$. In every case the post-bid gain on $\mathcal{G}_i \cap \mathcal{E}_{\text{mon}}$ is at most $\Lambda(1+2b_n(n)+R_K(n))+2N_n(2n^3)^{-1}$, and the Stage-1 gain is at most $L_n(\varphi_i(M_i)-\varphi_i(0)) \le L_n$.

Total. Per-pull utility lies in $[0,2]$, so the bad events $\mathcal{G}_i^c$, $\mathcal{E}_{\text{mon}}^c$ and the baseline's false switch cost at most $2n\big(n^{-4}+n^{-3}+(2n^3)^{-1}\big) \le 5$. Integrating the conditional bounds over $\mathcal{F}_{\text{bid}}$ and over the deviator's private coins gives, for every agent i, top or not,

$$\sup_{S'_{i,n}\in\mathcal{B}_{i,n}} \big[u_i(n;S'_{i,n},S^\star_{-i,n})-u_i(n;S^\star_n)\big] \le L_n+\Lambda\big(1+2b_n(n)+R_K(n)\big)+5.$$

No capacity gap and no other instance quantity enters, and the only cost quantity is the Lipschitz constant Λ. Since $L_n \le n^{2/3}(\log(en))^{1/3}+1$, $b_n(n) = \sqrt{n\log(2n^4)/2}$ and $R_K(n) = 3\sqrt{Kn\log K}+2K\log K$, the right side is $O_{K,\Lambda}(n^{2/3}(\log n)^{1/3})$, which is part (i). The price slack s_n enters the revenue bound below but not the deviation gain, because both coupled executions face the same τ_i. □

F.6 Equilibrium revenue

Revenue part of Theorem 3.2. At $S^\star$ every agent delivers its capacity in Stage 1, so the event $\mathcal{G} := \{\max_i|\beta_i - M_i| \le s_n\}$ has probability at least $1-2K(2Kn^4)^{-1} = 1-n^{-4}$ by Hoeffding's inequality. The second order statistic of a vector is 1-Lipschitz in the supremum norm, so on $\mathcal{G}$ we have $|\beta_{(2)} - M_{(2)}| \le s_n$ and hence

$$(M_{(2)}-2s_n)_+ \le \tau \le M_{(2)}, \qquad \tau \le (\beta_W - s_n)_+ \le M_W,$$

the second because $\beta_{(2)} \le \beta_W \le M_W + s_n$. Hence, on $\mathcal{G}$, the winner's primary target $\min\{\tau, M_W\}$, or $\max\{\mu_W, \min\{\tau, M_W\}\}$ if honest, is at least τ on every primary pull.

Let $\mathcal{F}_{\text{bid}}$ be the Stage-1 history, q the number of primary pulls, and m_r the winner's target on primary pull r. Conditional on $\mathcal{F}_{\text{bid}}$, the differences $Y_r - m_r$ are bounded martingale differences for the primary filtration, and q is a bounded stopping time for it, because the switch decision after pull r is a function of $Z_1,\dots,Z_r$ and $q \le N_n$. The optional stopping theorem gives $\mathbb{E}[\sum_{r\le q}(Y_r - m_r) \mid \mathcal{F}_{\text{bid}}] = 0$, so on $\mathcal{G}$ the expected primary revenue is $\mathbb{E}[A_q \mid \mathcal{F}_{\text{bid}}] \ge \tau\mathbb{E}[q \mid \mathcal{F}_{\text{bid}}]$. In fallback every agent delivers its capacity, so an agent j with $M_j = \max_k M_k \ge M_{(2)} \ge \tau$ is a comparator with $r_{j,t} = M_j$, and Lemma F.3 gives expected fallback revenue at least $\tau H - R_K(H)$ conditionally on every fallback-start history. Since $q+H = N_n$ and R_K is nondecreasing, on $\mathcal{G}$ the expected post-bid revenue conditional on $\mathcal{F}_{\text{bid}}$ is at least $\tau N_n - R_K(n) \ge (M_{(2)}-2s_n)N_n - R_K(n)$. Revenue is nonnegative on $\mathcal{G}^c$ and in Stage 1, and $N_n \Pr(\mathcal{G}^c) \le n^{-3} \le 1$. Therefore

$$\begin{aligned} P(n;S^\star) &\ge (M_{(2)}-2s_n)N_n - R_K(n) - 1 \\ &\ge M_{(2)}n - \big(KL_n + 2ns_n + R_K(n) + 1\big), \end{aligned}$$

using $N_n = n - KL_n$ and $M_{(2)} \le 1$. For fixed K the remainder is $O(n^{2/3}(\log n)^{1/3})$, since $ns_n = n\sqrt{\log(2Kn^4)/(2L_n)}$ is of that order when $L_n = \lceil n^{2/3}(\log(en))^{1/3}\rceil$ and is $O(\sqrt{Kn\log(2Kn^4)})$ at the finitely many horizons where $L_n = \lfloor n/(2K)\rfloor$. This is part (ii); capacity ties are included, since nothing above distinguishes the agents attaining $M_{(2)}$. □

F.7 Honest-benchmark robustness

Robustness part of Theorem 3.2. Fix any feasible behavioural profile S and an honest agent h with $\mu_h = \mu_H^\star$. Honesty means $m_{h,r} \ge \mu_H^\star$ on every pull after every history. On h's Stage-1 pulls the differences $Y_{h,r} - m_{h,r}$ are martingale differences in a predictable interval of length one, so the Azuma-Hoeffding inequality gives

$$\begin{aligned} \Pr\{\beta_h < \mu_H^\star - s_n\} &\le \Pr\Big\{\sum_{r\le L_n}(Y_{h,r}-m_{h,r}) < -L_n s_n\Big\} \\ &\le \exp(-2L_n s_n^2) = (2Kn^4)^{-1}. \end{aligned}$$

Call $\mathcal{G}_h := \{\beta_h \ge \mu_H^\star - s_n\}$ the good event and condition on the Stage-1 history $\mathcal{F}_{\text{bid}}$.

Primary phase. Let q be the number of primary pulls and A_q the winner's primary target sum. As in part (ii), q is a bounded stopping time and the expected primary revenue conditional on $\mathcal{F}_{\text{bid}}$ equals $\mathbb{E}[A_q \mid \mathcal{F}_{\text{bid}}]$. If $W = h$, honesty gives $A_q \ge q\mu_H^\star$. If $W \ne h$, then h is not the winner, so $\beta_{(2)} = \max_{j\ne W}\beta_j \ge \beta_h$, and on $\mathcal{G}_h$ the price satisfies $\tau \ge (\mu_H^\star - 2s_n)_+$; the winner's targets are arbitrary predictable targets, and Lemma F.2(2), applied with $\mathcal{G}_0 = \mathcal{F}_{\text{bid}}$, gives $A_q \ge q\tau - 1 - 2b_n(q) \ge q(\mu_H^\star - 2s_n) - 1 - 2b_n(n)$ off an event of probability at most n^{-3}, on which still $A_q \ge 0 \ge q(\mu_H^\star - 2s_n) - n$. In both cases $\mathbb{E}[A_q \mid \mathcal{F}_{\text{bid}}] \ge (\mu_H^\star - 2s_n)\mathbb{E}[q \mid \mathcal{F}_{\text{bid}}] - 1 - 2b_n(n) - n^{-2}$ on $\mathcal{G}_h$.

Fallback. Conditional on any fallback-start history, h is a comparator with $r_{h,t} \ge \mu_H^\star$ for every t, by honesty, so Lemma F.3 gives expected fallback revenue at least $\mu_H^\star H - R_K(H) \ge (\mu_H^\star - 2s_n)H - R_K(n)$.

Total. With $q+H = N_n$, on $\mathcal{G}_h$ the expected post-bid revenue conditional on $\mathcal{F}_{\text{bid}}$ is at least $(\mu_H^\star - 2s_n)N_n - 1 - 2b_n(n) - R_K(n) - n^{-2}$. Revenue is nonnegative on $\mathcal{G}_h^c$, whose probability is at most $(2Kn^4)^{-1}$, and in Stage 1. Hence

$$P(n;S) \ge \mu_H^\star n - \big(KL_n + 2ns_n + 2b_n(n) + R_K(n) + 2\big),$$

uniformly over feasible joint behavioural profiles, which is part (iii) with the stated rate. □

The fallback is essential for this robustness statement: after a statistically triggered switch it keeps every honest agent available as a comparator, even when that agent changes its target across phases. The result does not imply high revenue at every equilibrium; for example, a price-setting agent may lower off-equilibrium revenue without obtaining a linear utility gain (Appendix I).

F.8 The bidding term is attained

Proposition F.4 (Stage-1 underbidding on a unique-top instance)**.** *Fix $K \ge 2$ and an instance with a unique top agent i, so $M_i > V_i$, with $\mu_i < M_i$ if i is honest. Let m' be $(M_i+V_i)/2$, or $\max\{(M_i+V_i)/2, \mu_i\}$ if i is honest, and let $S'_{i,n}$ deliver m' on every Stage-1 pull and follow $S_i^\star$ afterwards. For every n with $4s_n < M_i - V_i$,*

$$u_i(n;S'_{i,n},S^\star_{-i,n}) - u_i(n;S^\star_n) \ge L_n(\varphi_i(M_i)-\varphi_i(m')) - 4n^{-3},$$

which is $\Theta(L_n)$ provided φ_i is strictly increasing on $[m', M_i]$. Linearity supplies that for free; a cost that is flat on $[m', M_i]$ makes the deviation save nothing, and then the elicitation term is not attained.

Proof. The target m' is feasible: $m' < M_i$ because $(M_i + V_i)/2 < M_i$ and $\mu_i < M_i$, and $m' \geq \mu_i$ when i is honest. Couple the baseline and the deviating execution through the same reward uniforms, indexed by agent and local pull index, the same priority, and the same EXP3 coins. In both executions every agent's Stage-1 target is a constant (M_j for $j \neq i$; M_i in the baseline and m' in the deviation), so the Stage-1 utility difference is the deterministic quantity $L_n(\varphi_i(M_i) - \varphi_i(m'))$ on every sample point. Let E be the event that in both executions every bid lies within s_n of its Stage-1 target; Hoeffding's inequality and a union bound over at most $2K$ agent-execution pairs with two tails each give $\Pr(E^c) \leq 2n^{-4}$. On E, the baseline bid satisfies $\beta_i \geq M_i - s_n > V_i + s_n \geq \max_{j \neq i} \beta_j$, and the deviating bid satisfies $\beta_i' \geq m' - s_n \geq (M_i + V_i)/2 - s_n > V_i + s_n$, both because $4s_n < M_i - V_i$; so i wins strictly in both executions. The other agents' bids coincide in the two executions, so the price does too. After Stage 1 the deviation follows $S_i^\star$, whose targets do not depend on Stage-1 rewards, the other agents' post-bid play is deterministic given the coupled randomness, and the same uniforms and coins are used, so the post-bid histories coincide on E. Writing the gain as $L_n(\varphi_i(M_i) - \varphi_i(m')) + D$, where D is the post-bid utility difference, we have $D = 0$ on E and $D \geq -2n$ off E, so $\mathbb{E}D \geq -2n \cdot 2n^{-4} = -4n^{-3}$. The upper bound of Theorem 3.2 shows the gain is $O(L_n)$, hence $\Theta(L_n)$ when $\varphi_i(M_i) > \varphi_i(m')$. As remarks: by convexity, no randomisation of the Stage-1 target with mean m' saves more than this constant target, and by (1) no Stage-1 deviation saves more than L_n. □

Hence the L_n term, and with it the order $n^{2/3}(\log n)^{1/3}$ of Theorem 3.2, is attained by MESP as specified, up to the instance factor $\varphi_i(M_i) - \varphi_i(m')$. The order is preserved under a general cost, but the matching constant is not: the upper bound of Theorem 3.2 charges a shading of ϵ at most $\Lambda\epsilon$, while the deviation here collects at least $\lambda_i \epsilon$, with λ_i a lower bound on the slope of φ_i on the relevant range, so the two match only up to the condition number Λ/λ_i, which is 1 in the linear model.

F.9 Contextual results for public capabilities

The three results below assume every agent's capability is public, so that all-maximum play is available to everyone, and assume i.i.d. arrivals. They are independent of every result in Section 4, which uses neither hypothesis.

They are also stated in the more general setting, in which each agent i carries its own context $x_{i,t} \in [0,1]^d$, a round is described by the profile $X_t = (x_{1,t}, \ldots, x_{K,t}) \in [0,1]^D$ with $D := Kd$, and the grid of Section 4.1 is laid on $[0,1]^D$ rather than on the space of tasks. Throughout this subsection $\mathcal{X}_F$ accordingly denotes a finite set of such profiles, not the finite set of common contexts used in the body. Taking every agent's context equal recovers the common-context model with $D = d$.

Notation for profiles and strategy classes. Profiles X_t are i.i.d. with a Borel law ν on $[0,1]^D$, independent of the reward seeds and of the mechanism's coins. For a profile $X = (x_1, \ldots, x_K)$ write $M_\star(X) := \max_j M_j(x_j)$ for the largest capacity, $\mathcal{T}(X) := \{j : M_j(x_j) = M_\star(X)\}$ for the set of top agents, and $b(X) := \max_{h \in \mathcal{K}_H} \mu_h(x_h)$ for the best honest intrinsic mean; the honest set is nonempty. For a measurable region $R \subseteq [0,1]^D$ and an agent i, $T_i(R;n) := \sum_{t \leq n} \mathbf{1}\{X_t \in R,\ I_t = i\}$ counts i's selections at profiles in R, V_C counts the visits to a cell C, and $G_{ij}(R) := \{X \in R : M_i(x_i) = M_j(x_j) = M_\star(X)\}$ is the part of R on which i and j share the top; $(\partial R)^\epsilon$ is the closed ϵ-neighbourhood of the boundary of R. The strategy classes are: $\mathcal{S}_{\mathrm{HI}}$, deterministic horizon-independent fixed targets, as in Section C; $\mathcal{S}_{\mathrm{HDS}}$, deterministic maps $X \mapsto m_i(X) \in [0, M_i(x_i)]$ fixed before play, which may depend on n, with $m_i(X) \geq \mu_i(x_i)$ for an honest i; and $\mathcal{S}_F$, the horizon-independent deterministic maps on a finite profile set. An independent ex ante mixture draws the map from a distribution before play, independently of everything else. The profile $S^{\max}$ is the one in which every agent delivers its capability at every profile.

The headline rule. Run an independent cell-local UCB1 instance in each cell, at the known-horizon width $\sqrt{4\log n / T_a}$ with the agent's local count in that cell, on a grid of side $h_n \asymp (\log(en)/n)^{1/(D+2)}$. Write

$$E_n := L\sqrt{d}nh_n + 4\sqrt{J_n n \log n} + 3\min\{J_n, n\} = O\left(n^{\frac{D+1}{D+2}} (\log n)^{\frac{1}{D+2}}\right)$$

and let E_n^K be the same expression with $8\sqrt{KJ_n n \log n} + 3K\min\{J_n, n\}$ in place of its last two terms.

Theorem F.5 (Static-grid UCB1 is contextual SAMF)**.** *Uniformly over $\mathcal{S}_{\mathrm{HDS}}$ and its independent ex ante mixtures, and for every $n \geq 2$: contextual sharp adaptivity holds for every measurable R with remainder $E_n + n\nu((\partial R)^{\sqrt{D}h_n})$; robustness holds with $P(n) \geq \mathbb{E}\sum_{t=1}^n b(X_t) - E_n^K$; contextual monotonicity is exact, with remainder zero, for every measurable R, every n, and every pointwise raise of one agent's map, with no regularity and no boundary condition; and expected contextual FATA holds on $G_{ij}(R)$ with remainder $n\nu((\partial G_{ij}(R))^{\sqrt{D}h_n})$. If capacities are L-Lipschitz then $P(n; S^{\max}) \geq \mathbb{E}\sum_{t=1}^n M_\star(X_t) - E_n^K$, and if in addition two agents attain the largest capacity at ν-almost every profile then every unilateral deviation gains at most ΛE_n.*

Cells are initialized lazily, so every count of cells above may be replaced by the number of cells that carry positive mass. The exploration terms therefore shrink on their own when the context law concentrates on a lower-dimensional set, with no knowledge of that set and no change to the rule. The exponent is set by the Lipschitz term $L\sqrt{d}nh_n$, which carries no cell count at all, so moving it instead requires the grid itself to be retuned to the lower dimension.

Proof idea. Conditional on the cell membership pattern, each cell is an ordinary fixed-mean instance whose means are the cell averages, and a cell-blind decision at one visit is independent of the profile at that visit, which turns a count over a region into the cell count times the conditional mass of the region. Monotonicity is then the same index race as in the scalar case, run cell by cell on the cell-average tapes, which is why it is exact and needs no boundary condition; the remaining claims combine the per-cell deficit bound with the Lipschitz comparison across a cell. Appendix G.3 gives the proofs.

An explore-then-commit variant. Static Grid-ETC attains a worse exponent but buys two things the rule above does not have: its monotonicity is pathwise rather than in expectation, and its equilibrium needs no smoothness of the capacities. It uses cubes of side

$h_n \asymp n^{-1/(D+3)}$ and $q_n := \lceil n^{2/(D+3)} \rceil$. In each lazily initialized cell, it explores through q_n independent uniform permutations of the K agents. After Kq_n visits it commits to the largest exploration empirical mean, using an independent uniform priority for ties and ignoring later rewards.

Theorem F.6 (Static Grid-ETC contextual SAMF)**.** *Uniformly over $\mathcal{S}_{\mathrm{HDS}}$, with constants depending only on fixed K, D, L, and for every fixed R with $\nu(\partial R) = 0$, Static Grid-ETC satisfies contextual sharp adaptivity with error $O\Big(n^{\frac{D+2}{D+3}} + n\nu\big((\partial R)^{\sqrt{D}h_n}\big)\Big) = o(n)$, and global robustness with error $O(n^{(D+2)/(D+3)})$. Its contextual monotonicity is exact and pathwise under the natural common-uniform coupling. If $\nu(\partial G_{ij}(R)) = 0$, it also satisfies expected contextual FATA on $G_{ij}(R)$. Hence it is contextual SAMF for any fixed region family satisfying these boundary conditions.*

The three core errors are Lipschitz bias nh_n, exploration $q_n h_n^{-D}$, and simple regret $n/\sqrt{q_n}$; the displayed choices balance them. Full details, including truncated cells, are in Appendix G.4.

For a cell C of positive mass write $\bar{M}_{j,C} := \mathbb{E}[M_j(x_j) \mid X \in C]$, $\eta_{i,C} := \big(\bar{M}_{i,C} - \max_{j \neq i} \bar{M}_{j,C}\big)_+$, $\beta_C := \mathbb{E}[M_\star(X) \mid X \in C] - \max_j \bar{M}_{j,C}$, and $c_1 := 2e^{-1/2}$.

Theorem F.7 (Continuous public restricted equilibrium)**.** *Suppose two agents attain the largest capacity at ν-almost every profile, and let $M_\star$ be public, so that all-maximum play is available to every agent. Then for every n, every agent i, and every deviation $m' \in \mathcal{S}_{\mathrm{HDS}}$, including independent ex ante mixtures,*

$$u_i(n; m', S_{-i}^{\max}) - u_i(n; S^{\max}) \le q_n J_n + \Lambda\big(c_1 n q_n^{-1/2} + n \textstyle\sum_C \nu(C)\eta_{i,C}\big),$$

and the last sum tends to zero, so $S^{\max}$ is an $\mathcal{S}_{\mathrm{HDS}}$-asymptotic equilibrium. If every M_j is L-Lipschitz then $\eta_{i,C} \le L\sqrt{d}h_n$ and the gain is at most $(2^{D+1} + \Lambda(c_1 + L\sqrt{d}))n^{(D+2)/(D+3)}$. With no top-set condition at all,

$$P(n; S^{\max}) \ge \mathbb{E}\sum_{t=1}^{n} M_\star(X_t) - \Big[Kq_nJ_n + (K-1)c_1 n q_n^{-1/2} + n\textstyle\sum_C \nu(C)\beta_C\Big],$$

whose bracket is $o(n)$, and at most $(K2^{D+1} + (K-1)c_1 + L\sqrt{d})n^{(D+2)/(D+3)}$ under L-Lipschitz capacities.

Proof idea. Lowering a deviator's map can only lower its exploration mean, so by the coupling above its commitment event shrinks, and its exploitation gain in a cell is at most its commitment probability times the cell-average shading. Either the shaded cell mean falls below some rival's, and the explore-then-commit tail bound caps that product by $c_1 q_n^{-1/2}$, or it does not, and the shading is at most $\eta_{i,C}$. Cell averages of a bounded measurable function converge in L^1 as the mesh shrinks, and two tops almost everywhere force the limit of $\eta_{i,C}$ to vanish; Lipschitz capacities replace that limit by a rate. Exploration costs at most q_n per cell, since one selection is worth at most one unit of effort by (1). Neither continuity, nor a null separator, nor a component structure, nor a constant top set is used. This is the one incentive argument in which the cost enters through its Lipschitz constant and not through its convexity: a deviator here shades at some contexts of a cell and not others, and what the proof needs is a bound on the chord slope of φ_i between the shaded target and the capacity, which is exactly Λ. A cost with a jump at the capacity and no Lipschitz bound defeats the claim, because the class $\mathcal{S}_{\mathrm{HDS}}$ is horizon dependent and a vanishing shading can then buy a constant saving.

The condition is sharp: an agent that is uniquely top on a set of positive mass, and has downward slack there if it is honest, gains a linear amount by shading on the cells in which that set has high density, so no such profile is an equilibrium (Proposition G.5 in Appendix G.5); the gain is linear as soon as its cost is strictly increasing on the shaded range, which linearity gives for free. Shading on the whole set instead can lose a linear amount, because in near-tie cells the forfeited commitment outweighs the effort saved.

The properties, and finitely many profiles. For a declared class $\mathcal{S}$ and fixed measurable R, *contextual sharp adaptivity with remainder r_n* means $\mathbb{E}_S \sum_{t \le n} \mathbf{1}\{X_t \in R, I_t = i\}[m_{i,t} - \mu_i(x_{i,t})] \ge \mathbb{E}_S \sum_{t \le n} \mathbf{1}\{X_t \in R, I_t = i\}[b(X_t) - \mu_i(x_{i,t})] - r_n$ uniformly over $S \in \mathcal{S}^K$. As in the scalar case, the condition is on the mean lift $m_{i,t} - \mu_i(x_{i,t})$ an agent delivers and not on what delivering it costs, so it refers to no φ_i; the two coincide under the linear cost. *Contextual monotonicity* means that, for every allowed map profile, $i \neq j$, and the profile replacing m_i pointwise by M_i, the latter gives $\mathbb{E}T_j(R; n)$ no increase beyond the stated remainder. *Contextual FATA* means that whenever $m_i(X) = m_j(X) = M_\star(X)$ on $G_{ij}(R)$, $|\mathbb{E}T_i(G_{ij}(R); n) - \mathbb{E}T_j(G_{ij}(R); n)|$ is at most the stated remainder. Uniformity is stated theorem by theorem; finite UCB1 monotonicity is pointwise in each fixed comparison, whereas the grid bounds are uniform after conditioning on realized ex ante maps and hence also for their independent mixtures.

Let $\mathcal{X}_F := \{z_1, \dots, z_Q\}$ be a fixed finite set of full profiles. Finite-Context UCB1 maintains an independent symmetric UCB1 instance for every exact z, each at the known-horizon width $\sqrt{4 \log n/T}$ with the agent's local count in that instance and the global horizon n; no statistics are shared. The global horizon is what keeps an agent's index a function of its own count and its own reward prefix, which the monotonicity argument needs.

Theorem F.8 (Finite-context UCB1)**.** *Assume exogenous profile arrivals independent of reward tapes and deterministic horizon-independent context maps. For every $R \subseteq \mathcal{X}_F$ and agent i, Finite-Context UCB1 satisfies*

$$\mathbb{E}\sum_{t \le n} \mathbf{1}\{X_t \in R, I_t = i\}[m_{i,t} - \mu_i(x_{i,t})]$$
$$\ge \mathbb{E}\sum_{t \le n} \mathbf{1}\{X_t \in R, I_t = i\}[b(X_t) - \mu_i(x_{i,t})] - O\big(\sqrt{|R| n \log(en)}\big).$$

Its contextual monotonicity is exact, with remainder zero for every R, it has exact expected FATA on $G_{ij}(R)$, and it is robust: $P(n) \ge \mathbb{E}\sum_{t=1}^n b(X_t) - O(K\sqrt{Qn \log(en)})$. If profiles are i.i.d. with positive fixed masses, $|\mathcal{T}(z)| \ge 2$ for every z, and $M_\star(z)$ is public, the all-maximum profile is an $\mathcal{S}_F$-asymptotic equilibrium and $P(n; S^{\max}) \ge \mathbb{E}\sum_{t=1}^n M_\star(X_t) - O(\log n)$.

Proof idea. Conditioning on the random number of visits to z produces an ordinary fixed-mean UCB1 instance. Gap-uniform deficit bounds and Cauchy–Schwarz sum over the fixed profiles; independent instances make shared-top exchangeability exact. The public result applies the scalar fixed-gap argument profile by profile. Appendix G.1 contains the proof.

G Contextual proofs

Local instances. Every local UCB1 instance in this appendix, at a profile or at a cell, runs for a random number $s \le n$ of visits, with the confidence width $\sqrt{4\log n/T}$ built from the global horizon n. Conditional on the number of visits being s, the fixed-gap bound $\mathbb{E}T_i \le 3 + 16\log n/\Delta^2$ used in the proof of Lemma D.1 holds unchanged, because the confidence level is still n^{-2} and the instance has at most n rounds. Splitting at $\Delta = 4\sqrt{\log n/s}$ as in that proof gives, for every comparator a at most the best local mean,

$$(a - m_i)_+ \,\mathbb{E}[T_i \mid s] \le 4\sqrt{s\log n} + 3, \tag{8}$$

and splitting the gaps at $\varepsilon = 4\sqrt{K\log n/s}$ as in the derivation of (2) gives

$$\mathbb{E}\Big[\sum_{r=1}^{s}(m^\star - m_{I_r}) \;\Big|\; s\Big] \le 8\sqrt{Ks\log n} + 3K. \tag{9}$$

G.1 Finite profiles

For a profile z, let $V_z := V_{\{z\}}(n)$ be its visit count and $T_i(z) := T_i(\{z\}; n)$ agent i's local pull count. Conditional on $V_z = s$, its independent local instance is exactly an s-round fixed-mean UCB1 execution. The context process may otherwise be arbitrary and exogenous.

Proof of Theorem F.8: sharp adaptivity and robustness. Choose an honest agent h_z attaining $b(z)$. Its fixed delivered mean is at least $b(z)$, so $b(z)$ is at most the best local mean, and (8), conditional on $V_z = s$, gives

$$(b(z) - m_i(z))_+ \mathbb{E}[T_i(z) \mid V_z = s] \le 4\sqrt{s\log n} + 3.$$

Jensen's inequality yields $4\sqrt{\mathbb{E}V_z \log n} + 3$ after averaging. Sum over $z \in R$, apply Cauchy–Schwarz, and use $m_i(z) - \mu_i(z_i) = b(z) - \mu_i(z_i) - (b(z) - m_i(z))$ to obtain the displayed mean-lift inequality. This is an identity between delivered means and holds for every cost function.

For robustness, apply (9) conditionally on V_z. The best delivered local mean is at least $b(z)$, so summing over the Q profiles, with Jensen's inequality and Cauchy–Schwarz, gives

$$\mathbb{E}\sum_{t=1}^{n}[b(X_t) - m_{I_t}(X_t)] \le 8\sqrt{KQn\log n} + 3KQ.$$

The additive $3KQ$ is absorbed because the claim is trivial when $Q \ge n$. □

Proof of Theorem F.8: monotonicity and FATA. Fix a profile z and condition on its visit sequence, which is a function of the contexts alone and so is unchanged by any map. Within that instance every agent's index is $\widehat{m}_a + \sqrt{4\log n/T_a}$, a function of the agent's own local count and its own reward prefix at z and of nothing else, because the width uses the global horizon rather than the local time. Realize the instance's rewards from tapes $\mathbf{1}\{U_{z,a,k} \le m_a(z)\}$ indexed by the profile, the agent, and the agent's local pull count; this is legitimate because the instance at z is an ordinary fixed-mean bandit conditional on its visit sequence, and the arrivals are exogenous.

Compare two maps differing only in that agent i's is raised pointwise. The raise raises $m_i(z)$ and leaves every $m_j(z)$ with $j \ne i$ unchanged, so it raises i's tape pointwise while no other agent's index sequence moves at all. Lemma D.2 then gives $T_j(z)$ weakly smaller for every $j \ne i$: on every sample point when the priority at z is drawn once before play, and in expectation under independent per-round tie coins. Every region $R \subseteq \mathcal{X}_F$ is a finite union of profiles, so $\mathbb{E}\,T_j(R) = \sum_{z\in R}\mathbb{E}\,T_j(z)$ is weakly smaller for every R, with remainder exactly zero. No regularity of R is used, and the raise need not be to capacity.

For $z \in G_{ij}(R)$, the premise gives identical reward laws to i and j in that independent instance. Symmetric initialization and tie-breaking imply equal expected local counts conditional on every $V_z = s$. Summing over the shared-top profiles proves exact expected FATA. □

Proof of Theorem F.8: public equilibrium. Fix profile z and write $q(z) := |\mathcal{T}(z)| \ge 2$. At all-maximum play, fixed-gap non-top agents receive $O(\log n)$ pulls, and symmetry gives each top agent $\mathbb{E}V_z/q(z) + o(n)$ pulls. A top agent's utility per such pull is $1 + \varphi_i(\mu_i(z_i)) - \varphi_i(M_\star(z)) \ge 0$.

Under a fixed deviation, a top agent either keeps mean $M_\star(z)$, leaving the local execution unchanged, or falls below another top agent and receives $O(\log n)$ pulls. A non-top agent remains below $M_\star(z)$ by its positive capacity gap under every feasible deviation. Hence no state contributes a positive linear utility gain. Summing over fixed Q proves restricted equilibrium. All non-top selections together incur only $O(\log n)$ benchmark loss, proving the revenue statement. □

G.2 A static-cell sampling lemma

Lemma G.1 (Cell-average law and future independence)**.** *Fix a deterministic cell C with $\nu(C) > 0$ and condition on any realized ex ante strategy map. Conditional on its membership-indicator sequence, the ordered exact profiles visiting C are i.i.d. with law $\nu(\cdot \mid C)$. If cell actions ignore the exact within-cell position, agent i's selected rewards used by the cell rule are, in the reward-tape sense, i.i.d. Bernoulli with mean $\bar{m}_{i,C} := \mathbb{E}[m_{i,n}(X) \mid X \in C]$. Data from the first r cell visits and any function of those data are independent of exact profiles and membership indicators after the rth cell visit.*

Proof. I.i.d. sampling implies the conditional statement about the ordered profiles and independence of past and future visits. Cell-blind assignments are independent of residual positions. Integrating a Bernoulli success probability $m_{i,n}(X)$ over $\nu(\cdot \mid C)$ gives a Bernoulli variable with parameter $\bar{m}_{i,C}$. Formally, the potential rewards $\mathbf{1}\{U_i^{(k)} \le m_{i,n}(X^{(k)})\}$ of the kth cell visit are i.i.d. across k with mean $\bar{m}_{i,C}$, and the cell-blind selection at visit k depends only on earlier cell data and independent coins, so the rewards agent i actually delivers in the cell are i.i.d. Bernoulli$(\bar{m}_{i,C})$ given the past. Coordinates are correlated across agents through the shared profile; the per-agent concentration arguments below never use independence across agents. Zero-mass cells are never visited and need no conditional mean. □

G.3 Static-grid UCB1

Cells are the $J_n = b_n^D$ equal cubes of side $h_n = 1/b_n$ tiling the closed cube; V_C counts visits to C and $T_i(C)$ counts agent i's selections there. Conditional on the membership sequence of C, Lemma G.1 makes the cell an ordinary fixed-mean instance with means $\bar{m}_{i,C}$,

so Lemma D.1 applies inside it with the global n in the confidence width.

Lemma G.2 (Region identity)**.** *For every cell C, every measurable R, and every agent j, $\mathbb{E}\,T_j(C \cap R; n) = \nu(R \mid C)\,\mathbb{E}\,T_j(C; n)$.*

Proof. Condition on the membership sequence of C. By Lemma G.1 the profiles at its visits are i.i.d. $\nu(\cdot \mid C)$, and the decision at the kth visit is a function of the first $k-1$ visits' data and of the cell's coins, which are drawn independently of the contexts. Hence the event that j is selected at a visit is independent of the profile at that visit, so its joint probability with $\{X \in R\}$ factorizes as $\nu(R \mid C)\Pr(I = j)$. Summing over visits and integrating over the membership sequence gives the claim; cells of zero mass are never visited, and cells visited fewer than K times are covered because the argument never uses how the decision is made. □

Proof of Theorem F.5: monotonicity. Fix a cell and condition on its membership sequence. Realize its rewards from tapes $\mathbf{1}\{U_{C,a,k} \le \bar{m}_{a,C}\}$ indexed by the cell, the agent, and the agent's local pull count, which is legitimate by Lemma G.1. Raising agent i's map pointwise raises $\bar{m}_{i,C}$ in every cell and leaves every other cell average unchanged, so it raises i's tape pointwise and hence its whole index sequence, while no other agent's index sequence moves, each being a function of its own local count and its own tape. Lemma D.2 then gives $T_j(C)$ weakly smaller for every $j \neq i$: on every sample point when the cell's priority is drawn once before play, and in expectation under independent per-round tie coins. Lemma G.2 converts this into $\mathbb{E}\,T_j(C \cap R)$ for every measurable R, and summing the finitely many cells gives the claim with remainder zero. No regularity of R is used, and the raise need not be to capacity. □

Proof of Theorem F.5: sharp adaptivity and robustness. Fix a cell C. Two profiles in C have block coordinates within $\sqrt{d}h_n$ of each other, so every μ_h oscillates by at most $L\sqrt{d}h_n$ on C, and $\mu_h(x_h) \le \bar{\mu}_{h,C} + L\sqrt{d}h_n$ at every profile of C, where $\bar{\mu}_{h,C} := \mathbb{E}[\mu_h(x_h) \mid X \in C]$. An honest agent delivers at least its intrinsic mean, so $\bar{\mu}_{h,C} \le \bar{m}_{h,C} \le \bar{m}^\star_C$ for every honest h; taking the maximum over honest h and averaging over C gives $\bar{b}_C \le \bar{m}^\star_C + L\sqrt{d}h_n$. Conditional on $V_C = s$, (8) with comparator $\bar{m}^\star_C$ gives $(\bar{m}^\star_C - \bar{m}_{i,C})_+ \mathbb{E}[T_i(C) \mid V_C = s] \le 4\sqrt{s \log n} + 3$, hence

$$(\bar{b}_C - \bar{m}_{i,C})_+ \mathbb{E}[T_i(C) \mid V_C = s] \le 4\sqrt{s \log n} + 3 + L\sqrt{d}h_n\,\mathbb{E}[T_i(C) \mid V_C = s].$$

The selection at a visit is independent of the profile at that visit, by the argument of Lemma G.2, so agent i's benchmark deficit in C, $\mathbb{E}\sum_t \mathbf{1}\{X_t \in C, I_t = i\}(b(X_t) - m_i(X_t))$, equals $(\bar{b}_C - \bar{m}_{i,C})\,\mathbb{E}\,T_i(C)$. Jensen and Cauchy–Schwarz over the cells turn $\sum_C \sqrt{\mathbb{E}V_C}$ into $\sqrt{J_n n}$, and at most $\min\{J_n, n\}$ cells are ever visited, so the first two terms sum to $4\sqrt{J_n n \log n} + 3\min\{J_n, n\}$; the last terms sum to at most $L\sqrt{d}nh_n$, because agent i's counts $T_i(C)$ sum to at most n. This gives the global statement, and for a region R the cells meeting both R and its complement lie in $(\partial R)^{\sqrt{D}h_n}$, which is charged one unit per visit. Using $m_i - \mu_i = b - \mu_i - (b - m_i)$ converts the deficit into sharp adaptivity. Robustness is the same computation with (9) in place of (8), which gives $8\sqrt{K J_n n \log n} + 3K\min\{J_n, n\}$ in place of the first two terms, that is, E^K_n. □

Proof of Theorem F.5: FATA. On a cell inside $G_{ij}(R)$ agents i and j have the same cell-average target, so exchanging their tapes, their positions in the initialization order, and their tie priorities leaves the local law invariant, and their expected local counts are equal; Lemma G.2 carries this to the region. Cells inside the complement contribute nothing, and the remaining cells meet $\partial G_{ij}(R)$ and are charged one unit per visit. □

Proof of Theorem F.5: revenue and equilibrium. At all-maximum play the deficit relative to $M_\star$ in a cell splits into $\beta_C \le L\sqrt{d}h_n$, by the one-step Lipschitz bound of Appendix G.5, and the cell instance's deficit relative to its best cell average, which (9) controls; summing gives E^K_n. For the equilibrium, let agent i deviate to m' and put $\delta_C := \mathbb{E}[M_i - m' \mid C]$, $\delta^\varphi_C := \mathbb{E}[\varphi_i(M_i) - \varphi_i(m') \mid C] \le \Lambda\delta_C$, and $a_C := \mathbb{E}[1 + \varphi_i(\mu_i) - \varphi_i(M_i) \mid C] \ge 0$. Lowering its map lowers $\bar{m}_{i,C}$, so by the monotonicity argument above its own expected count weakly falls, and the gain in the cell is at most $\mathbb{E}[T'_i(C)]\delta^\varphi_C \le \Lambda\,\mathbb{E}[T'_i(C)]\delta_C$. The remaining estimates bound $\mathbb{E}[T'_i(C)]\delta_C$, a quantity in mean space, and the factor Λ is carried through to the end. Two tops almost everywhere and the Lipschitz bound give a rival whose cell average is at least $\bar{M}_{i,C} - L\sqrt{d}h_n$, so the deviator's shaded cell average trails that rival by at least $\delta_C - L\sqrt{d}h_n$. Where that is positive, (8) bounds the product of it with the deviator's expected count by $4\sqrt{s \log n} + 3$ conditional on $V_C = s$; where it is not, $\delta_C \le L\sqrt{d}h_n$. Either way $\mathbb{E}[T'_i(C)]\delta_C \le 4\sqrt{\mathbb{E}V_C \log n} + 3 + L\sqrt{d}h_n\mathbb{E}[T'_i(C)]$, and summing over cells with $\sum_C \mathbb{E}[T'_i(C)] \le n$ gives ΛE_n, uniformly over deviations. Conditioning on the realized map extends this to independent ex ante mixtures. □

G.4 Static Grid-ETC

Let h_n be the largest dyadic width with $h_n \le n^{-1/(D+3)}$, so that $n^{-1/(D+3)}/2 < h_n \le n^{-1/(D+3)}$ and the $J_n = h_n^{-D} < 2^D n^{D/(D+3)}$ dyadic cubes of side h_n tile the cube, and let $q_n := \lceil n^{2/(D+3)} \rceil$. In each cell sample q_n independent uniform permutations. The first Kq_n visits follow these blocks, with the last observed block possibly truncated. If all blocks finish, use a further independent uniform priority permutation to choose among the largest exploration empirical means. Cells initialize lazily and exploitation rewards are ignored.

Lemma G.3 (ETC simple regret)**.** *If two empirical means use q i.i.d. Bernoulli observations with means $a < b$ and $\Delta := b - a$, then*

$$\Pr(\widehat{a} \ge \widehat{b}) \le 2e^{-q\Delta^2/2}, \qquad \Delta\Pr(\widehat{a} \ge \widehat{b}) \le c_1 q^{-1/2},$$

where $c_1 := 2e^{-1/2}$. Consequently, for K agents and empirical winner $\widehat{I}$, $\mathbb{E}[\bar{m}^\star - \bar{m}_{\widehat{I}}] \le c_1 K q^{-1/2}$.

Proof. If $\widehat{a} \ge \widehat{b}$ then $\widehat{a} - a \ge \Delta/2$ or $b - \widehat{b} \ge \Delta/2$, since otherwise $\widehat{a} < (a+b)/2 < \widehat{b}$. Hoeffding bounds each by $e^{-q\Delta^2/2}$ and a union bound gives the first inequality. For the second, $xe^{-qx^2/2}$ has derivative $e^{-qx^2/2}(1 - qx^2)$, so its supremum is $q^{-1/2}e^{-1/2}$ at $x = q^{-1/2}$. Sum this product bound over possible empirical winners against one best-mean agent. The observations of the two agents need not be independent of one another. □

Proof of Theorem F.6: sharp adaptivity and robustness. Exploration uses at most Kq_nJ_n rounds, pathwise. In a cell C, the Lipschitz comparison from the UCB1 proof gives $\bar{b}_C \le \bar{m}^\star_C + 2L\sqrt{D}h_n$. Lemmas G.1 and G.3 show that the expected benchmark deficit on each exploitation visit is at most $2L\sqrt{D}h_n + CKq_n^{-1/2}$. The exploration winner is independent of later cell visits by Lemma G.1, so

$$\mathbb{E}\sum_{t=1}^{n}\big[b(X_t) - m_{I_t,n}(X_t)\big] \le Kq_nJ_n + 2L\sqrt{D}\,nh_n + CKnq_n^{-1/2}. \quad (10)$$

For a fixed agent i, put $\Delta_{i,C} := (\bar{m}^\star_C - \bar{m}_{i,C})_+$. If i wins, it beats a best-average agent, so $\Delta_{i,C}\Pr(i \text{ wins}) \le Cq_n^{-1/2}$. The same calculation bounds the selected sum of $b(X_t) - m_{i,n}(X_t)$ in every cell contained in R. A cube meeting both R and R^c lies in $(\partial R)^{\sqrt{D}h_n}$, and charging one per visit adds at most $n\nu((\partial R)^{\sqrt{D}h_n})$. Using the mean lift $m_i - \mu_i$ converts this deficit bound to contextual sharp adaptivity.

Finally, nh_n, q_nJ_n, and $nq_n^{-1/2}$ are all $O(n^{(D+2)/(D+3)})$. Equation (10) proves global robustness; null boundary mass gives the local $o(n)$ term by continuity from above. □

Proof of Theorem F.6: monotonicity. Couple two executions with identical contexts, exploration permutations, tie priorities, other-agent reward uniforms, and reward uniforms on common exploration pulls of agent i. Feasibility gives $\mathbf{1}\{U \le M_i(X)\} \ge \mathbf{1}\{U \le m_{i,n}(X)\}$ on each such pull. Exploration choices are identical, so only i's empirical score can rise. With fixed priority, raising one score can leave the winner unchanged or replace it by i; it cannot make $j \ne i$ newly win. Exploration counts of j are equal and every exploitation indicator of j weakly decreases. Therefore $T_j(R; n; M_i, m_{-i}) \le T_j(R; n; m_i, m_{-i})$ pathwise for every R. □

Proof of Theorem F.6: FATA. Let $G := G_{ij}(R)$. In every cell $C \subseteq G$, agents i, j have the same target function $M_\star(X)$. Uniform exploration blocks are label-symmetric even when truncated, and the commitment priority is symmetric. Conditional on every local visit count, swapping i, j leaves the local law unchanged, so their expected counts are equal. Cells contained in G^c contribute no shared-top counts. Every remaining cell crosses ∂G, hence

$$|\mathbb{E}T_i(G; n) - \mathbb{E}T_j(G; n)| \le n\nu((\partial G)^{\sqrt{D}h_n}) = o(n).$$

This is expected fairness, not pathwise equality. □

G.5 Continuous public incentives

Write $\bar{M}_{j,C}$, $\eta_{i,C}$, β_C and $c_1 = 2e^{-1/2}$ as in Section F.9, let $N_C := (V_C - Kq_n)_+$ count the exploitation visits of a cell C, and let $p_{j,C}$ be the probability that j is committed there.

Lemma G.4 (Cell averages converge in L^1). *Let ν be a Borel probability measure on $[0,1]^D$ and let $\mathcal{P}_n$ be finite Borel partitions whose mesh tends to zero. For bounded Borel f let $f^{(n)}$ equal $\mathbb{E}[f \mid C]$ on each positive-mass $C \in \mathcal{P}_n$ and zero on null cells. Then $\|f^{(n)} - f\|_{L^1(\nu)} \to 0$.*

Proof. Fix $\varepsilon > 0$ and pick g continuous with $\|f - g\|_{L^1(\nu)} \le \varepsilon$, possible because continuous functions are dense in L^1 of a finite Borel measure on a compact metric space. Conditional expectation is an L^1 contraction, so $\|f^{(n)} - g^{(n)}\|_1 \le \varepsilon$. For X in a positive-mass cell C, $|g^{(n)}(X) - g(X)| \le \sup_{Y\in C}|g(Y) - g(X)| \le \omega_g(\operatorname{mesh}\mathcal{P}_n)$, where ω_g is the modulus of continuity of g, uniform by compactness. The triangle inequality gives $\limsup_n \|f^{(n)} - f\|_1 \le 2\varepsilon$, and ε was arbitrary. □

The maps $u \mapsto (u_i - \max_{j\ne i} u_j)_+$ and $(v, u) \mapsto v - \max_j u_j$ change by at most $\sum_l |u_l - u'_l|$, so Lemma G.4 applied to each M_j and to $M_\star$ gives

$$\sum_C \nu(C)\eta_{i,C} \to \mathbb{E}\big[(M_i(x_i) - \max_{j\ne i} M_j(x_j))_+\big], \qquad \sum_C \nu(C)\beta_C \to 0, \quad (11)$$

the second because $M_\star(X) = \max_j M_j(x_j)$ pointwise. Two tops at almost every profile says exactly that the first limit vanishes.

Proof of Theorem F.7: equilibrium. Fix i and a deterministic deviation m'; independent mixtures follow by conditioning because the bound below does not depend on the map. Exploration schedules are functions of the contexts and the cell permutations, so both executions explore the same rounds and their exploration utilities differ by at most one per pull, hence by at most q_nJ_n in total. Fix a positive-mass cell and put $a_C := \mathbb{E}[1 + \varphi_i(\mu_i) - \varphi_i(M_i) \mid C] \ge 0$, $\delta_C := \mathbb{E}[M_i - m' \mid C] \in [0,1]$ and $\delta^\varphi_C := \mathbb{E}[\varphi_i(M_i) - \varphi_i(m') \mid C]$. The coupling in the monotonicity proof gives $p'_{i,C} \le p_{i,C}$, so the exploitation gain in the cell is at most

$$\mathbb{E}[N_C]\big[p'_{i,C}(a_C + \delta^\varphi_C) - p_{i,C}a_C\big] \le \mathbb{E}[N_C]\,p'_{i,C}\delta^\varphi_C \le \Lambda\,\mathbb{E}[N_C]\,p'_{i,C}\delta_C,$$

the last step by the chord bound $\varphi_i(M_i) - \varphi_i(m') \le \Lambda(M_i - m')$. Only Lipschitzness of φ_i is used here; the deviator varies its target across the contexts of a cell rather than across pulls at one context, so there is no Jensen step and convexity plays no part. Let $\Delta' := \max_{j\ne i}\bar{M}_{j,C} - (\bar{M}_{i,C} - \delta_C)$. If $\Delta' > 0$, committing requires beating a best rival, whose exploration mean averages q_n independent observations of mean $\bar{M}_{j,C}$ by Lemma G.1; Lemma G.3 then gives $p'_{i,C}\Delta' \le c_1 q_n^{-1/2}$, whence $p'_{i,C}\delta_C \le c_1 q_n^{-1/2} + \eta_{i,C}$. If $\Delta' \le 0$ then $\delta_C \le \eta_{i,C}$ and the same bound holds. Summing over positive-mass cells with $\sum_C \mathbb{E}[N_C] \le n$ and $\mathbb{E}[N_C] \le n\nu(C)$, and multiplying the exploitation terms by Λ, gives the displayed bound, which (11) makes $o(n)$ uniformly over deviations. The exploration charge q_nJ_n is not multiplied by Λ, since it is bounded by one unit per pull through (1). If every M_j is L-Lipschitz, then $|x_j - x'_j| \le \sqrt{d}h_n$ for X, X' in one cell, so $M_j(x_j) \le \bar{M}_{j,C} + L\sqrt{d}h_n$ throughout C; almost every $X \in C$ satisfies $M_i(x_i) \le \max_{j\ne i} M_j(x_j)$, and integrating gives $\eta_{i,C} \le L\sqrt{d}h_n$. Finally $q_nJ_n \le 2^{D+1}n^{(D+2)/(D+3)}$ and $nh_n \le n^{(D+2)/(D+3)}$. □

Proof of Theorem F.7: revenue. Exploration costs at most Kq_n per cell. On exploitation visits Lemma G.1 factorizes a cell's expected deficit as $\mathbb{E}[N_C]\big(\beta_C + \sum_j p_{j,C}(\max_l \bar{M}_{l,C} - \bar{M}_{j,C})\big)$; an agent that is not cell-best must beat a cell-best agent to commit, so Lemma G.3 bounds its term by $c_1 q_n^{-1/2}$ and at most $K - 1$ agents contribute. Summing gives the display, which (11) makes $o(n)$; under Lipschitz capacities the one-step bound above gives $\beta_C \le L\sqrt{d}h_n$. No top-set condition enters. □

Proposition G.5 (Sharpness of the two-top condition). *For agent i let $s_i := M_i$ if i is strategic and $s_i := M_i - \mu_i$ if it is honest, and let U_i be the set of profiles at which i is the unique top. If $\nu(U_i \cap \{s_i > 0\}) > 0$, there are $g_0 > 0$, a Borel $A \subseteq U_i$ with $\nu(A) > 0$ on which*

$M_i - \max_{j\neq i} M_j \geq g_0$ and $s_i \geq g_0$, deviations $m'_n \in \mathcal{S}_{\mathrm{HDS}}$, and an n_0 with

$$u_i(n; m'_n, S^{\max}_{-i}) - u_i(n; S^{\max}) \geq (\lambda_i g_0 \nu(A)/32)\, n \qquad (n \geq n_0),$$

where λ_i is the right derivative of φ_i at $g_0/2$. The bound is linear, and so the two-top condition is sharp, provided $\lambda_i > 0$, which holds whenever φ_i is strictly increasing above $g_0/2$ and gives $\lambda_i = 1$ in the linear model. Conversely, if $\nu(U_i \cap \{s_i > 0\}) = 0$ for every i, then $S^{\max}$ is an $\mathcal{S}_{\mathrm{HDS}}$-asymptotic equilibrium, by Theorem F.7 with $\eta_{i,C}$ replaced by $\min\{\eta_{i,C}, \mathbb{E}[s_i \mathbf{1}\{X \notin U_i\} \mid C]\}$.

Proof. Choose g_0 and A by continuity from below. Call a cell good if $\nu(A \mid C) \geq 1/2$ and $\bar{M}_{i,C} - \max_{j\neq i} \bar{M}_{j,C} \geq 3g_0/4$, and let $m'_n := M_i - (g_0/2)\mathbf{1}_A$ on the good cells and M_i elsewhere; this is feasible, and honest-feasible when i is honest because $s_i \geq g_0$ on A. On a good cell $\delta_C \in [g_0/4, g_0/2]$ and the deviating cell mean still exceeds every rival's by at least $g_0/4$, so by Lemma G.3 the deviator fails to commit with probability at most $\rho_n := 2(K-1)e^{-q_n g_0^2/32}$. On A we have $M_i \geq s_i \geq g_0$, so the shaded range $[M_i - g_0/2, M_i]$ lies to the right of $g_0/2$, and convexity makes every chord slope of φ_i there at least λ_i; the cost saved per shaded context is therefore at least $\lambda_i g_0/2$, and the cost saved per exploitation visit to a good cell at least $\lambda_i g_0/4$. The forfeited baseline utility is at most one per visit by (1), so the deviator gains at least $(1-\rho_n)\lambda_i g_0/4 - \rho_n \geq \lambda_i g_0/8$ per exploitation visit once n is large; exploration only helps. By Lemma G.4 and Markov's inequality the good cells carry mass at least $\nu(A)/2$ for large n, while $Kq_nJ_n \leq n\nu(A)/4$ eventually, so the total gain is at least $(\lambda_i g_0/8)(n\nu(A)/4)$. For the converse, honesty and feasibility force $m' = M_i$ almost everywhere on U_i, so $\delta_C \leq \mathbb{E}[s_i \mathbf{1}\{X \notin U_i\} \mid C]$; the refined per-cell bound follows from $\min\{s, c+e\} \leq c + \min\{s, e\}$, and its remainder vanishes because at every profile one of the two limits of Lemma G.4 is zero. □

Shading on all of U_i instead of on the good cells is not a proof of the first part and can lose a linear amount: where A has small relative cell mass the effort saved is proportional to that mass, while the commitment probability drops by a constant, so the forfeited baseline utility dominates.

The argument fails for phase-dependent strategies, which can perform well during exploration and lower the mean after commitment. This is why the theorem is restricted to maps fixed before play.

H Private contextual mechanism: proofs

This appendix proves the finite-context result (Theorem H.1 and its exact form, Proposition H.2, in Appendix H.1), the noisy-label results (Appendix H.2), and Theorem 4.2 (Appendix H.3). Each mechanism runs one independent copy of MESP per region, so the scalar lemmas of Appendix F reappear in the forms the contextual case needs: the monitor is written as a global-time martingale with a predictable indicator, and the fallback lemma is proved for a prefix, because the number of remaining visits to a region is not known when its fallback starts.

Behavioural classes. We write $\mathcal{B}^{\mathrm{ctx}}_{i,n}$ for the contextual behavioural class: after being selected at round t, agent i chooses $m_{i,t} \in [0, M_i(x_t)]$ as an arbitrary measurable function of n, the current context x_t, its own entire history across all regions (its selection rounds, the contexts there, its targets, its realised rewards, and every message the mechanism sent it, including region phase labels and region prices), and its independent private randomisation; an honest agent satisfies $m_{i,t} \geq \mu_i(x_t)$ after every history. We write $\mathcal{B}^{\mathrm{ctx}+}_{i,n}$ for the same class enlarged by allowing the agent to condition in addition on the entire arrival sequence $(x_1, \ldots, x_n)$, hence on all future arrivals, on every visit count and on the exact global round index. Then $\mathcal{B}^{\mathrm{ctx}}_{i,n} \subseteq \mathcal{B}^{\mathrm{ctx}+}_{i,n}$.

Fresh draws given the path. The enlargement is free under (A2) but not under exogeneity and (A3) alone, for the following reason. (A3) fixes the law of each of a round's own draws conditionally on everything realised before it, that round's own context included. It says nothing about later contexts, which are realised after the round's seed. A later context may therefore be a fixed function of that seed while no agent is able to move it, so that exogeneity and (A3) both hold; a target in $\mathcal{B}^{\mathrm{ctx}+}_{i,n}$ that reads the next context then reads the current seed through it, and the seed clause, which is stated conditionally on the target, fails. The second half of (A2) excludes this: it fixes the law of every draw conditionally on the entire context sequence. We name that half separately, in the notation of this appendix, only to mark the places where it is used.

> **(A3-P), fresh draws given the path.** At every round the mechanism's randomisations are drawn in the order the mechanism specifies them and then the reward seed, and each of them has its specified law conditionally on the field generated by everything realised before it *together with the entire arrival sequence* $(x_1, \ldots, x_n)$.

(A3-P) is the second half of (A2) of Section 2.3, so (A2) implies it; it implies (A3) and is strictly stronger. It holds whenever the arrival sequence is deterministic, and whenever it is generated by randomness that the rewards and the mechanism do not touch. Under (A3-P) a target in $\mathcal{B}^{\mathrm{ctx}+}_{i,n}$ is measurable with respect to the path-augmented pre-seed field on which the seed clause is stated, and every visit count, every Stage-1 pull round and the length of every local fallback are measurable at the start of the phase in which they are used, being functions of the arrival sequence alone. Part (i) of Proposition H.2 is stated for $\mathcal{B}^{\mathrm{ctx}+}_{i,n}$ and uses (A3-P) only to license the standing conditioning on the arrival path described below. Part (i) of Proposition H.30 does not use it over $\mathcal{B}^{\mathrm{ctx}}_{i,n}$: its proof uses exogeneity and (A3) only and conditions on no arrival path, and it extends verbatim to $\mathcal{B}^{\mathrm{ctx}+}_{i,n}$ through (A3-P).

H.1 Finitely many contexts

The mechanism. Write CL-MESP for the context-local monitored empirical second price. The platform knows n and K and nothing else: not Q, not $\mathcal{X}_F$, not the arrival law, not any $\mu_i(\cdot)$, any $M_i(\cdot)$, or $\mathcal{K}_H$. For $n \geq 2K$ put

$$L_n := \min\left\{\left\lfloor \frac{n}{2K} \right\rfloor, \left\lceil n^{2/3} (\log(en))^{1/3} \right\rceil\right\}, \qquad \ell_n := \log(2n^4),$$

$$s_n := \sqrt{\frac{\log(2Kn^4)}{2L_n}}, \qquad b_n(r) := \sqrt{\frac{r\ell_n}{2}}, \quad b_n(0) := 0,$$

$$\gamma_n := \min\left\{\tfrac{1}{2}, \sqrt{\frac{K\log K}{n}}\right\}, \qquad \eta_n := \frac{\gamma_n}{K},$$
$$R_K(H) := 3\sqrt{KH\log K} + 2K\log K, \qquad R_K(0) := 0.$$

For $n < 2K$ the platform allocates cyclically; these finitely many horizons are absorbed into instance constants. All parameters are functions of n and K alone, which is what makes each local guarantee hold uniformly over whatever a context's visit count turns out to be.

Contexts are instantiated lazily on first arrival. For each context z met so far the platform keeps the local visit count v_z, the Stage-1 reward sums, the announced price τ_z, the identity W_z of the local winner, the local primary count r_z and reward sum Z_z, the local EXP3 weights, and one independent uniform priority permutation Π_z of $[K]$ drawn when z is first met. At round t the platform observes $x_t = z$, increments v_z, and acts according to z's local phase.

Stage 1 (bidding), while $v_z \le KL_n$. Pull agent $((v_z-1) \bmod K)+1$; the schedule is a deterministic round robin in label order, so agent i's rth Stage-1 pull at z is the $((r-1)K+i)$th visit to z, and it reads no reward, no bid and no coin. When $v_z = KL_n$, form the empirical bids $\beta_{i,z} := L_n^{-1}\sum_{r\le L_n} Y_{i,z,r}$, let W_z be a largest bidder with ties resolved by the highest Π_z priority, announce $\tau_z := (\beta_{(2),z} - s_n)_+$ with $\beta_{(2),z}$ the second order statistic with multiplicity, and set $r_z \leftarrow 0$, $Z_z \leftarrow 0$. A context that receives fewer than KL_n visits in the whole horizon has every visit a bidding pull and never awards, never prices and never monitors.

Stage 2 (monitored primary), while no switch has occurred at z. Pull W_z, set $r_z \leftarrow r_z + 1$ and $Z_z \leftarrow Z_z + Y_t$, and declare a local switch if $Z_z < r_z\tau_z - b_n(r_z)$. The test uses only rewards collected at z.

Stage 3 (local fallback), after a switch at z. Run explicit-exploration EXP3 over all K agents, restricted to visits to z, with fresh weights $w_{i,1} = 1$ and the parameters γ_n, η_n above, the local clock counting visits to z since the switch. The fallback is tuned once to n, not to the number of remaining visits to z, which the platform cannot know. It is never restarted at z and no doubling schedule is used.

Messages. A selected agent is told the current context's phase and, once Stage 1 at that context is over, the price τ_z. This is what makes $S^\star$ implementable, since a selected agent cannot otherwise distinguish a primary pull from a fallback pull; all guarantees below are uniform over the whole deviation class, so the extra information cannot help a deviator.

The proposed profile $S^\star$. At a visit to z, agent i delivers M_i^z in Stage 1 and in Stage 3; as the Stage-2 winner it delivers $\min\{\tau_z, M_i^z\}$ if strategic and $\max\{\mu_i^z, \min\{\tau_z, M_i^z\}\}$ if honest. This is feasible on every history and implementable from the messages.

Notation. Write $\mu_i^z := \mu_i(z)$ and $M_i^z := M_i(z)$, $M_\star(z) := \max_j M_j^z$, $\mathcal{T}(z) := \{j : M_j^z = M_\star(z)\}$, $V_i^z := \max_{j\ne i} M_j^z$, $\delta_i(z) := V_i^z - M_i^z$, $M_{(2)}(z)$ for the second largest of $(M_1^z,\dots,M_K^z)$ counting multiplicity, and $b(z) := \max_{h\in\mathcal{K}_H} \mu_h^z$; thus $i \in \mathcal{T}(z)$ if and only if $\delta_i(z) \le 0$, and $M_{(2)}(z) = V_i^z$ for any $i \in \mathcal{T}(z)$. Let V_z be the number of visits to z over the horizon, $N_z := (V_z - KL_n)_+$ its number of post-bidding visits, and $l_i(z)$ the number of Stage-1 pulls agent i receives at z, that is $l_i(z) = \#\{r \le L_n : (r-1)K + i \le V_z\}$, a deterministic function of V_z. Then $\sum_z l_i(z) \le \min\{QL_n, n\}$ and $\sum_z \min\{V_z, KL_n\} \le \min\{QKL_n, n\}$. As in Appendix F, $b(\cdot)$ with a context argument is the best honest intrinsic mean and $b_n(\cdot)$ with an integer argument is the monitoring boundary; the argument type disambiguates.

The result. The finite-context result is stated here in two forms: a rate form, and an exact form with every constant written out.

Theorem H.1 (Context-local monitored empirical second price)**.** *Assume* (A2), *$K \ge 2$, $n \ge 2K$, a finite context set of size Q, and a nonempty honest set, and run* CL-MESP *with the profile $S^\star$. For fixed K, Q and provider characteristics, the following hold with the error $O(Qn^{2/3}(\log n)^{1/3}) = o(n)$: for every agent i, every deviation in $\mathcal{B}_{i,n}^{\mathrm{ctx}+}$ gains at most that error once $4s_n$ is below the smallest positive capability gap $\delta_i(z)$ separating i from the top at the contexts where it is not top; $P(n;S^\star) \ge \mathbb{E}\sum_{t\le n} M_{(2)}(x_t)$ minus that error; and every feasible behavioural profile S satisfies $P(n;S) \ge \mathbb{E}\sum_{t\le n} b(x_t)$ minus that error. Only the incentive bound depends on the cost function, through its Lipschitz constant Λ, and only it carries the instance quantity $\sum_{z:i\notin\mathcal{T}(z)}\left[\log n/\delta_i(z)^2 + \sqrt{Kn\log K}/\delta_i(z)\right]$. Exact constants are in Proposition H.2.*

Exact constants. Part (i) of the proposition below is stated for the enlarged class $\mathcal{B}_{i,n}^{\mathrm{ctx}+}$ and uses (A3-P), the second half of (A2), which licenses the standing conditioning on the arrival path used throughout this subsection; once it is assumed, the enlargement of the class costs nothing. The continuum subsection dispenses with both, at no cost in the bound.

Proposition H.2 (Context-local MESP, exact form)**.** *Assume* (A2), *$K \ge 2$, $n \ge 2K$, and a finite $\mathcal{X}_F$ with $|\mathcal{X}_F| = Q$; assume a nonempty honest set, which is used in part (iii) only. Run* CL-MESP. *Then:*

(i) *for every agent i and every $S'_{i,n} \in \mathcal{B}_{i,n}^{\mathrm{ctx}+}$,*

$$u_i(n;S'_{i,n},S^\star_{-i,n}) - u_i(n;S^\star_n) \le \mathbb{E}\Big[\sum_z l_i(z)\Big] + \Lambda\sum_{z:i\in\mathcal{T}(z)}\mathbb{E}\big[1 + 2b_n(N_z) + R_K(n)\big] + \sum_{z:i\notin\mathcal{T}(z)}\mathbb{E}[D_i(z)] + 8n^{-2},$$

where, for a context z with $i \notin \mathcal{T}(z)$, so that $\delta_i(z) > 0$, $D_i(z) := 8(1+\sqrt{2\ell_n})^2/\delta_i(z)^2 + 2R_K(n)/\delta_i(z)$ if $4s_n \le \delta_i(z)$ and $D_i(z) := 2N_z$ otherwise; consequently, for every n with $4s_n \le \min\{\delta_i(z) : z \in \mathcal{X}_F,\ i \notin \mathcal{T}(z)\}$,

$$\sup_{S'_{i,n}\in\mathcal{B}_{i,n}^{\mathrm{ctx}+}}\big[u_i(n;S'_{i,n},S^\star_{-i,n}) - u_i(n;S^\star_n)\big] \le \min\{QL_n, n\} + \Lambda\big[Q + \sqrt{2\ell_n Qn} + QR_K(n)\big] + \sum_{z:i\notin\mathcal{T}(z)}\left[\frac{8(1+\sqrt{2\ell_n})^2}{\delta_i(z)^2} + \frac{2R_K(n)}{\delta_i(z)}\right] + 8n^{-2};$$

(ii) $P(n;S^\star) \ge \mathbb{E}\sum_{t\le n} M_{(2)}(x_t) - \big[\min\{QKL_n, n\} + 2s_n n + QR_K(n) + n^{-3}\big]$;

(iii) *for every feasible behavioural profile $S = (S_1,\dots,S_K)$ with $S_j \in \mathcal{B}_{j,n}^{\mathrm{ctx}+}$, in which every honest agent respects pointwise honesty and every agent, up to all K of them, may otherwise*

deviate,

$$P(n;S) \geq \mathbb{E}\sum_{t\leq n} b(x_t) - \left[\min\{QKL_n, n\} + 2s_n n + Q + \sqrt{2\ell_n Qn} + QR_K(n) + 3n^{-2}\right].$$

The constants are absolute except where $\delta_i(z)$ and the Lipschitz constant Λ appear; part (i) is uniform over deviations for each fixed instance but not over instance sequences whose positive capacity gaps vanish with n. Only part (i) refers to the cost, and within it only the top-winning sum carries Λ: the elicitation term $\sum_z l_i(z)$ and the whole non-top sum do not, since the first is bounded through (1) *and the second through the range $[0,2]$ and a mean-space feasibility bound. Parts (ii) and (iii) hold for an arbitrary cost function.*

Since $L_n \leq n^{2/3}(\log(en))^{1/3} + 1$, $\ell_n = \log 2 + 4\log n$, $2s_n n = O(n^{2/3}(\log n)^{1/3})$ and $R_K(n) = O(\sqrt{Kn\log K})$, the three brackets are $O_K\big(QL_n + \sqrt{Qn\log n} + Q\sqrt{Kn\log K} + QK\log K\big)$ plus, in (i), an instance constant of order $O\big(\sum_{z:i\notin\mathcal{T}(z)}[\log n/\delta_i(z)^2 + \sqrt{Kn\log K}/\delta_i(z)]\big)$. For a fixed K, a fixed Q and a fixed instance all three are $O(Qn^{2/3}(\log n)^{1/3}) = o(n)$, which is the rate of Theorem H.1; sublinearity requires $Q = o(n^{1/3}(\log n)^{-1/3})$, and Proposition H.7 below shows that the restriction is not an artefact.

Probability space and the standing conditioning. Five families carry everything: the arrival sequence $A := (x_1,\dots,x_n)$; the reward seeds $(U_t)_{t\leq n}$; the priorities (Π_z); the mechanism's fallback selection coins; and the agents' private coin arrays. Under (A3-P) we condition on A throughout and integrate over it only at the end: by (A3-P) every seed and every coin still has its stated law conditionally on everything realised before it together with the whole of A. Appendix H.3 shows that the device can be dispensed with, at the cost of writing every fallback statement with a conditional expectation of the local horizon rather than with the horizon itself. All private coins are placed in $\mathcal{F}_0$ together with $\sigma(A)$, and $\mathcal{F}_t$ is generated by $\mathcal{F}_0$, the seeds $U_1,\dots,U_t$ and the coins used at rounds $1,\dots,t$; coins not yet used are excluded, because the fallback lemma needs the selection at a fallback round to be conditionally random. Write $\mathcal{F}_t^- := \sigma(\mathcal{F}_{t-1}, x_t, I_t, m_{I_t,t})$ for the pre-seed field, the information present after the target has been chosen and before U_t is used. The only probabilistic input used below is

$$\mathbb{E}[Y_t \mid \mathcal{F}_t^-] = m_{I_t,t}, \qquad \text{with } Y_t - m_{I_t,t} \in [-m_{I_t,t},\ 1 - m_{I_t,t}], \quad (*)$$

an interval of length 1 with $\mathcal{F}_t^-$-measurable endpoints. This is the content of (A3-P) and not of (A3), since $\mathcal{F}_t^-$ contains $\sigma(A)$ and therefore the contexts of every round after t. Conditioning on A only enlarges the deviation class, and every bound below is uniform in A, so integrating recovers the unconditional claim. Under this conditioning every visit count V_z, every Stage-1 pull round and the length of every local fallback are deterministic. Two pathwise facts are used repeatedly: the per-pull agent utility $1 + \varphi_i(\mu_i^z) - \varphi_i(m)$ lies in $[0,2]$ by (1), so two executions differ in total utility by at most $2n$; and an agent's utility in any phase is nonnegative, which is what lets every losing branch below be discarded.

Context locality.

Lemma H.3 (Context locality)**.** *In* CL-MESP*, for every context z, the local phase at z, the bids $\beta_{\cdot,z}$, the price τ_z, the winner W_z, the local visit index at which a switch is declared, and the conditional selection distribution used at each Stage-3 visit to z are measurable functions of the visit count v_z, the rewards realised at visits to z, the priority Π_z, and the fallback coins used at visits to z. No quantity computed by the mechanism at z depends on anything realised at a visit to any $z' \neq z$.*

Proof. Every branch of the round rule reads only z's own state, and z's state is updated only at visits to z. □

Two consequences are used repeatedly. A monitor switch at one context cannot alter a price, a phase, a statistic or a selection law at another context. And τ_z is a function of the Stage-1 rewards delivered at z alone, so when a deviator i wins at z, $\tau_z = \tau_{i,z} := (\max_{j\neq i}\beta_{j,z} - s_n)_+$ does not involve i's own bid. The deviator is not context-local: its target at a visit to z may depend on everything it has seen anywhere, and Lemmas H.4, H.5 and H.6 are proved under exactly that generality. Agents may correlate their behaviour across contexts; the mechanism cannot.

The monitor in the global filtration. Call round s a *primary pull at z* if $x_s = z$ and z's local phase at s is Stage 2, and let ψ_s^z be the indicator of that event. Let q_z be the local primary tenure, the number of primary pulls at z over the horizon, and for $r \geq 1$ let Z_r^z and A_r^z be the sums of realised rewards and of chosen targets over the first $\min\{r, q_z\}$ primary pulls at z, with $Z_0^z = A_0^z := 0$; thus both sums are constant in r beyond q_z.

Lemma H.4 (Anytime local monitor)**.** *Whatever feasible nonanticipating targets the Stage-2 winners choose at all contexts, conditionally on A:*

(a) $\Pr\{\exists z \in \mathcal{X}_F,\ \exists r \leq V_z : |Z_r^z - A_r^z| > b_n(r)\} \leq n^{-3}$;
(b) *on the complement of the event in* (a)*, for every z with $q_z \geq 1$, $A_{q_z}^z \geq q_z\tau_z - 1 - 2b_n(q_z)$, and hence the winner's realised primary utility at z is at most $q_z(1 + \varphi_{W_z}(\mu_{W_z}^z) - \varphi_{W_z}(\tau_z)) + \Lambda(1 + 2b_n(q_z))$;*
(c) *if at context z every primary target is at least τ_z, the probability that a switch occurs at z is at most $V_z(2n^4)^{-1}$; summed over contexts, the probability that any such switch occurs anywhere is at most $(2n^3)^{-1}$;*
(d) $\mathbb{E}[Z_{q_z}^z - A_{q_z}^z] = 0$*, and the same identity holds conditionally on the Stage-1 data at z;*
(e) *the per-context statements inside* (a) *and* (c) *and the identity* (d) *hold verbatim conditionally on any $\mathcal{F}_t$ with $t \leq t_1(z)$, where $t_1(z)$ is the round of the KL_nth visit to z.*

Proof. *The martingale.* The local phase at s is determined by the data collected at visits to z strictly before s (Lemma H.3) and by x_s, which lies in $\sigma(A) \subseteq \mathcal{F}_0$; hence ψ_s^z is $\mathcal{F}_{s-1}$-measurable. Put $M_t^z := \sum_{s\leq t}\psi_s^z(Y_s - m_{I_s,s})$ and $M_0^z := 0$. The increment at round s is $\Delta_s := \psi_s^z(Y_s - m_{I_s,s})$. Conditionally on $\mathcal{F}_s^-$ it has mean zero by $(*)$ and lies in the interval $\psi_s^z[-m_{I_s,s}, 1 - m_{I_s,s}]$, whose endpoints are $\mathcal{F}_s^-$-measurable and whose length is $\psi_s^z \leq 1$. Hoeffding's lemma therefore gives $\mathbb{E}[e^{\lambda\Delta_s} \mid \mathcal{F}_s^-] \leq \exp(\lambda^2\psi_s^z/8)$ for every real λ, and since ψ_s^z is $\mathcal{F}_{s-1}$-measurable and $\mathcal{F}_{s-1} \subseteq \mathcal{F}_s^-$, the tower property gives $\mathbb{E}[e^{\lambda\Delta_s} \mid \mathcal{F}_{s-1}] \leq \exp(\lambda^2\psi_s^z/8)$. In particular M^z is an $(\mathcal{F}_t)$-martingale, and $\exp\big(\lambda M_t^z - \lambda^2\sum_{s\leq t}\psi_s^z/8\big)$ is a nonnegative supermartingale started at 1.

(a). Fix z and $r \geq 1$, and let T_r be the round of the rth primary pull at z, with $T_r := +\infty$ if $q_z < r$. Since ψ^z is predictable, T_r is a stopping

time, and $\sum_{s\le t\wedge T_r}\psi_s^z \le r$ pathwise. Hence $\exp(\lambda M^z_{t\wedge T_r} - \lambda^2 r/8)$ is a supermartingale started at 1, and Markov's inequality with $\lambda := 4b/r$ gives $\Pr(M^z_{n\wedge T_r} > b) \le \exp(-\lambda b + \lambda^2 r/8) = \exp(-2b^2/r)$; the lower tail is the same with $-\lambda$. On $\{q_z \ge r\}$ we have $M^z_{n\wedge T_r} = Z^z_r - A^z_r$, and for $r > q_z$ the sums $Z^z_r - A^z_r = Z^z_{q_z} - A^z_{q_z}$ are constant while $b_n(r) \ge b_n(q_z)$, so the event in (a) is the event that $|Z^z_r - A^z_r| > b_n(r)$ for some z and some $r \le q_z$. With $b = b_n(r)$, $2b^2/r = \ell_n$ and each tail has probability at most $e^{-\ell_n} = (2n^4)^{-1}$. The index set $\{(z,r) : z \in \mathcal{X}_F,\ r \le V_z\}$ has cardinality $\sum_z V_z = n$ given A, so the union over it and over both tails costs at most $n \cdot 2 \cdot (2n^4)^{-1} = n^{-3}$, uniformly in A. This is the step that makes knowledge of Q unnecessary.

(b). Work on the complement of the event in (a). If no switch occurred at z, the test did not fire at the last primary pull, so $Z^z_{q_z} \ge q_z\tau_z - b_n(q_z)$, while $Z^z_{q_z} \le A^z_{q_z} + b_n(q_z)$; hence $A^z_{q_z} \ge q_z\tau_z - 2b_n(q_z)$. If the switch occurred at the pull $q_z \ge 2$, apply the two inequalities at $q_z - 1$ to get $A^z_{q_z-1} \ge (q_z-1)\tau_z - 2b_n(q_z-1)$, and use $A^z_{q_z} \ge A^z_{q_z-1}$, $\tau_z \le 1$ and the monotonicity of b_n; for $q_z = 1$ the bound $A^z_1 \ge 0 \ge \tau_z - 1$ suffices. This proves the first claim, which concerns delivered means alone. For the utility statement write $U^{\mathrm{pri}} = q_z(1 + \varphi_{W_z}(\mu^z_{W_z})) - \sum_{r\le q_z}\varphi_{W_z}(m_r)$, where m_r is the target at the rth primary pull, and apply Lemma F.1(a) with $\theta = \tau_z$ and $\Delta = 1 + 2b_n(q_z)$, which bounds the subtracted sum below by $q_z\varphi_{W_z}(\tau_z) - \Lambda(1 + 2b_n(q_z))$.

(c). The hypothesis is pathwise: on the event that every primary target at z is at least τ_z, a switch declared at the rth primary pull means $Z^z_r < r\tau_z - b_n(r) \le A^z_r - b_n(r)$, that is $M^z_{T_r} < -b_n(r)$. So the switch event is contained in $\bigcup_{r\le V_z}\{M^z_{n\wedge T_r} < -b_n(r)\}$, whose probability is at most $V_z(2n^4)^{-1}$ by the lower-tail bound of (a); summing over contexts with $\sum_z V_z = n$ gives $(2n^3)^{-1}$.

(d). By construction $M^z_n = Z^z_{q_z} - A^z_{q_z}$, and M^z is a martingale with bounded increments, so $\mathbb{E}[M^z_n] = \mathbb{E}[M^z_0] = 0$. No optional stopping theorem is used, which matters because a local primary phase can also end because the context stops arriving, which is not a stopping time of any filtration generated by the local rewards.

(e). For $t \le t_1(z)$ no primary pull at z has occurred, so $M^z_t = 0$; the supermartingale of (a) may therefore be started at round t instead of at 0, and the Stage-1 data at z are $\mathcal{F}_{t_1(z)}$-measurable. □

The only place a deviator's global-history adaptivity could enter is (∗), and it does not: $\mathcal{F}_s^-$ already contains everything the deviator saw anywhere, and the seed is fresh. A deviator may therefore time its play across contexts however it likes without weakening the bound.

The fallback with an unknown local horizon. The local fallback uses explicit-exploration EXP3. Index the Stage-3 visits to a fixed context z after its switch by $t = 1, 2, \ldots$; with weights $w_{i,1} := 1$ and $q_{i,t} := w_{i,t}/\sum_k w_{k,t}$, the selection law is $p_{i,t} := (1-\gamma_n)q_{i,t} + \gamma_n/K$, the agent I_t is drawn from p_t with the fresh coin of that round, and after Y_t is observed $\widehat{Y}_{i,t} := Y_t\mathbf{1}\{I_t = i\}/p_{i,t}$ and $w_{i,t+1} := w_{i,t}\exp(\eta_n\widehat{Y}_{i,t})$.

Lemma H.5 (Local selected-reactive EXP3, prefix form)**.** *Fix a context z and a fallback start there, and let $\mathcal{F}_{\mathrm{fb}}$ be the field generated by A and everything realised up to and including the switching round at z; the number H of visits to z after the switch is $\mathcal{F}_{\mathrm{fb}}$-measurable and $H \le n$. Suppose there are an agent j and a value $v \in [0,1]$ such that at every Stage-3 visit to z at which j is selected, j's conditional expected reward given the past is at least v. The other agents, including a deviator, may choose arbitrary feasible nonanticipating targets after selection, depending on their entire global histories. Then, with the parameters γ_n, η_n tuned to n and not to H,*

$$\mathbb{E}\Big[\sum_{\text{the } H \text{ Stage-3 visits to } z} Y \,\Big|\, \mathcal{F}_{\mathrm{fb}}\Big] \ge vH - R_K(n).$$

Proof. Condition on $\mathcal{F}_{\mathrm{fb}}$; H is then a constant, and there is nothing to prove if $H = 0$. Let $\mathcal{J}_{t-1}$ be the global field just before the selection at the tth Stage-3 visit to z. The weights, hence $p_{\cdot,t}$, are $\mathcal{J}_{t-1}$-measurable, I_t has law p_t given $\mathcal{J}_{t-1}$ because its coin is fresh, and $r_{i,t} := \mathbb{E}[Y_t \mid \mathcal{J}_{t-1}, I_t = i]$ is well defined, with $r_{j,t} \ge v$ by hypothesis. No counterfactual reward of an unselected agent is needed.

The potential inequality. Write $W_t := \sum_i w_{i,t}$. Since $p_{i,t} \ge \gamma_n/K = \eta_n$, we have $0 \le \eta_n\widehat{Y}_{i,t} \le 1$, and $e^x \le 1 + x + x^2$ on $[0,1]$ gives

$$\frac{W_{t+1}}{W_t} = \sum_i q_{i,t}e^{\eta_n\widehat{Y}_{i,t}} \le 1 + \eta_n\sum_i q_{i,t}\widehat{Y}_{i,t} + \eta_n^2\sum_i q_{i,t}\widehat{Y}^2_{i,t}.$$

Taking logarithms, using $\log(1+x) \le x$, summing over $t \le H$, and bounding $\log(W_{H+1}/W_1) \ge \log(w_{j,H+1}/K) = \eta_n\sum_{t\le H}\widehat{Y}_{j,t} - \log K$ below, we obtain the pathwise inequality

$$\sum_{t\le H}\widehat{Y}_{j,t} - \frac{\log K}{\eta_n} \le \sum_{t\le H}\sum_i q_{i,t}\widehat{Y}_{i,t} + \eta_n\sum_{t\le H}\sum_i q_{i,t}\widehat{Y}^2_{i,t}, \tag{12}$$

which holds for the H rounds actually played, whatever η_n is. This is the only change from Lemma F.3, which fixes η from the exact H.

Expectations. Conditionally on $\mathcal{J}_{t-1}$, $\mathbb{E}\widehat{Y}_{j,t} = r_{j,t} \ge v$, $\mathbb{E}\sum_i q_{i,t}\widehat{Y}_{i,t} = \sum_i q_{i,t}r_{i,t}$, and $\mathbb{E}\sum_i q_{i,t}\widehat{Y}^2_{i,t} = \sum_i q_{i,t}r_{i,t}/p_{i,t} \le \sum_i q_{i,t}/p_{i,t} \le K/(1-\gamma_n)$, because $p_{i,t} \ge (1-\gamma_n)q_{i,t}$. Moreover $\mathbb{E}[Y_t \mid \mathcal{J}_{t-1}] = \sum_i p_{i,t}r_{i,t} \ge (1-\gamma_n)\sum_i q_{i,t}r_{i,t}$. Taking expectations in (12), multiplying by $1-\gamma_n$ and using $\eta_n K = \gamma_n$,

$$\mathbb{E}\sum_{t\le H}Y_t \ge (1-\gamma_n)vH - (1-\gamma_n)\frac{\log K}{\eta_n} - \gamma_n H \ge vH - 2\gamma_n H - \frac{K\log K}{\gamma_n}.$$

The two cases of γ_n. If $\gamma_n = \sqrt{K\log K/n} \le 1/2$ then, since $H \le n$, $2\gamma_n H + K\log K/\gamma_n \le 3\sqrt{Kn\log K}$. Otherwise $\gamma_n = 1/2$ and $n < 4K\log K$, so $2\gamma_n H + K\log K/\gamma_n \le n + 2K\log K \le 3\sqrt{Kn\log K} + 2K\log K$, because $n < 4K\log K$ gives $n \le 2\sqrt{Kn\log K}$. Either way the loss is at most $R_K(n)$. □

Two facts make the unknown local horizon harmless. First, (12) is a prefix inequality, so a run tuned to n and stopped after $H \le n$ rounds still pays only $2\gamma_n H + K\log K/\gamma_n$. Second, the fallback at z is never restarted, so no anytime or doubling device is needed. The rounds of the lemma are separated by arbitrarily many rounds at other contexts, during which the deviator acts and learns; nothing in the proof cares, because $\mathcal{J}_{t-1}$ contains all of it. The crude charge $R_K(n)$ per context is of lower order than L_n, so the simplification costs only a lower-order term.

Bid domination. Fix agent i and a deviation $S'_{i,n}$. Under (A2) the law of the arrival sequence does not depend on the agents' strategies, so both executions can be built on one space with the same A, the same seeds, the same priorities, the same fallback coins and the same private coins; in execution 1 agent i plays $S'_{i,n}$, in execution 2 (the baseline) it plays $S^\star_i$, and all other agents play $S^\star$ in both.

Lemma H.6 (Bid domination with round-indexed seeds). (a) *For every z, the set of rounds constituting Stage 1 at z and the identity of the agent pulled at each of them are the same in both executions and are deterministic given A.*

(b) *For every $j \neq i$ and every z, $\beta_{j,z}$ is the same random variable in both executions and, whenever Stage 1 at z completes, is an average of L_n i.i.d.* Bernoulli(M_j^z) *variables.*

(c) $\beta'_{i,z} \leq \beta_{i,z}^{\max}$ *pathwise, the prime denoting execution 1 and the maximum execution 2.*

(d) $\{i$ *wins at z in execution* $1\} \subseteq \{i$ *wins at z in execution* $2\}$, *and on the former event both executions announce the same price* $\tau_{i,z} = (\max_{j\neq i} \beta_{j,z} - s_n)_+$.

(e) *Agent i's Stage-1 utility at z in execution 1 exceeds its Stage-1 utility at z in execution 2 by at most $l_i(z) \leq L_n$.*

Proof. (a) Stage 1 at z occupies the first $\min\{V_z, KL_n\}$ visits to z and assigns the kth visit to agent $((k-1) \bmod K) + 1$; both the visit rounds and this assignment are functions of A alone. No seed relabelling is used: the seed at a given Stage-1 pull is the seed U_t of that round, literally the same variable in both executions.

(b) Agent $j \neq i$ plays $S_j^\star$, which delivers M_j^z at every Stage-1 visit to z whatever the history, so its Stage-1 rewards at z are $\mathbf{1}\{U_t \leq M_j^z\}$ at the same rounds in both executions. Given A these rounds are deterministic, and by (A3-P) each U_t is uniform conditionally on everything realised before it and on A, so the seeds at these rounds are i.i.d. uniform given A and the indicators are i.i.d. Bernoulli(M_j^z).

(c) At the round t carrying agent i's rth Stage-1 pull at z, feasibility gives $m'_{i,t} \leq M_i^z$, so $Y'_t = \mathbf{1}\{U_t \leq m'_{i,t}\} \leq \mathbf{1}\{U_t \leq M_i^z\} = Y_t^{\max}$; the deviator's target may depend on everything it has seen at every context, and the inequality is pathwise regardless.

(d) If i wins in execution 1 then $\beta'_{i,z} \geq \beta_{j,z}$ for all $j \neq i$ and i has the highest Π_z priority in the tie set. In execution 2, $\beta_{i,z}^{\max} \geq \beta'_{i,z} \geq \beta_{j,z}$ by (b) and (c), so i is again a largest bidder; if $\beta_{i,z}^{\max} > \beta'_{i,z}$ then no $j \neq i$ ties with i and i wins outright, and otherwise the tie set and the priority are unchanged. When i wins, the second order statistic with multiplicity equals $\max_{j\neq i} \beta_{j,z}$, whether or not some j ties with i, and by (b) this is the same in both executions.

(e) At each of the $l_i(z)$ Stage-1 pulls of i at z the per-pull utilities differ by $\varphi_i(M_i^z) - \varphi_i(m'_{i,t}) \in [0,1]$, by monotonicity and (1). □

Part (d) is the only pathwise statement carried past Stage 1. After Stage 1 the two executions may diverge completely at z, and they are compared only through conditional expectations given the Stage-1 data at z, exactly as in the scalar proof. Bids live on the grid $\{0, 1/L_n, \ldots, 1\}$, so exact ties have positive probability, and (d) is written without a strict inequality for that reason.

Proof of Proposition H.2(i). Fix i and $S'_{i,n} \in \mathcal{B}_{i,n}^{\mathrm{ctx}+}$, build the two coupled executions of Lemma H.6, and condition on the arrival path. Let $U_i(z)$ be i's realised utility on the visits to z, with a prime for execution 1 and a star for execution 2. The rounds $[n]$ are partitioned by the contexts, so

$$u_i(n; S'_{i,n}, S^\star_{-i,n}) - u_i(n; S_n^\star) = \sum_{z \in \mathcal{X}_F} \big(\mathbb{E}[U'_i(z)] - \mathbb{E}[U_i^\star(z)]\big). \quad (13)$$

This is an identity, not an estimate, and it is where the cross-context deviator is neutralised: no independence across contexts is used or needed, only additivity of expectation. Split $U_i(z) = U_i^{\mathrm{bid}}(z) + U_i^{\mathrm{post}}(z)$ into the Stage-1 part and the post-bidding part.

Stage 1. By Lemma H.6(e) the Stage-1 parts contribute at most $l_i(z)$ per context, pathwise. A context with $V_z < KL_n$ has every visit a bidding pull, so $U_i^{\mathrm{post}}(z) = 0$ in both executions and the whole context contributes at most $l_i(z)$: a deviator cannot buy itself extra pulls at a low-mass context, because the Stage-1 schedule is reward independent.

The good event and the price. Fix z with $V_z \geq KL_n$ and put $G(z) := \{|\beta_{j,z} - M_j^z| \leq s_n \text{ for every } j \neq i\}$. By Lemma H.6(b) and Hoeffding's inequality, $\Pr(G(z)^c) \leq 2(K-1)\exp(-2L_n s_n^2) = (K-1)/(Kn^4) \leq n^{-4}$, because $2L_n s_n^2 = \log(2Kn^4)$. On $G(z)$, $\max_{j\neq i} \beta_{j,z} \in [V_i^z - s_n, V_i^z + s_n]$, and since $(\cdot)_+$ is nondecreasing and 1-Lipschitz,

$$(V_i^z - 2s_n)_+ \leq \tau_{i,z} \leq V_i^z. \quad (14)$$

Let Mon′ and Mon* be the complements of the event of Lemma H.4(a) in the two executions, each of probability at least $1 - n^{-3}$. Post-bidding utilities lie in $[0, 2N_z]$ in both executions, and the baseline post-bidding utility is nonnegative; these two facts absorb all bad events. Write win′ for the event that i wins at z in execution 1; it lies in $\mathcal{F}_{t_1(z)}$, as does $G(z)$.

The losing branch. On $G(z) \setminus \text{win}'$ the local winner $W \neq i$ plays $S^\star$, and $\tau_z = (\beta_{(2),z} - s_n)_+ \leq (\beta_{W,z} - s_n)_+ \leq M_W^z$, because W is a largest bidder and its bid is within s_n of M_W^z. So W's primary target is at least τ_z on this event, and by the pathwise inclusion of Lemma H.4(c) a switch at z on this event has probability at most $V_z(2n^4)^{-1} \leq (2n^3)^{-1}$. This rules out the attack in which the deviator bids so as to raise the announced price above the winner's capability and provoke a fallback in which the deviator is itself selected: whenever i loses, $\beta_{(2),z} \leq \beta_{W,z}$, so the price is at most M_W^z on $G(z)$ and the winner meets it. Without a switch, i receives no post-bidding pull at z, so $U_i'^{\mathrm{post}}(z) = 0$; hence, the baseline post-bidding utility being nonnegative,

$$\mathbb{E}\big[(U_i'^{\mathrm{post}}(z) - U_i^{\star\mathrm{post}}(z))\mathbf{1}_{(\text{win}')^c}\big] \leq 2N_z\big(n^{-4} + (2n^3)^{-1}\big).$$

The winning branch with $i \in \mathcal{T}(z)$: the deviating side. Work on $G(z) \cap \text{win}'$. By Lemma H.6(d) both executions face the price $\tau_{i,z}$, and by (14) $\tau_{i,z} \leq V_i^z \leq M_i^z$. Put $c_z := 1 + \varphi_i(\mu_i^z) - \varphi_i(\tau_{i,z})$, an $\mathcal{F}_{t_1(z)}$-measurable quantity in $[0,2]$, and let q be the deviating local primary length and $H_z = N_z - q$ the deviating local fallback length. On Mon′, Lemma H.4(b) gives $U_i'^{\mathrm{pri}}(z) \leq c_z q + \Lambda(1 + 2b_n(q)) \leq c_z q + \Lambda(1 + 2b_n(N_z))$. For the fallback, let $\mathcal{F}_{\mathrm{fb}}$ be the field up to the switching round at z, which contains $\mathcal{F}_{t_1(z)}$, and let $\bar{Q}_i$ and $\bar{A}_i$ be the conditional expectations given $\mathcal{F}_{\mathrm{fb}}$ of the number of i's fallback pulls at z and of the sum of its targets there. Lemma H.5 with the comparator $j \neq i$ attaining $M_j^z = V_i^z$, which exists since $K \geq 2$ and which plays $S^\star$, gives $\mathbb{E}[\sum Y \mid \mathcal{F}_{\mathrm{fb}}] \geq V_i^z H_z - R_K(n)$, the sum running over the H_z fallback visits; since every agent other than i delivers at most V_i^z in the fallback, (∗) and the tower property give $\mathbb{E}[\sum Y \mid \mathcal{F}_{\mathrm{fb}}] \leq \bar{A}_i + V_i^z(H_z - \bar{Q}_i)$. Hence $\bar{A}_i \geq V_i^z \bar{Q}_i - R_K(n)$. If $\bar{Q}_i = 0$ the deviator's fallback utility vanishes; otherwise let π be the measure on $[0,1]$ given by $\pi(B) := \mathbb{E}[\sum \mathbf{1}\{m_{i,t} \in B\} \mid \mathcal{F}_{\mathrm{fb}}]$, the sum running over i's fallback pulls at z, which has mass $\bar{Q}_i$ and $\int m \, d\pi = \bar{A}_i$. Lemma F.1(b) with $\theta = V_i^z$ and $\Delta = R_K(n)$, the inequality $1 + \varphi_i(\mu_i^z) - \varphi_i(V_i^z) \geq 0$ (normalisation), $\bar{Q}_i \leq H_z$ and

$\tau_{i,z} \le V_i^z$ give

$$\begin{aligned}\mathbb{E}[U_i'^{\mathrm{fb}}(z) \mid \mathcal{F}_{\mathrm{fb}}] &= (1+\varphi_i(\mu_i^z))\bar{Q}_i - \int \varphi_i \,\mathrm{d}\pi \\ &\le (1+\varphi_i(\mu_i^z) - \varphi_i(V_i^z))H_z + \Lambda R_K(n) \\ &\le c_z H_z + \Lambda R_K(n).\end{aligned}$$

Adding the two phases, with $q + H_z = N_z$ and c_z measurable with respect to $\mathcal{F}_{\mathrm{fb}}$, and charging $2N_z$ on $(\mathrm{Mon}')^c$,

$$\begin{aligned}\mathbb{E}\big[U_i'^{\mathrm{post}}(z)\mathbf{1}_{G(z)\cap \mathrm{win}'}\big] \le \mathbb{E}\big[c_z N_z \mathbf{1}_{G(z)\cap \mathrm{win}'}\big] \\ + \Lambda\big(1 + 2b_n(N_z) + R_K(n)\big) + 2N_z n^{-3}.\end{aligned} \tag{15}$$

The winning branch with $i \in \mathcal{T}(z)$: the baseline side. On $G(z) \cap \mathrm{win}'$ the baseline also has i winning at z at the price $\tau_{i,z}$. If i is strategic, or honest with $\tau_{i,z} > \mu_i^z$, its baseline primary target is exactly $\tau_{i,z}$, so all baseline primary targets are at least $\tau_{i,z}$ and Lemma H.4(c) bounds the baseline switch probability by $(2n^3)^{-1}$; without a switch the baseline post-bidding utility at z is exactly $c_z N_z$, the leading term of (15), and the two cancel. If i is honest with $\tau_{i,z} \le \mu_i^z$, the baseline primary target is $\mu_i^z \ge \tau_{i,z}$, the same containment bounds the baseline switch probability by $(2n^3)^{-1}$, and without a switch the baseline post-bidding utility is exactly N_z; on the other side honesty forces every deviating target at z to be at least μ_i^z, so $1+\varphi_i(\mu_i^z)-\varphi_i(m) \le 1$ by monotonicity of φ_i, every deviating post-bidding selection is worth at most 1, and there are at most N_z of them; this branch needs no monitoring and no fallback term at all. In both cases the baseline false switch costs at most $2N_z(2n^3)^{-1}$. Adding the piece $2N_z n^{-4}$ for $G(z)^c$ and the losing branch,

$$\mathbb{E}[U_i'(z)] - \mathbb{E}[U_i^\star(z)] \le l_i(z) + \Lambda\big(1 + 2b_n(N_z) + R_K(n)\big) + 8N_z n^{-3}, \tag{16}$$

the bad events being n^{-3} (Mon'), $(2n^3)^{-1}$ (baseline false switch), n^{-4} ($G(z)^c$ on the winning branch), and $n^{-4} + (2n^3)^{-1}$ (losing branch), in total at most $4n^{-3}$, each charged $2N_z$.

The cancellation is worth naming. The deviator's total post-bidding utility at z is at most $c_z N_z$ plus Λ times the monitor slack $1 + 2b_n(N_z)$ plus Λ times the fallback regret $R_K(n)$, and the baseline collects $c_z N_z$ exactly. Triggering the fallback therefore gains the deviator nothing of order N_z: the fallback is worth at most $1 + \varphi_i(\mu_i^z) - \varphi_i(V_i^z)$ per remaining round, which on $G(z)$ is at most c_z, the baseline rate. The comparison is made at the common price $\tau_{i,z}$ and never at $V_i^z - 2s_n$; were it made at the latter each context would contribute $2s_n N_z$ and the total would acquire a $2s_n n$ term, which would not change the rate but would be avoidable slack. Parts (ii) and (iii) do pay $2s_n n$, because there the price itself is the quantity being collected.

The winning branch with $i \notin \mathcal{T}(z)$. Put $\delta := \delta_i(z) > 0$ and suppose $4s_n \le \delta$. On $G(z)$, (14) and $V_i^z = M_i^z + \delta \ge \delta \ge 4s_n$ give $\tau_{i,z} \ge V_i^z - 2s_n$, hence $\tau_{i,z} - M_i^z \ge \delta - 2s_n \ge \delta/2$. On Mon', Lemma H.4(b) gives $A_q^z \ge q\tau_{i,z} - 1 - 2b_n(q)$ while feasibility gives $A_q^z \le M_i^z q$; subtracting, with $b_n(q) = \sqrt{q\ell_n/2}$, $q\delta/2 \le 1 + \sqrt{2q\ell_n} \le \sqrt{q}(1+\sqrt{2\ell_n})$ for $q \ge 1$, so

$$q \le \frac{4(1+\sqrt{2\ell_n})^2}{\delta_i(z)^2} = O\left(\frac{\log n}{\delta_i(z)^2}\right), \tag{17}$$

and $U_i'^{\mathrm{pri}}(z) \le 2q \le 8(1+\sqrt{2\ell_n})^2/\delta_i(z)^2$. In the fallback, $\bar{A}_i \le M_i^z \bar{Q}_i$ by feasibility and $\bar{A}_i \ge V_i^z \bar{Q}_i - R_K(n)$ as above, so $\bar{Q}_i \le R_K(n)/\delta$ and $\mathbb{E}[U_i'^{\mathrm{fb}}(z) \mid \mathcal{F}_{\mathrm{fb}}] \le 2\bar{Q}_i \le 2R_K(n)/\delta_i(z)$. The baseline post-bidding utility is nonnegative. This branch acquires no factor Λ and uses nothing about φ_i beyond (1): its tenure bound comes from the feasibility inequality $A_q^z \le M_i^z q$, which is in mean space, and its utility charges are the crude range bound $[0,2]$. Adding the same bad events,

$$\mathbb{E}[U_i'(z)] - \mathbb{E}[U_i^\star(z)] \le l_i(z) + \frac{8(1+\sqrt{2\ell_n})^2}{\delta_i(z)^2} + \frac{2R_K(n)}{\delta_i(z)} + 8N_z n^{-3}. \tag{18}$$

If instead $4s_n > \delta_i(z)$, use $U_i'^{\mathrm{post}}(z) \le 2N_z$ and a nonnegative baseline, giving $l_i(z) + 2N_z$. The case $\delta_i(z) = 0$ is never handed to the branch whose bound divides by $\delta_i(z)$: a capacity tie at the top of a context puts both tied agents in $\mathcal{T}(z)$, hence in the branch of (16).

Summation. Summing (16) and (18) over $z \in \mathcal{X}_F$ through (13), and using $\sum_z N_z \le \sum_z V_z = n$, gives the first display of (i). The right-hand side is a deterministic function of the arrival path and does not depend on $S'_{i,n}$, so averaging over the path and then taking the supremum over the deviation class is legitimate. For the second display, assume $4s_n \le \delta_i(z)$ at every context at which i is not top and use: $\sum_z l_i(z) \le \min\{QL_n, n\}$ pathwise; $\mathbb{E}[2b_n(N_z)] = \sqrt{2\ell_n}\,\mathbb{E}\sqrt{N_z} \le \sqrt{2\ell_n}\sqrt{\mathbb{E}N_z}$ by Jensen's inequality, and $\sum_z \sqrt{\mathbb{E}N_z} \le \sqrt{Q\sum_z \mathbb{E}N_z} \le \sqrt{Qn}$ by the Cauchy–Schwarz inequality, so $\sum_z \mathbb{E}[2b_n(N_z)] \le \sqrt{2\ell_n Qn}$; the constants 1 contribute at most Q; and $R_K(n)$ contributes at most $QR_K(n)$. These three are exactly the terms that carry the factor Λ; $\min\{QL_n, n\}$ and the non-top sum do not. Since $s_n \to 0$, the hypothesis $4s_n \le \min_z \delta_i(z)$ holds for every n beyond an instance threshold depending only on K and that minimum, and below it the trivial bound $2n$ applies and is an instance constant. □

Proof of Proposition H.2(ii). All agents play $S^\star$. Since $P(n;S) = \mathbb{E}\sum_t m_{I_t,t}$ by the tower property applied to $(*)$ at each fixed t, revenue is additive over contexts exactly as utility is, and we may bound delivered targets instead of rewards; no optional stopping is needed. If $V_z < KL_n$, all visits are bidding pulls and we charge $V_z M_{(2)}(z) \le \min\{KL_n, V_z\}$. Let $V_z \ge KL_n$; the KL_n bidding visits deliver nonnegative targets and we charge $KL_n M_{(2)}(z) \le KL_n$. Put $G^{\mathrm{all}}(z) := \{|\beta_{j,z} - M_j^z| \le s_n \text{ for every } j \in [K]\}$; every bid is now the mean of L_n independent Bernoulli(M_j^z) variables, so Hoeffding's inequality gives $\Pr(G^{\mathrm{all}}(z)^c) \le 2K\exp(-2L_n s_n^2) = n^{-4}$. On $G^{\mathrm{all}}(z)$, order statistics being 1-Lipschitz in the supremum norm,

$$\begin{aligned}M_{(2)}(z) - 2s_n \le \tau_z \le M_{(2)}(z), \\ \tau_z = (\beta_{(2),z} - s_n)_+ \le (\beta_{W_z,z} - s_n)_+ \le M_{W_z}^z,\end{aligned}$$

so the winner's $S^\star$ primary target is τ_z if strategic and at least τ_z if honest. Every primary target at z is then at least τ_z, hence pathwise $\sum_{r\le q} m_r \ge q\tau_z$. For the fallback, Lemma H.5 with the comparator $j \in \mathcal{T}(z)$, which delivers $M_\star(z) \ge M_{(2)}(z) \ge \tau_z$ at every Stage-3 visit, gives expected fallback delivery at least $\tau_z H_z - R_K(n)$ conditionally on the switch. Since $q + H_z = N_z$ and $\tau_z \ge M_{(2)}(z) - 2s_n$ on $G^{\mathrm{all}}(z)$, while $M_{(2)}(z) \le 1$ and delivery is nonnegative off $G^{\mathrm{all}}(z)$,

$$\begin{aligned}\mathbb{E}[\text{delivered targets at } z] \ge V_z M_{(2)}(z) \\ - \big[\min\{KL_n, V_z\} + 2s_n N_z + R_K(n) + N_z n^{-4}\big].\end{aligned}$$

Summing over contexts with $\sum_z \min\{KL_n, V_z\} \le \min\{QKL_n, n\}$, $\sum_z N_z \le n$ and $\sum_z N_z n^{-4} \le n^{-3}$, and averaging over the arrival path, gives (ii). Ties in capacity at a context are included because $M_{(2)}(z)$ counts multiplicity; if all K capacities at z coincide then $M_{(2)}(z) = M_\star(z)$ and the mechanism collects the full capacity there up to the same error. The $2s_n n$ term is part of the mechanism's cost rather than of any deviation gain: the announced price sits up to $2s_n$ below $M_{(2)}$, and the mechanism charges the winner only the announced price. □

Proof of Proposition H.2(iii). Every agent may now play an arbitrary strategy in its class, subject to pointwise honesty for $j \in \mathcal{K}_H$. Fix z and choose an honest $h = h_z$ with $\mu_h^z = b(z)$. If $V_z < KL_n$ charge $V_z b(z) \le \min\{KL_n, V_z\}$; let $V_z \ge KL_n$ and charge the KL_n bidding visits at $b(z) \le 1$ each.

The honest bid. Agent h is pulled exactly L_n times in local Stage 1 at z, at rounds that are deterministic given A, because the Stage-1 schedule is reward independent; this holds under every behavioural profile. At each of these rounds t, honesty forces $m_{h,t} \ge \mu_h^z = b(z)$, and by $(*)$ the differences $Y_t - m_{h,t}$ have conditional mean zero given $\mathcal{F}_t^-$ and range 1. The Azuma–Hoeffding inequality for these L_n martingale differences gives $\Pr\{\sum_t (Y_t - m_{h,t}) < -L_n s_n\} \le \exp(-2L_n s_n^2) = (2Kn^4)^{-1}$, and since $\sum_t m_{h,t} \ge L_n b(z)$,

$$\Pr\{\beta_{h,z} < b(z) - s_n\} \le (2Kn^4)^{-1}.$$

Call the complement $B(z)$.

Two cases on $B(z)$. If $W_z = h$, honesty makes every primary target at z at least $b(z)$, so pathwise $\sum_{r \le q} m_r \ge b(z)q \ge (b(z) - 2s_n)q$. If $W_z \ne h$, then $\beta_{h,z}$ is not the largest bid, so $\beta_{(2),z} \ge \beta_{h,z} \ge b(z) - s_n$ and $\tau_z \ge b(z) - 2s_n$; on the complement of the event of Lemma H.4(a), of probability at least $1 - n^{-3}$, part (b) of that lemma gives $\sum_{r \le q} m_r = A_q^z \ge q\tau_z - 1 - 2b_n(q) \ge (b(z) - 2s_n)q - 1 - 2b_n(N_z)$.

The fallback. Whenever a switch occurs, Lemma H.5 with the comparator h, whose conditional mean when selected is at least $b(z)$ by honesty whatever its strategy and whatever the other agents do, gives expected fallback delivery at least $b(z)H_z - R_K(n) \ge (b(z) - 2s_n)H_z - R_K(n)$. This is the step for which the fresh restart over all agents is essential: it keeps an honest comparator eligible after a statistically triggered switch, even though the platform cannot identify the honest agents.

Accounting. Adding the primary and fallback bounds with $q + H_z = N_z$, and charging at most N_z on each of the two bad events $B(z)^c$ and the monitor event,

$$\begin{aligned}\mathbb{E}[\text{delivered targets at } z] \ge V_z b(z) \\ - \big[\min\{KL_n, V_z\} + 2s_n N_z + 1 + 2b_n(N_z) \\ + R_K(n) + N_z((2Kn^4)^{-1} + n^{-3})\big].\end{aligned}$$

Summing over contexts, with the Jensen and Cauchy–Schwarz steps of the proof of (i) for $\sum_z \mathbb{E}[2b_n(N_z)] \le \sqrt{2\ell_n Qn}$, with $\sum_z N_z((2Kn^4)^{-1} + n^{-3}) \le 3n^{-2}$, and averaging over the arrival path gives (iii). The bound is uniform over the joint feasible behavioural profile: the argument never assumed that any agent other than h does anything in particular, and h itself is only required to be honest, so no independence across agents is used anywhere. □

Proposition H.7 (The elicitation term is attained)**.** *Fix $K \ge 2$, a finite $\mathcal{X}_F$ with $|\mathcal{X}_F| = Q$, and an instance in which one agent i is the unique top agent at every context, with $\mu_i^z < M_i^z$ for every z if i is honest. Let the arrival sequence be any deterministic sequence with $V_z \ge KL_n$ for every z, which exists only when $n \ge QKL_n$. Put $m'(z) := \max\{(M_i^z + V_i^z)/2,\ \mu_i^z \mathbf{1}\{i \text{ honest}\}\}$ and $c_\varphi := \min_z(\varphi_i(M_i^z) - \varphi_i(m'(z)))$. For every n with $4s_n < \min_z(M_i^z - V_i^z)$, the deviation $S'_{i,n}$ that delivers $m'(z)$ on every Stage-1 pull at every z and follows $S_i^\star$ afterwards lies in $\mathcal{B}_{i,n}^{\mathrm{ctx}}$ and satisfies*

$$u_i(n; S'_{i,n}, S^\star_{-i,n}) - u_i(n; S_n^\star) \ge c_\varphi QL_n - 4Qn^{-3},$$

which is $\Theta(QL_n)$ provided $c_\varphi > 0$, that is, provided φ_i is strictly increasing on $[m'(z), M_i^z]$ at every z. Linearity gives $c_\varphi = \min_z(M_i^z - m'(z)) > 0$ for free.

Proof. The deviation is measurable with respect to the context and the local phase, which the mechanism announces, hence lies in the class; feasibility holds because $(M_i^z + V_i^z)/2 < M_i^z$ and, in the honest case, $\mu_i^z < M_i^z$, and honesty holds by the definition of m' as a maximum. Couple the two executions by the same arrivals, seeds, priorities, fallback coins and private coins.

Stage 1 is deterministic. Both executions pull i at the same L_n Stage-1 rounds of every context (Lemma H.6(a)), and all Stage-1 targets are constants, so the Stage-1 utility difference is the deterministic number $\sum_z L_n(\varphi_i(M_i^z) - \varphi_i(m'(z))) \ge c_\varphi QL_n$ on every path.

The post-bidding phases coincide on a good event. Each bid is the mean of L_n independent Bernoulli variables with a constant mean; there are at most $2KQ$ distinct bid variables, the bids of $j \ne i$ being shared by the two executions, and each leaves its band of half-width s_n with probability at most $2\exp(-2L_n s_n^2) = (Kn^4)^{-1}$, so the event E that all of them concentrate has $\Pr(E^c) \le 2Qn^{-4}$. On E, at every context z: $\beta_{j,z} \le V_i^z + s_n$ for $j \ne i$; in the baseline $\beta^\star_{i,z} \ge M_i^z - s_n > V_i^z + s_n$; and in the deviation $\beta'_{i,z} \ge m'(z) - s_n \ge (M_i^z + V_i^z)/2 - s_n > V_i^z + s_n$, both because $4s_n < M_i^z - V_i^z$. So i is the strict largest bidder at every z in both executions and both face the same price $(\max_{j \ne i} \beta_{j,z} - s_n)_+$, which by Lemma H.6(b) is the same random variable in the two. After local Stage 1 at z both executions run $S^\star$ with the same winner, the same price, the same local phase, the same priority, the same fallback coins and, since the winner's rth primary pull at z is at the same round in both, the same seeds; an induction on the visits to z shows that the two post-bidding local executions coincide pathwise, so on E the post-bidding utilities are equal.

Accounting. Off E the post-bidding utilities differ by at most $2n$, contributing at most $2n \cdot 2Qn^{-4} = 4Qn^{-3}$. Adding the deterministic Stage-1 saving proves the claim. □

The witness is simply not bidding at any context, and it shows that the Q in the elicitation term is not an artefact: a deviator that is uniquely top at many contexts collects the elicitation saving once per context. Hence the restriction $Q = o(n^{1/3}(\log n)^{-1/3})$ is sharp up to the instance constant c_φ and the constraint $QKL_n \le n$ needed for the construction. It says nothing about whether a different mechanism could avoid paying L_n per context, and here the primitives at different contexts are unrelated, so no hypothesis permits elicitation effort to be shared across them. That last sentence is specific to the finite case and does not carry to the continuum of Subsection H.3, where Lipschitz continuity makes neighbouring

cubes informative about one another. There the only sharing this mechanism performs is through the cube side h_n, which the three-way balance already optimises, and the corresponding tightness statement is a lower bound against CG-MESP rather than against every mechanism.

Remark H.8 (Two-top conditions and regularity are not used). *Nothing in Proposition H.2 requires $|\mathcal{T}(z)| \geq 2$, positive context masses, continuity, or i.i.d. arrivals, in contrast with Theorem F.8 and Theorem F.7. The price of this is that the benchmark is the second capability $M_{(2)}(z)$ rather than the first, which is the same trade the scalar Theorem 3.2 makes: the two-top condition is absorbed into the benchmark rather than removed. If the platform is in addition given an upper bound $\bar{Q} \geq Q$, taking $L_n := \min\{\lfloor n/(2K)\rfloor, \lceil (n/(\bar{Q}K))^{2/3}(\log(en))^{1/3}\rceil\}$ balances the two leading terms QKL_n and $2s_n n$, making both of order $(\bar{Q}K)^{1/3}n^{2/3}(\log n)^{1/3}$; with Q unknown the statement above keeps QKL_n.*

Remark H.9 (Exogeneity is used once, and is not removable). *Part (i) uses the exogeneity half of* (A2) *at exactly one place, the coupling of Lemma H.6(a): the two executions must have the same Stage-1 rounds at every context and pull the same agent at each. It uses* (A3-P) *at one further place, the standing conditioning on A, which is a device of this subsection and not of the result; Appendix H.3 proves the corresponding statement without it. Parts (ii) and (iii) couple nothing and use the exogeneity half nowhere, but as written they run under the standing conditioning on A and therefore use* (A3-P) *as well; their continuum analogues in Appendix H.3 need only clauses of* (A3). *If arrivals may react to a realised reward, part (i) is false and not merely unproved: the two executions separate at the first round at which the deviator's target differs from its capacity and the seed falls between them, and thereafter they visit different contexts at different rounds, so there is nothing to compare. The monitor argument of Lemma H.4 is not what needs exogeneity, contrary to what its use of $\sigma(A) \subseteq \mathcal{F}_0$ suggests; written at the pre-seed field with the indicator stopped, as in Lemma H.27 below, it uses no property of the arrival path at all.*

H.2 When the platform sees a label and not the context

This subsection proves Theorem A.1 and Proposition A.2. The platform no longer observes the context of a round: it observes a label produced by a classifier, and it runs CL-MESP of Appendix H.1 on the labels, unchanged and with the label in the role of the context everywhere. No quantity the mechanism computes refers to a context, and the platform is given nothing about the noise: not the confusion matrix, not the error rate, not the posterior. Two things do change. The proposed winner meets the announced price in the mean in which the price was elicited, which is the mean over the contexts consistent with the label and not the value at the realised context; and the hypothesis on arrivals is strengthened, because the label of a round no longer determines the primitives of that round. The first change is in the profile $S^\star_{\mathcal{L}}$ below, the second in condition (N).

Labels, contexts and mixture primitives. The context set is a finite $\mathcal{X}_F$, unknown to the platform, with all the notation of Appendix H.1: μ_i^z, M_i^z, $M_\star(z)$, $\mathcal{T}(z)$, V_i^z, $\delta_i(z)$, $M_{(2)}(z)$ and $b(z)$. The label set is a finite $\mathcal{L}$ with $Q_{\mathcal{L}} := |\mathcal{L}|$, also unknown to the platform; a generic label is $a \in \mathcal{L}$, the label of round t is α_t, and $\alpha := (\alpha_1, \ldots, \alpha_n)$ is the label sequence, which plays throughout the role played by the arrival sequence A in Appendix H.1. A *posterior kernel* $\pi = (\pi_a)_{a\in\mathcal{L}}$ is fixed: each π_a is a probability on $\mathcal{X}_F$, the same at every round. In the classifier reading $\pi_a(z) = \Pr(x = z \mid \alpha = a)$, but no result below uses more than the kernel. A bar denotes the π_a-mixture of a context primitive:

$$\bar{M}_j(a) := \sum_z \pi_a(z) M_j^z, \qquad \bar{\mu}_j(a) := \sum_z \pi_a(z)\mu_j^z,$$
$$\bar{\varphi}_j(a) := \sum_z \pi_a(z)\varphi_j(\mu_j^z), \qquad \bar{V}_i(a) := \max_{j\neq i} \bar{M}_j(a),$$
$$\bar{\delta}_i(a) := \bar{V}_i(a) - \bar{M}_i(a), \qquad \bar{b}(a) := \max_{h\in\mathcal{K}_H} \bar{\mu}_h(a),$$

and $\bar{M}_{(2)}(a)$ is the second largest of $(\bar{M}_1(a), \ldots, \bar{M}_K(a))$ counting multiplicity. Visit counts and elicitation counts are those of a label: V_a is the number of rounds with $\alpha_t = a$, $N_a := (V_a - KL_n)_+$, and $l_i(a)$ is the number of Stage-1 pulls agent i receives at a, so $\sum_a l_i(a) \leq \min\{Q_{\mathcal{L}}L_n, n\}$ and $\sum_a \min\{V_a, KL_n\} \leq \min\{Q_{\mathcal{L}}KL_n, n\}$. Write τ_a, W_a, $\beta_{j,a}$ and q_a for the price, the winner, the bids and the primary tenure at label a, and T_a for the round of the KL_nth visit to a. When the labels name contexts, so that $\mathcal{L} \subseteq \mathcal{X}_F$, write $\bar{\varepsilon} := n^{-1}\sum_{t\leq n} \Pr(x_t \neq \alpha_t)$ for the average per-round label error rate.

The round, and all its draws. A round t runs as follows. The label α_t is revealed to the platform; the mechanism draws its round randomisation ζ_t and selects I_t, which in Stages 1 and 2 is the deterministic branch and in Stage 3 is $I_t := \mathrm{inv}(p_t, \zeta_t)$, where $\mathrm{inv}(p, u) := \min\{l : p_1 + \cdots + p_l \geq u\}$ and p_t is the label's own EXP3 selection law; the selected agent then reads the task, that is, it observes the true context x_t, and chooses $m_{I_t,t} \in [0, M_{I_t}(x_t)]$, an honest agent subject to $m_{I_t,t} \geq \mu_{I_t}(x_t)$; the seed U_t is drawn and $Y_t = \mathbf{1}\{U_t \leq m_{I_t,t}\}$. The platform never observes x_t. The selected agent's utility is $1 + \varphi_i(\mu_i(x_t)) - \varphi_i(m_{i,t}) \in [0, 2]$ by (1), and revenue is $P(n; S) = \mathbb{E}\sum_t Y_t$ as before. The complete list of draws, in the order in which they are made, is: the label sequence α; the agents' private coin arrays $\xi_1, \ldots, \xi_K$; the priorities $(\Pi_a)_{a\in\mathcal{L}}$; and then, for $t = 1, \ldots, n$ in this order, ζ_t, x_t, U_t. Every other random quantity of an execution, that is every selection, bid, price, switch, message, target and reward, is a measurable function of these draws and of the strategies: the mechanism has no other randomness, Stages 1 and 2 select deterministically from the label's state, the water-filling level of the profile below is fixed by a deterministic rule, and an agent that randomises at round t reads a coordinate of its own array ξ_i, which is why the arrays are drawn before round 1 and are therefore before every context and every seed. The fields of round t are

$$\mathcal{F}_t^\circ := \sigma\big(\mathcal{F}_{t-1}, \zeta_t\big), \qquad \mathcal{F}_t^- := \sigma\big(\mathcal{F}_t^\circ, x_t, m_{I_t,t}\big), \qquad \mathcal{F}_t := \sigma\big(\mathcal{F}_t^-, U_t\big),$$

with $\mathcal{F}_0 := \sigma\big(\alpha, \xi_1, \ldots, \xi_K, (\Pi_a)_{a\in\mathcal{L}}\big)$. So $\mathcal{F}_t^\circ$ is the field before the round's context, every one of these fields contains the whole label sequence, and I_t is $\mathcal{F}_t^\circ$-measurable.

The proposed profile $S^\star_{\mathcal{L}}$, and water filling. Put $\mathrm{lo}_j(z) := \mu_j^z$ if j is honest and $\mathrm{lo}_j(z) := 0$ otherwise, and $\overline{\mathrm{lo}}_j(a) := \sum_z \pi_a(z)\mathrm{lo}_j(z)$.

For $y \in [\overline{\mathrm{lo}}_j(a), \bar{M}_j(a)]$ let

$$\Psi_{j,a}(y) := \min\Big\{\sum_z \pi_a(z)\varphi_j(m(z)) : \mathrm{lo}_j(z) \le m(z) \le M_j^z \text{ for every } z,\ \sum_z \pi_a(z)m(z) = y\Big\} \quad (19)$$

be the cheapest way of delivering the label mean y at a, and for $c \in [0,1]$ let $m_c(z) := \min\{\max\{c, \mathrm{lo}_j(z)\}, M_j^z\}$ be the water-filling allocation at level c. In $S^\star_{\mathcal{L}}$, agent j delivers $M_j(x_t)$ at every Stage-1 and every Stage-3 visit to every label, and as the Stage-2 winner at a label a carrying the price τ it delivers $m_c(x_t)$ with c the smallest level whose π_a-mean is

$$y_j^\star(a,\tau) := \max\big\{\overline{\mathrm{lo}}_j(a),\ \min\{\tau, \bar{M}_j(a)\}\big\}. \quad (20)$$

Such a level exists because $c \mapsto \sum_z \pi_a(z)m_c(z)$ is continuous and nondecreasing from $\overline{\mathrm{lo}}_j(a)$ at $c = 0$ to $\bar{M}_j(a)$ at $c = 1$, and taking the smallest one makes the rule deterministic. The profile is feasible and honest at every context, is implementable from the messages of CL-MESP together with π_a and the realised context, and its target at a visit to a is a function of $(a$, the phase at $a, \tau_a, j, x_t)$ alone, so the locality of Lemma H.3 is unaffected. The variant in which no one knows the kernel is Corollary H.17.

The behavioural class. Let $\mathcal{B}^{\mathcal{L}}_{i,n}$ be the class of behaviours in which, after being selected at round t, agent i chooses $m_{i,t} \in [0, M_i(x_t)]$ as an arbitrary measurable function of everything realised before it in its execution: of n, the whole label sequence α and hence every visit count and the exact global round index, the current context x_t, all past contexts, targets, rewards and seeds at every label, all mechanism randomisations already made, all the coin arrays, and all messages. An honest agent satisfies $m_{i,t} \ge \mu_i(x_t)$ after every history. The class does not contain behaviours that read a context not yet realised, and Remark H.10 explains why that exclusion is the same statement as (N) rather than an extra restriction. It contains $\mathcal{B}^{\mathrm{ctx}+}_{i,n}$ with the label read for the context, and it contains the smaller class in which the deviator sees only its own past selections and the contexts there; every bound below is proved against a response measurable with respect to the field just before the round's context, so it holds verbatim for all three.

The hypothesis. Condition (N) is stated once and is shared by the two results of this subsection: Proposition H.11 assumes it, and Proposition H.18 is about two context paths of which one satisfies it and one does not.

> **(N), fresh draws given the label sequence.** No agent's behaviour can change the realised sequence of labels, and the joint law of the label sequence and of the agents' private coin arrays is the same under every strategy profile. Each remaining draw of the list above has its specified law conditionally on the field generated by everything realised before it *together with the entire label sequence*: each priority Π_a is uniform on the permutations of $[K]$ given $\sigma(\alpha, \xi_1, \dots, \xi_K)$ and the priorities drawn before it; the mechanism's round-t randomisation ζ_t is uniform on $[0,1]$ given $\mathcal{F}_{t-1}$; the context x_t has law π_{α_t} given $\mathcal{F}_t^\circ$; and the seed U_t is uniform on $[0,1]$ given $\mathcal{F}_t^-$. The selected agent's target is chosen between x_t and U_t and is unconstrained: whatever its strategy specifies, it is part of everything realised before the seed, and the seed clause is stated conditionally on it.

Read with the label sequence in the role of the context sequence, the two halves of (N) are the two halves of (A2) of Section 2.3, together with one clause more, the clause on x_t, for which (A2) has no counterpart because there the context of a round is observed and is not a draw. Nothing else is assumed: the label sequence may have an arbitrary law, it may be adversarial and chosen in advance, agents may know all of it, and the coin arrays may have an arbitrary joint law given it. That the joint law of the label sequence and the coin arrays is profile independent is stated explicitly rather than treated as automatic, because the coupling of Lemma H.14(f) needs one fixed array while a deviator's own randomisation changes with its strategy; any behavioural strategy can be read off a fixed uniform array, so the clause costs nothing.

Remark H.10 (What (N) forbids, and why the class excludes future contexts). *The clause on x_t is* posterior stationarity*: given the whole label sequence, the coin arrays and the entire past, the context of a round is a fresh draw from its label's posterior. It forbids a bursty or drifting context order even behind a classifier whose errors are independent, since then the context at t predicts the context at $t+1$ given the labels; a classifier whose error depends on the round or on what the agents delivered; contexts that react to a past reward; and a coin array that carries information about a future context. It is implied by the following, which is the reading Theorem A.1 has in mind: the pairs (x_t, α_t) are i.i.d. with a law $p(z)C(z,a)$, the coin arrays are a function of α and of randomness independent of the pairs, and the priorities, the mechanism randomisations and the seeds are fresh independent uniforms; then $\pi_a(z) = p(z)C(z,a)/\sum_{z'} p(z')C(z',a)$. It is strictly stronger than the hypothesis of Proposition H.2, which assumes nothing about how contexts arrive beyond* (A2)*, and Proposition H.18 shows that the strengthening cannot be dispensed with. The exclusion of future contexts from $\mathcal{B}^{\mathcal{L}}_{i,n}$ is not a separate restriction: if a target at round t could read $x_{t'}$ for some $t' > t$, then $x_{t'}$ would be realised before a quantity on which the clause for $x_{t'}$ conditions, so that clause would be false in the execution, which is the same statement as the clause itself.*

Exact constants. The following is Theorem A.1 with every constant written out: part (i) is the incentive guarantee, parts (ii) and (iii) the revenue and robustness guarantees, and part (iv) the statement that a label-level compliant winner is not convicted. The clauses of (N) are used as follows: the clause on x_t and the seed clause everywhere; the clause on ζ_t in the fallback only; the exogeneity half and the clause on the coin arrays in part (i) only, through the coupling of two executions; and the uniformity of the priorities nowhere, since every statement about a winner holds for whoever wins.

Proposition H.11 (Label-keyed MESP, exact form). *Assume* (N), *$K \ge 2$, $n \ge 2K$, a finite $\mathcal{X}_F$, a finite $\mathcal{L}$ with $|\mathcal{L}| = Q_{\mathcal{L}}$, and a fixed posterior kernel; assume a nonempty honest set, which is used in part (iii) only. Run* CL-MESP *keyed on the label with the profile $S^\star_{\mathcal{L}}$, and put $\mathcal{W}_i := \{a \in \mathcal{L} : N_a \ge 1,\ \bar{\delta}_i(a) \le 2s_n\}$. Then:*

(i) *for every agent i and every $S'_{i,n} \in \mathcal{B}^{\mathcal{L}}_{i,n}$,*

$$u_i(n; S'_{i,n}, S^\star_{\mathcal{L},-i}) - u_i(n; S^\star_{\mathcal{L}}) \le \mathbb{E}\Big[\sum_a l_i(a)\Big] + \Lambda\,\mathbb{E}\Big[\sum_{a\in W_i}\big(1 + b_n(N_a) + R_K(n)\big)\Big] + n^{-2} + 2n^{-3},$$

and consequently

$$\sup_{S'_{i,n}\in\mathcal{B}^{\mathcal{L}}_{i,n}}\big[u_i(n; S'_{i,n}, S^\star_{\mathcal{L},-i}) - u_i(n; S^\star_{\mathcal{L}})\big] \le \min\{Q_{\mathcal{L}}L_n, n\} + \Lambda\Big[Q_{\mathcal{L}} + \sqrt{\tfrac{1}{2}\ell_n Q_{\mathcal{L}} n} + Q_{\mathcal{L}}R_K(n)\Big] + n^{-2} + 2n^{-3};$$

(ii)

$$P(n; S^\star_{\mathcal{L}}) \ge \mathbb{E}\sum_{t\le n}\bar{M}_{(2)}(\alpha_t) - \big[\min\{Q_{\mathcal{L}}KL_n, n\} + 2s_n n + n^{-3} + n^{-2}/2\big];$$

(iii) *for every feasible behavioural profile $S = (S_1, \dots, S_K)$ with $S_j \in \mathcal{B}^{\mathcal{L}}_{j,n}$, in which every honest agent respects pointwise honesty and every agent, up to all K of them, may otherwise deviate,*

$$P(n; S) \ge \mathbb{E}\sum_{t\le n}\bar{b}(\alpha_t) - \Big[\min\{Q_{\mathcal{L}}KL_n, n\} + 2s_n n + \mathbb{E}\sum_{a:N_a\ge 1}\big(1 + b_n(N_a) + R_K(n)\big) + n^{-3}\Big];$$

(iv) *fix an agent j and a label a, and suppose j delivers $M_j(x_t)$ at its Stage-1 pulls at a and, at every primary pull at a, a target whose π_a-mean is at least $\min\{\tau_a, \bar{M}_j(a)\}$, as $S^\star_{\mathcal{L}}$ does. Then, whatever the other $K-1$ agents do within their classes,*

$$\Pr\big(W_a = j \text{ and the monitor at } a \text{ switches} \,\big|\, \alpha\big) \le (2Kn^4)^{-1} + N_a(2n^4)^{-1},$$

so $\Pr(j$ is switched out at some label $\mid \alpha) \le Q_{\mathcal{L}}(2Kn^4)^{-1} + (2n^3)^{-1}$; and if all agents play $S^\star_{\mathcal{L}}$ then $\Pr(a$ switch occurs anywhere $\mid \alpha) \le \#\{a : V_a \ge KL_n\}(2n^4)^{-1} + (2n^3)^{-1}$.

Moreover, if the labels name contexts, so that $\mathcal{L} \subseteq \mathcal{X}_F$, then both benchmarks degrade by at most the label error rate,

$$\mathbb{E}\sum_{t\le n}M_{(2)}(x_t) - \mathbb{E}\sum_{t\le n}\bar{M}_{(2)}(\alpha_t) \le \bar{\varepsilon}n, \qquad \mathbb{E}\sum_{t\le n}b(x_t) - \mathbb{E}\sum_{t\le n}\bar{b}(\alpha_t) \le \bar{\varepsilon}n,$$

the second being an equality at 0 when exactly one agent is honest, and the constant 1 in front of $\bar{\varepsilon}n$ is attained in both (Lemma H.16).

With $L_n \asymp n^{2/3}(\log n)^{1/3}$, $\ell_n = \log 2 + 4\log n$, $2s_n n = O(n^{2/3}(\log n)^{1/3})$ and $R_K(n) = O(\sqrt{Kn\log K})$, every bracket is $O_K\big(Q_{\mathcal{L}}L_n + \sqrt{Q_{\mathcal{L}}n\log n} + Q_{\mathcal{L}}\sqrt{Kn\log K}\big)$, which is the rate of Proposition H.2 with the number of labels in place of the number of contexts, and all of them are $o(n)$ when $Q_{\mathcal{L}} = o(n^{1/3}(\log n)^{-1/3})$. Only that direction is proved here; the analogue of Proposition H.7 for labels is not checked in this appendix. Two differences from Proposition H.2 are differences in the bound and not in the setting, and both are sharpenings. Part (i) carries no capability-gap term and no instance threshold on n: the good event $G(a)$ below controls the deviator's own elicitation bid as well as the others', which rules out a win at a label with $\bar{\delta}_i(a) > 2s_n$ outright, and at a label with $0 < \bar{\delta}_i(a) \le 2s_n$ the top-winning argument applies with $\min\{\bar{V}_i(a), \bar{M}_i(a)\}$ in place of $\bar{V}_i(a)$. And the monitor slack is $1 + b_n(N_a)$ rather than $1 + 2b_n(N_a)$, because it is taken in expectation against a martingale identity rather than pathwise on a two-sided event. Both sharpenings apply to Proposition H.2(i) itself, with a point-mass posterior; they are recorded here and are not applied there.

Frozen responses and the effective mean. The single probabilistic input of Appendix H.1, the identity $(*)$, is not enough here, because a price is a label quantity while a target is chosen at a context. What replaces it is a second conditional mean, taken before the context is drawn.

Lemma H.12 (Frozen responses and the effective mean)**.** *Assume* (N) *and fix a round t, writing $a := \alpha_t$. Then:*

(a) *there is a map $f_t : \mathcal{X}_F \to [0,1]$ with $f_t(z)$ $\mathcal{F}_t^\circ$-measurable and $\mathrm{lo}_{I_t}(z) \le f_t(z) \le M^z_{I_t}$ for every z, such that $m_{I_t,t} = f_t(x_t)$;*

(b) *for every bounded $g : \mathcal{X}_F \times [0,1] \to \mathbb{R}$, $\mathbb{E}[g(x_t, m_{I_t,t}) \mid \mathcal{F}_t^\circ] = \sum_z \pi_a(z) g(z, f_t(z))$;*

(c) *$\mathbb{E}[Y_t \mid \mathcal{F}_t^-] = m_{I_t,t}$ and $\mathbb{E}[Y_t \mid \mathcal{F}_t^\circ] = \bar{m}_t := \sum_z \pi_a(z) f_t(z)$, the* effective mean *of the round, with $Y_t - \bar{m}_t$ lying in $[-\bar{m}_t, 1-\bar{m}_t]$, an interval of length 1 with $\mathcal{F}_t^\circ$-measurable endpoints, and $\overline{\mathrm{lo}}_{I_t}(a) \le \bar{m}_t \le \bar{M}_{I_t}(a)$;*

(d) *on $\{I_t = j\}$, $\mathbb{E}\big[1 + \varphi_j(\mu_j(x_t)) - \varphi_j(m_{j,t}) \,\big|\, \mathcal{F}_t^\circ\big] \le 1 + \bar{\varphi}_j(a) - \Psi_{j,a}(\bar{m}_t)$, where $\bar{m}_t$ lies in the domain $[\overline{\mathrm{lo}}_j(a), \bar{M}_j(a)]$ of $\Psi_{j,a}$ by (c); equality holds when f_t is a water-filling allocation.*

Proof. (a) The set $\mathcal{X}_F$ is finite and a strategy is a measurable function of its inputs, so let $f_t(z)$ be the target the selected agent's strategy chooses on the realised pre-context history when the context is z; every input of that function other than x_t is $\mathcal{F}_t^\circ$-measurable, the selection I_t included. Feasibility and honesty hold at every z and not only at the realised one, because a strategy maps every history into the feasible set.

(b) Write $g(x_t, f_t(x_t)) = \sum_z \mathbf{1}\{x_t = z\} g(z, f_t(z))$; each factor $g(z, f_t(z))$ is $\mathcal{F}_t^\circ$-measurable, so it comes out of the conditional expectation, and the clause of (N) for x_t gives $\Pr(x_t = z \mid \mathcal{F}_t^\circ) = \pi_a(z)$. The indicator here multiplies a quantity measurable before the context, never after.

(c) The first identity is the seed clause of (N), $m_{I_t,t}$ being $\mathcal{F}_t^-$-measurable; the second is the first together with (b) applied to $g(z,m) = m$ and the tower property at $\mathcal{F}_t^\circ$. The range statement holds because $Y_t \in \{0,1\}$ and $\bar{m}_t$ is $\mathcal{F}_t^\circ$-measurable, and the two bounds on $\bar{m}_t$ are the π_a-averages of the bounds in (a).

(d) Apply (b) with $g(z,m) = 1 + \varphi_j(\mu_j^z) - \varphi_j(m)$, which gives $1 + \bar{\varphi}_j(a) - \sum_z \pi_a(z)\varphi_j(f_t(z))$; by (a) the map f_t is a feasible allocation of π_a-mean $\bar{m}_t$, so the subtracted sum is at least $\Psi_{j,a}(\bar{m}_t)$ by (19), with equality when f_t attains that minimum, which a water-filling allocation does by Lemma H.13(b). □

This is the one place where (N) is used essentially. A price is set from elicitation rewards collected at queries of several contexts, so it is a label quantity; if the conditional law of the context given everything realised before it were not the label's posterior, the effective mean of a round would not be a function of the allocation and the label alone, and the quantity the monitor compares with

the price would not be the quantity the price was built from. Note also what the lemma makes harmless: the deviator's knowledge of past contexts, its own and everyone else's, is inside $\mathcal{F}_t^\circ$, the field against which every bound below is stated, and the clause on x_t says precisely that this knowledge carries no information about the current context beyond π_{α_t}.

Lemma H.13 (The effective cost of a label mean)**.** *Fix j and a and write $\pi, \mathrm{lo}, M, \overline{\mathrm{lo}}, \bar{M}, \Psi, \bar{\varphi}$ without indices. Then:*

(a) *the minimum in* (19) *is attained, and Ψ is convex, nondecreasing and Λ-Lipschitz on $[\overline{\mathrm{lo}}, \bar{M}]$, with $0 \le 1 + \bar{\varphi} - \Psi(y) \le 2$ there;*

(b) *for every level $c \in [0,1]$, the water-filling allocation m_c minimises $\sum_z \pi(z)\varphi_j(m(z))$ over all feasible allocations m with the same π-mean as m_c, so it attains Ψ at that mean; and for every $y \in [\overline{\mathrm{lo}}, \bar{M}]$ there is a level c with $\sum_z \pi(z) m_c(z) = y$. Both statements hold for every convex nondecreasing φ_j and every π;*

(c) *if j is strategic and $y = \min\{\tau, \bar{M}\}$, then $m_c(z) \ge \min\{\tau, M^z\}$ at every z of positive posterior mass; for an honest j this pointwise domination can fail.*

Proof. (a) The feasible set is a nonempty compact polytope, nonempty because y is a convex combination of $\overline{\mathrm{lo}}$ and $\bar{M}$ and the same combination of lo and M is feasible, and φ_j is continuous, so the minimum is attained. Convexity: a convex combination of optimal allocations for y_1 and y_2 is feasible for the combined mean, and its cost is at most the combined cost by convexity of φ_j. Monotonicity: from an optimum m at $y > y'$, the allocation $\mathrm{lo} + \theta(m - \mathrm{lo})$ with $\theta = (y' - \overline{\mathrm{lo}})/(y - \overline{\mathrm{lo}}) \in [0,1)$ is feasible with mean y' and is pointwise below m, so $\Psi(y') \le \Psi(y)$ because φ_j is nondecreasing. Lipschitz: from an optimum m at $y < y''$, the allocation $m + \theta(M - m)$ with $\theta = (y'' - y)/(\bar{M} - y) \in (0,1]$ is feasible with mean y'', and $\sum_z \pi(z)[\varphi_j(m(z) + \theta(M^z - m(z))) - \varphi_j(m(z))] \le \Lambda\theta \sum_z \pi(z)(M^z - m(z)) = \Lambda(y'' - y)$, so $\Psi(y'') \le \Psi(y) + \Lambda(y'' - y)$. For the range, at an optimum m we have $1 + \bar{\varphi} - \Psi(y) = 1 + \sum_z \pi(z)[\varphi_j(\mu_j^z) - \varphi_j(m(z))]$, and each bracket lies in $[-1, 1]$ by (1).

(b) Fix c and let g be the right derivative of φ_j at c if $c < 1$ and its left derivative if $c = 1$; then $g \in [0, \Lambda]$ and $\varphi_j(x) \ge \varphi_j(c) + g(x - c)$ on $[0,1]$, since one-sided derivatives of a convex function are subgradients. We claim that $\varphi_j(x) - \varphi_j(m_c(z)) \ge g(x - m_c(z))$ for every z and every feasible $x \in [\mathrm{lo}(z), M^z]$. If $m_c(z) = c$ this is the supporting line. If $m_c(z) = M^z < c$ then $x \le M^z$ and the chord slope $(\varphi_j(M^z) - \varphi_j(x))/(M^z - x)$ is at most the left derivative of φ_j at M^z, hence at most g, because one-sided derivatives of a convex function are nondecreasing. If $m_c(z) = \mathrm{lo}(z) > c$, which forces $c < 1$, then $x \ge \mathrm{lo}(z)$ and the chord slope is at least the right derivative at $\mathrm{lo}(z)$, hence at least g. Weighting by π and using that x and m_c have the same mean gives $\sum_z \pi(z)\varphi_j(x(z)) \ge \sum_z \pi(z)\varphi_j(m_c(z))$, which is the first statement. The existence of a level with mean y is the continuity and monotonicity recorded after (20).

(c) If $\tau < \bar{M}$ then $y = \tau$, and since $\sum_z \pi(z) \min\{c, M^z\} \le c$ while this mean equals τ, we get $c \ge \tau$ and $m_c(z) = \min\{c, M^z\} \ge \min\{\tau, M^z\}$ at every z. If $\tau \ge \bar{M}$ then $y = \bar{M}$, and a level whose mean is $\bar{M}$ has $\min\{c, M^z\} = M^z$ at every z of positive posterior mass, so $m_c(z) = M^z \ge \min\{\tau, M^z\}$ there. For the failure in the honest case take $\pi = (\frac{1}{2}, \frac{1}{2})$, $\mu = (0.9, 0)$, $M = (1,1)$ and $\tau = 0.5$: the level target is $\max\{0.45, 0.5\} = 0.5$, water filling delivers $(0.9, 0.1)$, and the pointwise honest rule $\max\{\mu, \min\{\tau, M\}\}$ delivers $(0.9, 0.5)$. □

Part (c) is why water filling is the rule and not merely one rule with the right label mean: for a strategic winner it delivers at least the pointwise rule $\min\{\tau_a, M_j^z\}$ of $S^\star$ at every context that can arrive, so it is at least as compliant in the pointwise sense as well. For an honest winner that comparison can go the other way, and nothing below depends on it. Under a linear cost every allocation with the same label mean costs the same, so the choice of rule matters only through this pointwise comparison and through part (b), which is what carries a general cost.

Elicitation at a label.

Lemma H.14 (Bids at a label)**.** *Condition on α and fix a label a with $V_a \ge KL_n$, so that the Stage-1 rounds at a and the agent pulled at each are deterministic. For each agent k let $\beta_{k,a}^{\max}$ and $\beta_{k,a}^{\min}$ be the averages over k's L_n Stage-1 rounds at a of $\mathbf{1}\{U_t \le M_k(x_t)\}$ and of $\mathbf{1}\{U_t \le \mu_k(x_t)\}$, which are functions of the draws alone. Then:*

(a) *$\beta_{k,a}^{\max}$ is an average of L_n i.i.d.* Bernoulli$(\bar{M}_k(a))$ *variables and $\beta_{k,a}^{\min}$ of L_n i.i.d.* Bernoulli$(\bar{\mu}_k(a))$ *variables;*

(b) *$G(a) := \bigcap_k \{|\beta_{k,a}^{\max} - \bar{M}_k(a)| \le s_n\}$ has $\Pr(G(a)^c) \le n^{-4}$; $G^{\max}(a) := \bigcap_k \{\beta_{k,a}^{\max} \le \bar{M}_k(a) + s_n\}$ has $\Pr(G^{\max}(a)^c) \le (2n^4)^{-1}$; for a single agent k, $\Pr(\beta_{k,a}^{\max} > \bar{M}_k(a) + s_n) \le (2Kn^4)^{-1}$; and, with h^* an honest agent attaining $\bar{b}(a)$, $B(a) := \{\beta_{h^*,a}^{\min} \ge \bar{b}(a) - s_n\}$ has $\Pr(B(a)^c) \le (2Kn^4)^{-1}$;*

(c) *pathwise and under every profile, $\beta_{k,a}^{\min} \mathbf{1}\{k \text{ honest}\} \le \beta_{k,a} \le \beta_{k,a}^{\max}$, with equality on the right for an agent following $S_{\mathcal{L}}^\star$;*

(d) *on $G^{\max}(a)$, under every profile, the winner satisfies $\tau_a \le \bar{M}_{W_a}(a)$;*

(e) *with every $j \ne i$ following $S_{\mathcal{L}}^\star$ and i arbitrary: on $G(a)$, if i wins at a then $\bar{\delta}_i(a) \le 2s_n$ and $\tau_a \le \min\{\bar{V}_i(a), \bar{M}_i(a)\}$;*

(f) *in the two executions coupled on the draws, if i wins at a in the deviating execution then it wins at a in the baseline, and both announce the same price $\tau_{i,a} = (\max_{j \ne i} \beta_{j,a} - s_n)_+$;*

(g) *under $S_{\mathcal{L}}^\star$, on $G(a)$, $\tau_a \ge \bar{M}_{(2)}(a) - 2s_n$.*

Proof. (a) The L_n rounds are distinct and deterministic given α. Applying the clauses of (N) draw by draw in the order of the list, the context at each of these rounds has law π_a given everything before it and the seed is uniform given everything before it, so the pairs (x_t, U_t) at those rounds are, conditionally on α, i.i.d. with law $\pi_a \otimes \mathrm{Unif}[0,1]$; the indicators are therefore i.i.d. Bernoulli with mean $\sum_z \pi_a(z) M_k^z = \bar{M}_k(a)$, and likewise with μ_k.

(b) Hoeffding's inequality with $2L_n s_n^2 = \log(2Kn^4)$ makes a single tail cost $(2Kn^4)^{-1}$; the union over the K agents and both tails costs n^{-4}, and over the K agents and one tail $(2n^4)^{-1}$. For $B(a)$, the agent h^* receives exactly L_n Stage-1 pulls at a under every profile, the schedule being reward independent, and $\beta_{h^*,a}^{\min}$ is the lower tail of (a).

(c) Stage-1 rewards are $\mathbf{1}\{U_t \le m\}$ with $\mathrm{lo}_k(x_t) \le m \le M_k(x_t)$ at the same (x_t, U_t), pointwise honesty giving the lower bound, and $S_{\mathcal{L}}^\star$ delivers $M_k(x_t)$ there.

(d) $\tau_a = (\beta_{(2),a} - s_n)_+ \le (\beta_{W_a,a} - s_n)_+ \le (\beta_{W_a,a}^{\max} - s_n)_+ \le \bar{M}_{W_a}(a)$, by (c) and the definition of $G^{\max}(a)$.

(e) On $G(a)$, if i wins then, by (c) for i, the winning condition, and (c) with equality for the others, $\bar{M}_i(a) + s_n \geq \beta^{\max}_{i,a} \geq \beta_{i,a} \geq \max_{j\neq i} \beta_{j,a} = \max_{j\neq i} \beta^{\max}_{j,a} \geq \bar{V}_i(a) - s_n$, which is $\bar{\delta}_i(a) \leq 2s_n$; moreover $\tau_a = (\max_{j\neq i} \beta_{j,a} - s_n)_+ \leq \bar{V}_i(a)$ and $\tau_a \leq (\beta_{i,a} - s_n)_+ \leq \bar{M}_i(a)$.

(f) Two facts carry over from Lemma H.6(c) and (d). The other agents' Stage-1 targets are $M_j(x_t)$ at the same rounds with the same contexts and the same seeds in both executions, so their bids coincide; and $\beta'_{i,a} \leq \beta^{\max}_{i,a}$ pathwise, by feasibility at the common context. The tie argument and the identification of the price are then unchanged.

(g) Order statistics are 1-Lipschitz in the supremum norm, so on $G(a)$, with every bid equal to its capacity bid, $\beta_{(2),a} \geq \bar{M}_{(2)}(a) - s_n$. □

Only part (f) is pathwise across two executions, and it is pathwise only because the two executions see the same context at every round. That is the one consequence of (N) that the coupling needs, and it is why part (i) of Proposition H.11 assumes the exogeneity half and the clause on the coin arrays while parts (ii) to (iv) do not. Part (b) differs from the good event $G(z)$ of Appendix H.1 in controlling every agent's capability bid and not only the bids of the agents other than the deviator, at the same total cost n^{-4}; that is what part (e) needs.

The label monitor. Lemma H.4 does not apply as it stands, because it centres the local martingale at the realised target whereas the price at a label has to be compared with the effective mean. The following is its label analogue. Fix a label a with $N := N_a \geq 1$, let $u_1 < \cdots < u_N$ be its post-elicitation visits, let $W := W_a$ and $\tau := \tau_a$, both measurable with respect to $\mathcal{F}^{(a)} := \mathcal{F}_{T_a}$, the field at the end of Stage 1 at a, let q_a be the number of primary pulls at a, so that the primary pulls are the visits $u_1, \ldots, u_{q_a}$, let $\bar{m}_k$ be the effective mean at u_k, and put

$$Z^a_r := \sum_{k\leq r} \mathbf{1}\{k \leq q_a\} Y_{u_k}, \quad \bar{A}^a_r := \sum_{k\leq r} \mathbf{1}\{k \leq q_a\} \bar{m}_k, \quad M^a_r := Z^a_r - \bar{A}^a_r.$$

Lemma H.15 (Anytime label monitor, centred at the effective mean)**.** *Condition on α and let every agent behave arbitrarily within its class. Then:*

(a) *$\{k \leq q_a\}$ is $\mathcal{F}^\circ_{u_k}$-measurable, the increments of M^a are martingale differences with respect to $(\mathcal{F}^\circ_{u_k})_{k\leq N}$ with conditional ranges of length at most 1, and $\mathbb{E}[M^a_N \mid \mathcal{F}^{(a)}] = 0$;*

(b) *pathwise, $\bar{A}^a_{q_a} \geq q_a\tau - 1 - b_n(q_a) - M^a_N$;*

(c) *$\Pr\big(\exists r \leq N : |M^a_r| > b_n(r)\big) \leq Nn^{-4} \leq n^{-3}$;*

(d) *for every event $\Gamma \in \mathcal{F}^{(a)}$ on which every effective mean at a primary pull at a is at least τ, $\Pr\big(\Gamma \cap \{a \text{ switch occurs at } a\}\big) \leq \Pr(\Gamma) N(2n^4)^{-1} \leq \Pr(\Gamma)(2n^3)^{-1}$.*

Proof. (a) The event $\{k \leq q_a\}$ says that no switch was declared at the primary pulls $1, \ldots, k-1$, which is a function of $Y_{u_1}, \ldots, Y_{u_{k-1}}$ and of τ, all measurable with respect to $\mathcal{F}_{u_{k-1}} \subseteq \mathcal{F}^\circ_{u_k}$. Hence, by Lemma H.12(c), $\mathbb{E}[\mathbf{1}\{k \leq q_a\}(Y_{u_k} - \bar{m}_k) \mid \mathcal{F}^\circ_{u_k}] = \mathbf{1}\{k \leq q_a\}\, \mathbb{E}[Y_{u_k} - \bar{m}_k \mid \mathcal{F}^\circ_{u_k}] = 0$, the indicator being measurable with respect to the conditioning field, and the conditional range is $\mathbf{1}\{k \leq q_a\}[-\bar{m}_k, 1 - \bar{m}_k]$, an interval of length at most 1 with $\mathcal{F}^\circ_{u_k}$-measurable endpoints; the increment at k is $\mathcal{F}_{u_k} \subseteq \mathcal{F}^\circ_{u_{k+1}}$-measurable. The identity follows by the tower property, since $\mathcal{F}^{(a)} \subseteq \mathcal{F}^\circ_{u_1}$.

(b) No switch was declared at the primary pull $q_a - 1$, which is vacuous for $q_a = 1$ because $Z^a_0 = 0$, so $Z^a_{q_a} \geq Z^a_{q_a-1} \geq (q_a - 1)\tau - b_n(q_a - 1) \geq q_a\tau - 1 - b_n(q_a)$, using $\tau \leq 1$ and the monotonicity of b_n; if no switch occurs at all the same bound holds a fortiori. Now subtract $M^a_N = Z^a_{q_a} - \bar{A}^a_{q_a}$, which holds because both sums are constant beyond q_a.

(c) By (a) and the Azuma–Hoeffding inequality, $\Pr(|M^a_r| > b_n(r)) \leq 2\exp(-2b_n(r)^2/r) = 2e^{-\ell_n} = n^{-4}$ for each r, and the index set $\{1, \ldots, N\}$ is deterministic given α, so the union over it costs $Nn^{-4} \leq n^{-3}$.

(d) A switch at the primary pull r requires $r \leq q_a$ and $Z^a_r < r\tau - b_n(r)$, and on Γ, where $\bar{A}^a_r \geq r\tau$ for $r \leq q_a$, this forces $M^a_r < -b_n(r)$. The one-sided form of (c), applied conditionally on $\mathcal{F}^{(a)}$ (the martingale starts at $\mathcal{F}^{(a)}$), gives $\Pr(M^a_r < -b_n(r) \mid \mathcal{F}^{(a)}) \leq e^{-\ell_n} = (2n^4)^{-1}$ for each r; multiplying by $\mathbf{1}_\Gamma$, which is legitimate because $\Gamma \in \mathcal{F}^{(a)}$, summing over $r \leq N$ and taking expectations gives the claim, with $N \leq n$. □

The centring is the whole of the repair that Theorem A.1 describes. A label-level compliant winner's realised target can lie below τ_a at some contexts of the label, and only its π_a-mean is at least τ_a; part (d) is therefore stated for the effective means, and nothing else in the monitor changes, the increments being bounded by 1 under either centring. Conversely, the pointwise rule $\min\{\tau_a, M_j(x_t)\}$ of $S^\star$ has effective mean $\tau_a - \sum_z \pi_a(z)(\tau_a - M^z_j)_+$ at a primary pull, which is strictly below τ_a whenever the winner is below the price at some context of positive posterior mass; parts (b) and (c) then show, heuristically, that a switch occurs after roughly $(1 + 2b_n)/\sum_z \pi_a(z)(\tau_a - M^z_j)_+$ primary pulls on the good event. That is the mechanism of the failure described in Section A, and it is a property of the rule and not of the monitor.

The fallback. Lemma H.5 is used unchanged, with the label sequence in the role of the arrival sequence. Its proof uses exactly three properties: that the weights and the selection law p_t at a Stage-3 visit are measurable with respect to the field just before the selection; that the selection is drawn from p_t with a fresh coin at that round; and that the conditional expected reward of the selected agent given that field and the selection is well defined and at least v for the comparator. Here the field just before the selection is $\mathcal{F}_{t-1}$, the selection is $\mathrm{inv}(p_t, \zeta_t)$ with ζ_t fresh by (N), and the conditional expected reward of the selected agent is its effective mean, because $\sigma(\mathcal{F}_{t-1}, I_t) \subseteq \mathcal{F}^\circ_t$ and Lemma H.12(c) applies; so the comparator value is a label quantity: an agent following $S^\star_{\mathcal{L}}$ has $v = \bar{M}_j(a)$, and an honest agent has $v = \bar{\mu}_j(a)$ whatever its strategy, every counterfactual value $f_t(z)$ being at least μ^z_j. The number H of visits to a after the switch is a function of α and of the switching round, hence measurable at the switch as the lemma requires. No counterfactual reward of an unselected agent is used anywhere in that proof, so both a deviator that reacts to being selected and a context that is drawn after the selection are harmless.

The benchmarks.

Lemma H.16 (The label benchmark against the context benchmark). *For every label a and every context z_0,*

$$\sum_z \pi_a(z) M_{(2)}(z) - \bar{M}_{(2)}(a) \le \sum_{z \neq z_0} \pi_a(z)\big[M_{(2)}(z) - \min\{M^z_{j_1}, M^z_{j_2}\}\big] \le 1 - \pi_a(z_0),$$

where $j_1 \neq j_2$ are two agents attaining the top two capabilities at z_0. The same holds with $b(\cdot)$ and $\bar{b}(\cdot)$ in place of $M_{(2)}$ and $\bar{M}_{(2)}$ and with $\min\{M^z_{j_1}, M^z_{j_2}\}$ replaced by μ^z_h for an honest h attaining $b(z_0)$. If $\mathcal{K}_H = \{h\}$ then $\bar{b}(a) = \sum_z \pi_a(z)\mu^z_h$ exactly.

(b) *Under* (N)*, for every map $\iota : \mathcal{L} \to \mathcal{X}_F$,*

$$\mathbb{E}\sum_{t\le n} M_{(2)}(x_t) - \mathbb{E}\sum_{t\le n} \bar{M}_{(2)}(\alpha_t) \le \sum_{t\le n} \Pr\big(x_t \neq \iota(\alpha_t)\big),$$

and likewise for b; the best choice of ι gives the Bayes error $\sum_{t\le n}(1 - \mathbb{E}\max_z \pi_{\alpha_t}(z))$. If $\mathcal{L} \subseteq \mathcal{X}_F$, taking ι to be the identity gives $\bar{\varepsilon} n$ in both.

(c) *The constant 1 in front of $\bar{\varepsilon} n$ is attained. Take $K = 3$, $\mathcal{X}_F = \mathcal{L} = \{z_1, z_2\}$, capability vectors $(1, 1, 0)$ at z_1 and $(0, 1, 1)$ at z_2, and $\pi_{z_1} = (1 - e, e)$, $\pi_{z_2} = (e, 1 - e)$ with $e \le \frac{1}{2}$: then $M_{(2)} \equiv 1$ while $\bar{M}_{(2)} \equiv 1 - e$, and $\bar{\varepsilon} = e$. With two honest agents and $\mu_1 = \mathbf{1}\{z_1\}$, $\mu_2 = \mathbf{1}\{z_2\}$ the same instance attains it for b.*

(d) *The inequality is one sided: mixing can also raise the label benchmark above the context benchmark.*

Proof. (a) Both $\bar{M}_{j_1}(a)$ and $\bar{M}_{j_2}(a)$ are at least $\sum_z \pi_a(z)\min\{M^z_{j_1}, M^z_{j_2}\}$, and the second order statistic with multiplicity is at least the minimum of any two entries, so $\bar{M}_{(2)}(a) \ge \sum_z \pi_a(z)\min\{M^z_{j_1}, M^z_{j_2}\}$; subtract this from $\sum_z \pi_a(z) M_{(2)}(z)$, use $\min\{M^{z_0}_{j_1}, M^{z_0}_{j_2}\} = M_{(2)}(z_0)$ so that the term $z = z_0$ cancels, note that every remaining bracket is nonnegative because $M_{(2)}(z)$ is at least the minimum of any two entries, and bound each by 1, the primitives lying in $[0, 1]$. For b, $\bar{b}(a) \ge \bar{\mu}_h(a) = \sum_z \pi_a(z)\mu^z_h$ and $\mu^{z_0}_h = b(z_0)$; with one honest agent the maximum defining $\bar{b}$ has a single term. (b) By the clause of (N) for x_t and the tower property, $\mathbb{E}[M_{(2)}(x_t) \mid \sigma(\alpha)] = \sum_z \pi_{\alpha_t}(z) M_{(2)}(z)$, since α_t is $\sigma(\alpha)$-measurable; apply (a) with $z_0 = \iota(\alpha_t)$ and note that $\mathbb{E}[1 - \pi_{\alpha_t}(\iota(\alpha_t))] = \Pr(x_t \neq \iota(\alpha_t))$. (c) is a direct computation: at the label z_1 the mixed capabilities are $(1 - e, 1, e)$, whose second largest is $1 - e$ for $e \le \frac{1}{2}$, and symmetrically at z_2. (d) follows because (a) bounds the difference in one direction only; the instances of Section 5 supply cases of either sign. □

The bracket in part (a) is at most 1, and it is also at most the range of M_j plus the range of $M_{(2)}$ over the contexts carrying positive posterior mass at a. Part (b) is exactly the degradation bound εn stated in Theorem A.1, and part (c) is the statement there that the coefficient one is attained.

Proof of Proposition H.11(i). Fix i and $S'_{i,n} \in \mathcal{B}^{\mathcal{L}}_{i,n}$ and build the two executions on one space carrying all the draws of the list, the deviating one under $(S'_{i,n}, S^\star_{\mathcal{L},-i})$ and the baseline under $S^\star_{\mathcal{L}}$; this is legitimate because (N) is a property of the draws and because, by (a) the exogeneity half and the clause on the coin arrays, the law of (α, ξ) is the same in both. Condition on α, so that every visit count, every Stage-1 round, every N_a and every post-elicitation visit round is deterministic, and every bound below is uniform in α. Rounds are partitioned by labels, so, exactly as in (13),

$$u_i(n; S'_{i,n}, S^\star_{\mathcal{L},-i}) - u_i(n; S^\star_{\mathcal{L}}) = \sum_{a\in\mathcal{L}} \big(\mathbb{E}[U'_i(a)] - \mathbb{E}[U^\star_i(a)]\big),$$

an identity that uses only additivity of expectation and no independence across labels. Split $U_i(a) = U^{\mathrm{bid}}_i(a) + U^{\mathrm{post}}_i(a)$. At each of i's Stage-1 pulls the two executions differ by $\varphi_i(M_i(x_t)) - \varphi_i(m'_{i,t}) \le \varphi_i(1) - \varphi_i(0) \le 1$ by (1), at the same context, so the elicitation parts contribute at most $l_i(a)$ per label, pathwise, and a label with $V_a < KL_n$ contributes nothing more.

Fix a with $N := N_a \ge 1$. Both executions' histories up to T_a are functions of draws realised by T_a, so $\mathcal{F}^{(a)}$ carries both, and the events $G(a)$ and $\mathcal{W}'_a := \{i \text{ wins at } a \text{ in the deviating execution}\}$ lie in it. We prove

$$\mathbb{E}\big[U'^{\mathrm{post}}_i(a) - U^{\star\mathrm{post}}_i(a) \,\big|\, \mathcal{F}^{(a)}\big] \le \mathbf{1}_{G(a)\cap\mathcal{W}'_a}\Lambda\big(1 + b_n(N) + R_K(n)\big) + N^2 n^{-4} + 2N\mathbf{1}_{G(a)^c}. \tag{21}$$

The winning branch: notation. On $G(a) \cap \mathcal{W}'_a$, Lemma H.14(e) gives $\bar{\delta}_i(a) \le 2s_n$, so $a \in \mathcal{W}_i$, and $\tau := \tau_{i,a} \le w := \min\{\bar{V}_i(a), \bar{M}_i(a)\}$, while by Lemma H.14(f) agent i wins at the same price in the baseline. Write $\Psi := \Psi_{i,a}$, $\bar{\varphi} := \bar{\varphi}_i(a)$, $\overline{\mathrm{lo}} := \overline{\mathrm{lo}}_i(a)$, $y^\star := \max\{\overline{\mathrm{lo}}, \tau\}$, which by $\tau \le \bar{M}_i(a)$ is the level $y^\star_i(a, \tau)$ of (20), and $c_0 := 1 + \bar{\varphi} - \Psi(y^\star) \in [0, 2]$. All of these are $\mathcal{F}^{(a)}$-measurable.

The winning branch: the baseline side. The baseline winner is i playing $S^\star_{\mathcal{L}}$, so by Lemma H.13(b) and Lemma H.12(d) its conditional per-pull utility at a primary pull, given $\mathcal{F}^\circ_{u_k}$, is exactly c_0, and its effective mean is exactly $y^\star \ge \tau$. Hence Lemma H.15(d) bounds the baseline switch probability by $N(2n^4)^{-1}$, the baseline fallback utility is nonnegative, and summing the per-pull identity over the primary pulls with the indicator $\mathbf{1}\{k \le q\}$, which is $\mathcal{F}^\circ_{u_k}$-measurable,

$$\mathbb{E}\big[U^{\star\mathrm{post}}_i(a) \,\big|\, \mathcal{F}^{(a)}\big] \ge c_0\,\mathbb{E}[q \mid \mathcal{F}^{(a)}] \ge c_0\big(N - N^2(2n^4)^{-1}\big) \ge c_0 N - \frac{N^2}{n^4},$$

using $q = N$ in the absence of a switch and $c_0 \le 2$.

The winning branch: the deviator's primary phase. Let $\bar{m}_k$ be i's effective mean at its kth primary pull at a, so that $\bar{m}_k \in [\overline{\mathrm{lo}}, \bar{M}_i(a)]$ by Lemma H.12(c), and let $g := 0$ if $y^\star = \overline{\mathrm{lo}}$ and $g := \Psi'_-(\tau) \in [0, \Lambda]$ otherwise, a subgradient of Ψ at $y^\star$ by Lemma H.13(a), so that $\Psi(\bar{m}) \ge \Psi(y^\star) + g(\bar{m} - y^\star)$ on $[\overline{\mathrm{lo}}, \bar{M}_i(a)]$, the case $g = 0$ holding by monotonicity. By Lemma H.12(d) the deviator's conditional per-pull utility at its kth primary pull is at most $1 + \bar{\varphi} - \Psi(\bar{m}_k)$. If $g > 0$ then $y^\star = \tau$ and Lemma H.15(b) gives $\sum_{k\le q}(\bar{m}_k - \tau) \ge -(1 + b_n(q) + M^a_N)$; since g and τ are $\mathcal{F}^{(a)}$-measurable, $\mathbb{E}[M^a_N \mid \mathcal{F}^{(a)}] = 0$ by Lemma H.15(a), and b_n is nondecreasing,

$$\mathbb{E}\Big[\sum_{k\le q} \Psi(\bar{m}_k) \,\Big|\, \mathcal{F}^{(a)}\Big] \ge \Psi(y^\star)\,\mathbb{E}[q \mid \mathcal{F}^{(a)}] - \Lambda\big(1 + b_n(N)\big),$$

and the same holds trivially if $g = 0$. So the deviator's primary utility is at most $c_0\,\mathbb{E}[q \mid \mathcal{F}^{(a)}] + \Lambda(1 + b_n(N))$ conditionally on $\mathcal{F}^{(a)}$.

The winning branch: the deviator's fallback. Condition on $\mathcal{F}_{\mathrm{fb}}$, the field up to and including the switching round, which contains $\mathcal{F}^{(a)}$, let $H = N - q$, and let $\bar{Q}_i$ be the conditional expected number of fallback pulls of i at a. Lemma H.5 with the comparator j^* attaining $\bar{M}_{j^*}(a) = \bar{V}_i(a)$, which exists since $K \geq 2$ and which plays $S^{\star}_{\mathcal{L}}$, gives $\mathbb{E}[\sum Y \mid \mathcal{F}_{\mathrm{fb}}] \geq \bar{V}_i(a)H - R_K(n)$, while every $j \neq i$ has effective mean $\bar{M}_j(a) \leq \bar{V}_i(a)$ at a fallback pull and the selection indicator is $\mathcal{F}_t^{\circ}$-measurable, so $\mathbb{E}[\sum_{I_t \neq i} Y \mid \mathcal{F}_{\mathrm{fb}}] \leq \bar{V}_i(a)(H - \bar{Q}_i)$; subtracting, $\mathbb{E}[\sum_{I_t=i} \bar{m}_t \mid \mathcal{F}_{\mathrm{fb}}] \geq w\bar{Q}_i - R_K(n)$. With $y^{\star\star} := \max\{\overline{\mathrm{lo}}, w\} \geq y^{\star}$, which lies in the domain of Ψ because $w \leq \bar{M}_i(a)$, and the subgradient of Ψ at $y^{\star\star}$, the same two lines give $\mathbb{E}[\sum_{I_t=i} \Psi(\bar{m}_t) \mid \mathcal{F}_{\mathrm{fb}}] \geq \Psi(y^{\star\star})\bar{Q}_i - \Lambda R_K(n)$, so the deviator's fallback utility is at most $(1 + \bar{\varphi} - \Psi(y^{\star\star}))\bar{Q}_i + \Lambda R_K(n) \leq c_0 H + \Lambda R_K(n)$, using $\bar{Q}_i \leq H$, $\Psi(y^{\star\star}) \geq \Psi(y^{\star})$ and $1 + \bar{\varphi} - \Psi(y^{\star\star}) \geq 0$.

The winning branch: cancellation. Adding the primary and the fallback and using $\mathbb{E}[q + H \mid \mathcal{F}^{(a)}] = N$, the deviating post-elicitation utility is at most $c_0 N + \Lambda(1 + b_n(N) + R_K(n))$ conditionally on $\mathcal{F}^{(a)}$, and the terms $c_0 N$ cancel against the baseline. This is the branch that would carry a capability gap in Proposition H.2(i): the fallback comparison is made at $w = \min\{\bar{V}_i(a), \bar{M}_i(a)\}$ and not at $\bar{V}_i(a)$, which is what lets a label with $0 < \bar{\delta}_i(a) \leq 2s_n$ be treated exactly like a top label.

The losing branch. On $G(a) \setminus \mathcal{W}'_a$ the winner $j \neq i$ plays $S^{\star}_{\mathcal{L}}$, and $G(a) \subseteq G^{\max}(a)$, so $\tau_a \leq \bar{M}_j(a)$ by Lemma H.14(d) and its effective mean at every primary pull is $y^{\star}_j(a, \tau_a) \geq \min\{\tau_a, \bar{M}_j(a)\} = \tau_a$; Lemma H.15(d) with $\Gamma = G(a) \setminus \mathcal{W}'_a$ bounds the probability of a switch at a on that event by $N(2n^4)^{-1}$. Without a switch i receives no post-elicitation pull at a, so, at a utility of at most 2 per pull, $\mathbb{E}[U_i'^{\mathrm{post}}(a) \mid \mathcal{F}^{(a)}] \leq 2N \cdot N(2n^4)^{-1}$ there, while the baseline is nonnegative. This is the step that rules out bidding so as to price the winner out: the price is at most the winner's label mean, and the winner meets its label mean on average, which is all the monitor asks. A pointwise capability M_j^z below τ_a at some context of the label is not merely harmless here, it is the normal case.

Off $G(a)$. Post-elicitation utilities lie in $[0, 2N]$ in both executions. This proves (21).

Summation. Taking expectations in (21), using $\Pr(G(a)^c) \leq n^{-4}$ and $\mathbf{1}_{G(a) \cap \mathcal{W}'_a} \leq \mathbf{1}\{a \in \mathcal{W}_i\}$, and summing over labels with $\sum_a N_a^2 \leq n^2$ and $\sum_a N_a \leq n$, the bad events contribute at most $n^{-2} + 2n^{-3}$ in total and the first display follows. Its right side is a function of α alone and does not depend on $S'_{i,n}$, so integrating over α and then taking the supremum is legitimate. For the second display, $\sum_a l_i(a) \leq \min\{Q_{\mathcal{L}} L_n, n\}$ pathwise, $|\mathcal{W}_i| \leq Q_{\mathcal{L}}$, and $\mathbb{E}[b_n(N_a)] \leq \sqrt{\ell_n/2}\sqrt{\mathbb{E}N_a}$ by Jensen's inequality with $\sum_a \sqrt{\mathbb{E}N_a} \leq \sqrt{Q_{\mathcal{L}} n}$ by the Cauchy–Schwarz inequality. □

Every place where an indicator multiplies a conditional expectation in that proof is one of the following, and in each the indicator is measurable with respect to the conditioning field: $\mathbf{1}\{k \leq q\}$ at a primary pull, measurable with respect to $\mathcal{F}^{\circ}_{u_k}$ because it depends on earlier rewards at a alone; the selection indicator at a fallback pull, measurable with respect to $\mathcal{F}_t^{\circ}$ because the selection precedes the context and the seed; the subgradient g and the price τ multiplying M_N^a, both measurable with respect to $\mathcal{F}^{(a)}$; and the events $G(a)$, $G^{\max}(a)$, $B(a)$ and $\mathcal{W}'_a$, all measurable with respect to $\mathcal{F}^{(a)}$. The union bounds are over the agents, a fixed index set, and over the primary indices $r \leq N_a$, an index set that is deterministic once α is fixed; labels are never unioned with a per-label failure probability, each bad event being charged $2N_a$ times its probability and summed with $\sum_a N_a \leq n$.

Proof of Proposition H.11(ii). All agents play $S^{\star}_{\mathcal{L}}$; condition on α. Since $P(n; S) = \mathbb{E}\sum_t Y_t$, revenue is additive over labels. A label with $V_a < KL_n$ is charged $V_a \bar{M}_{(2)}(a) \leq \min\{V_a, KL_n\}$, and for $V_a \geq KL_n$ the KL_n elicitation visits are charged KL_n. On $G(a)$, Lemma H.14(g) and (d) give $\bar{M}_{(2)}(a) - 2s_n \leq \tau_a \leq \bar{M}_{W_a}(a)$, so the winner's effective mean at every primary pull is the $\mathcal{F}^{(a)}$-measurable constant $y^{\star}_{W_a}(a, \tau_a) \geq \tau_a$ and Lemma H.15(d) bounds the switch probability by $N_a(2n^4)^{-1}$. By Lemma H.12(c) and the tower property at $\mathcal{F}^{\circ}_{u_k}$, the indicator $\mathbf{1}\{k \leq q\}$ being measurable there, $\mathbb{E}[\sum_{k \leq q} Y_{u_k} \mid \mathcal{F}^{(a)}] = y^{\star}_{W_a}(a, \tau_a)\,\mathbb{E}[q \mid \mathcal{F}^{(a)}] \geq \tau_a \mathbb{E}[q \mid \mathcal{F}^{(a)}] \geq \tau_a N_a - N_a^2(2n^4)^{-1}$, using $\tau_a \leq 1$ and $q = N_a$ in the absence of a switch; the fallback revenue is nonnegative. Hence the post-elicitation delivery is at least $N_a(\bar{M}_{(2)}(a) - 2s_n) - N_a^2(2n^4)^{-1}$ on $G(a)$ and at least $N_a \bar{M}_{(2)}(a) - N_a$ off it. Summing over labels with $\sum_a \min\{V_a, KL_n\} \leq \min\{Q_{\mathcal{L}} KL_n, n\}$, $\sum_a N_a \leq n$, $\sum_a N_a^2(2n^4)^{-1} \leq n^{-2}/2$ and $\sum_a N_a n^{-4} \leq n^{-3}$, and integrating over α, gives (ii), since $\sum_a V_a \bar{M}_{(2)}(a) = \sum_t \bar{M}_{(2)}(\alpha_t)$. The fallback contributes no $R_K(n)$ term here because at $S^{\star}_{\mathcal{L}}$ a switch has probability at most $N_a(2n^4)^{-1}$ and its revenue may be charged at zero. The benchmark comparison is Lemma H.16. □

Proof of Proposition H.11(iii). Every agent may play arbitrarily within its class, subject to pointwise honesty for $j \in \mathcal{K}_H$; condition on α. Fix a with $N := N_a \geq 1$ and an honest h^* with $\bar{\mu}_{h^*}(a) = \bar{b}(a)$; a label with $V_a < KL_n$ is charged $V_a \bar{b}(a) \leq \min\{V_a, KL_n\}$ and the KL_n elicitation visits of a label with $V_a \geq KL_n$ are charged KL_n. By Lemma H.12(c) and the tower property, $\mathbb{E}[\sum_{k \leq q} Y_{u_k} \mid \mathcal{F}^{(a)}] = \mathbb{E}[\bar{A}^a_q \mid \mathcal{F}^{(a)}]$, and $\bar{A}^a_q \geq 0$ always, so it is enough to bound that quantity below on $B(a)$, which lies in $\mathcal{F}^{(a)}$ and has $\Pr(B(a)^c) \leq (2Kn^4)^{-1}$ by Lemma H.14(b). On $B(a) \cap \{W_a = h^*\}$, pointwise honesty makes every effective mean at a primary pull at least $\bar{b}(a)$ by Lemma H.12(a) and (c), so $\bar{A}^a_q \geq \bar{b}(a)q$. On $\Gamma := B(a) \setminus \{W_a = h^*\}$, an $\mathcal{F}^{(a)}$-measurable event, the bid $\beta_{h^*,a}$ is not the largest, so $\beta_{(2),a} \geq \beta_{h^*,a} \geq \bar{b}(a) - s_n$ and $\tau_a \geq \bar{b}(a) - 2s_n$, and Lemma H.15(a) and (b) give $\mathbb{E}[\mathbf{1}_\Gamma \bar{A}^a_q \mid \mathcal{F}^{(a)}] \geq \mathbf{1}_\Gamma(\tau_a \mathbb{E}[q \mid \mathcal{F}^{(a)}] - 1 - b_n(N))$. In the fallback, h^* has effective mean at least $\bar{b}(a)$ whenever selected, whatever its strategy and whatever the other agents do, so Lemma H.5 with the comparator h^* gives expected fallback delivery at least $\bar{b}(a)H - R_K(n)$. Adding, with $q + H = N$, and charging N off $B(a)$,

$$\mathbb{E}[\text{delivered at } a] \geq V_a \bar{b}(a) - \big[\min\{V_a, KL_n\} + 2s_n N + 1 + b_n(N) + R_K(n) + N(2Kn^4)^{-1}\big].$$

Summing over labels with $\sum_a N_a(2Kn^4)^{-1} \leq n^{-3}$ and integrating over α gives (iii). No independence across agents is used, and no property of any agent other than the honesty of h^*. □

Proof of Proposition H.11(iv). Condition on α, fix j and a label a with $V_a \geq KL_n$, and let the others behave arbitrarily within their classes. Let $E_j := \{\beta^{\max}_{j,a} \leq \bar{M}_j(a) + s_n\}$, of complementary probability at most $(2Kn^4)^{-1}$ by Lemma H.14(b). On $\{W_a = j\} \cap E_j$,

an event in $\mathcal{F}^{(a)}$, Lemma H.14(c) and the hypothesis that j bids at capacity give $\tau_a = (\beta_{(2),a} - s_n)_+ \le (\beta_{j,a} - s_n)_+ = (\beta_{j,a}^{\max} - s_n)_+ \le \bar{M}_j(a)$, so the hypothesis on the primary targets gives effective means at least $\min\{\tau_a, \bar{M}_j(a)\} = \tau_a$ at every primary pull and Lemma H.15(d) applies with $\Gamma = \{W_a = j\} \cap E_j$; adding $\Pr(E_j^c)$ gives the first display. Summing over the labels with $V_a \ge KL_n$, a set that is deterministic given α and of size at most $Q_{\mathcal{L}}$, and using $\sum_a N_a \le n$, gives the second. If all agents play $S^\star_{\mathcal{L}}$ then $\Pr(\text{a switch at } a) \le \sum_j \Pr(E_j^c) + \Pr\big(\{\text{a switch at } a\} \cap \bigcup_j (\{W_a = j\} \cap E_j)\big) \le K(2Kn^4)^{-1} + N_a(2n^4)^{-1}$, the second term being one application of Lemma H.15(d) to the single event $\bigcup_j (\{W_a = j\} \cap E_j) \in \mathcal{F}^{(a)}$, on which the winner's effective means are at least τ_a, and not K applications; summing over labels gives the third. □

Part (iv) is the formal content of the sentence that a compliant agent is convicted with probability $O(n^{-3})$ per label. What is compliant is a label-level notion: a winner that meets the announced price in the mean over the queries carrying its label. It is not a statement about the pointwise rule of $S^\star$, for which the paragraph after Lemma H.15 shows the opposite.

Corollary H.17 (The agent need not know the posterior). *Let $\widehat{S}_{\mathcal{L}}$ be the profile that follows $S^\star_{\mathcal{L}}$ except that the Stage-2 winner j at a label a replaces π_a by the empirical law $\widehat{\pi}_{j,a}$ of the contexts at its own L_n Stage-1 pulls at a, which it observed, replaces τ_a by $\tau_a + \chi_n$, where*

$$\chi_n := \sqrt{\frac{|\mathcal{X}_F| \log 2 + \log(Kn^4)}{2L_n}},$$

and delivers the water-filling allocation at the largest level whose $\widehat{\pi}_{j,a}$-mean equals $\max\{\widehat{\mathrm{lo}}_j, \min\{\tau_a + \chi_n, \widehat{M}_j\}\}$, where $\widehat{\mathrm{lo}}_j$ and $\widehat{M}_j$ are the $\widehat{\pi}_{j,a}$-means of lo_j and M_j. Then $\widehat{S}_{\mathcal{L}}$ requires no knowledge of the kernel by anyone, and Proposition H.11 holds with $S^\star_{\mathcal{L}}$ replaced by $\widehat{S}_{\mathcal{L}}$, part (i) acquiring the additional term $2\Lambda\chi_n \mathbb{E}\sum_{a \in \mathcal{W}_i} N_a + 2n^{-3} \le 2\Lambda\chi_n n + 2n^{-3}$, part (ii) an additional n^{-3}, part (iii) nothing, and part (iv) an additional n^{-4} per label. Since $\chi_n = O(\sqrt{(|\mathcal{X}_F| + \log n)/L_n})$, the extra term in part (i) is $O\big(\sqrt{|\mathcal{X}_F| + \log n}\, n^{2/3} (\log n)^{-1/6}\big)$, of the order of $2s_n n$ for a fixed context set.

Proof. *The estimation event.* Put $T(a) := \bigcap_j \{\mathrm{TV}(\widehat{\pi}_{j,a}, \pi_a) \le \chi_n\}$ with $\mathrm{TV}(p, p') := \max_{S \subseteq \mathcal{X}_F} (p(S) - p'(S))$, which equals $\max_S (p'(S) - p(S))$ by passing to complements. By Lemma H.14(a) the contexts at j's Stage-1 pulls at a are, given α, L_n i.i.d. draws from π_a, so Hoeffding's inequality for each of the $2^{|\mathcal{X}_F|}$ subsets and a union over them and over the K agents give $\Pr(T(a)^c) \le K 2^{|\mathcal{X}_F|} e^{-2L_n \chi_n^2} = n^{-4}$; the event lies in $\mathcal{F}^{(a)}$, and it is the same in both coupled executions because the Stage-1 schedule and the contexts there are. For $f : \mathcal{X}_F \to [0,1]$ we have $|\sum_z (\pi_a - \widehat{\pi}_{j,a})(z) f(z)| \le \mathrm{TV}(\widehat{\pi}_{j,a}, \pi_a)$, by writing $f = \int_0^1 \mathbf{1}\{f \ge v\}\, \mathrm{d}v$.

The true effective mean of the estimated rule. Let $\widehat{y}$ be the level target above and $\bar{m}$ the true, that is π_a, mean of the resulting allocation, and work on $T(a) \cap G^{\max}(a)$, where $\tau_a \le \bar{M}_j(a)$ by Lemma H.14(d). Three cases. If $\widehat{y} = \tau_a + \chi_n \le \widehat{M}_j$ then the allocation has $\widehat{\pi}_{j,a}$-mean $\tau_a + \chi_n$, so $\bar{m} \ge \widehat{y} - \chi_n = \tau_a$ and $\bar{m} \le \tau_a + 2\chi_n$. If $\widehat{y} = \widehat{M}_j < \tau_a + \chi_n$ then the largest level with $\widehat{\pi}_{j,a}$-mean $\widehat{M}_j$ is 1, so the allocation is M_j and $\bar{m} = \bar{M}_j(a) \ge \tau_a$, while $\bar{M}_j(a) \le \widehat{M}_j + \chi_n < \tau_a + 2\chi_n$. If $\widehat{y} = \widehat{\mathrm{lo}}_j$ exceeds $\min\{\tau_a + \chi_n, \widehat{M}_j\}$, then, since $\widehat{\mathrm{lo}}_j \le \widehat{M}_j$ always, either the minimum is $\widehat{M}_j = \widehat{\mathrm{lo}}_j$, which is the previous case, or $\widehat{\mathrm{lo}}_j > \tau_a + \chi_n$, and then the allocation has $\widehat{\pi}_{j,a}$-mean $\widehat{\mathrm{lo}}_j$, so $\tau_a < \widehat{\mathrm{lo}}_j - \chi_n \le \bar{m} \le \widehat{\mathrm{lo}}_j + \chi_n \le \overline{\mathrm{lo}}_j(a) + 2\chi_n$. In every case $\bar{m} \ge \overline{\mathrm{lo}}_j(a)$, being the true mean of a feasible allocation, and $y_j^\star(a, \tau_a) \le \bar{m} \le y_j^\star(a, \tau_a) + 2\chi_n$, where $y_j^\star(a, \tau_a) = \max\{\overline{\mathrm{lo}}_j(a), \tau_a\}$ because $\tau_a \le \bar{M}_j(a)$.

Its cost. The allocation is a water-filling allocation, hence cost optimal for the π_a-mean it produces by Lemma H.13(b), whose first statement holds for every level and every kernel. So its conditional per-pull utility is $1 + \bar{\varphi}_j(a) - \Psi_{j,a}(\bar{m}) \ge c_0 - 2\Lambda\chi_n$ by the Lipschitz property of $\Psi_{j,a}$, with $c_0 := 1 + \bar{\varphi}_j(a) - \Psi_{j,a}(y_j^\star(a, \tau_a))$.

Rereading the four proofs. The deviating side of part (i) never used the baseline profile beyond the Stage-1 and Stage-3 rules of the non-deviators, which are unchanged, so only the baseline side moves: on $G(a) \cap T(a) \cap \mathcal{W}'_a$ the baseline winner's effective mean is at least τ_a, so Lemma H.15(d) still applies and its per-pull utility is at least $c_0 - 2\Lambda\chi_n$, which leaves $2\Lambda\chi_n N_a$ uncancelled on that event; off $T(a)$ charge $2N_a$, contributing at most $2n^{-3}$ in total. In parts (ii) and (iv) the winner's effective mean is again at least τ_a on $T(a) \cap G^{\max}(a)$, which costs the stated extra failure mass $\Pr(T(a)^c) \le n^{-4}$ per label, at most N_a each in (ii). Part (iii) does not involve the profile. □

Labels hide shading. Condition (N) is stronger than the hypothesis of Proposition H.2, and the next proposition says that the strengthening is not an artefact of this analysis. It concerns two exogenous context paths that differ only in the order of the contexts, one of which satisfies (N) and one of which does not, and its parts (a) and (b) hold for every mechanism that sees only labels.

Proposition H.18 (Labels hide shading, exact form). *Fix $\varepsilon \in (0, 1]$ and an even $n \ge 2K$. Let $\mathcal{X}_F = \{z_1, z_2, z_3\}$ and $\mathcal{L} = \{a_1, a_2\}$, the correct label of z_1 being a_1 and that of z_2 and z_3 being a_2, and let the classifier draw $\alpha_t \sim C(x_t, \cdot)$, conditionally independently across rounds given the contexts, with*

$$C(z_1, a_1) = 1, \qquad C(z_2, a_1) = \varepsilon, \quad C(z_2, a_2) = 1 - \varepsilon, \qquad C(z_3, a_2) = 1,$$

so that the three per-context misclassification probabilities are 0, ε and 0. Let $K = 3$ with all agents strategic, $\mu \equiv 0$ and linear costs, let $M_W(z_1) = 1$ and $M_W(z_2) = M_W(z_3) = 0$ for one agent W, and let $M_j \equiv \frac{1}{2}$ at every context for the other two. Let the two exogenous context paths be

$$P_1: \quad x_t = z_1 \ (t \le n/2), \qquad x_t = z_2 \ (t > n/2);$$
$$P_2: \quad x_t = z_1 \ (t \le n/2), \qquad x_t \text{ i.i.d. with } \Pr(z_1) = \varepsilon,\ \Pr(z_3) = 1 - \varepsilon \ (t > n/2),$$

both independent of everything else. Let $\mathcal{M}$ be any label-measurable *mechanism: its selection at round t is a measurable function of $\alpha_1, \ldots, \alpha_t$, of its past selections and rewards and of its own coins, and its messages are functions of the same. Let agents 2 and 3 play any strategies that do not read contexts. Let σ be any strategy of W that is feasible on P_1, and let $\widetilde{\sigma}$ be the strategy on P_2 that at every round delivers what σ would deliver if the contexts so far, the current one included, were those of P_1; it reads the round index. Then:*

(a) *the joint law of labels, mechanism coins, selections, messages, targets and rewards is the same under (P_1, σ) and under*

$(P_2, \widetilde{\sigma})$, *and* $u_W(P_1;\sigma) = u_W(P_2;\widetilde{\sigma})$*; on* P_2 *the strategy* $\widetilde{\sigma}$ *is feasible and delivers* 0 *at every round* $t > n/2$*;*

(b) *suppose in addition that* σ *and* $\widetilde{\sigma}$ *coincide on* P_2 *at the rounds* $t \le n/2$*, which holds whenever* σ *reads contexts only through the current one, and that* σ *delivers* 0 *at every label-*a_2 *round* $t > n/2$ *on* P_2*, which is forced because the context there is* z_3*. Write* N_W^{II} *for the number of rounds* $t > n/2$ *allocated to* W *and put*

$$\mathrm{G} := u_W(P_2;\widetilde{\sigma}) - u_W(P_2;\sigma),$$
$$\mathrm{P} := \mathbb{E}_{P_2,\sigma}[N_W^{\mathrm{II}}] - \mathbb{E}_{P_1,\sigma}[N_W^{\mathrm{II}}],$$
$$D := \mathbb{E}_{P_2,\sigma}\Big[\sum_{t>n/2,\ \alpha_t=a_1,\ I_t=W} m_{W,t}\Big].$$

Then $\mathrm{G} + \mathrm{P} = D$ *exactly;*

(c) *the path* P_2 *satisfies* (N) *with* $\pi_{a_1} = \delta_{z_1}$ *and* $\pi_{a_2} = \delta_{z_3}$*, while* P_1 *violates the clause of* (N) *on the context. Let* $\mathcal{M}$ *be* CL-MESP *keyed on the label, let* $\sigma = S^\star$ *be the pointwise profile of Appendix H.1 with the other two agents also on* $S^\star$*, and let* n *satisfy* $4s_n \le \frac{1}{2}$*. Then* $D \ge (\frac{1}{2} - 2s_n)\varepsilon n/2 - n^{-2}$ *and* $\mathrm{G} \le \Delta_n := \min\{2L_n, n\} + 2 + \sqrt{\ell_n n} + 2R_K(n) + n^{-2} + 2n^{-3}$*, the latter by Proposition H.11(i) applied on* P_2 *with* $Q_{\mathcal{L}} = 2$ *and* $\Lambda = 1$*, so that*

$$\begin{aligned}\mathbb{E}_{P_2,S^\star}[N_W^{\mathrm{II}}] - \mathbb{E}_{P_1,S^\star}[N_W^{\mathrm{II}}] &\ge \big(\tfrac{1}{2} - 2s_n\big)\frac{\varepsilon n}{2} - \Delta_n - n^{-2} \\ &= \frac{\varepsilon n}{4} - O\big(n^{2/3}(\log n)^{1/3}\big);\end{aligned}$$

(d) *for* CL-MESP *on* P_1 *with* $\sigma = S^\star$*, the agent* W *is switched out at label* a_1 *with probability at least* $1 - \Pr(R < k_n) - 2n^{-3}$*, where* $R \sim \mathrm{Bin}(n/2, \varepsilon)$ *is the number of rounds* $t > n/2$ *with* $\alpha_t = a_1$ *and* $k_n := 1 + \sqrt{2n\ell_n}/(\frac{1}{2} - 2s_n)$*. For a fixed* $\varepsilon > 0$ *that probability tends to* 1*.*

Proof. (a) Build one space carrying the mechanism's coins, the agents' coin arrays, the seeds (U_t) and an independent i.i.d. uniform sequence (V_t). Put $\alpha_t := a_1$ for $t \le n/2$ and, for $t > n/2$, $\alpha_t := a_1$ if $V_t \le \varepsilon$ and a_2 otherwise. In world 1 put $x_t := z_1$ for $t \le n/2$ and $x_t := z_2$ after; in world 2 put $x_t := z_1$ for $t \le n/2$ and, for $t > n/2$, $x_t := z_1$ if $V_t \le \varepsilon$ and z_3 otherwise. In world 1 the labels then have the law prescribed by C along P_1, and in world 2 the contexts have the law of P_2 and $\alpha_t = a_1$ exactly when $x_t = z_1$, which is the law prescribed by C along P_2; in both the pair of sequences is independent of the coins and the seeds, so each world's execution has its model law here. Now induct on t. If the observables of the rounds before t coincide in world 1 under (σ, others) and in world 2 under $(\widetilde{\sigma}, \text{others})$, then the labels at t coincide by construction; the mechanism's selection and message are the same function of the same observables and the same coin; an agent other than W chooses the same target, its strategy reading no context; and W chooses the same target, because $\widetilde{\sigma}$ is by definition σ evaluated at the world-1 context path and, by the induction hypothesis, at the world-1 history. The seed is the same, so the reward is. Feasibility of $\widetilde{\sigma}$ on P_2: at $t \le n/2$ the two paths agree, and at $t > n/2$ feasibility of σ on P_1 at the context z_2 forces it to deliver 0, which is feasible at every context. Finally W's realised utility is $\sum_t \mathbf{1}\{I_t = W\}(1 - m_{W,t})$, a function of the observables, so its expectations agree.

(b) Compare $(P_2, \widetilde{\sigma})$ with (P_2, σ). They deliver the same targets as functions of the observables at the rounds $t \le n/2$, so the same induction, now with both executions in world 2, gives pathwise equality of the first-half observables and of W's first-half utility. In the second half of world 2: under $\widetilde{\sigma}$ the agent W delivers 0, so its second-half utility is exactly N_W^{II}; under σ a label-a_2 round carries the context z_3, where $M_W = 0$, so W delivers 0 and collects 1, while a label-a_1 round carries z_1, where it delivers $m_{W,t}$ and collects $1 - m_{W,t}$, so its second-half utility is $N_W^{\mathrm{II}} - \sum_{t>n/2,\alpha_t=a_1,I_t=W} m_{W,t}$. Taking expectations and using (a), which gives $\mathbb{E}_{P_2,\widetilde{\sigma}}[N_W^{\mathrm{II}}] = \mathbb{E}_{P_1,\sigma}[N_W^{\mathrm{II}}]$ because N_W^{II} is a function of the observables, $\mathrm{G} = \mathbb{E}_{P_1,\sigma}[N_W^{\mathrm{II}}] - \mathbb{E}_{P_2,\sigma}[N_W^{\mathrm{II}}] + D = -\mathrm{P} + D$.

(c) On P_2, given the label sequence the context is the deterministic function $x_t = z_1$ if $\alpha_t = a_1$ and z_3 otherwise, which is the stated kernel, and the seeds, priorities, mechanism coins and coin arrays are fresh and independent of the path, so (N) holds. On P_1 the label a_1 carries the context z_1 for $t \le n/2$ and z_2 for $t > n/2$, both with positive probability when $\varepsilon > 0$, so no fixed kernel serves. On P_2 the profile $S^\star$ coincides with $S^\star_{\mathcal{L}}$ for every agent, each posterior being a point mass: at a_1 the posterior is δ_{z_1} and water filling at the level $\min\{\tau, M_W(z_1)\}$ delivers that value, and at a_2 the level is $\min\{\tau, 0\} = 0$; and the hypothesis of (b) holds because $S^\star$ reads only the current context. The deviation $\widetilde{\sigma}$ lies in $\mathcal{B}^{\mathcal{L}}_{W,n}$. For the bound on G: Proposition H.11(i) applies because its part (i) does not use the honest set, which is empty here; at a_1, $\bar{M}_W(a_1) = 1 > \frac{1}{2} = \bar{V}_W(a_1)$, so $\bar{\delta}_W(a_1) \le 0 \le 2s_n$ and $a_1 \in \mathcal{W}_W$; at a_2, $\bar{M}_W(a_2) = 0$ and $\bar{V}_W(a_2) = \frac{1}{2} > 2s_n$, so $a_2 \notin \mathcal{W}_W$ and that label contributes only its elicitation term. With $Q_{\mathcal{L}} = 2$ and $\Lambda = 1$ the second display of Proposition H.11(i) is Δ_n. For the bound on D: since $L_n \le \lfloor n/(2K) \rfloor$, Stage 1 at a_1 ends within the first half; W's elicitation rewards at a_1 are $\mathbf{1}\{U_t \le 1\} = 1$, so $\beta_{W,a_1} = 1$ and, on $G(a_1)$, of probability at least $1 - n^{-4}$, the agent W is the strict largest bidder because $s_n < \frac{1}{2}$, and $\tau_{a_1} \in [\frac{1}{2} - 2s_n, \frac{1}{2}]$. Agent W delivers τ_{a_1} at every primary pull at a_1, the context there being z_1, so its effective mean is τ_{a_1} and Lemma H.15(d) bounds the probability of a switch at a_1 by $(2n^3)^{-1}$. On the intersection with the event that no switch occurs, every second-half label-a_1 round is a primary pull of W with target $\tau_{a_1} \ge \frac{1}{2} - 2s_n$, and the number of such rounds has mean $\varepsilon n/2$ and is at most $n/2$, so $D \ge (\frac{1}{2} - 2s_n)\varepsilon n/2 - (n/2)(n^{-4} + (2n^3)^{-1}) \ge (\frac{1}{2} - 2s_n)\varepsilon n/2 - n^{-2}$. Then $\mathrm{P} = D - \mathrm{G}$ by (b), and both $2s_n\varepsilon n/2 \le s_n n$ and Δ_n are $O(n^{2/3}(\log n)^{1/3})$.

(d) On P_1 the first half is as in (c), a deterministic context path being the degenerate kernel at the realised context, for which Lemmas H.12 and H.15 hold as stated, neither of them using more than the conditional law of x_t given $\mathcal{F}_t^\circ$. So on $G(a_1)$ the agent W wins a_1 at $\tau_{a_1} \in [\frac{1}{2} - 2s_n, \frac{1}{2}]$ and delivers τ_{a_1} at each of the $r_1 := n/2 - KL_n$ first-half primary pulls, while in the second half its label-a_1 targets are $\min\{\tau_{a_1}, M_W(z_2)\} = 0$, so $\bar{A}_r^{a_1} = r_1\tau_{a_1}$ for every $r \ge r_1$. On $G(a_1)$ intersected with the event of Lemma H.15(c), of total complementary probability at most $n^{-4} + n^{-3} \le 2n^{-3}$, if no switch has occurred by the primary index $r = r_1 + k$ then $r\tau_{a_1} - b_n(r) \le Z_r^{a_1} \le \bar{A}_r^{a_1} + b_n(r) = r_1\tau_{a_1} + b_n(r)$, so $k\tau_{a_1} \le 2b_n(r) \le \sqrt{2n\ell_n}$ and $k \le \sqrt{2n\ell_n}/(\frac{1}{2} - 2s_n)$. Hence a switch occurs whenever $R \ge k_n$, which is the claim. For a fixed $\varepsilon > 0$, $k_n = O(\sqrt{n \log n}) = o(\varepsilon n)$, so $\Pr(R < k_n) \to 0$ exponentially by Hoeffding's inequality. □

Remark H.19 (What Proposition H.18 does and does not give). *Part (b) is an identity and not an inequality, and the identity is what holds mechanism independently. The corollary one would like, that a label-measurable mechanism which withholds none of the cooperative winner's second-half label-a_1 rounds leaves a shading gain of order εn, does not follow from it and is false in that generality, because* P *counts allocations at both labels and a mechanism may punish at the other label. An explicit mechanism: label a_1 always goes to W, and label a_2 goes to W in the second half as long as W's second-half label-a_1 reward rate is at least $\frac{1}{4}$, and to agent 2 otherwise. Take the cooperative W to deliver $\frac{1}{2}$ at z_1, the price of $S^\star$. On P_1 no label-a_1 round is withheld from the cooperative W, but every label-a_2 round of the second half is, because at z_2 the cooperative W has capability 0 and its label-a_1 reward rate is $0 < \frac{1}{4}$; on P_2 the shading $\widetilde{\sigma}$ forfeits the same label-a_2 rounds, each worth 1 to W, while the cooperative W keeps them. Hence $\mathrm{P} = (1-\varepsilon)n/2 + O(\sqrt{n})$, $D = \varepsilon n/4 + O(\sqrt{n})$, and the shading gain is $\mathrm{G} = D - \mathrm{P} = \varepsilon n/4 - (1-\varepsilon)n/2 + O(\sqrt{n})$, which is negative for $\varepsilon < \frac{2}{3}$. The dichotomy becomes quantitative exactly when D is of order εn and* P *is small, that is when the mechanism lets its compliant top agent deliver on a constant share of the second-half label-a_1 rounds on the path that satisfies* (N) *and does not punish it at the other label; parts (c) and (d) establish that for* CL-MESP*, which by Lemma H.3 cannot punish across labels. A mechanism that never allocates W in the second half has $D = 0$ and is excluded by nothing here. Second, what is lost on P_1 is the protection of* pointwise *compliance and not platform efficiency: the withheld rounds are rounds at which the cooperative W has capability 0 while its competitors have $\frac{1}{2}$, so withholding them is efficient and costs W only rounds for which it was being paid nothing. The loss of protection is nevertheless real and permanent for* CL-MESP*, whose switch is never undone, so after the conviction of part (d) the agent W serves label a_1 through the fallback only. This appendix does not prove a quantitative form of that last sentence: the statement to prove is that on a path carrying a further block of z_1 rounds, at which W is pointwise compliant and is nevertheless confined to the fallback, W loses a linear amount, and it is not proved here.*

Remark H.20 (How the statements in Section A relate to this appendix). *Theorem A.1 states the benchmark degradation as εn with coefficient one attained, which is Lemma H.16(b) and (c) with the average per-round misclassification probability $\bar{\varepsilon}$ in the role of ε. The comparison requires the labels to name contexts; for a coarser label set, as in Proposition H.18, only the label benchmark is defined, and no comparison with the context benchmark is available without a further hypothesis relating the primitives to the labelling. Proposition A.2 states its allocation loss for* CL-MESP *only and says explicitly that it is not a universal lower bound; that is the content of Proposition H.18(b) to (d), and Remark H.19 gives the mechanism that shows why the universal version would be false.*

H.3 A continuum of contexts

The mechanism. Write CG-MESP for the continuous-grid monitored empirical second price of Section 4.1. The platform knows n, K and d; it does not know the arrival law, L, any μ_i, any M_i, or $\mathcal{K}_H$. Put

$$g_n := \lceil n^{1/(d+3)} \rceil, \qquad h_n := 1/g_n, \qquad J_n := g_n^d,$$
$$L_n := \lceil n^{2/(d+3)} \rceil, \qquad \ell_n := \log(2Kn^5), \qquad s_n := \sqrt{\frac{\ell_n}{2L_n}},$$
$$\rho_n := \sqrt{d}\, h_n \log(en), \qquad \sigma_n := 3s_n + \rho_n,$$
$$b_n(r) := \sqrt{\frac{r\ell_n}{2}}, \qquad b_n(0) := 0,$$
$$R_K(H) := 3\sqrt{KH\log K} + 2K\log K,$$
$$R_K^{\mathrm{dbl}}(V) := 11\sqrt{K(V+1)\log K} + 2K\log K \log_2(2(V+1)),$$

with $R_K(0) := R_K^{\mathrm{dbl}}(0) := 0$; R_K is the scalar fallback bound of Appendix F, recalled for comparison, and R_K^{dbl} is its doubling-epoch analogue proved in Lemma H.28. The grid consists of the J_n half-open cubes of side h_n that tile $[0,1)^d$. For $x \in [0,1)^d$, $C(x)$ denotes the cube containing x; the position of x inside its cube is used only to locate the cube. Cubes are instantiated on their first visit, so J_n never has to be enumerated. Each cube C keeps a visit counter V_C, its Stage-1 reward sums, its price τ_C, its winner W_C, its monitor count and reward sum, its local fallback clock and EXP3 weights, and one uniform random priority permutation Π_C of $[K]$, drawn when C is first visited.

At round t the platform observes x_t, sets $C := C(x_t)$, increments V_C and acts according to the local phase of C.

Stage 1 (bidding), while $V_C \le KL_n$. Pull agent $((V_C - 1) \bmod K) + 1$. Thus agent i's rth bidding pull at C is the $((r-1)K + i)$th visit to C, and the schedule reads no reward, no bid and no coin. When $V_C = KL_n$, form the bids $\beta_{i,C} := L_n^{-1} \sum_{r \le L_n} Y_{i,C,r}$, where $Y_{i,C,r}$ is the reward at agent i's rth bidding pull at C; let W_C be a largest bidder, ties broken in favour of the highest Π_C priority; let $\beta_{(2),C}$ be the second order statistic of the bids counting multiplicity; and announce

$$\tau_C := \left(\beta_{(2),C} - \sigma_n\right)_+ = \left(\beta_{(2),C} - 3s_n - \rho_n\right)_+.$$

A cube that receives fewer than KL_n visits over the whole horizon never awards, never prices and never monitors.

Stage 2 (monitored service), until a local switch. Pull W_C; with r the number of Stage-2 pulls at C so far and Z_r their reward sum, switch at the first r with $Z_r < r\tau_C - b_n(r)$.

Stage 3 (local fallback), after a switch at C. Run explicit-exploration EXP3 over all K agents on the visits to C, restarted in doubling epochs on the local fallback clock: epoch $k \ge 0$ covers the local fallback times $2^k, \ldots, 2^{k+1} - 1$, starts with fresh weights $w_j = 1$, and uses

$$\gamma_k := \min\{1/2, \sqrt{K\log K / 2^k}\}, \qquad \eta_k := \gamma_k / K,$$

with the sampling distribution $p_{j,u} := (1-\gamma_k)q_{j,u} + \gamma_k/K$, where $q_{j,u}$ is the normalised weight of agent j at round u, the importance-weighted estimate $\widehat{Y}_{j,u} := Y_u \mathbf{1}\{I_u = j\}/p_{j,u}$, and the update $w_j \leftarrow w_j \exp(\eta_k \widehat{Y}_{j,u})$, exactly as in Appendix F. No local horizon is used.

A selected agent is told the current cube's phase and, in Stage 2 or 3, τ_C. The proposed profile $S^\star$ is that of Appendix H.1 with the cube in place of the context: agent i delivers $M_i(x_t)$ at a bidding pull and at a fallback pull, and, as the Stage-2 winner, $\min\{\tau_C, M_i(x_t)\}$ if strategic and $\max\{\mu_i(x_t), \min\{\tau_C, M_i(x_t)\}\}$ if honest.

Three design choices carry the proofs. The shading $\sigma_n = 3s_n + \rho_n$ replaces the scalar mechanism's s_n. Of it, the proofs below use one bid radius s_n, which absorbs the estimation error of the realised bid of whoever won, and one oscillation allowance $\rho_n/3$, which makes a price derived from averages over the cube attainable at every context of the cube (Lemmas H.23 and H.24); the remaining $2s_n + \frac{2}{3}\rho_n$ is slack and enters only the lower bounds on the price (Lemma H.25 and part (iii) of Proposition H.30). The Stage-1 schedule is reward independent and determined by the arrival sequence alone, which is what makes the bid-domination coupling in the proof of Proposition H.30(i) pathwise. The fallback restarts in doubling epochs because a cube's remaining number of visits is unknown when its fallback starts.

Hypotheses. Throughout this subsection $K \geq 2$; one context $x_t \in [0,1)^d$ arrives per round, observed by the platform before it selects and by the selected agent before it chooses its target, so that agent i's primitives at round t are $\mu_i(x_t)$ and $M_i(x_t)$; the functions $\mu_i, M_i : [0,1)^d \to [0,1]$ satisfy $\mu_i \leq M_i$ and are L-Lipschitz in the Euclidean norm for a constant $L \geq 0$ unknown to the platform; the honest set $\mathcal{K}_H$ is fixed and nonempty; and n is known to the platform, with $n \geq 2K$ and

$$\log(en) \geq 3L, \tag{22}$$

which holds in particular when $n \geq e^{3L}$, the form used in Theorem 4.2. All guarantees are for the fixed horizon n; no anytime claim is made.

The form of (A3) *that is used.* Fix a round u and define the fields

$$\begin{aligned}
\mathcal{G}_u^0 &:= \sigma(\mathcal{G}_{u-1}, x_u),\\
\mathcal{G}_u^k &:= \sigma\big(\mathcal{G}_u^{k-1},\ k\text{th mechanism randomisation at } u\big),\\
\mathcal{G}_u^- &:= \sigma\big(\mathcal{G}_u^{\kappa},\ \text{selected agent's coin at } u,\ m_{I_u,u}\big),\\
\mathcal{G}_u &:= \sigma(\mathcal{G}_u^-, U_u),
\end{aligned}$$

for $k = 1, \ldots, \kappa$, where $\mathcal{G}_0$ is the trivial field (the primitives μ_i, M_i, φ_i and $\mathcal{K}_H$ are fixed constants), κ is the number of randomisations the mechanism performs at a round, indexed in the order the mechanism performs them, and U_u is the round's reward seed.

> **(A3), in the form used here.** At every round u the mechanism's randomisations are drawn in the order the mechanism specifies them, and then the reward seed. Each has its specified law conditionally on the field generated by everything realised before it: the kth randomisation of the mechanism has its specified law conditionally on $\mathcal{G}_u^{k-1}$, and the seed U_u is uniform on $[0,1]$ conditionally on $\mathcal{G}_u^-$, with $Y_u = \mathbf{1}\{U_u \leq m_{I_u,u}\}$. The selected agent's private randomisation and its target are realised between the last mechanism randomisation and the seed and are otherwise unconstrained; they belong to $\mathcal{G}_u^-$, and the seed clause is stated conditionally on them.

We call the two parts of this condition the *mechanism clauses* (one for each k) and the *seed clause*. The quantifier runs over the mechanism's own randomisations, whatever they are. For CG-MESP, $\kappa = 2$: (f1) the priority permutation $\Pi_{C(x_u)}$, uniform on the permutations of $[K]$, if $C(x_u)$ is visited for the first time at round u; (f2) the selection I_u, which in Stages 1 and 2 is the deterministic function of the cube's state specified above and in Stage 3 is a draw with the law p_u computed from the cube's own EXP3 state; and then (f3) the seed. Nothing is assumed about how the contexts are generated: x_u may be an arbitrary measurable function of the entire past, including past rewards and past randomisations, and of fresh randomness, provided it is realised before the round's own draws.

Two consequences of the seed clause are used repeatedly. First, $\mathbb{E}[Y_u \mid \mathcal{G}_u^-] = m_{I_u,u}$, so for every $\mathcal{G}_u^-$-measurable indicator ϕ_u, $\mathbb{E}[\phi_u Y_u] = \mathbb{E}[\phi_u m_{I_u,u}]$; in particular $P(n;S) = \mathbb{E}_S \sum_{t\leq n} m_{I_t,t}$ for every profile S. Second, for a $\mathcal{G}_u^-$-measurable $\phi_u \in \{0,1\}$, the increment $\phi_u(Y_u - m_{I_u,u})$ has conditional mean zero and conditional range at most one given $\mathcal{G}_u^-$, so Hoeffding's lemma gives $\mathbb{E}[\exp(\lambda\phi_u(Y_u - m_{I_u,u})) \mid \mathcal{G}_u^-] \leq \exp(\lambda^2\phi_u/8)$ for every real λ.

(A3) constrains a round's own draws given that round's context and the past, not given later contexts, which are realised after the round's seed. Adjoining $\sigma(x_1, \ldots, x_n)$ to the fields above need not preserve the seed clause: if x_{u+1} is a fixed function of U_u, which no agent can influence and which exogeneity and (A3) permit, then U_u is not uniform conditionally on $\sigma(\mathcal{G}_u^-, x_{u+1})$. Condition (A3-P) of the preamble of this appendix is (A3) with every field so enlarged; it is the second half of (A2). Part (i) of Theorem 4.2 is stated over $\mathcal{B}_{i,n}^{\mathrm{ctx}+}$, whose targets may depend on contexts not yet realised, and for that class (A3-P) is needed. Over $\mathcal{B}_{i,n}^{\mathrm{ctx}}$ it is not used: the proof of part (i) places the arrival sequence in no conditioning field.

Which clause each result uses is recorded with the result. Lemmas H.21, H.23, H.24, H.25 and parts (i) and (ii) of Lemma H.26 are deterministic. Lemmas H.22, H.26(iii) and H.27 use the seed clause only. Lemma H.28 uses the Stage-3 selection clause as well, and this is essential: without it, an arrival path that is a function of the round's own fallback coin can prevent the honest comparator from ever being selected at a cube, and the lemma fails by a linear amount. Part (i) of Proposition H.30 uses all of (A3) and, at exactly one place (the coupling of two executions), the exogeneity half of (A2); over $\mathcal{B}_{i,n}^{\mathrm{ctx}+}$ it uses (A3-P) in place of (A3).

Notation and the oscillation allowance. For a cube C write V_C for its visit count over the horizon, $n_C := (V_C - KL_n)_+$ for its number of post-bidding visits, and J_n' for the number of cubes visited at least once, so that $\sum_C V_C = n$, $\sum_C n_C \leq n$ and $J_n' \leq \min\{J_n, n\}$. Since $g_n \leq n^{1/(d+3)} + 1 \leq 2n^{1/(d+3)}$,

$$J_n \leq 2^d n^{d/(d+3)},$$

so that $J_n \leq n$ whenever $n \geq n_1(d) := 2^{d(d+3)/3}$, which is 2.5 at $d = 1$, 10.1 at $d = 2$ and 64 at $d = 3$. The condition $J_n \leq n$ is used only to write union bounds over the J_n cubes in the forms n^{-4} and n^{-3}/K; where it is used, this is said. Put $M_j^{\inf}(C) := \inf_{x\in C} M_j(x)$, $M_j^{\sup}(C) := \sup_{x\in C} M_j(x)$, and $\mu_j^{\inf}(C)$, $\mu_j^{\sup}(C)$ likewise. For an agent j and a cube C that completes bidding, let $m_{j,C,r}$ and $Y_{j,C,r}$ be the target and the reward at agent j's rth Stage-1 pull at C, which takes place at round $t_{j,C,r}$, and set

$$\beta_{j,C} := \frac{1}{L_n}\sum_{r\leq L_n} Y_{j,C,r}, \qquad \bar{A}_{j,C} := \frac{1}{L_n}\sum_{r\leq L_n} m_{j,C,r},$$

$$\bar{M}_{j,C} := \frac{1}{L_n}\sum_{r\leq L_n} M_j(x_{t_{j,C,r}}).$$

Feasibility gives $m_{j,C,r} \le M_j(x_{t_{j,C,r}})$ at every pull, hence

$$\bar{A}_{j,C} \le \bar{M}_{j,C} \qquad \text{pathwise, for every agent and every behaviour,} \tag{23}$$

with equality when j bids at capacity, as it does under $S^\star$.

Lemma H.21 (Within-cube oscillation)**.** *Write $\omega := L\sqrt{d}\,h_n$ and $M_\star(x) := \max_j M_j(x)$. For every agent j and every cube C, the oscillation $M_j^{\sup}(C) - M_j^{\inf}(C)$ is at most ω, and the same holds for μ_j, for $M_\star$, for $M_{(2)}$ and for b. Moreover $\omega \le \rho_n/3$, and for every $x \in C$,*

$$M_j(x) - \omega \le \bar{M}_{j,C} \le M_j(x) + \omega.$$

Proof. Two points of the same half-open cube of side h_n differ by less than h_n in each of the d coordinates, hence by less than $\sqrt{d}\,h_n$ in Euclidean norm. Each μ_j and each M_j is L-Lipschitz, so each varies by at most $L\sqrt{d}\,h_n = \omega$ over a cube. For two contexts $x, y \in C$, the vectors $(M_1(x), \dots, M_K(x))$ and $(M_1(y), \dots, M_K(y))$ differ by at most ω in every coordinate; a maximum and an order statistic are 1-Lipschitz with respect to the supremum norm, so $M_\star$ and $M_{(2)}$ vary by at most ω over C, and so does $b(x) = \max_{h \in \mathcal{K}_H} \mu_h(x)$. Next, (22) gives $L \le \frac{1}{3}\log(en)$, so $\omega = L\sqrt{d}\,h_n \le \frac{1}{3}\sqrt{d}\,h_n \log(en) = \rho_n/3$; the factor $\sqrt{d}$ cancels, which is why (22) carries no dimension. Finally, $\bar{M}_{j,C}$ is an average of values of M_j at contexts of C, so it lies in $[M_j^{\inf}(C), M_j^{\sup}(C)]$, an interval of length at most ω that also contains $M_j(x)$. □

This is the only place where L and (22) enter: L appears in no bound, only in the hypothesis.

Lemma H.22 (Predictable bid concentration, uniform over all behaviour)**.** *This lemma uses the seed clause of* (A3) *only. For a cube C and an agent j, let $E_{j,C}$ be the event that either C does not complete bidding or $|\beta_{j,C} - \bar{A}_{j,C}| \le s_n$, and put $E_C := \bigcap_{j \in [K]} E_{j,C}$ and $E := \bigcap_C E_C$, the intersection over all J_n cubes of the grid. Then, whatever strategies the K agents use,*

$$\Pr(E_{j,C}^c) \le 2e^{-\ell_n} = \frac{1}{Kn^5}, \qquad \Pr(E_C^c) \le 2Ke^{-\ell_n} = n^{-5},$$
$$\Pr(E^c) \le 2KJ_n e^{-\ell_n} = J_n n^{-5},$$

so that $\Pr(E^c) \le n^{-4}$ whenever $J_n \le n$. The one-sided events $\{\beta_{j,C} \ge \bar{A}_{j,C} - s_n\}$ and $\{\beta_{j,C} \le \bar{A}_{j,C} + s_n\}$, each again taken to hold if C does not complete bidding, obey the same bounds with the factor 2 removed.

Proof. Fix C and j. Let ϕ_s be the indicator that round s is agent j's Stage-1 pull at C with local index at most L_n. Whether round s has this description is determined by x_s and by the number of visits to C among rounds $1, \dots, s$, hence by $x_1, \dots, x_s$ alone; so ϕ_s is $\mathcal{G}_s^0$-measurable and a fortiori $\mathcal{G}_s^-$-measurable. For a real λ put

$$M_t := \sum_{s \le t} \phi_s (Y_s - m_{I_s,s}), \qquad c_t := \sum_{s \le t} \phi_s,$$
$$N_t := \exp\bigl(\lambda M_t - \lambda^2 c_t/8\bigr), \qquad N_0 := 1.$$

Since N_{t-1} and c_t are $\mathcal{G}_t^-$-measurable, the Hoeffding consequence of the seed clause recorded in the hypotheses paragraph gives $\mathbb{E}[N_t \mid \mathcal{G}_t^-] \le N_{t-1}$, hence $\mathbb{E}[N_n] \le 1$. Always $c_n \le L_n$. On the event that C completes bidding, $c_n = L_n$ exactly and $M_n = L_n(\beta_{j,C} - \bar{A}_{j,C})$, so on that event $\{\beta_{j,C} - \bar{A}_{j,C} > s_n\} \subseteq \{N_n > \exp(\lambda L_n s_n - \lambda^2 L_n/8)\}$. Markov's inequality with $\lambda = 4s_n$ bounds the probability of the latter event by $\exp(-2L_n s_n^2) = e^{-\ell_n}$, using $s_n^2 = \ell_n/(2L_n)$. The lower tail is identical with $\lambda = -4s_n$. This proves the first bound, and the one-sided forms are the single tails. The second and third bounds are union bounds over the K agents and over the deterministic set of the J_n cubes of the grid, so no measurability of the set of visited cubes is needed; and $J_n n^{-5} \le n^{-4}$ when $J_n \le n$. No claim is made that ϕ_s is $\mathcal{G}_{s-1}$-measurable, and none is needed: the supermartingale argument conditions on $\mathcal{G}_s^-$, which contains x_s. □

Two remarks. First, E does not say that a bid concentrates around a capability; it says that a bid concentrates around the average of the targets actually delivered during bidding. For an agent following $S^\star$ the two coincide, since then $\bar{A}_{j,C} = \bar{M}_{j,C}$; for a deviator they need not, and E does not claim that they do. What E and (23) give for a deviator is the one-sided bound $\beta_{j,C} \le \bar{M}_{j,C} + s_n$, which is the only direction any argument below needs, and it requires no counterfactual capacity bid. Second, E_C is determined by the rewards at C's Stage-1 rounds, hence by the information available at the start of Stage 2 at C and a fortiori at the start of C's fallback; the global event E is not, so the per-cube events are used wherever a conditional expectation is taken.

The three price lemmas. Throughout this paragraph, fix a cube C that completes bidding, let $W = W_C$ be its winner, which is a largest bidder, and recall $\tau_C = (\beta_{(2),C} - \sigma_n)_+$ with $\sigma_n = 3s_n + \rho_n$. Since W is a largest bidder, $\beta_{(2),C} \le \beta_{W,C}$.

Lemma H.23 (The realised winner can meet the price at every context of its cube)**.** *On E_C, for every $x \in C$,*

$$\tau_C \le \bigl(M_W(x) - 2s_n - \tfrac{2}{3}\rho_n\bigr)_+ \le M_W(x).$$

This holds whoever W is, whether or not W deviates, and whatever any agent bid.

Proof. On E_C, $\beta_{W,C} \le \bar{A}_{W,C} + s_n$; by (23), $\bar{A}_{W,C} \le \bar{M}_{W,C}$; and by Lemma H.21, $\bar{M}_{W,C} \le M_W(x) + \omega \le M_W(x) + \rho_n/3$ for every $x \in C$. Hence $\beta_{(2),C} \le \beta_{W,C} \le M_W(x) + s_n + \rho_n/3$, so that $\beta_{(2),C} - \sigma_n \le M_W(x) - 2s_n - \frac{2}{3}\rho_n$. Taking positive parts preserves the inequality, and $(M_W(x) - 2s_n - \frac{2}{3}\rho_n)_+ \le M_W(x)$ because $M_W(x) \ge 0$. □

The point of Lemma H.23 is that the mechanism never needs an elicited value to equal a capability; it needs the shaded second bid to lie *below* the capability of whoever won, at every context of the cube. The first is false on a continuum, while the second is a one-sided statement that survives both the averaging over the cube, paid for by $\rho_n/3$, and arbitrary deviation, paid for by s_n. Two consequences are used repeatedly. On E_C, at every Stage-2 pull at C the winner's $S^\star$ target $\min\{\tau_C, M_W(x_t)\}$ equals τ_C, and the honest variant $\max\{\mu_W(x_t), \tau_C\}$ is at least τ_C; so under $S^\star$ the cube's monitor is a test against a constant that is met at every pull. And on E_C, if the deviator i is the winner, then $\tau_C \le M_i(x)$ at every $x \in C$, so $S_i^\star$ is feasible against the announced price everywhere in the cube: no capability gap and no threshold on n of the form $4s_n \le \delta_i$ is needed.

Lemma H.24 (A non-deviating competitor dominates the price)**.** *Fix an agent i and suppose that every $j \ne i$ plays $S^\star$. Put $v_i(C) := \max_{j \ne i} M_j^{\inf}(C)$. On E_C, $\tau_C \le v_i(C)$.*

Proof. Let $j^* \ne i$ attain $\max_{j \ne i} \beta_{j,C}$. If i is a largest bidder, the second order statistic counting multiplicity is $\beta_{(2),C} = \max_{j \ne i} \beta_{j,C} =$

$\beta_{j^*,C}$. If i is not a largest bidder, then $W \neq i$ and $\beta_{(2),C} \le \beta_{W,C} \le \beta_{j^*,C}$. In both cases $\beta_{(2),C} \le \beta_{j^*,C}$. Since j^* plays $S^\star$, $\bar{A}_{j^*,C} = \bar{M}_{j^*,C}$, so on E_C, by Lemma H.21,

$$\beta_{j^*,C} \le \bar{M}_{j^*,C} + s_n \le M^{\sup}_{j^*}(C) + s_n \le M^{\inf}_{j^*}(C) + \omega + s_n.$$

Therefore $\beta_{(2),C} - \sigma_n \le M^{\inf}_{j^*}(C) + \omega + s_n - 3s_n - \rho_n \le M^{\inf}_{j^*}(C) \le v_i(C)$, and since $v_i(C) \ge 0$, also $\tau_C = (\beta_{(2),C} - \sigma_n)_+ \le v_i(C)$. □

Lemma H.25 (The price is close to the second capability at $S^\star$). *Suppose every agent plays $S^\star$. On E_C, for every $x \in C$, $\tau_C \ge M_{(2)}(x) - 4s_n - \frac{4}{3}\rho_n$.*

Proof. Under $S^\star$, $\bar{A}_{j,C} = \bar{M}_{j,C}$ for every j, so on E_C Lemma H.21 gives $\beta_{j,C} \ge \bar{M}_{j,C} - s_n \ge M_j(x) - \omega - s_n$ for every j and every $x \in C$. The second order statistic is nondecreasing in each argument, so $\beta_{(2),C} \ge M_{(2)}(x) - \omega - s_n$. Hence $\tau_C \ge \beta_{(2),C} - \sigma_n \ge M_{(2)}(x) - \omega - 4s_n - \rho_n \ge M_{(2)}(x) - 4s_n - \frac{4}{3}\rho_n$, using $\omega \le \rho_n/3$. The claim is trivially true when its right side is negative. □

Lemma H.26 (Honest bid floor). *Fix a cube C and put*

$$v_C := \max_{h \in \mathcal{K}_H} \mu^{\inf}_h(C);$$

let $h_C \in \mathcal{K}_H$ attain this maximum, with a fixed tie break. Then:

(i) *$b(x) \le v_C + \omega \le v_C + \rho_n/3$ for every $x \in C$;*
(ii) *under any behaviour that respects pointwise honesty, $m_{h_C,t} \ge v_C$ at every round t with $x_t \in C$ at which h_C is selected;*
(iii) *let E^H_C be the event that either C does not complete bidding or $\beta_{h_C,C} \ge \bar{A}_{h_C,C} - s_n$. Then $\Pr((E^H_C)^c) \le e^{-\ell_n}$, by the one-sided form of Lemma H.22; and if C completes bidding, then on E^H_C the honest bid satisfies $\beta_{h_C,C} \ge v_C - s_n$.*

Proof. (i) Let $h \in \mathcal{K}_H$ attain $b(x) = \mu_h(x)$. By Lemma H.21, $\mu_h(x) \le \mu^{\inf}_h(C) + \omega \le v_C + \omega$, and $\omega \le \rho_n/3$. (ii) Honesty gives $m_{h_C,t} \ge \mu_{h_C}(x_t) \ge \mu^{\inf}_{h_C}(C) = v_C$. (iii) By (ii), every Stage-1 target of h_C at C is at least v_C, so $\bar{A}_{h_C,C} \ge v_C$, and on E^H_C therefore $\beta_{h_C,C} \ge \bar{A}_{h_C,C} - s_n \ge v_C - s_n$. □

Monitor and fallback.

Lemma H.27 (Anytime cube monitor). *This lemma uses the seed clause of* (A3) *only, and no property of the arrival path. Fix a cube C of the grid. Let Y^C_r and m^C_r be the reward and the target at the rth Stage-2 pull at C, which takes place at round t_r, put $Z_r := \sum_{q \le r} Y^C_q$ and $A_r := \sum_{q \le r} m^C_q$, both extended with zero increments beyond the number of Stage-2 pulls actually taken, and define*

$$\mathcal{H}_C := \{|Z_r - A_r| \le b_n(r) \text{ for every } r \le n\}, \qquad \mathcal{H} := \bigcap_C \mathcal{H}_C,$$

the intersection over all J_n cubes of the grid. Then, for any behaviour of every agent:

(a) *$\Pr(|Z_r - A_r| > b_n(r)) \le 2e^{-\ell_n}$ for each $r \le n$; hence $\Pr(\mathcal{H}^c_C) \le 2ne^{-\ell_n} = K^{-1}n^{-4}$ and $\Pr(\mathcal{H}^c) \le 2J_n ne^{-\ell_n} = J_n K^{-1} n^{-4}$, which is at most n^{-3}/K whenever $J_n \le n$;*
(b) *on $\mathcal{H}_C$, if the Stage-2 phase at C lasted $q \ge 1$ pulls, counting its failing pull when there is one, then $A_q \ge q\tau_C - 1 - 2b_n(q)$;*
(c) *on $\mathcal{H}_C$, the Stage-2 winner's realised utility over that phase is at most $\sum_{r \le q}\big(1 + \varphi_W(\mu_W(x_{t_r})) - \varphi_W(\tau_C)\big) + \Lambda(1 + 2b_n(q))$;*
(d) *on $\mathcal{H}_C$, if every Stage-2 target at C is at least τ_C, then no switch occurs at C;*
(e) *$\mathbb{E}[Z_r - A_r] = 0$ for r the total number of Stage-2 pulls at C; more generally, $\mathbb{E}[(Z_r - A_r)\mathbf{1}_\Gamma] = 0$ for every event Γ determined by the information available at C's award round.*

Proof. (a) Fix $r \ge 1$. Let ψ_s be the indicator that round s is a Stage-2 pull at C, let T_r be the round of the rth such pull, with $T_r := n$ if there are fewer than r, and put $\phi_s := \psi_s \mathbf{1}\{s \le T_r\}$. Whether round s is a Stage-2 pull at C is determined by x_s, by the number of visits to C among rounds $1, \dots, s$, and by whether C has awarded and not yet switched, which depends on the price τ_C and on rewards at rounds strictly before s; and $\{s \le T_r\}$ is the event that fewer than r Stage-2 pulls at C occurred before round s. Hence ϕ_s is $\mathcal{G}^0_s$-measurable, and a fortiori $\mathcal{G}^-_s$-measurable. Define M_t, c_t and N_t from this ϕ exactly as in the proof of Lemma H.22; the same argument gives $\mathbb{E}[N_n] \le 1$. Under the zero extension, $M_n = Z_r - A_r$ and $c_n \le r$, so $\mathbb{E}[\exp(\lambda(Z_r - A_r))] \le \exp(\lambda^2 r/8)$, and Markov's inequality with $\lambda = 4b_n(r)/r$ gives $\Pr(Z_r - A_r > b_n(r)) \le \exp(-2b_n(r)^2/r) = e^{-\ell_n}$; the lower tail is symmetric. The union bound runs over $r \in \{1, \dots, n\}$ and over the deterministic set of the J_n cubes of the grid, so no measurability of the set of visited cubes is needed, and $2J_n ne^{-\ell_n} = J_n/(Kn^4) \le n^{-3}/K$ when $J_n \le n$.

(b) If no switch occurred through pull q, the test was passed at q, so $Z_q \ge q\tau_C - b_n(q)$, and on $\mathcal{H}_C$, $A_q \ge Z_q - b_n(q) \ge q\tau_C - 2b_n(q)$. If the switch occurred at pull $q \ge 2$, the test was passed at $q - 1$, so the same argument gives $A_{q-1} \ge (q-1)\tau_C - 2b_n(q-1)$, and $A_q \ge A_{q-1} \ge q\tau_C - 1 - 2b_n(q)$ because $\tau_C \le 1$ and b_n is nondecreasing. If the switch occurred at pull $q = 1$, then $A_1 \ge 0 \ge \tau_C - 1 - 2b_n(1)$.

(c) The winner's realised Stage-2 utility is

$$\sum_{r \le q}\big(1 + \varphi_W(\mu_W(x_{t_r}))\big) - \sum_{r \le q} \varphi_W(m^C_r).$$

By (b), the targets $m^C_1, \dots, m^C_q \in [0,1]$ satisfy $\sum_{r \le q} m^C_r \ge q\tau_C - (1 + 2b_n(q))$, so Lemma F.1(a) with $\theta = \tau_C$ and $\Delta = 1 + 2b_n(q)$ bounds the subtracted sum below by $q\varphi_W(\tau_C) - \Lambda(1 + 2b_n(q))$.

(d) The hypothesis gives $A_r \ge r\tau_C$ for every r up to the number of Stage-2 pulls, so on $\mathcal{H}_C$, $Z_r \ge A_r - b_n(r) \ge r\tau_C - b_n(r)$, which is the negation of the switch test at r.

(e) Put $\phi_s := \psi_s \mathbf{1}_\Gamma$. Since Γ is determined at C's award round, which precedes every Stage-2 pull at C, ϕ_s is $\mathcal{G}^0_s$-measurable, and $M_t := \sum_{s \le t} \phi_s (Y_s - m_{I_s,s})$ is a martingale with respect to $(\mathcal{G}_t)_{t \le n}$ by the seed clause and the tower property. Hence $\mathbb{E}[M_n] = 0$, and $M_n = (Z_r - A_r)\mathbf{1}_\Gamma$ for r the total number of Stage-2 pulls at C. Taking Γ to be the whole space gives the first claim. No optional stopping is involved, since the time n is deterministic. □

Lemma H.28 (Cube fallback with a local horizon that is not known in advance). *This lemma uses the Stage-3 selection clause and the seed clause of* (A3). *Fix a cube C, let T_C be the round at which C switches, with $T_C := n$ if it never switches, and let $\mathcal{G}^{\mathrm{fb}}_C := \mathcal{G}_{T_C}$ be the stopped field, the information present at C's fallback start. Let H be the number of fallback visits to C over the horizon; H is* not *assumed to be $\mathcal{G}^{\mathrm{fb}}_C$-measurable. Suppose that for some deterministic $v \in [0,1]$ and some agent j^*, the target of j^* satisfies $m_{j^*,u} \ge v$ at every fallback visit u to C at which j^* is selected. Then, almost surely,*

$$\mathbb{E}\Big[\sum_{\text{fallback visits } u \text{ to } C} Y_u \,\Big|\, \mathcal{G}^{\mathrm{fb}}_C\Big] \ge v\,\mathbb{E}\big[H \mid \mathcal{G}^{\mathrm{fb}}_C\big] - \mathbb{E}\big[R^{\mathrm{dbl}}_K(H) \mid \mathcal{G}^{\mathrm{fb}}_C\big],$$

and taking expectations gives the unconditional form

$$\mathbb{E}\Big[\sum_{\text{fallback visits } u \text{ to } C} Y_u\Big] \ge v\,\mathbb{E}[H] - \mathbb{E}[R_K^{\mathrm{dbl}}(H)].$$

Proof. *Measurability.* The event $\{T_C \le u\}$ is determined by τ_C and by the rewards at C's Stage-2 rounds up to round u, so T_C is a stopping time of $(\mathcal{G}_u)_{u\le n}$ and $\mathcal{G}_C^{\mathrm{fb}}$ is well defined. Index by rounds, not by local times. Let ϕ_u be the indicator that round u is a fallback visit to C, and let $k(u)$ be the epoch index of C's local fallback clock at round u. Both are $\mathcal{G}_u^0$-measurable: the switch is determined by earlier rounds, the local clock counts previous fallback visits to C, and x_u tells whether round u is a visit to C. The EXP3 weights at round u, hence q_u and p_u, are functions of $k(u)$ and of the cube's own earlier fallback selections and rewards, so they are $\mathcal{G}_u^0$-measurable as well. For an event $B \in \mathcal{G}_C^{\mathrm{fb}}$, the indicator $\mathbf{1}_B\phi_u$ is $\mathcal{G}_u^0$-measurable: $\phi_u = 1$ forces $u > T_C$, and $B \cap \{T_C < u\} \in \mathcal{G}_{u-1}$ because T_C is a stopping time. We prove that for every such B,

$$\mathbb{E}\Big[\mathbf{1}_B \sum_u \phi_u(v - Y_u)\Big] \le \mathbb{E}\big[\mathbf{1}_B R_K^{\mathrm{dbl}}(H)\big], \tag{†}$$

which is the conditional claim; B equal to the whole space gives the unconditional form.

One round. Fix u and write $\gamma := \gamma_{k(u)}$, $\eta := \eta_{k(u)}$. By the Stage-3 selection clause, I_u has law p_u conditionally on the field preceding the selection, which contains $\mathcal{G}_u^0$; since p_u is $\mathcal{G}_u^0$-measurable, the tower property gives $\Pr(I_u = j \mid \mathcal{G}_u^0) = p_{j,u}$ on $\{\phi_u = 1\}$. Put $a_{j,u} := \mathbb{E}[Y_u\mathbf{1}\{I_u = j\} \mid \mathcal{G}_u^0] \ge 0$, so that $\mathbb{E}[Y_u \mid \mathcal{G}_u^0] = \sum_j a_{j,u}$. By the seed clause and the hypothesis on j^*, $a_{j^*,u} = \mathbb{E}[m_{I_u,u}\mathbf{1}\{I_u = j^*\} \mid \mathcal{G}_u^0] \ge v\,p_{j^*,u}$ on $\{\phi_u = 1\}$. Hence, on $\{\phi_u = 1\}$,

$$\begin{aligned}
\mathbb{E}[\widehat{Y}_{j^*,u} \mid \mathcal{G}_u^0] &= \frac{a_{j^*,u}}{p_{j^*,u}} \ge v,\\
\mathbb{E}\Big[\sum_j q_{j,u}\widehat{Y}_{j,u} \,\Big|\, \mathcal{G}_u^0\Big] &= \sum_j \frac{q_{j,u}}{p_{j,u}} a_{j,u} \le \frac{\mathbb{E}[Y_u \mid \mathcal{G}_u^0]}{1-\gamma},\\
\mathbb{E}\Big[\sum_j q_{j,u}\widehat{Y}_{j,u}^2 \,\Big|\, \mathcal{G}_u^0\Big] &= \sum_j \frac{q_{j,u}}{p_{j,u}^2}\mathbb{E}[Y_u^2\mathbf{1}\{I_u = j\} \mid \mathcal{G}_u^0]\\
&\le \sum_j \frac{q_{j,u}}{p_{j,u}} \le \frac{K}{1-\gamma},
\end{aligned}$$

using $p_{j,u} \ge (1-\gamma)q_{j,u}$, $Y_u^2 \le 1$ and $\Pr(I_u = j \mid \mathcal{G}_u^0) = p_{j,u}$. Also $p_{j,u} \ge \gamma/K = \eta$, so $0 \le \eta\widehat{Y}_{j,u} \le 1$.

One epoch. Fix an epoch index k and let $T_k := \sum_u \phi_u\mathbf{1}\{k(u) = k\}$ be its realised length, so that $T_k \le 2^k$, and $T_k \ge 1$ only if $H \ge 2^k$. Within epoch k the weights start at one and are updated with the fixed η_k, so the exponential-potential calculation ($e^z \le 1 + z + z^2$ for $z \in [0,1]$, as in Lemma F.3) gives, pathwise and for the rounds of the epoch actually played, writing $\sum_{(k)}$ for the sum over the rounds u with $\phi_u = 1$ and $k(u) = k$,

$$\begin{aligned}
&\sum\nolimits_{(k)} \widehat{Y}_{j^*,u} - \frac{\log K}{\eta_k}\mathbf{1}\{T_k \ge 1\}\\
&\qquad \le \sum\nolimits_{(k)}\Big(\sum_j q_{j,u}\widehat{Y}_{j,u} + \eta_k \sum_j q_{j,u}\widehat{Y}_{j,u}^2\Big).
\end{aligned}$$

Multiply by $(1-\gamma_k)\mathbf{1}_B$ and take expectations. Every indicator $\mathbf{1}_B\phi_u\mathbf{1}\{k(u) = k\}$ is $\mathcal{G}_u^0$-measurable, so each term may be replaced by its conditional expectation given $\mathcal{G}_u^0$ by the tower property, and the one-round bounds give

$$\begin{aligned}
&(1-\gamma_k)\,v\,\mathbb{E}[\mathbf{1}_B T_k] - \frac{\log K}{\eta_k}\Pr(B \cap \{T_k \ge 1\})\\
&\qquad \le \mathbb{E}\Big[\mathbf{1}_B \sum\nolimits_{(k)} Y_u\Big] + \eta_k K\,\mathbb{E}[\mathbf{1}_B T_k].
\end{aligned}$$

With $\eta_k = \gamma_k/K$ and $v \le 1$, this rearranges to

$$\mathbb{E}\Big[\mathbf{1}_B \sum\nolimits_{(k)}(v - Y_u)\Big] \le \mathbb{E}\Big[\mathbf{1}_B\Big(2\gamma_k T_k + \frac{K\log K}{\gamma_k}\mathbf{1}\{T_k \ge 1\}\Big)\Big].$$

If $2^k \ge 4K\log K$, then $\gamma_k = \sqrt{K\log K/2^k} \le \frac{1}{2}$, and since $T_k \le 2^k$, $2\gamma_k T_k + (K\log K/\gamma_k)\mathbf{1}\{T_k \ge 1\} \le 3\sqrt{K2^k\log K}\,\mathbf{1}\{T_k \ge 1\}$. Otherwise $\gamma_k = \frac{1}{2}$ and $2\gamma_k T_k + 2K\log K\,\mathbf{1}\{T_k \ge 1\} \le (2^k + 2K\log K)\mathbf{1}\{T_k \ge 1\} \le (3\sqrt{K2^k\log K} + 2K\log K)\mathbf{1}\{T_k \ge 1\}$, because $2^k < 4K\log K$ gives $3\sqrt{K2^k\log K} > \frac{3}{2}\cdot 2^k$. In both cases the bracket is at most $(3\sqrt{K2^k\log K} + 2K\log K)\mathbf{1}\{T_k \ge 1\}$.

Summing the epochs. Sum the last display over $k \ge 0$. Pathwise, $\mathbf{1}\{T_k \ge 1\} = 0$ unless $2^k \le H$, so with $k_{\max} := \lfloor \log_2 H \rfloor$ when $H \ge 1$,

$$\begin{aligned}
\sum_{k \le k_{\max}} 3\sqrt{K2^k\log K} &= 3\sqrt{K\log K}\,\frac{2^{(k_{\max}+1)/2} - 1}{\sqrt{2} - 1}\\
&\le \frac{3\sqrt{2}}{\sqrt{2}-1}\sqrt{KH\log K} = 3(2+\sqrt{2})\sqrt{KH\log K}\\
&\le 11\sqrt{K(H+1)\log K},
\end{aligned}$$

using $2^{k_{\max}+1} \le 2H$; and the number of epochs with $T_k \ge 1$ is $k_{\max} + 1 \le \log_2(2H) \le \log_2(2(H+1))$, each contributing at most $2K\log K$. The sum is therefore at most $R_K^{\mathrm{dbl}}(H)$ pathwise, also when $H = 0$, where both sides vanish. This proves (†).

Nothing in the argument requires H to be known at the fallback start, or at all: H enters only as $\sum_u \phi_u$, with each ϕ_u already $\mathcal{G}_u^0$-measurable, which is also why the mechanism needs no local horizon. The filtration may carry every agent's cross-cube history and, under (A3-P), the entire arrival path; the argument needs only that p_u is measurable before the selection and that the selection and the seed are fresh conditionally on the fields that precede them. □

Lemma H.29 (The deviator's fallback utility at a cube). *Let agent i deviate while every $j \ne i$ plays $S^\star$, fix a cube C, and let H be the number of fallback visits to C. With $\mathcal{G}_C^{\mathrm{fb}}$ the field of Lemma H.28, write $\bar H := \mathbb{E}[H \mid \mathcal{G}_C^{\mathrm{fb}}]$ and $\bar R := \mathbb{E}[R_K^{\mathrm{dbl}}(H) \mid \mathcal{G}_C^{\mathrm{fb}}]$, and let $U_i^{\mathrm{fb}}(C)$ be the realised utility of i over its fallback pulls at C. Then, almost surely,*

$$\mathbb{E}\big[U_i^{\mathrm{fb}}(C) \mid \mathcal{G}_C^{\mathrm{fb}}\big] \le \big(1 + \varphi_i(\mu_i^{\sup}(C)) - \varphi_i(\nu_i(C))\big)\bar H + \Lambda\big(\bar R + \tfrac{1}{3}\rho_n\bar H\big),$$

and if i is honest, also $\mathbb{E}[U_i^{\mathrm{fb}}(C) \mid \mathcal{G}_C^{\mathrm{fb}}] \le \bar H$. Neither H nor $R_K^{\mathrm{dbl}}(H)$ is asserted to be $\mathcal{G}_C^{\mathrm{fb}}$-measurable.

Proof. Let $j \ne i$ attain $\nu_i(C) = \max_{j'\ne i} M_{j'}^{\inf}(C)$. Under $S^\star$, j delivers $M_j(x_u) \ge \nu_i(C)$ at every fallback pull at C, so Lemma H.28 applies with $j^* = j$ and the deterministic $v = \nu_i(C)$:

$$\mathbb{E}\Big[\sum_{\text{fallback visits } u \text{ to } C} Y_u \,\Big|\, \mathcal{G}_C^{\mathrm{fb}}\Big] \ge \nu_i(C)\bar H - \bar R.$$

Let Q_i and A_i be the number of fallback pulls of i at C and the sum of its targets there, and $\bar Q_i$, $\bar A_i$ their conditional expectations

given $\mathcal{G}_C^{\mathrm{fb}}$. For a fallback round u at C, an agent j' and an event $B \in \mathcal{G}_C^{\mathrm{fb}}$, the indicator $\mathbf{1}_B \phi_u \mathbf{1}\{I_u = j'\}$ is $\mathcal{G}_u^-$-measurable, so by the seed clause the fallback reward collected by i at C has conditional expectation $\bar{A}_i$, and the fallback reward collected by the other agents has conditional expectation equal to that of their target sum. Every $j' \neq i$ delivers $M_{j'}(x_u) \le M_{j'}^{\sup}(C) \le M_{j'}^{\inf}(C) + \omega \le \nu_i(C) + \rho_n/3$ by Lemma H.21, so the others' target sum has conditional expectation at most $(\nu_i(C) + \rho_n/3)(\bar{H} - \bar{Q}_i)$. Combining,

$$\nu_i(C)\bar{H} - \bar{R} \le \bar{A}_i + (\nu_i(C) + \rho_n/3)(\bar{H} - \bar{Q}_i),$$
$$\text{hence} \qquad \bar{A}_i \ge \nu_i(C)\bar{Q}_i - \bar{R} - \tfrac{1}{3}\rho_n \bar{H}.$$

Now $U_i^{\mathrm{fb}}(C) = \sum\big(1 + \varphi_i(\mu_i(x_u)) - \varphi_i(m_{i,u})\big)$, the sum running over i's fallback pulls at C, so $\mathbb{E}[U_i^{\mathrm{fb}}(C) \mid \mathcal{G}_C^{\mathrm{fb}}] \le (1 + \varphi_i(\mu_i^{\sup}(C)))\bar{Q}_i - \int \varphi_i \, d\pi$, where π is the measure on $[0,1]$ defined by $\pi(D) := \mathbb{E}[\sum \mathbf{1}\{m_{i,u} \in D\} \mid \mathcal{G}_C^{\mathrm{fb}}]$, again over i's fallback pulls at C; it has total mass $\bar{Q}_i$ and $\int m \, d\pi = \bar{A}_i \ge \nu_i(C)\bar{Q}_i - (\bar{R} + \frac{1}{3}\rho_n \bar{H})$. If $\bar{Q}_i = 0$ there is nothing to prove. Otherwise Lemma F.1(b) with $\theta = \nu_i(C)$ and $\Delta = \bar{R} + \frac{1}{3}\rho_n \bar{H}$ gives $\int \varphi_i \, d\pi \ge \bar{Q}_i \varphi_i(\nu_i(C)) - \Lambda\Delta$, hence

$$\mathbb{E}[U_i^{\mathrm{fb}}(C) \mid \mathcal{G}_C^{\mathrm{fb}}] \le \big(1 + \varphi_i(\mu_i^{\sup}(C)) - \varphi_i(\nu_i(C))\big)\bar{Q}_i + \Lambda\big(\bar{R} + \tfrac{1}{3}\rho_n \bar{H}\big).$$

The coefficient of $\bar{Q}_i$ is nonnegative, because $\varphi_i(\nu_i(C)) \le \varphi_i(1) \le 1 + \varphi_i(0) \le 1 + \varphi_i(\mu_i^{\sup}(C))$ by monotonicity and (1), so $\bar{Q}_i \le \bar{H}$ gives the first claim. If i is honest, every fallback pull yields $1 + \varphi_i(\mu_i(x_u)) - \varphi_i(m_{i,u}) \le 1$ because $m_{i,u} \ge \mu_i(x_u)$, so $\mathbb{E}[U_i^{\mathrm{fb}}(C) \mid \mathcal{G}_C^{\mathrm{fb}}] \le \bar{Q}_i \le \bar{H}$. □

The term $\bar{R}$ is the fallback's full regret, used here as a bound on what the deviator can gain from having triggered a switch. It is loose in the regime that makes it largest: a deviator that triggers a switch and then delivers nothing is down-weighted by the local EXP3 and collects only its exploration share, of order $\sqrt{H \log K / K}$ pulls, against a charge of order $\sqrt{KH \log K}$. We keep the regret bound because it holds against every deviation, because it is the same object the revenue part needs, and because the fallback error term e_{fb} below sits on a strictly smaller power of n than the leading term, so tightening it would not change the rate.

Error terms. Every bound below is first proved pathwise in J_n', the number of cubes visited at least once, and in the per-cube counts n_C; expectations are taken at the end. Put

$$e_{\mathrm{bid}} := \min\{\mathbb{E}[J_n']L_n, n\}, \qquad e_{\mathrm{stage1}} := \min\{K\,\mathbb{E}[J_n']L_n, n\},$$
$$e_{\mathrm{osc}} := n\rho_n, \qquad e_{\mathrm{price}} := n\big(4s_n + \tfrac{4}{3}\rho_n\big),$$
$$e_{\mathrm{mon}} := \mathbb{E}[J_n'] + 2\sqrt{\mathbb{E}[J_n']\, n\ell_n/2},$$
$$e_{\mathrm{fb}} := 11\sqrt{2K\,\mathbb{E}[J_n']\, n \log K} + 2K \log K\, \mathbb{E}[J_n']\, \log_2(2(n+1)).$$

Three pathwise facts and Jensen's inequality justify these forms. The elicitation counts satisfy $\sum_C \min\{V_C, KL_n\} \le \min\{KJ_n'L_n, n\}$ and, for a single agent i, $\sum_C l_i(C) \le \min\{J_n'L_n, n\}$, where $l_i(C) \le L_n$ is the number of Stage-1 pulls that agent i receives at C; the function $x \mapsto \min\{xL_n, n\}$ is concave, so the expectations are at most e_{stage1} and e_{bid}. The monitoring charge satisfies, by the Cauchy–Schwarz inequality and $\sum_C n_C \le n$,

$$\sum_{C \text{ visited}} \big(1 + 2b_n(n_C)\big) \le J_n' + 2\sqrt{\ell_n/2}\sum_C \sqrt{n_C} \le J_n' + 2\sqrt{J_n' n \ell_n/2},$$

and the fallback charge satisfies, using $\sum_{C:\, n_C \ge 1}(n_C + 1) \le n + J_n' \le 2n$,

$$\sum_C R_K^{\mathrm{dbl}}(n_C) \le 11\sqrt{K \log K}\sum_{C:\, n_C \ge 1}\sqrt{n_C + 1} + 2K \log K\, J_n' \log_2(2(n+1)) \le 11\sqrt{2KJ_n' n \log K} + 2K \log K\, J_n' \log_2(2(n+1)).$$

Both right-hand sides are concave in J_n', so Jensen's inequality bounds their expectations by e_{mon} and e_{fb}. Since $\mathbb{E}[J_n'] \le \min\{J_n, n\}$, each error term is at most the same expression with $\min\{J_n, n\}$ in place of $\mathbb{E}[J_n']$, so the forms above are never worse than the worst-case forms and are strictly better on an arrival path that concentrates in few cubes.

Proposition H.30 (Continuous-grid MESP, exact form)**.** *Assume $K \ge 2$, L-Lipschitz capabilities and intrinsic means with L unknown to the platform,* (22)*, a nonempty honest set, and $J_n \le n$, which holds whenever $n \ge 2^{d(d+3)/3}$. Run* CG-MESP*. Then:*

(i) *under* (A2)*, for every agent i and every $S_{i,n}' \in \mathcal{B}_{i,n}^{\mathrm{ctx}+}$, and, under exogeneity of the arrival path and* (A3) *alone, for every $S_{i,n}' \in \mathcal{B}_{i,n}^{\mathrm{ctx}}$,*

$$u_i(n; S_{i,n}', S_{-i,n}^{\star}) - u_i(n; S_n^{\star}) \le e_{\mathrm{bid}} + \Lambda\big(e_{\mathrm{osc}} + e_{\mathrm{mon}} + e_{\mathrm{fb}}\big) + 4n^{-3} + 4K^{-1}n^{-2};$$

(ii) *under the seed clause of* (A3) *alone, with every agent playing $S^{\star}$,*

$$P(n; S_n^{\star}) \;\ge\; \mathbb{E}\sum_{t \le n} M_{(2)}(x_t) - \big(e_{\mathrm{stage1}} + e_{\mathrm{price}} + O(n^{-2})\big);$$

(iii) *under the Stage-3 selection clause and the seed clause of* (A3)*, for every feasible behavioural profile S in which each agent plays some strategy in $\mathcal{B}_{j,n}^{\mathrm{ctx}}$ and every honest agent respects pointwise honesty, the set of deviating agents being all of $[K]$ if it likes,*

$$P(n; S) \;\ge\; \mathbb{E}\sum_{t \le n} b(x_t) - \big(e_{\mathrm{stage1}} + e_{\mathrm{price}} + e_{\mathrm{mon}} + e_{\mathrm{fb}} + O(n^{-2})\big).$$

The $O(n^{-2})$ terms are $n^{-3} + K^{-1}n^{-2}$ in (ii) and $(2K)^{-1}n^{-3} + K^{-1}n^{-2}$ in (iii). Parts (ii) and (iii) place no condition on the arrival process beyond (A3) *and refer to no cost function: they are statements about delivered means and hold for arbitrary φ_i. No condition on a top set, on the boundary of a cube or on the mass of a cube is used, and no capability gap and no property of the arrival law appears in any error term; the only instance-dependent hypotheses are* (22) *and, in part (i), the Lipschitz constant Λ of the cost. The elicitation charge e_{bid} does not carry Λ, being bounded by one unit of effort per Stage-1 pull through* (1)*.*

Proof of Proposition H.30(i). Fix an agent i and a deviation $S_{i,n}' \in \mathcal{B}_{i,n}^{\mathrm{ctx}}$; the enlarged class is treated in Step 7.

Step 1: the coupling. Build two executions on one probability space: the *deviating* execution, in which i plays $S_{i,n}'$ (its quantities carry a prime), and the *baseline*, in which i plays $S_i^{\star}$ (its quantities carry a star); every $j \neq i$ plays $S_j^{\star}$ in both. The two executions share the arrival sequence $A := (x_1, \ldots, x_n)$, the seed array $(U_t)_{t \le n}$, the priorities (Π_C), the EXP3 coins and the private coins of i. Exogeneity is used here and nowhere else: because no agent's behaviour

can influence the sequence of contexts, A is the same random sequence in the two executions. Consequently the visit counts V_C, the post-bidding counts $n_C = (V_C - KL_n)_+$, the set of rounds that constitute Stage 1 at each cube, the agent pulled at each of them, and the counts $l_i(C)$ are the same random variables in both executions, all being functions of A alone. This is an identity between random variables on one space, not a measurability statement, and no conditioning is involved. Each execution satisfies (A3) in its own filtration, so Lemmas H.22, H.27, H.28 and H.29 apply to each. The arrival sequence is placed in no conditioning field anywhere in this proof.

Write $U'_{i,C}$ and $U^\star_{i,C}$ for the realised utility of i over the visits to cube C in the two executions. The rounds are partitioned by the cubes, so

$$u_i(n; S'_{i,n}, S^\star_{-i,n}) - u_i(n; S^\star_n) = \sum_C \mathbb{E}[U'_{i,C} - U^\star_{i,C}],$$

a decomposition of a random variable and not an independence claim; the deviator may correlate its behaviour across cubes freely. Write $E_C, \mathcal{H}_C$ for the events of Lemmas H.22 and H.27 in the deviating execution and $E^\star_C, \mathcal{H}^\star_C$ for the same events in the baseline, and put

$$F_C := E_C \cap \mathcal{H}_C \cap E^\star_C.$$

Every per-pull utility lies in $[0, 2]$ by (1), so $U'_{i,C} \le 2V_C$ and $U^\star_{i,C} \ge 0$.

Step 2: bid domination, pathwise. Fix a cube C that completes bidding. The two executions pull the same agent at the same Stage-1 rounds of C. For $j \ne i$, $S^\star_j$ delivers the deterministic target $M_j(x_t)$, so $Y_t = \mathbf{1}\{U_t \le M_j(x_t)\}$ in both executions and $\beta_{j,C}$ is the same random variable in both. At a Stage-1 round t of i, feasibility gives $m'_{i,t} \le M_i(x_t)$, so with the shared seed $\mathbf{1}\{U_t \le m'_{i,t}\} \le \mathbf{1}\{U_t \le M_i(x_t)\}$, and hence $\beta'_{i,C} \le \beta^\star_{i,C}$ pathwise. If i wins at C under the deviation, then $\beta'_{i,C} \ge \beta_{j,C}$ for every $j \ne i$ and i has the highest Π_C priority among the bidders tied with it. In the baseline, $\beta^\star_{i,C} \ge \beta'_{i,C} \ge \beta_{j,C}$ for all $j \ne i$; if the first inequality is strict, i wins outright, and otherwise the tie set and the shared priority are unchanged. Therefore

$$\{i \text{ wins at } C \text{ under the deviation}\} \subseteq \{i \text{ wins at } C \text{ in the baseline}\}, \tag{24}$$

and on the left event $\beta_{(2),C} = \max_{j \ne i} \beta_{j,C}$ in both executions, so τ_C is the same number in both and is a function of the other agents' bids alone. This is a pathwise domination and not a concentration statement, and it is the only use made of the baseline bid.

Step 3: Stage 1, pathwise. At each of the $l_i(C)$ Stage-1 pulls of i at C, the two per-pull utilities differ by $\varphi_i(M_i(x_t)) - \varphi_i(m'_{i,t}) \in [0, 1]$, by monotonicity and (1). So the Stage-1 part of $U'_{i,C} - U^\star_{i,C}$ is at most $l_i(C)$, pathwise, and $\sum_C l_i(C) \le \min\{J'_n L_n, n\}$. A cube with $V_C < KL_n$ never awards and contributes nothing else.

Step 4: cubes where i does not win, on F_C. Suppose C completes bidding and i is not its winner under the deviation. The winner $W \ne i$ plays $S^\star$, so on E_C its Stage-2 target is at least τ_C at every context of C by Lemma H.23, and on $\mathcal{H}_C$ no switch occurs at C by Lemma H.27(d). Hence there is no fallback at C, i receives no post-Stage-1 pull at C under the deviation, and the post-Stage-1 part of $U'_{i,C} - U^\star_{i,C}$ is at most 0 on F_C.

Step 5: cubes where i wins, on F_C. Suppose i wins at C under the deviation; by (24) it also wins in the baseline, at the same price τ_C. Write $n_C = q + H$, where q is the Stage-2 tenure under the deviation, counting the failing pull if there is one, and H is the number of fallback visits to C. Put

$$w_C := \begin{cases} 1 + \varphi_i(\mu_i^{\inf}(C)) - \varphi_i(\tau_C) & \text{if } i \text{ is strategic,} \\ \min\{1,\ 1 + \varphi_i(\mu_i^{\inf}(C)) - \varphi_i(\tau_C)\} & \text{if } i \text{ is honest.} \end{cases}$$

Since $\tau_C \le 1$, $\varphi_i(\tau_C) \le \varphi_i(1) \le 1 + \varphi_i(\mu_i^{\inf}(C))$ by (1), so $0 \le w_C \le 2$. Both w_C and the event $\{i \text{ wins at } C\}$ are determined at C's award round.

Stage 2 of the deviating execution. On $\mathcal{H}_C$, Lemma H.27(c) gives $U'^{\mathrm{pri}} \le q\big(1 + \varphi_i(\mu_i^{\sup}(C)) - \varphi_i(\tau_C)\big) + \Lambda(1 + 2b_n(q))$. By Lemma H.21, $\mu_i^{\sup}(C) \le \mu_i^{\inf}(C) + \rho_n/3$, so the Lipschitz property of φ_i gives $\varphi_i(\mu_i^{\sup}(C)) \le \varphi_i(\mu_i^{\inf}(C)) + \Lambda\rho_n/3$, and in the strategic case

$$U'^{\mathrm{pri}} \le q w_C + \Lambda q \rho_n/3 + \Lambda(1 + 2b_n(q)).$$

In the honest case every target is at least $\mu_i(x_t)$, so each per-pull utility is at most 1 and $U'^{\mathrm{pri}} \le q$; the display then holds as well, because if $w_C = 1$ its right side is at least q, and otherwise $w_C = 1 + \varphi_i(\mu_i^{\inf}(C)) - \varphi_i(\tau_C)$ and the strategic bound applies verbatim. This oscillation step is where the Lipschitz constant of the cost is indispensable: the monitor controls delivered means, the two ends of a cube differ in intrinsic mean, and convexity alone says nothing about the resulting difference in cost.

Fallback of the deviating execution. By Lemma H.24, $\tau_C \le \nu_i(C)$ on E_C, so $\varphi_i(\nu_i(C)) \ge \varphi_i(\tau_C)$ there, and Lemma H.29 gives, almost surely on $E_C \cap \{i \text{ wins at } C\}$,

$$\begin{aligned} \mathbb{E}\big[U'^{\mathrm{fb}} \mid \mathcal{G}^{\mathrm{fb}}_C\big] &\le \big(1 + \varphi_i(\mu_i^{\sup}(C)) - \varphi_i(\tau_C)\big)\bar{H} + \Lambda\big(\bar{R} + \tfrac{1}{3}\rho_n \bar{H}\big) \\ &\le w_C \bar{H} + \Lambda\big(\tfrac{2}{3}\rho_n \bar{H} + \bar{R}\big), \end{aligned}$$

with $\bar{H} = \mathbb{E}[H \mid \mathcal{G}^{\mathrm{fb}}_C]$ and $\bar{R} = \mathbb{E}[R^{\mathrm{dbl}}_K(H) \mid \mathcal{G}^{\mathrm{fb}}_C]$, using the same Lipschitz step; in the honest case take the smaller of this bound and $\bar{H}$, which again yields the display.

The baseline. On $E^\star_C$, $S^\star_i$ delivers exactly τ_C, or $\max\{\mu_i(x_t), \tau_C\}$ if honest, at every Stage-2 pull at C, both at least τ_C, so on $E^\star_C \cap \mathcal{H}^\star_C$ the baseline never switches at C by Lemma H.27(d) and i receives all n_C post-Stage-1 pulls, each with utility at least w_C: in the strategic case the utility is $1 + \varphi_i(\mu_i(x_t)) - \varphi_i(\tau_C) \ge w_C$, and in the honest case it is $\min\{1, 1 + \varphi_i(\mu_i(x_t)) - \varphi_i(\tau_C)\} \ge w_C$. Since $U^{\star,\mathrm{post}}_{i,C} \ge 0$ always and $w_C \le 2$, pathwise on $E^\star_C$,

$$U^{\star,\mathrm{post}}_{i,C} \ \ge\ n_C w_C - 2n_C \mathbf{1}_{(\mathcal{H}^\star_C)^c}.$$

Assembling the winning case. On $F_C \cap \{i \text{ wins at } C\}$, the three displays combine to

$$U'^{\mathrm{post}}_{i,C} - U^{\star,\mathrm{post}}_{i,C} \le \big(U'^{\mathrm{fb}} - H w_C\big) + \Lambda\big(q\rho_n/3 + 1 + 2b_n(q)\big) + 2n_C \mathbf{1}_{(\mathcal{H}^\star_C)^c}.$$

The event $G_C := F_C \cap \{i \text{ wins at } C\}$ belongs to $\mathcal{G}^{\mathrm{fb}}_C$: E_C and $E^\star_C$ are functions of the seeds and contexts at C's Stage-1 rounds, which precede C's switch; the winning event is determined at the award round; and $\mathcal{H}_C$ is determined at the switch round, because the Stage-2 sums at C are frozen once the switch occurs. Both $\mathbf{1}_{G_C}$ and w_C are therefore $\mathcal{G}^{\mathrm{fb}}_C$-measurable, and the tower property gives

$$\begin{aligned} \mathbb{E}\big[(U'^{\mathrm{fb}} - H w_C)\mathbf{1}_{G_C}\big] &= \mathbb{E}\big[\mathbf{1}_{G_C}\big(\mathbb{E}[U'^{\mathrm{fb}} \mid \mathcal{G}^{\mathrm{fb}}_C] - w_C \bar{H}\big)\big] \\ &\le \Lambda\, \mathbb{E}\big[\mathbf{1}_{G_C}\big(\tfrac{2}{3}\rho_n \bar{H} + \bar{R}\big)\big] \\ &= \Lambda\, \mathbb{E}\big[\mathbf{1}_{G_C}\big(\tfrac{2}{3}\rho_n H + R^{\mathrm{dbl}}_K(H)\big)\big], \end{aligned}$$

the middle step by the fallback display, which is valid on $G_C \subseteq E_C$. At no point is a conditional expectation multiplied by an event outside the field it is taken on, and at no point is H required to be known at C's fallback start. The global events E and $\mathcal{H}$ are not $\mathcal{G}_C^{\mathrm{fb}}$-measurable, because they involve cubes whose bidding has not finished when C switches; this is why the per-cube events are used. Nor is $\mathcal{H}_C^{\star}$: in the winning case the baseline's Stage-2 phase at C runs through all n_C post-Stage-1 visits, in general beyond the round at which the deviating execution switched, which is why $(\mathcal{H}_C^{\star})^c$ was charged pathwise above rather than placed in G_C. Hence, with $H \le n_C$, $q \le n_C$, and b_n and R_K^{dbl} nondecreasing,

$$\begin{aligned}\mathbb{E}\big[(U_{i,C}^{\prime\mathrm{post}} - U_{i,C}^{\star,\mathrm{post}})\mathbf{1}_{G_C}\big] &\le \Lambda\,\mathbb{E}\big[\mathbf{1}_{\{i \text{ wins at } C\}}\big(n_C\rho_n + 1 + 2b_n(n_C) + R_K^{\mathrm{dbl}}(n_C)\big)\big] \\ &\quad + 2\,\mathbb{E}\big[n_C\mathbf{1}_{(\mathcal{H}_C^{\star})^c}\big].\end{aligned}$$

Step 6: off F_C, and summation. On F_C^c the post-Stage-1 part of $U'_{i,C} - U^{\star}_{i,C}$ is at most $U'_{i,C}$ restricted to the post-Stage-1 rounds, hence at most $2n_C$. Summing Steps 3 to 6 over the cubes, and using $\sum_C n_C \le n$ in the form $\sum_C \mathbb{E}[n_C\mathbf{1}_{D_C}] \le n\Pr(\bigcup_C D_C)$, valid for any events D_C,

$$\begin{aligned}\sum_C \mathbb{E}[U'_{i,C} - U^{\star}_{i,C}] \le \mathbb{E}\big[\min\{J'_nL_n, n\}\big] &+ \Lambda\,\mathbb{E}\Big[\sum_C\big(n_C\rho_n + \mathbf{1}_{\{i \text{ wins at } C\}}(1 + 2b_n(n_C)) \\ &\qquad + R_K^{\mathrm{dbl}}(n_C)\big)\Big] \\ &+ 2n\Pr\Big(\bigcup_C F_C^c\Big) + 2n\Pr\Big(\bigcup_C (\mathcal{H}_C^{\star})^c\Big).\end{aligned}$$

By Lemmas H.22 and H.27(a), applied to each execution and with $J_n \le n$, $\Pr(\bigcup_C F_C^c) \le \Pr(E^c) + \Pr(\mathcal{H}^c) + \Pr(E^{\star c}) \le 2n^{-4} + K^{-1}n^{-3}$ and $\Pr(\bigcup_C(\mathcal{H}_C^{\star})^c) \le K^{-1}n^{-3}$, so the last two terms total $4n^{-3} + 4K^{-1}n^{-2}$. The first term is at most e_{bid} by concavity, $\sum_C n_C\rho_n \le e_{\mathrm{osc}}$, and the two Cauchy–Schwarz bounds of the error-terms paragraph followed by Jensen's inequality give e_{mon} and e_{fb}. The factor Λ multiplies e_{osc}, e_{mon} and e_{fb} and neither e_{bid}, which was bounded pathwise through (1), nor the bad-event charges, which use the range $[0, 2]$. The right-hand side depends on n, K, d and the law of the arrival process alone and not on $S'_{i,n}$, so the supremum over the deviation class obeys the same bound.

Step 7: the enlarged class. Under (A2), hence under (A3-P), the same proof applies verbatim to $S'_{i,n} \in \mathcal{B}_{i,n}^{\mathrm{ctx}+}$: enlarge every field of the hypotheses paragraph by $\sigma(x_1, \dots, x_n)$, on which such a target is measurable. (A3-P) is precisely the statement that the seed clause and the mechanism clauses survive this enlargement; every indicator that was $\mathcal{G}_u^0$-measurable remains so; and no other step of the proof sees the difference. Over $\mathcal{B}_{i,n}^{\mathrm{ctx}+}$, exogeneity and (A3) alone would not suffice, for the reason given in the hypotheses paragraph: a target that reads a later context may read the current seed through it. □

Proof of Proposition H.30(ii). All agents play $S^{\star}$, so $\bar{A}_{j,C} = \bar{M}_{j,C}$ for every j and C. By the seed clause, $P(n; S^{\star}) = \mathbb{E}\sum_{t\le n} m_{I_t,t}$. Split the rounds into Stage-1 rounds and post-Stage-1 rounds. At a Stage-1 round the target is nonnegative and $M_{(2)} \le 1$, so the Stage-1 rounds contribute at most $\min\{KJ'_nL_n, n\}$ to $\sum_t(M_{(2)}(x_t) - m_{I_t,t})$, pathwise, and at most e_{stage1} in expectation by concavity.

Fix a cube C that completes bidding and consider the event $E_C \cap \mathcal{H}_C$. By Lemma H.23, on E_C the winner's $S^{\star}$ target at every Stage-2 pull at C is $\min\{\tau_C, M_{W_C}(x_t)\} = \tau_C$, or $\max\{\mu_{W_C}(x_t), \tau_C\} \ge \tau_C$ if the winner is honest; so by Lemma H.27(d), on $E_C \cap \mathcal{H}_C$ no switch occurs at C, every post-Stage-1 round of C is a Stage-2 round, and Lemma H.28 is never needed. By Lemma H.25, at every such round

$$m_{I_t,t} \;\ge\; \tau_C \;\ge\; M_{(2)}(x_t) - 4s_n - \tfrac{4}{3}\rho_n.$$

Hence, pathwise, the post-Stage-1 rounds of C contribute at most $n_C(4s_n + \frac{4}{3}\rho_n)$ to $\sum_t(M_{(2)}(x_t) - m_{I_t,t})$ on $E_C \cap \mathcal{H}_C$, and at most n_C on its complement, since $M_{(2)} \le 1$ and targets are nonnegative. Summing over the cubes and using $\sum_C n_C \le n$,

$$\sum_{t\le n}\big(M_{(2)}(x_t) - m_{I_t,t}\big) \le \min\{KJ'_nL_n, n\} + n\big(4s_n + \tfrac{4}{3}\rho_n\big) + n\mathbf{1}_{E^c\cup\mathcal{H}^c}$$

pathwise. Taking expectations, Lemmas H.22 and H.27(a) with $J_n \le n$ give $\Pr(E^c \cup \mathcal{H}^c) \le n^{-4} + K^{-1}n^{-3}$, so the remainder is $n^{-3} + K^{-1}n^{-2} = O(n^{-2})$. No fallback regret appears, because on the good events $S^{\star}$ never triggers a switch; nothing is conditioned on a future event, no two executions are coupled, and no conditional expectation is multiplied by an indicator, which is why the mechanism's own randomisation never enters. Lemmas H.23 and H.25 hold for whoever wins, so the tie break needs no probabilistic property. □

Proof of Proposition H.30(iii). Every agent may now play any feasible strategy, subject for $h \in \mathcal{K}_H$ to pointwise honesty. By the seed clause, $P(n; S) = \mathbb{E}\sum_t m_{I_t,t}$, and more generally $\mathbb{E}[\phi_u Y_u] = \mathbb{E}[\phi_u m_{I_u,u}]$ for every $\mathcal{G}_u^-$-measurable indicator ϕ_u. As in part (ii), the Stage-1 rounds contribute at most e_{stage1} in expectation to $\sum_t(b(x_t) - m_{I_t,t})$, since $b \le 1$ and targets are nonnegative.

Fix a cube C that completes bidding and take v_C and h_C as in Lemma H.26. Let q be the Stage-2 tenure at C, A_q the Stage-2 target sum, and H the number of fallback visits, so that $n_C = q + H$. Define the award-time events

$$\Gamma_1 := E_C^H \cap \{W_C = h_C\}, \qquad \Gamma_2 := E_C^H \cap \{W_C \ne h_C\},$$

both determined by the Stage-1 rewards and the priority at C, hence by the information available at C's award round, which precedes every post-Stage-1 round of C. In particular both belong to $\mathcal{G}_C^{\mathrm{fb}}$, and $\mathbf{1}_\Gamma\phi_u$ is $\mathcal{G}_u^-$-measurable for every post-Stage-1 round u of C and every $\mathcal{G}_u^-$-measurable indicator ϕ_u. Let S_C denote the sum of the targets $m_{I_t,t}$ over the post-Stage-1 rounds of C, so that $S_C = A_q + (\text{fallback target sum at } C)$.

The fallback at C. By Lemma H.26(ii), $m_{h_C,u} \ge v_C$ whenever h_C is selected at a visit to C, pathwise and whatever the other agents do. Lemma H.28 with $j^* = h_C$ and $v = v_C$, in its conditional form with $B = \Gamma \in \mathcal{G}_C^{\mathrm{fb}}$, together with the seed clause, gives for $\Gamma \in \{\Gamma_1, \Gamma_2\}$

$$\mathbb{E}\big[\mathbf{1}_\Gamma\,(\text{fallback target sum at } C)\big] \;\ge\; \mathbb{E}\big[\mathbf{1}_\Gamma\big(v_CH - R_K^{\mathrm{dbl}}(H)\big)\big].$$

No good event is needed for this step.

Case Γ_1: the honest comparator wins. Every Stage-2 target at C is at least $\mu_{h_C}(x_t) \ge v_C$ by honesty, so $A_q \ge v_Cq$ pathwise, and therefore $\mathbb{E}[\mathbf{1}_{\Gamma_1}S_C] \ge \mathbb{E}[\mathbf{1}_{\Gamma_1}(v_Cn_C - R_K^{\mathrm{dbl}}(H))]$.

Case Γ_2: another agent wins. Since $h_C \ne W_C$, the second order statistic counting multiplicity satisfies $\beta_{(2),C} \ge \beta_{h_C,C} \ge v_C - s_n$ on

E_C^H by Lemma H.26(iii), so $\tau_C \ge v_C - s_n - \sigma_n = v_C - 4s_n - \rho_n$. On $\mathcal{H}_C$, Lemma H.27(b) gives $A_q \ge q\tau_C - 1 - 2b_n(q) \ge q(v_C - 4s_n - \rho_n) - 1 - 2b_n(q)$; off $\mathcal{H}_C$ we use only $A_q \ge 0$. Since the right side is at most $q \le n$, pathwise

$$\mathbf{1}_{\Gamma_2} A_q \ge \mathbf{1}_{\Gamma_2}\big(q(v_C - 4s_n - \rho_n) - 1 - 2b_n(q)\big) - n\mathbf{1}_{\mathcal{H}_C^c},$$

and with the fallback bound, using $b_n(q) \le b_n(n_C)$, $R_K^{\mathrm{dbl}}(H) \le R_K^{\mathrm{dbl}}(n_C)$ and $v_C H \ge (v_C - 4s_n - \rho_n)H$,

$$\mathbb{E}[\mathbf{1}_{\Gamma_2} S_C] \ge \mathbb{E}\big[\mathbf{1}_{\Gamma_2}\big(n_C(v_C - 4s_n - \rho_n) - 1 - 2b_n(n_C) - R_K^{\mathrm{dbl}}(n_C)\big)\big] - n\Pr(\mathcal{H}_C^c).$$

The benchmark at C. By Lemma H.26(i), $b(x_t) \le v_C + \rho_n/3$ for every $x_t \in C$, so the benchmark over the post-Stage-1 rounds of C is at most $n_C(v_C + \rho_n/3)$ pathwise. Since $\Gamma_1 \cup \Gamma_2 = E_C^H$ and Case Γ_2 gives the weaker bound,

$$\mathbb{E}\Big[\mathbf{1}_{E_C^H} \sum_{\text{post-Stage-1 } t \text{ at } C} \big(b(x_t) - m_{I_t,t}\big)\Big] \le \mathbb{E}\Big[n_C\big(4s_n + \tfrac{4}{3}\rho_n\big) + 1 + 2b_n(n_C) + R_K^{\mathrm{dbl}}(n_C)\Big] + n\Pr(\mathcal{H}_C^c),$$

and on $(E_C^H)^c$ the same sum is at most $n_C \le n$, since $b \le 1$ and targets are nonnegative, so it contributes at most $n\Pr((E_C^H)^c)$.

Summation. Sum over the cubes: $\sum_C \mathbb{E}[n_C](4s_n + \frac{4}{3}\rho_n) \le e_{\mathrm{price}}$; the Cauchy–Schwarz and Jensen steps of the error-terms paragraph bound $\mathbb{E}\sum_C(1 + 2b_n(n_C))$ over the awarding cubes by e_{mon} and $\mathbb{E}\sum_C R_K^{\mathrm{dbl}}(n_C)$ by e_{fb}; and, over the deterministic set of the J_n cubes, Lemmas H.26(iii) and H.27(a) give

$$\sum_C n\big[\Pr((E_C^H)^c) + \Pr(\mathcal{H}_C^c)\big] \le nJ_n\big(e^{-\ell_n} + 2ne^{-\ell_n}\big) = \frac{J_n}{2Kn^4} + \frac{J_n}{Kn^3} \le \frac{1}{2Kn^3} + \frac{1}{Kn^2}$$

when $J_n \le n$. Adding the Stage-1 charge proves (iii), uniformly over the joint behavioural profile.

Two bookkeeping points deserve to be stated. First, the proof never multiplies a conditional expectation by an event outside the field it is taken on: the fallback step is applied with the award-time events Γ_1, Γ_2, which lie in $\mathcal{G}_C^{\mathrm{fb}}$, and the Stage-2 target sum is never equated with a reward sum on $\mathcal{H}_C$; the monitor event is handled through the pathwise inequality $A_q \ge 0$ and the charge $n\Pr(\mathcal{H}_C^c)$. Second, the identity between expected rewards and expected target sums holds within each case, that is $\mathbb{E}[\mathbf{1}_\Gamma \sum \phi_u Y_u] = \mathbb{E}[\mathbf{1}_\Gamma \sum \phi_u m_{I_u,u}]$, because $\mathbf{1}_\Gamma \phi_u$ is $\mathcal{G}_u^-$-measurable; this is the seed clause and the tower property, as in Lemma H.27(e). □

Proof of Theorem 4.2. It remains to read off the rate from Proposition H.30. The finitely many horizons with $J_n > n$, all below $2^{d(d+3)/3}$, are absorbed into the constants. With $g_n = \lceil n^{1/(d+3)} \rceil$, $h_n = 1/g_n \le n^{-1/(d+3)}$, $L_n = \lceil n^{2/(d+3)} \rceil \le 2n^{2/(d+3)}$, $\ell_n = \log(2Kn^5)$, and $\mathbb{E}[J_n'] \le J_n \le 2^d n^{d/(d+3)}$:

$$e_{\mathrm{bid}} \le 2^{d+1} n^{\frac{d+2}{d+3}}, \qquad e_{\mathrm{stage1}} \le K 2^{d+1} n^{\frac{d+2}{d+3}},$$
$$e_{\mathrm{osc}} = \sqrt{d}\, nh_n \log(en) \le \sqrt{d}\, n^{\frac{d+2}{d+3}} \log(en),$$
$$ns_n = n\sqrt{\ell_n/(2L_n)} \le \sqrt{\ell_n/2}\; n^{\frac{d+2}{d+3}},$$
$$e_{\mathrm{price}} \le \Big(4\sqrt{\ell_n/2} + \tfrac{4}{3}\sqrt{d}\log(en)\Big) n^{\frac{d+2}{d+3}},$$
$$e_{\mathrm{mon}} \le 2^d n^{\frac{d}{d+3}} + 2^{\frac{d}{2}+1}\sqrt{\ell_n/2}\; n^{\frac{2d+3}{2d+6}},$$
$$e_{\mathrm{fb}} \le 11\sqrt{2K\log K}\, 2^{\frac{d}{2}} n^{\frac{2d+3}{2d+6}} + 2^{d+1} K \log K\, n^{\frac{d}{d+3}} \log_2(2(n+1)).$$

The three exponents are $(d+2)/(d+3) = (2d+4)/(2d+6)$, $(2d+3)/(2d+6)$ and $d/(d+3) = 2d/(2d+6)$, and the first is the largest. Since $\ell_n \le 5\log(eKn)$, $\log(en) \le \log(eKn)$ and $\log_2(2(n+1)) \le 3\log(en)$ for $n \ge 1$, every error term is at most a constant depending on d, K and, in part (i), Λ, times $n^{(d+2)/(d+3)}\log(eKn) = R_{d,n}$, which is the rate claimed in Theorem 4.2.

The exponent is unchanged by the cost: in part (i) the factor Λ multiplies e_{osc}, e_{mon} and e_{fb} and so enters the constant only. The number of agents is absent from the exponent, and it is worth recording where it sits in the constants. On the dominant power $n^{(d+2)/(d+3)}$, the bound of part (i) carries the coefficient $2^{d+1} + \Lambda\sqrt{d}\log(en)$, which is free of K: e_{bid} and e_{osc} carry no K, and the two terms of part (i) that do, e_{mon} and e_{fb}, sit on the smaller powers $n^{(2d+3)/(2d+6)}$ and $n^{d/(d+3)}$ with factors $\sqrt{K\log K}$ and $K\log K$. The bounds of parts (ii) and (iii) carry on the dominant power the coefficient $K2^{d+1} + 4\sqrt{\ell_n/2} + \frac{4}{3}\sqrt{d}\log(en)$, which is linear in K through e_{stage1} alone, since every agent must be elicited in every cube that awards; the only other K on that power is the $\sqrt{\log K}$ inside ℓ_n. The balance that produces the exponent is three way: elicitation $J_n L_n \asymp n^{(d+2)/(d+3)}$, price slack $ns_n \asymp n^{(d+2)/(d+3)}\sqrt{\log n}$ and oscillation $nh_n \log n \asymp n^{(d+2)/(d+3)} \log n$. It is the balance $h_n = n^{-1/(D+3)}$, $q_n = n^{2/(D+3)}$ of Theorem F.6, with L_n in the role of q_n, read at the dimension of the space on which the grid is laid: here the task space, of dimension d, whereas that theorem is stated in the general model of Appendix F.9, where the grid is laid on a profile of K contexts and $D = Kd$. Both mechanisms spend L_n or q_n observations per cell on a J_n-cell grid and pay a Lipschitz discretisation error. □

Remark H.31 (The benchmarks are endogenous under the weaker hypothesis). *Parts (ii) and (iii) compare revenue with $\mathbb{E}\sum_t M_{(2)}(x_t)$ and $\mathbb{E}\sum_t b(x_t)$. Under* (A3) *the arrival path may react to past rewards, so the contexts at which $M_{(2)}$ and b are evaluated are partly determined by what the mechanism did. The inequalities are true exactly as stated and remain meaningful as one-sided guarantees, but the sentence proved is not that the mechanism attains the second capability of a path fixed in advance. The accurate reading is: on the realised arrival path, whatever it was, the mechanism collects within the stated error of the second capability, respectively of the best honest intrinsic mean, evaluated along that same path. Under* (A2) *the path is not influenced by the mechanism and the two readings coincide, which is why part (i) is unaffected.*

Remark H.32 (What part (i) needs, and why it is not weakened). *Part (i) uses the seed and mechanism clauses wherever the other two parts do, and in addition it uses, at exactly one place, that the two coupled executions have the same Stage-1 rounds at every*

cube and pull the same agent at each. That is what makes (24)*, the identity of the price across the two executions, and the comparison of* $U'^{\text{post}}_{i,C}$ *with* $n_C w_C$ *over the same number of pulls true pathwise. The accurate statement of what is needed is that the arrival path is the same random sequence in the two coupled executions, that is, that a deviation cannot move the path; exogeneity delivers this and is formally stronger. The requirement is necessary, and the following witness is an instance of the model of this subsection with one context per round. Take* $d = 1$*,* $K = 2$*, linear costs* $\varphi_1(m) = \varphi_2(m) = m$*,* $M_1 \equiv 1$ *and* $\mu_1 \equiv 0$*, so that agent* 1 *is strategic, and* $M_2(x) = \mu_2(x) = x$ *with* $\mathcal{K}_H = \{2\}$*; both primitives are* 1*-Lipschitz, so* (22) *holds for* $n \geq e^2$*, and we take* $n \geq 20$*, so that* $g_n \geq 3$ *and the cubes of* 0*, of* $1/2$ *and of* $1 - h_n/2$ *are distinct. Let* $x_1 = 1/2$ *and, for* $t \geq 2$*,* $x_t = 0$ *if* $Y_1 = 0$ *and* $x_t = 1 - h_n/2$ *if* $Y_1 = 1$*. Each context is realised before its round's own draws, so* (A3) *holds, and the path is not exogenous. Round* 1 *is the first visit to the cube of* $1/2$*, hence a Stage-1 pull of agent* 1*, and that cube is visited once and never awards. The deviation that delivers* 0 *at round* 1 *and follows* $S_1^\star$ *afterwards forces* $Y_1 = 0$ *and moves the whole remaining horizon to the context* 0*, in the cube* $[0, h_n)$*, where the honest agent's capability and intrinsic mean are both* 0*; its bid there is* 0*, so* $\beta_{(2)} = 0$ *and* $\tau = 0$*, agent* 1 *wins with* $\beta_1 = 1$*, never triggers a switch, and earns* $1 + \varphi_1(0) - \varphi_1(0) = 1$ *per Stage-2 pull. The baseline forces* $Y_1 = 1$ *and sits at the context* $1 - h_n/2$ *in the top cube, where the honest agent delivers* $1 - h_n/2$*, so on the good events* $\tau \geq 1 - h_n/2 - 4s_n - \rho_n$ *and agent* 1*, which delivers* τ*, earns at most* $1 - \tau \leq h_n/2 + 4s_n + \rho_n$ *per pull, which at* $d = 1$ *is* $O(n^{-1/4} \log n)$*; off the good events, of probability* $O(n^{-3})$*, it earns at most* 2 *per pull. The gain is therefore* $n(1 - o(1))$*. The natural weakening, arrivals exogenous until every cube has completed Stage 1 and reactive afterwards, preserves* (24) *but destroys the accounting, because* n_C *then differs between the two executions and the comparison of* $U'^{\text{post}}_{i,C}$ *with* $n_C w_C$ *is between two different numbers of pulls, a difference of order* n*. No refinement of a filtration repairs this: it is not a measurability statement.*

Remark H.33 (Where the guarantee begins to bite). *The mechanism prices at zero whenever* $\sigma_n \geq 1$*, and the error terms exceed the trivial bound* n *until* n *is large. We computed both thresholds from the constants of Proposition H.30 with* $K = 2$*, with* $\Lambda = 1$ *in part (i), and with* $\mathbb{E}[J'_n]$ *at its worst case* $\min\{J_n, n\}$*, scanning* $\log_{10} n$ *in steps of* 0.005*; the trivial value is* 1 *for* σ_n *and* n *for the bound of part (i) and for the slack of part (ii). The smallest* n *at which each quantity first drops below its trivial value is: for* $d = 1$*,* $10^{6.1}$ *for* $\sigma_n < 1$*,* $10^{5.2}$ *for the bound of (i), and* $10^{6.8}$ *for the slack of (ii); for* $d = 2$*,* $10^{8.5}$*,* $10^{7.5}$*,* $10^{9.3}$*; for* $d = 3$*,* $10^{11.1}$*,* $10^{10.0}$*,* $10^{12.0}$*; for* $d = 4$*,* $10^{13.7}$*,* $10^{12.7}$*,* $10^{14.9}$*; for* $d = 6$*,* $10^{19.4}$*,* $10^{18.4}$*,* $10^{20.8}$*; and for* $d = 10$*,* $10^{31.7}$*,* $10^{30.7}$*,* $10^{33.7}$*. At* $K = 3$ *each figure moves by at most* 0.2 *decades, whereas each unit of* d *costs about* 2.5 *decades, which is the numerical face of a rate whose exponent carries the dimension of a task and whose constants alone carry the number of agents. The threshold grows like* $\exp(cd)$ *rather than polynomially, because* J_n *carries a factor* 2^d *and* ρ_n *a factor* $\sqrt{d}$*. Two consequences should be stated. The theorem is asymptotic in a strong sense: at* $d = 3$ *the predicted thresholds lie beyond* 10^{10}*, and Appendix B.8 reaches them only with a dedicated constant-work-per-round simulation. And the comparison with the public contextual results is about exponents and not constants: the error* E_n *of Theorem F.5, stated there in the general model with a grid on a profile of* K *contexts, has no factor* 2^D *multiplying an elicitation block of length* L_n*, so at any given* n *it is very much smaller.*

I Equilibrium selection and collusion: proofs

This appendix proves Theorem I.1 and the collusion statements below, first for the scalar mechanism MESP and then for its context-local contextual form. One discipline is kept throughout and should be read before the statements. The *negative* half, that a low-revenue profile is an additive behavioural equilibrium, is proved for deviations ranging over the full behavioural class $\mathcal{B}_{i,n}$, including adaptive elicitation targets, bait and switch, and arbitrary fallback play; it is a strong statement and needs no restriction. The *positive* half, that with two top agents every equilibrium earns the second capacity, is proved only inside the declared family of elicitation profiles defined below, and we do not claim it outside that family. Self-enforcement of a pair deviation is likewise always tested against arbitrary behavioural deviations by each member, while the pair deviations themselves range over the family.

I.1 The scalar mechanism

Notation. All notation is that of Section 3.2, Appendix C and Appendix F: L_n, N_n, s_n, $b_n(\cdot)$, the price $\tau = (\beta_{(2)} - s_n)_+$, the monitor test $Z_r < r\tau - b_n(r)$, the fallback constant $R_K(H) = 3\sqrt{KH \log K} + 2K \log K$, the top set $\mathcal{K}_{\text{top}}$ with $K_{\text{top}} = |\mathcal{K}_{\text{top}}|$ and M_{top}, $V_i = \max_{j \neq i} M_j$, and the proposed profile $S^\star$. Write $M_{(3)}$ for the third largest capacity counting ties, with $M_{(3)} := 0$ if $K = 2$. We use Lemma F.2, Lemma F.3 and the elicitation coupling $\beta'_i \leq \beta_i^{\max}$ with the price $\tau_i = (\max_{j \neq i} \beta_j - s_n)_+$, which does not depend on i's own bid, all as proved in Appendix F, and prove nothing about them here. As there, the uniform tie break is realised through one priority order drawn before play, so that coupled executions share it.

Nondegeneracy. Throughout this subsection assume

$$\kappa_\varphi := 1 - \max_i \big(\varphi_i(M_{\text{top}}) - \varphi_i(\mu_i)\big) > 0. \tag{ND}$$

Condition (ND) guarantees that a selection is worth an amount bounded away from zero: every feasible target $m \leq M_{\text{top}}$ gives per-pull utility $1 + \varphi_i(\mu_i) - \varphi_i(m) \geq \kappa_\varphi$. Under the standing assumptions of Section 2.1 it follows from $M_{\text{top}} < 1$ together with φ_i strictly increasing, since then $\varphi_i(M_{\text{top}}) - \varphi_i(\mu_i) < \varphi_i(1) - \varphi_i(0) \leq 1$, and in the linear model it reads $\kappa_\varphi = 1 - M_{\text{top}} + \min_i \mu_i > 0$. It is used only in the necessity direction; the sufficiency direction, which is the negative result, needs no condition beyond feasibility. The knife edge $M_{\text{top}} = 1$ with $\mu_i = 0$ is the only case in which winning can be worthless, and there the characterisation degenerates.

The family of elicitation profiles. Call $c = (c_1, \ldots, c_K)$ a *feasible bid vector* if $c_i \in [0, M_i]$ for every i and $c_h \geq \mu_h$ for every honest h. Let S^c be the profile that agrees with $S^\star$ everywhere except in Stage 1, where agent i delivers the constant target c_i on each of its L_n elicitation pulls: in the primary phase the winner delivers $\min\{\tau, M_i\}$, or $\max\{\mu_i, \min\{\tau, M_i\}\}$ if honest, and in the fallback every agent delivers its capacity. Then $S^c \in \prod_i \mathcal{B}_{i,n}$ and is feasible on every history, exactly as $S^\star = S^M$ is. Write $\Sigma := \{S^c : c \text{ feasible}\}$. This is the family in which the question lives: it varies precisely the quantity that sets the price and leaves everything Appendix F

proves about the monitor and the fallback untouched. Deviations are never restricted to Σ.

Theorem I.1 (Equilibrium selection for MESP, scalar part)**.** *Assume* (ND). *If $K_{\text{top}} = 1$, then for every fixed $c_W \in (M_{(2)}, M_{\text{top}}]$ of the unique top agent W and every fixed feasible choice of the other bids, S^c is an additive behavioural asymptotic equilibrium of* MESP*, and and their per-round revenues fill the interval $[\mu_H^\star, \max\{\mu_H^\star, M_{(2)}\}]$ up to $O(n^{-1/3}(\log n)^{1/3})$. If $K_{\text{top}} \geq 2$, every profile in Σ that is an additive behavioural asymptotic equilibrium earns $M_{(2)}n - O(n^{2/3}(\log n)^{1/3})$. The contextual counterpart is Proposition I.7 with Corollary I.8.*

Proposition I.2 (The equilibrium set of MESP inside the elicitation family)**.** *Assume* (ND) *and fix a feasible bid vector c that does not depend on n. Then S^c is an additive behavioural asymptotic equilibrium of* MESP *if and only if one of the following holds:*

- (A) *there is an agent W with $c_W > V_W$, in which case necessarily $M_W > V_W$, so $K_{\text{top}} = 1$, W is the unique top agent, $V_W = M_{(2)}$ and $c_W \in (M_{(2)}, M_{\text{top}}]$, the other $K-1$ bids being unrestricted;*
- (B) *$K_{\text{top}} \geq 2$ and $c_i = M_{\text{top}}$ for every $i \in \mathcal{K}_{\text{top}}$, the bids of non-top agents being unrestricted.*

In case (A) the deviation bound is that of Theorem 3.2: for every agent i,

$$\sup_{S'_{i,n} \in \mathcal{B}_{i,n}} \left[u_i(n; S'_{i,n}, S^c_{-i,n}) - u_i(n; S^c_n)\right] \leq L_n + \Lambda\big(1 + 2b_n(n) + R_K(n)\big) + O(1),$$

where the $O(1)$ depends on K and on $c_W - V_W$ through the smallest n with $2s_n < c_W - V_W$, and the revenue is $P(n; S^c) = \nu(c)n + O(KL_n + ns_n)$ with $\nu(c) := \max\{\bar{c},\ \mu_W \mathbf{1}\{W \text{ honest}\}\}$ and $\bar{c} := \max_{k \neq W} c_k$. In case (B), $P(n; S^c) \geq M_{(2)}n - O(KL_n + ns_n + R_K(n))$.

Sufficiency. For a set A of agents playing S^c, let $G_A := \{|\beta_j - c_j| \leq s_n \text{ for all } j \in A\}$. Each β_j, $j \in A$, averages L_n independent Bernoulli(c_j) variables, so Hoeffding's inequality with $\exp(-2L_n s_n^2) = (2Kn^4)^{-1}$ and a union bound over at most $2K$ tails give $\Pr(G_A^c) \leq n^{-4}$. This is the good event of Appendix F with c in place of M, and no other property of c is used.

Case (A), a loser. Let W have $\Delta := c_W - V_W > 0$, put $\bar{c} := \max_{k \neq W} c_k \leq V_W$, so $c_W - \bar{c} \geq \Delta$, and take n large enough that $2s_n < \Delta$; for smaller n any gain is at most $2n$, which is the instance-dependent $O(1)$ of the statement. Fix $k \neq W$ and $S'_k \in \mathcal{B}_{k,n}$. Couple the deviating Stage-1 rewards to capacity-target rewards, so $\beta'_k \leq \beta_k^{\max}$ pathwise, and let $E := G_{[K]\setminus\{k\}} \cap \{\beta_k^{\max} \leq M_k + s_n\}$, of probability at least $1 - 2n^{-4}$, whose description does not involve S'_k. On E, $\beta'_k \leq M_k + s_n \leq V_W + s_n < c_W - s_n \leq \beta_W$, so k loses the elicitation; also every bid other than W's is at most $V_W + s_n$, so $\beta_{(2)} \leq V_W + s_n$ and $\tau \leq V_W < M_W$. Hence W's primary target is at least τ, and Lemma F.2(3) bounds the probability of a switch by $(2n^3)^{-1}$. A losing agent is selected after Stage 1 only in fallback, so on E and on the no-switch event its post-bid utility is 0; since per-pull utility is at most 2 and at most n pulls occur, its expected post-bid utility under the deviation is at most $2n(2n^{-4} + (2n^3)^{-1}) \leq 2n^{-2}$ for $n \geq 4$. Its deviating Stage-1 utility is at most $L_n(1 + \varphi_k(\mu_k) - \varphi_k(0))$, its baseline Stage-1 utility is exactly $L_n(1 + \varphi_k(\mu_k) - \varphi_k(c_k))$, and its baseline post-bid utility is nonnegative, so $u_k(n; S'_k, S^c_{-k}) - u_k(n; S^c) \leq L_n(\varphi_k(c_k) - \varphi_k(0)) + 2n^{-2} \leq L_n + 2n^{-2}$, the last step by (1).

Case (A), the winner. Fix $S'_W \in \mathcal{B}_{W,n}$. The price $\tau_W := (\max_{j \neq W} \beta_j - s_n)_+$ is a function of the other agents' Stage-1 rewards alone, hence identical in the two coupled executions, and on $G_{-W} := G_{[K]\setminus\{W\}}$ we have $(\bar{c} - 2s_n)_+ \leq \tau_W \leq \bar{c} \leq V_W < M_W$; in particular the price is always feasible for W, and no concentration of W's own bid is needed for that. On the branch where the deviation wins, Lemma F.2(2) at $\tau = \tau_W$ gives primary utility at most $q(1 + \varphi_W(\mu_W) - \varphi_W(\tau_W)) + \Lambda(1 + 2b_n(q))$ outside an event of probability n^{-3}; the non-deviating agents deliver their capacities in the fallback under S^c exactly as under $S^\star$, so the comparator mean is V_W and (7) gives $\mathbb{E}[U_W^{\text{fb}} \mid \mathcal{F}_{\text{fb}}] \leq (1 + \varphi_W(\mu_W) - \varphi_W(V_W))H + \Lambda R_K(H) \leq (1 + \varphi_W(\mu_W) - \varphi_W(\tau_W))H + \Lambda R_K(H)$, the last step by $\tau_W \leq V_W$. With $H = N_n - q$ the deviating post-bid utility is at most $(1 + \varphi_W(\mu_W) - \varphi_W(\tau_W))N_n + \Lambda(1 + 2b_n(n) + R_K(n))$. On the branch where the deviation loses, the actual winner j satisfies $\tau \leq (\beta_j - s_n)_+ \leq c_j \leq M_j$ on G_{-W}, so it targets at least τ, the switch probability is at most $(2n^3)^{-1}$ and the deviator's post-bid utility is at most $2n^{-2}$. In the baseline, on $G_{[K]}$, $\beta_W \geq c_W - s_n > \bar{c} + s_n \geq \max_{k \neq W} \beta_k$, so W wins; its primary targets are all at least τ_W, a false switch has probability at most $(2n^3)^{-1}$ and costs at most $2N_n$, and on the complement its post-bid utility is exactly $N_n(1 + \varphi_W(\mu_W) - \varphi_W(\tau_W))$ when W is strategic, and likewise when W is honest with $\tau_W > \mu_W$. When W is honest and $\tau_W \leq \mu_W$, the baseline earns exactly 1 per primary pull, which is the largest per-pull utility any honest strategy can earn, and any deviation has at most N_n post-bid selections, so its post-bid gain is at most $2N_n(2n^3)^{-1}$. Charging Stage 1, at most $L_n(\varphi_W(c_W) - \varphi_W(0)) \leq L_n$ by (1), and the bad events, at most $2n(n^{-4} + n^{-3})$, gives $u_W(n; S'_W, S^c_{-W}) - u_W(n; S^c) \leq L_n + \Lambda(1 + 2b_n(n) + R_K(n)) + O(n^{-2})$. Both bounds are $O(n^{2/3}(\log n)^{1/3}) = o(n)$. This is the equilibrium proof of Appendix F with c substituted for M; the only inputs that change are the centre of the concentration event and the numerical value of $\bar{c}$.

Case (B). A non-top agent k has $M_k < M_{\text{top}}$, so for n with $2s_n < M_{\text{top}} - M_k$, on $G_{[K]\setminus\{k\}} \cap \{\beta_k^{\max} \leq M_k + s_n\}$ it loses to any top agent, whose bid is at least $M_{\text{top}} - s_n$, and the actual winner j has $\tau \leq (\beta_j - s_n)_+ \leq c_j \leq M_j$; the loser computation applies verbatim and gives a gain of at most $L_n + 2n^{-2}$. A top agent i has $c_i = M_{\text{top}} = M_i$, so its baseline bid is its capacity bid, and the equilibrium proof of Appendix F applies with the other agents' Stage-1 targets c_j in place of M_j: the coupling $\beta'_i \leq \beta_i^{\max}$ holds pathwise, on every coupled history where the deviation wins the baseline also wins at the same price τ_i, and on the histories where the deviation loses the baseline's post-bid utility is nonnegative. On G_{-i} another top agent bids M_{top} and no bid exceeds $M_{\text{top}} + s_n$, so $(M_{\text{top}} - 2s_n)_+ \leq \tau_i \leq M_{\text{top}} = M_i = V_i$; the fallback comparator has mean $V_i = M_{\text{top}}$, and the losing-branch winner j has $\tau \leq c_j \leq M_j$. Every step of that proof then goes through unchanged, giving a gain of at most $L_n + \Lambda(1 + 2b_n(n) + R_K(n)) + O(1)$. Case (B) with all non-top bids at capacity is exactly $S^\star$, so this reproves Theorem 3.2's equilibrium claim as a special case.

Revenue. On $G_{[K]}$ in case (A), W wins, $\tau \in [(\bar{c} - 2s_n)_+, \bar{c}]$, the winner delivers $\max\{\mu_W \mathbf{1}\{W \text{ honest}\}, \tau\} \in [\nu(c) - 2s_n, \nu(c)]$ on every primary pull, and a switch has probability at most $(2n^3)^{-1}$. Stage-1 revenue is exactly $L_n \sum_i c_i = KL_n$, fallback revenue lies in $[0, n]$, and $\Pr(G_{[K]}^c) \leq n^{-4}$, so $P(n; S^c) = \nu(c)n + O(KL_n + ns_n + 1)$, where the O is two-sided. In case (B) at least two bids concentrate at M_{top}, so $\tau \geq M_{\text{top}} - 2s_n = M_{(2)} - 2s_n$ on $G_{[K]}$, the winner W has

$\tau \le (\beta_W - s_n)_+ \le c_W \le M_W$, and the revenue proof of Appendix F applies unchanged. □

Necessity. Assume (ND), let c be feasible and fixed, and suppose neither (A) nor (B) holds. Put $c^{\max} := \max_i c_i$ and $T := \{i : c_i = c^{\max}\}$. In each case below we exhibit one agent and one deviation whose gain is at least a fixed positive multiple of n for all large n, contradicting the $o(n)$ requirement. All deviations deliver a constant m' on every elicitation pull and then follow S_i^c, so they lie in $\mathcal{B}_{i,n}$ and are honest whenever the agent is, since $m' = M_i \ge \mu_i$ in every case. "The good event" below is $G_{[K]}$ evaluated with the deviator's Stage-1 target in place of its c_i; it has probability at least $1 - n^{-4}$. Two standing estimates are used, the first for the deviating execution and the second for the baseline.

(E1) Winning value. If an agent i wins the elicitation and on the good event faces $\tau_i \le \theta$ with $\theta \le M_{\text{top}}$ and $\theta \le M_i$, then its primary target $\min\{\tau_i, M_i\} = \tau_i$, all its primary targets are at least τ_i, a switch has probability at most $(2n^3)^{-1}$ by Lemma F.2(3), and its post-bid utility is at least $N_n(1 + \varphi_i(\mu_i) - \varphi_i(\theta)) - 2N_n(2n^3)^{-1}$, hence at least $N_n\kappa_\varphi - 1$ by (ND); for an honest i the same holds with $1 + \varphi_i(\mu_i) - \varphi_i(\theta)$ replaced by $\min\{1, 1 + \varphi_i(\mu_i) - \varphi_i(\theta)\} \ge \kappa_\varphi$.

(E2) Losing value. If on the good event agent i loses the elicitation and the actual winner j plays S_j^c with $\tau \le M_j$, its post-bid utility is at most $2n^{-2}$, as computed in the sufficiency proof, so its baseline utility is at most $L_n(1 + \varphi_i(\mu_i) - \varphi_i(0)) + 2n^{-2} \le 2L_n + 1$ by (1).

Case 1: $|T| = 1$, say $T = \{W\}$, and $c_W \le V_W$, so (A) fails. Let k attain $M_k = V_W$; then $k \ne W$ and $c_k < c^{\max} = c_W$. In the baseline, W wins on the good event once $2s_n < c_W - \max_{j \ne W} c_j$, with $\tau \le (\beta_W - s_n)_+ \le c_W \le M_W$, so (E2) bounds k's baseline utility by $2L_n + 1$. Let k bid M_k throughout Stage 1. If $M_k > c_W$, then for n with $2s_n < M_k - c_W$, on the good event k outbids everyone and $\tau_k = (\max_{j \ne k} \beta_j - s_n)_+ \le c_W \le V_W = M_k$, so (E1) applies with $\theta = c_W$; its deviating Stage-1 utility is nonnegative, so the gain is at least $N_n\kappa_\varphi - 2L_n - 2 - 4n^{-3} \ge (\kappa_\varphi/2)n - 2L_n - 3$, which exceeds $(\kappa_\varphi/4)n$ for all large n since $L_n = o(n)$. If $M_k = c_W$, then β_k and β_W are independent with the common law $\mathrm{Bin}(L_n, c_W)/L_n$, so by exchangeability and the uniform priority $\Pr(k \text{ beats } W) = 1/2$ exactly; on the good event $G_{[K]}$ every other bid is at most $\max_{j \notin \{k,W\}} c_j + s_n < c_W - s_n \le \min\{\beta_k, \beta_W\}$, so k wins with probability at least $1/2 - n^{-4}$ at a price $\tau_k \le c_W = M_k$, and (E1) with (E2) give a gain of at least $(1/2 - n^{-4})(N_n\kappa_\varphi - 1) - 2L_n - 2 \ge (\kappa_\varphi/8)n$ for all large n.

Case 2: $|T| \ge 2$ at the common level $c^{\max}$. In the baseline each $j \in T$ wins with probability at most $1/|T|$, because on the event that j wins its bid is in particular the largest, with priority, of the $|T|$ exchangeable bids $\{\beta_l : l \in T\}$. (2a) If some $j \in T$ has $M_j > c^{\max}$, let j bid M_j; for $2s_n < M_j - c^{\max}$, on the good event the deviation wins outright and faces $\tau_j \le c^{\max} \le M_j$, so by (E1) its post-bid utility is at least $N_n(1 + \varphi_j(\mu_j) - \varphi_j(c^{\max})) - 1$. In the baseline, when j wins it delivers exactly τ_j on every primary pull, so its primary utility is $q(1 + \varphi_j(\mu_j) - \varphi_j(\tau_j))$ and, with (7) and $\tau_j \le c^{\max} \le V_j$ (another member of T has capacity at least $c^{\max}$), its post-bid utility is at most $N_n(1 + \varphi_j(\mu_j) - \varphi_j(\tau_j)) + \Lambda(1 + 2b_n(n) + R_K(n))$; when it loses, (E2) gives at most $2n^{-2}$. On the good event $\tau_j \ge (c^{\max} - 2s_n)_+$, so $\varphi_j(\tau_j) \ge \varphi_j(c^{\max}) - 2\Lambda s_n$. The baseline post-bid utility is therefore at most $(1/|T|)[N_n(1 + \varphi_j(\mu_j) - \varphi_j(c^{\max}) + 2\Lambda s_n) + \Lambda(1 + 2b_n(n) + R_K(n))] + 2n^{-2}$, the Stage-1 difference is at most L_n, and since $|T| \ge 2$ the gain is at least $(1/2)N_n(1 + \varphi_j(\mu_j) - \varphi_j(c^{\max})) - \Lambda N_n s_n - L_n - \Lambda(1 + 2b_n(n) + R_K(n)) - O(1)$. With $1 + \varphi_j(\mu_j) - \varphi_j(c^{\max}) \ge \kappa_\varphi$, $N_n s_n = O(n^{2/3}(\log n)^{1/3})$ and $b_n(n) + R_K(n) = O(\sqrt{n \log n})$, the gain is at least $(\kappa_\varphi/8)n$ for all large n. (2b) If every $j \in T$ has $M_j = c^{\max}$ and some k outside T has $M_k > c^{\max}$, let k bid M_k; it wins outright on the good event at a price at most $c^{\max} \le M_k$, while in the baseline it loses to a member of T, whose price is at most $c^{\max} = M_l$; by (E1) and (E2) the gain is at least $N_n\kappa_\varphi - 2L_n - 3 \ge (\kappa_\varphi/4)n$ for large n. (2c) If $M_i \le c^{\max}$ for every i and $M_j = c^{\max}$ for every $j \in T$, then $c^{\max} = M_{\text{top}}$ and $T \subseteq \mathcal{K}_{\text{top}}$; if some $k \in \mathcal{K}_{\text{top}} \setminus T$, then $c_k < M_{\text{top}} = M_k$ and k bidding M_{top} ties with the $|T|$ members of T, so by exchangeability it wins with probability at least $1/(|T|+1) - n^{-4} \ge 1/K - n^{-4}$ at a price at most $M_{\text{top}} = M_k$, and by (E1) and (E2) the gain is at least $(1/K)N_n\kappa_\varphi - 2L_n - 3 \ge \kappa_\varphi n/(4K)$ for large n. Hence $\mathcal{K}_{\text{top}} = T$, $|T| \ge 2$ and $c_i = M_{\text{top}}$ on $\mathcal{K}_{\text{top}}$, which is (B), contradicting the assumption that (B) fails. This exhausts the cases. □

Corollary I.3 (The revenue range of the equilibrium set)**.** *Assume* (ND). *If $K_{\text{top}} \ge 2$, every equilibrium in Σ has revenue $M_{(2)}n - O(n^{2/3}(\log n)^{1/3})$. If $K_{\text{top}} = 1$, the set of equilibrium revenues in Σ is exactly $\{vn + O(n^{2/3}(\log n)^{1/3}) : v \in [\mu_H^\star, \max\{\mu_H^\star, M_{(2)}\}]\}$. In particular the smallest equilibrium revenue is $\mu_H^\star n + O(n^{2/3}(\log n)^{1/3})$, it is attained, and it coincides with the bound that the robustness part of Theorem 3.2 already proves against arbitrary behavioural play. The gap between the revenue of $S^\star$ and the worst equilibrium revenue is $(M_{(2)} - \mu_H^\star)_+ n$, linear whenever the best honest intrinsic mean is below the runner-up capacity.*

Proof. If $K_{\text{top}} \ge 2$, every equilibrium in Σ is of type (B) by Proposition I.2, whose revenue bound gives the lower bound; the upper bound $P \le M_{\text{top}} n = M_{(2)} n$ is trivial. If $K_{\text{top}} = 1$ with unique top W, the equilibria in Σ are exactly the c with $c_W > M_{(2)}$ and arbitrary feasible other bids, and their revenue is $v(c)n + O(KL_n + ns_n)$ with $v(c) = \max\{\bar{c}, \mu_W \mathbf{1}\{W \text{ honest}\}\}$. As c_{-W} ranges over its feasible set, $\bar{c} = \max_{k \ne W} c_k$ ranges over the interval $[\max_{h \in \mathcal{K}_H, h \ne W} \mu_h,\ M_{(2)}]$, with the convention that a maximum over the empty set is 0: its smallest value is attained when every honest loser bids μ_h and every strategic loser bids 0; its largest value is attained when a loser of capacity $M_{(2)}$ bids $M_{(2)}$, since every loser has $c_k \le M_k \le M_{(2)}$; and every intermediate value is attained by moving that loser's bid, a continuous parameter. Hence $v(c)$ ranges over the interval with endpoints $\max\{\mu_W \mathbf{1}\{W \text{ honest}\}, \max_{h \ne W} \mu_h\} = \mu_H^\star$ and $\max\{M_{(2)}, \mu_W \mathbf{1}\{W \text{ honest}\}\} = \max\{M_{(2)}, \mu_H^\star\}$, the last equality because every honest $h \ne W$ has $\mu_h \le M_h \le M_{(2)}$, so if $\mu_W \mathbf{1}\{W \text{ honest}\} > M_{(2)}$ then $\mu_H^\star = \mu_W$. No profile whatever can go below the lower endpoint by more than $O(KL_n + ns_n + b_n(n) + R_K(n))$, by the robustness part of Theorem 3.2. □

Corollary I.4 (The proposed profile is not selected by anyone's interest)**.** *Assume $K_{\text{top}} = 1$ and let k be an agent with $M_k = M_{(2)}$, the agent whose bid sets the price at $S^\star$. Lowering only c_k, from $M_{(2)}$ to any feasible $c_k' < M_{(2)}$, raises k's own utility by exactly $L_n(\varphi_k(M_{(2)}) - \varphi_k(c_k')) - O(n^{-2})$ and lowers platform revenue by $(M_{(2)} - c_k')n - O(KL_n + ns_n)$. The move from the proposed profile to the worst equilibrium is therefore a sequence of strictly improving unilateral moves, not a coordinated one.*

Proof. Compare S^c and $S^{c'}$ differing only in the kth coordinate. Couple the two executions by the same reward uniforms indexed by agent and local pull index, the same priority and the same EXP3 coins. In both executions every Stage-1 target is a constant, so k's Stage-1 utilities differ by exactly $L_n(\varphi_k(M_{(2)}) - \varphi_k(c'_k))$ on every sample point. Let E be the event that every bid is within s_n of its target in both executions and that no switch occurs in either; the bid part has probability at least $1 - 2n^{-4}$, and on it W wins both executions at a price it can meet, so by Lemma F.2(3) each switch has probability at most $(2n^3)^{-1}$, whence $\Pr(E^c) \le 2n^{-4} + n^{-3}$. On E, agent k is never selected after Stage 1 in either execution, so its post-bid utility is 0 in both; off E the post-bid utilities differ by at most $2n$, contributing at most $2n(2n^{-4} + n^{-3}) = O(n^{-2})$. This is the utility clause. The revenue statement is the revenue formula of Proposition I.2 applied to the two profiles. Agent k's gain does not depend on whether W is honest and requires no coordination with W. This is the computation of Proposition F.4 applied to the runner-up instead of to the winner, and the runner-up's version is simpler, because its post-bid histories are unaffected by its own bid on the good event, so no coupling of the post-bid phase is needed. □

Proof of the scalar part of Theorem I.1. Immediate from Proposition I.2 and Corollary I.3: with a unique top capacity the family contains the continuum of additive behavioural equilibria indexed by $c_W \in (M_{(2)}, M_{\text{top}}]$ and arbitrary losing bids, whose revenues fill $[\mu_H^\star, \max\{\mu_H^\star, M_{(2)}\}]$ per round up to $O(n^{2/3}(\log n)^{1/3})$; with at least two top agents, case (B) is the only possibility and every such profile earns $M_{(2)}n - o(n)$. □

Remark I.5 (Three readings of the characterisation, and the hypothesis each carries). *First, under S^c the price is set by an agent with no stake in it: the announced price is the highest losing bid, shaded, and a loser receives nothing in the primary phase whatever it bids, so the only consequence of its bid for itself is the Stage-1 effort $L_n(\varphi_k(M_k) - \varphi_k(c_k)) \le L_n = O(n^{2/3}(\log n)^{1/3})$, inside the additive slack; the bound is (1) alone, so no cost function of bounded range can make one Stage-1 block worth a linear amount. The qualifier "under S^c" is needed. The price is announced to every agent, so a winner's continuation may condition on it, and then a loser's bid decides which continuation it faces; there are profiles outside Σ, in which the winner tanks above a price threshold, at which a loser moves its own utility by $\Theta(n)$ through its own bid. Such a profile is not a counterexample to Proposition I.2, which is scoped to Σ, and it need not be an equilibrium; but the first reading is a property of the profile and not of the mechanism. Second, mutual underbidding in the usual sense does not occur: at price τ a winner earns $1 + \varphi_i(\mu_i) - \varphi_i(\tau) \ge \kappa_\varphi > 0$ per round, so lowering the price raises the prize, and any agent that can outbid the current winner takes it for a linear gain. The low-revenue equilibria are asymmetric, with one agent bidding above every other capacity and the rest bidding low because they have already lost. Third, the two-top condition is what pins the price: with $K_{\text{top}} \ge 2$ the price setter is itself a potential winner, its bid carries the linear stake $(1/K_{\text{top}})N_n(1 + \varphi_i(\mu_i) - \varphi_i(M_{\text{top}}))$, and the price is forced to $M_{\text{top}} = M_{(2)}$. This is the role the condition plays in Theorem C.3, one level down: in the public setting a unique top agent shades its delivery; here it shades nothing, and instead the runner-up shades the price. Like every equilibrium statement, the construction presumes the agents know the game: a loser lowers its bid only if it knows it cannot win, since lowering a bid that would have won costs a linear amount, and that knowledge requirement falls on the losers rather than on the winner.*

Coalitions of two. A *pair deviation* from a profile S by $\{i, j\}$ is a pair (S'_i, S'_j) in $\mathcal{B}_{i,n} \times \mathcal{B}_{j,n}$ with the other $K-2$ strategies unchanged. Call it *profitable* if there is $\epsilon > 0$ such that, for all large n, both members gain at least ϵn; call it *self-enforcing* if from the resulting profile neither member has a unilateral deviation in its own behavioural class gaining $\Omega(n)$, outsiders not being required to be content. Call S *pair-proof* if no profitable pair deviation exists and *pair-coalition-proof* if no profitable self-enforcing pair deviation exists. The model has no transfers, so a coalitional deviation must make both members better off on its own.

Proposition I.6 (Pairs at the scalar mechanism). *Assume* (ND) *and consider pair deviations from $S^\star$ in which the two members change only their elicitation targets; self-enforcement is tested against arbitrary behavioural deviations.*

(i) Unique top capacity. *Let W be the top agent and k an agent with $M_k = M_{(2)}$. The pair $\{W, k\}$ deviating to $c_W = M_W$ and c_k equal to its lowest feasible bid, μ_k if k is honest and 0 otherwise, is profitable and self-enforcing: W gains $(\varphi_W(M_{(2)}) - \varphi_W(\theta))_+ N_n - O(\Lambda n s_n + \Lambda\sqrt{n \log n})$ and k gains $L_n(\varphi_k(M_{(2)}) - \varphi_k(c_k)) - O(n^{-2})$, positive whenever φ_k is strictly increasing on $[c_k, M_{(2)}]$, where $\theta := \max\{c_k, M_{(3)}, \mu_W \mathbf{1}\{W \text{ honest}\}\}$, and platform revenue falls by $(M_{(2)} - \theta)_+ n - O(KL_n + ns_n)$. The resulting profile satisfies case (A) of Proposition I.2, so no agent whatsoever, member or outsider, has a unilateral deviation gaining more than $L_n + \Lambda(1 + 2b_n(n) + R_K(n)) + O(1)$. Hence $S^\star$ is neither pair-proof nor pair-coalition-proof, and the sharp per-round transfer from the platform to the pair is $M_{(2)} - \max\{M_{(3)}, \mu_k \mathbf{1}\{k \text{ honest}\}, \mu_W \mathbf{1}\{W \text{ honest}\}\}$, which is never below zero and never takes revenue below $\mu_H^\star$.*

(ii) Exactly two top agents. *Let $\{i, j\} = \mathcal{K}_{\text{top}}$. The joint deviation to a common bid $\theta \in (\max\{M_{(3)}, \mu_i \mathbf{1}\{i \text{ honest}\}, \mu_j \mathbf{1}\{j \text{ honest}\}\}, M_{\text{top}})$ is profitable, provided the members' costs are strictly increasing on $[\theta, M_{\text{top}}]$: each member l gains $((1/2)N_n + L_n)(\varphi_l(M_{\text{top}}) - \varphi_l(\theta)) - O(\Lambda n s_n + \Lambda\sqrt{n \log n})$, and revenue falls by $(M_{\text{top}} - \theta)n - O(KL_n + ns_n)$. But no profitable pair deviation of this form is self-enforcing: the only self-enforcing pair bid vector is $(M_{\text{top}}, M_{\text{top}})$, which is $S^\star$ itself. Hence $S^\star$ is not pair-proof but is pair-coalition-proof, and every self-enforcing pair deviation leaves revenue at $M_{(2)}n - O(n^{2/3}(\log n)^{1/3})$.*

(iii) Three or more top agents. *No pair deviation of this form is profitable, so $S^\star$ is pair-proof.*

Proof. Throughout, "the good event" is the event that every bid is within s_n of its Stage-1 target in the profiles under comparison; it has probability at least $1 - n^{-4}$ per profile, and off it utilities and revenue change by at most $2n$, which is charged as $O(n^{-3})$.

(i) Under $S^\star$, on the good event, W wins and $\tau \in [M_{(2)} - 2s_n, M_{(2)}]$. Under the deviation, on the good event, W still wins, its bid being unchanged and only one other bid having moved down, and the second order statistic concentrates

at $\max\{c_k, \max_{l\neq W,k} M_l\} = \theta_0$ with $\theta_0 := \max\{c_k, M_{(3)}\}$, so $\tau' \in [(\theta_0 - 2s_n)_+, \theta_0]$. Agent W's Stage-1 utility is unchanged and its primary target moves from $\max\{\mu_W \mathbf{1}\{W \text{ honest}\}, \tau\}$ to $\max\{\mu_W \mathbf{1}\{W \text{ honest}\}, \tau'\}$, so its per-round primary utility rises by $\varphi_W(M_{(2)}) - \varphi_W(\theta)$ up to $2\Lambda s_n$, with $\theta = \max\{\theta_0, \mu_W \mathbf{1}\{W \text{ honest}\}\}$, and by 0 when $\mu_W \mathbf{1}\{W \text{ honest}\} \geq M_{(2)}$, which is the positive part; multiply by N_n and charge the false switch, at most $2N_n(2n^3)^{-1}$, and the bad events. Agent k's gain is the utility clause of Corollary I.4. Self-enforcement and the stronger unilateral statement are case (A) of Proposition I.2, which applies because $c_W = M_W > M_{(2)} = V_W$. The revenue statement is the revenue formula of that proposition evaluated at the two profiles: per-round revenue moves from $\max\{M_{(2)}, \mu_W \mathbf{1}\{W \text{ honest}\}\}$ to θ. Finally $\theta \geq \mu_H^\star$, because every honest agent other than W and k has $\mu_h \leq M_h \leq M_{(3)}$.

(ii) At $S^\star$ the two top bids are exchangeable and the other bids are below both on the good event, so each member wins with probability $1/2$ up to n^{-4}, at a price in $[M_{\text{top}} - 2s_n, M_{\text{top}}]$; after the deviation each still wins with probability $1/2$ up to n^{-4}, because the two bids remain exchangeable and the other capacities are below θ, at a price in $[(\theta - 2s_n)_+, \theta]$, which exceeds μ_l for an honest member once $2s_n < \theta - \mu_l$, so the winner's primary target is the price in both profiles. Every other term of the utility is unchanged under the coupling, so each member l gains $((1/2)N_n + L_n)(\varphi_l(M_{\text{top}}) - \varphi_l(\theta))$ up to $2\Lambda N_n s_n$, the false switch and the bad events. For non-self-enforcement, let (c_i, c_j) be the pair's bids with $\min(c_i, c_j) < M_{\text{top}}$. If $c_i = c_j = \theta < M_{\text{top}}$, then i unilaterally bidding M_{top} wins outright on the good event instead of with probability $1/2$, at the same price, since the price it faces is set by j's bid, which has not moved; by (E1) and the computation above it gains at least $(1/2)N_n\kappa_\varphi - L_n - O(\sqrt{n \log n})$. If $c_i > c_j$ then j loses on the good event, and bidding $M_{\text{top}} \geq c_i$ it wins with probability at least $1/2 - n^{-4}$ at a price at most $M_{\text{top}} = M_j$, while its current post-bid utility is at most $2n^{-2}$ by (E2), so it gains at least $(1/2)N_n\kappa_\varphi - 2L_n - O(1)$; the case $c_j > c_i$ is symmetric. At $(M_{\text{top}}, M_{\text{top}})$ the price is $M_{\text{top}} - O(s_n)$ and revenue is preserved.

(iii) With $K_{\text{top}} \geq 3$, at least one top agent l lies outside any pair, and it bids M_{top}. *Who can win.* On the good event a member with bid below M_{top} has bid at most that value plus s_n, which is below $\beta_l \geq M_{\text{top}} - s_n$ for large n; so a member wins only if it bids M_{top}, which requires $M_m = M_{\text{top}}$. *At what price.* When a member wins, l's bid enters the second order statistic and every bid is at most $M_{\text{top}} + s_n$, so the price is $M_{\text{top}} - O(s_n)$. It is not true that the price is $M_{\text{top}} - O(s_n)$ whatever the pair does: if the pair contains two of exactly three top agents and both members underbid, the outside top agent wins alone and the second order statistic is the larger of the two members' bids, which can be anything. *With what probability.* If a top member bids M_{top}, it wins with probability $1/|B|$ up to n^{-4}, where B is the set of agents bidding M_{top}, all of capacity M_{top} and hence exchangeable; if it bids below, it wins with probability at most n^{-4}. *The two member types.* If $M_m < M_{\text{top}}$, then m loses on the good event whatever it bids, so by (E2) every change of its bid moves its utility by at most $L_n + 2n^{-2}$, which is not $\Omega(n)$. If $M_m = M_{\text{top}}$, its per-round value of winning is $1 + \varphi_m(\mu_m) - \varphi_m(M_{\text{top}}) + O(\Lambda s_n)$ whatever the pair does, so the only effect of the pair's bids on m is its winning probability. If both members bid M_{top}, both get exactly their $S^\star$ values and neither gains; if a member bids below, its probability drops from $1/K_{\text{top}}$ to at most n^{-4} and it loses $(1/K_{\text{top}})N_n\kappa_\varphi - O(L_n)$, a linear amount, while only its partner can gain. So in every case at least one member is not better off by $\Omega(n)$. □

In (i) and (ii) the monitor is irrelevant: the winner's target $\min\{\tau, M_W\}$ meets the announced price exactly, so Lemma F.2 gives a false-switch probability of at most $(2n^3)^{-1}$ and the coalition is never detected. Part (i) needs no coordination at all, because each member's part of the deviation is already individually profitable by Corollary I.4; in (ii) the agreement is not self-enforcing, so the two-top condition protects the mechanism against pairs exactly as it protects it against equilibrium selection. One caveat on (iii): in the configuration just described the pair can destroy $\Omega(n)$ of revenue while both members strictly lose, so pair-proofness protects revenue only against pairs that are trying to gain, which is all the definition claims.

I.2 The contextual mechanism

We now analyse CL-MESP of Appendix H.1, under its hypotheses, with fixed K and Q, and condition throughout on the arrival path so that every V_z is a constant. Two structural facts from Appendix H.1 are the load-bearing inputs and are used as given: context locality (Lemma H.3), and additive accounting (13), which needs no independence across contexts. Assume the local nondegeneracy

$$\kappa_\varphi := 1 - \max_i \max_z \big(\varphi_i(M_\star(z)) - \varphi_i(\mu_i^z)\big) > 0. \qquad \text{(ND-ctx)}$$

As in the scalar case this says only that a selection is worth an amount bounded away from zero at every context; it follows from $M_\star(z) < 1$ at every context together with φ_i strictly increasing, and in the linear model it reduces to $1 - \max_z M_\star(z) + \min_{i,z} \mu_i^z > 0$.

The family Σ^{ctx} is the set of profiles S^c indexed by an array $c = (c_{i,z})$ with $c_{i,z} \in [0, M_i^z]$ and $c_{h,z} \geq \mu_h^z$ for honest h: agent i delivers the constant $c_{i,z}$ on each of its Stage-1 pulls at z and otherwise follows $S^\star$, so $S^\star = S^M$. Deviations always range over the full class $\mathcal{B}_{i,n}^{\text{ctx}}$, including strategies that couple across contexts. At a context z say that $c_{\cdot,z}$ is

- (A_z) if some agent W has $c_{W,z} > V_W^z$, which forces $|\mathcal{T}(z)| = 1$ with W the local top agent and $c_{W,z} > M_{(2)}(z)$, since $c_{W,z} \leq M_W^z$;
- (B_z) if $|\mathcal{T}(z)| \geq 2$ and $c_{i,z} = M_\star(z)$ for every $i \in \mathcal{T}(z)$.

Call z *compliant* if (A_z) or (B_z) holds and *violating* otherwise; these are the scalar conditions (A) and (B) of Proposition I.2 evaluated at the local primitives.

The stake an agent has in its own elicitation at z is exactly what it was in the scalar mechanism. By context locality, agent i's Stage-1 targets at z enter $\beta_{i,z}$, hence the local winner and the local price, and no other object the mechanism holds; winning at z' changes neither. So the stake is $L_n(\varphi_i(M_i^z) - \varphi_i(c_{i,z}))$ of Stage-1 effort plus whatever its chance of winning at z is worth, and the first is at most L_n per context by (1) and $O(QL_n) = o(n)$ in total. The contextual mechanism does not couple the auctions; it runs Q of them side by side. What does change is the scale at which each auction is judged: the equilibrium definition tolerates $o(n)$ in total, but the prize at z is worth N_z and not n, so a deviation confined to z can gain at most $2V_z$, and at a context visited $o(n)$ times nothing is pinned at all.

Two quantities from the scalar necessity proof are used below. Write

$$\epsilon_n := 2L_n + \Lambda\big(ns_n + 1 + 2b_n(n) + R_K(n)\big) + 4 = O_{K,\Lambda}\big(n^{2/3}(\log n)^{1/3}\big)$$

for the lower-order term of that proof: inspecting its five cases, the exhibited deviation at a violating scalar instance gains at least $(\kappa_\varphi/K)N_n - \epsilon_n$ in every case, the terms $\Lambda(ns_n + 1 + 2b_n(n) + R_K(n))$ arising in case (2a) from the price shading and the monitor and fallback slacks of the baseline winner.

Proposition I.7 (The equilibrium set of CL-MESP)**.** *Assume the hypotheses of Proposition H.2,* (ND-ctx)*, fixed K and Q, and a fixed array c independent of n, and condition on the arrival path. Put* $\mathrm{Viol}(c) := \sum_{z\ violating} V_z$*. Then:*

(i) Sufficiency, per context. *For every agent i and every n with $2s_n < \min\{c_{W,z} - V_W^z : z \text{ satisfies } (\mathrm{A}_z)\}$ (a vacuous condition when no context satisfies (A_z)),*

$$\sup_{S'_{i,n}\in\mathcal{B}^{\mathrm{ctx}}_{i,n}} \big[u_i(n; S'_{i,n}, S^c_{-i,n}) - u_i(n; S^c_n)\big] \le QL_n + \Lambda\big[Q(1 + R_K(n)) + \sqrt{2Qn\ell_n}\big] + 2\,\mathrm{Viol}(c) + O(Qn^{-2}),$$

the middle term being $\sum_z 2b_n(N_z) \le \sqrt{2Q\ell_n \sum_z N_z}$ by the Cauchy–Schwarz inequality.

(ii) Necessity, per context. *If z is violating and $N_z > 0$, some agent has a deviation confined to z whose gain is at least $(\kappa_\varphi/K)N_z - \epsilon_n$. Consequently some single agent has a deviation whose gain is at least $(\kappa_\varphi/K^2)\sum_{z\ violating,\, N_z>0} N_z - Q\epsilon_n$.*

(iii) Characterisation. *S^c is an additive behavioural asymptotic equilibrium if and only if* $\mathrm{Viol}(c) = o(n)$*. In particular the local condition must hold at every context of positive visit frequency and need not hold anywhere else.*

Proof. (i) By additive accounting (13) the total gain is the sum over contexts of the local gains. A deviation's local gain at z is evaluated against the local mechanism, which by context locality is a scalar MESP with the global constants L_n, s_n, $b_n(\cdot)$ and the local primary length N_z, and whose other agents bid the constants $c_{j,z}$ and otherwise play $S^\star$. At a compliant context the sufficiency proof of Proposition I.2 applies word for word with N_n replaced by N_z, V_W by V_W^z, M_j by M_j^z, and the fallback bound replaced by the prefix bound of Lemma H.5, which is at most $R_K(n)$ uniformly in the local length. Its inputs are the following five facts, each of which holds for a deviator that couples across contexts: Lemma H.4, which is stated for targets predictable in the global history; the local good event $\{|\beta_{j,z} - c_{j,z}| \le s_n \text{ for all } j \ne i\}$, of probability at least $1 - n^{-4}$ by Lemma H.6(b) with $c_{j,z}$ in place of M_j^z; the local price identity $\tau_z = (\max_{j\ne i}\beta_{j,z} - s_n)_+$ on the event that i wins at z; the local bid domination of Lemma H.6(c), (d); and, in case (A_z), the threshold $2s_n < c_{W,z} - V_W^z$, which is the hypothesis. At a violating context bound the local gain by $2V_z$. Summing gives the display; the error terms are those of Proposition H.2.

(ii) At a violating z, the necessity proof of Proposition I.2 applied to the local primitives exhibits an agent i and a constant Stage-1 target at z whose local gain is at least $(\kappa_\varphi/K)N_z - \epsilon_n$; the deviation that delivers that constant at every Stage-1 pull at z and follows S_i^c everywhere else is a legitimate element of $\mathcal{B}^{\mathrm{ctx}}_{i,n}$, being a function of the current context and the local messages alone. Assign each violating context with $N_z > 0$ to one such agent and let i^* be an agent assigned contexts with the largest total N_z; by the pigeonhole principle that total is at least $1/K$ of the whole. Let i^* use the violating target at every context assigned to it and $S_{i^*}^c$ elsewhere. The local execution at an assigned context z has the same law as under the deviation confined to z: the mechanism at z reads only z's data (context locality), every agent's behaviour at z, the deviator's included, is a function of the local phase, the local price and the current context, and the seeds at visits to z are fresh. Hence, by additive accounting, the gain of i^* is the sum of the per-context gains, at least $(\kappa_\varphi/K^2)\sum_{z\ \text{violating},N_z>0} N_z - Q\epsilon_n$.

(iii) If $\mathrm{Viol}(c) = o(n)$ then (i) is $o(n)$, because $QL_n = o(n)$ for fixed Q and the threshold of (i) holds for all large n. If $\mathrm{Viol}(c) \ne o(n)$ then along a subsequence $\mathrm{Viol}(c) \ge \epsilon n$, and since $N_z \ge V_z - KL_n$ the contexts concerned have $\sum N_z \ge \epsilon n - QKL_n \ge (\epsilon/2)n$ eventually, so (ii) gives an agent a gain of at least $(\kappa_\varphi\epsilon/(2K^2))n - Q\epsilon_n$, which is not $o(n)$. □

Corollary I.8 (The revenue of the equilibrium set is a context-by–context selection)**.** *Under the hypotheses of Proposition I.7, at every additive equilibrium in Σ^{ctx},*

$$P(n; S^c) \ge \sum_z V_z g(z) - \mathrm{Viol}(c) - O\big(QKL_n + ns_n + QR_K(n)\big),$$

$$g(z) := \begin{cases} M_{(2)}(z) & \text{if } |\mathcal{T}(z)| \ge 2, \\ b(z) & \text{if } |\mathcal{T}(z)| = 1, \end{cases}$$

where $\mathrm{Viol}(c) = o(n)$ by part (iii) of that proposition, and the bound is attained: taking (B_z) at every two-top context and (A_z) with every loser at its lowest feasible bid at every unique-top context gives an additive equilibrium with $\mathrm{Viol}(c) = 0$ whose revenue equals $\sum_z V_z g(z)$ up to the same error. Taking expectations over the arrival path, the minimum over equilibria of $P(n; S)$ equals $\mathbb{E}\sum_{t\le n}[M_{(2)}(x_t)\mathbf{1}\{|\mathcal{T}(x_t)| \ge 2\} + b(x_t)\mathbf{1}\{|\mathcal{T}(x_t)| = 1\}]$ up to $o(n)$.

Proof. Per context, as in Corollary I.3. At a compliant two-top context two bids concentrate at $M_\star(z) = M_{(2)}(z)$, so on the local good event the price is at least $M_{(2)}(z) - 2s_n$, the winner delivers at least the price, and the revenue proof of Proposition H.2(ii) gives local revenue $V_z M_{(2)}(z) - O(KL_n + N_z s_n + R_K(n))$. At a compliant unique-top context the winner W delivers $\max\{\tau_z, \mu_W^z\mathbf{1}\{W \text{ honest}\}\}$ with τ_z within $2s_n$ of $\max_{k\ne W} c_{k,z}$, so the local revenue rate is $\max\{\max_{k\ne W} c_{k,z},\ \mu_W^z\mathbf{1}\{W \text{ honest}\}\}$ up to $O(s_n)$; as the losers' bids range over their feasible sets this rate ranges over $[b(z), \max\{b(z), M_{(2)}(z)\}]$, because honest losers bid at least their intrinsic means, strategic losers may bid 0, and some loser has capacity $M_{(2)}(z)$. Violating contexts have nonnegative revenue and are charged V_z each, in total $\mathrm{Viol}(c)$. Summing and charging the QKL_n Stage-1 rounds, the $2ns_n$ price shading and the $QR_K(n)$ fallback terms gives the display. The attaining profile has $\mathrm{Viol}(c) = 0$ and local revenue rate exactly $b(z)$ at each unique-top context, since every honest agent's intrinsic mean at z is then either the winner's or a loser's lowest feasible bid. Part (iii) of Proposition H.2 shows that no profile of any kind can go below $\sum_z V_z b(z)$ by more than the same order. □

The quantity g is a *selection* between the paper's two guarantees, made context by context by the local two-top condition, and

not their pointwise maximum. The distinction is not cosmetic: at a unique-top context the equilibrium-set floor is $b(z)$, which is normally strictly below $M_{(2)}(z)$, so a pointwise maximum would overstate the guarantee at exactly the contexts at which the result is negative. The two-top condition does not have to hold everywhere; it has to hold at the contexts that carry a positive fraction of the visits, and where it fails the mechanism falls back to the honest benchmark at that context rather than globally. No coarser statement is available: both endpoints are attained at genuine equilibria of the same instance.

Remark I.9 (What "at almost every context" means). *Under i.i.d. arrivals with law ν on a finite $\mathcal{X}_F$, $V_z/n \to \nu(z)$, so the positive-frequency contexts are exactly the support of ν, and for an instance with all masses positive the condition of Proposition I.7(iii) is the same as at every context. The relaxation bites exactly in the regime in which the contextual theorem is stronger than the grid results, namely the non-i.i.d. exogenous arrivals* (A2) *allows: a context whose share of the visits vanishes, for example one visited $n^{3/4}$ times, is completely unconstrained, its local price may be anything down to the local honest floor, and its contribution to revenue is $o(n)$ either way. We therefore say "at every context carrying a positive fraction of the visits" rather than "almost everywhere", the second phrase suggesting a measure-zero relaxation that is unavailable when all masses are positive.*

Collusion: the mechanism has no punishment technology. Every cartel needs two things, a signal that a member defected and an action that hurts the defector more than the deal pays it. CL-MESP supplies neither, and the second failure is exact.

Lemma I.10 (The harshest action is the proposed one)**.** *Fix agents $i \neq j$ and a context z, and suppose that at z every agent other than j plays a strategy of Σ^{ctx} type, a constant Stage-1 target followed by $S^\star$. Over all $S'_j \in \mathcal{B}^{\mathrm{ctx}}_{j,n}$, agent i's expected utility at visits to z is minimised, up to $O(V_z n^{-2})$, by j playing $S^\star_j$ at z.*

Proof. Let $c_{i,z} \in [0, M_i^z]$ be i's Stage-1 target at z. Couple the executions under S'_j and under $S^\star_j$ on the same seeds, priorities and coins, and condition on the arrival path.

Stage 1. Agent i's Stage-1 utility at z is $L_n(1 + \varphi_i(\mu_i^z) - \varphi_i(c_{i,z}))$, a function of its own target only, which j cannot touch.

Bid domination. By feasibility and the seed coupling, j's bid under $S^\star_j$, which delivers M_j^z at every Stage-1 pull, dominates its bid under S'_j pathwise, exactly as in Lemma H.6(c); the other bids are the same in both executions, because every agent other than i and j plays a constant Stage-1 target at z. Hence, as in Lemma H.6(d), if i wins at z against $S^\star_j$ then it wins at z against S'_j, and the price i faces, $\tau_{i,z} = (\max_{l \neq i} \beta_{l,z} - s_n)_+$, which does not involve i's own bid, is pathwise no larger against S'_j than against $S^\star_j$.

The value of winning is nonincreasing in the price. On i's own good event $\{\beta_{i,z} \leq c_{i,z} + s_n\}$, of probability at least $1 - (2Kn^4)^{-1}$, a win by i forces $\tau_{i,z} \leq (\beta_{i,z} - s_n)_+ \leq c_{i,z} \leq M_i^z$, so i's $S^\star$ primary target is $\max\{\mu_i^z \mathbf{1}\{i \text{ honest}\}, \tau_{i,z}\} \geq \tau_{i,z}$, a false switch has probability at most $(2n^3)^{-1}$ by Lemma H.4(c), and without a switch i's post-bidding utility at z is exactly $N_z f(\tau_{i,z})$, where $f(\tau) := 1 + \varphi_i(\mu_i^z) - \varphi_i(\tau)$ if i is strategic and $f(\tau) := \min\{1, 1 + \varphi_i(\mu_i^z) - \varphi_i(\tau)\}$ if i is honest; in both cases f is nonincreasing. When i loses at z its post-bidding utility is nonnegative, and it is zero unless a local switch occurs.

Comparison. Let win* and win$'$ be the events that i wins at z against $S^\star_j$ and against S'_j, so win$^\star \subseteq$ win$'$. Up to the bad events, which cost at most $2N_z((2Kn^4)^{-1} + (2n^3)^{-1}) = O(V_z n^{-2})$, i's post-bidding utility against $S^\star_j$ is $N_z f(\tau^\star_{i,z})\mathbf{1}_{\text{win}^\star}$ and against S'_j it is at least $N_z f(\tau'_{i,z})\mathbf{1}_{\text{win}'} \geq N_z f(\tau^\star_{i,z})\mathbf{1}_{\text{win}^\star}$, because $\tau'_{i,z} \leq \tau^\star_{i,z}$ pathwise and f is nonincreasing. Taking expectations proves the claim.

Post-bidding behaviour of j. Nothing above depends on what j does after Stage 1 at z: when i wins, j is not selected at z before a switch, and a switch by i has the probability bounded above whatever j does; when i loses, j can only create switches, never prevent i from being selected in a fallback, so every post-bidding choice of j weakly adds to i's payoff. Hence j's harshest action at z is to bid capacity, which is $S^\star_j$ in Stage 1, and its post-bidding behaviour is irrelevant to the bound. □

The lemma holds verbatim in the scalar mechanism, the case $Q = 1$, and explains Proposition I.6(ii) in one line: the reason the two-top joint underbid is not self-enforcing is not a detail of that computation, it is that the only threat available to a partner is competitive bidding, and competitive bidding is the proposed profile. A cartel here can agree only to be softer than $S^\star$; it can never agree to be harsher, so it has no way to pay for a concession. The signal is weak too. No observable of any agent other than j depends on j's Stage-1 targets at z before $t_1(z)$, the round of the KL_nth visit to z, because the mechanism reads them only when it forms $\beta_{j,z}$; any reaction to a concession or a defection at z must therefore be delivered through behaviour at a context whose Stage 1 is still open at that moment.

Proposition I.11 (Pairs at the contextual mechanism)**.** *Assume the hypotheses of Proposition I.7 and consider pair deviations from $S^\star$ inside Σ^{ctx}, the two members changing only their Stage-1 targets, at any contexts they like; self-enforcement is tested against arbitrary deviations in $\mathcal{B}^{\mathrm{ctx}}$ by each member.*

(i) Within a context. *Proposition I.6 applies at each context with n replaced by V_z: at a unique-top context the local top agent and the local runner-up have a jointly profitable and self-enforcing concession that lowers the local revenue rate to $\max\{M_{(3)}(z), b(z)\}$; at a context with $|\mathcal{T}(z)| = 2$ a joint underbid is profitable but not self-enforcing; at a context with $|\mathcal{T}(z)| \geq 3$ no pair gains.*

(ii) Across contexts, market division can be profitable. *Suppose two agents i, j satisfy $\mathcal{T}(z_1) = \mathcal{T}(z_2) = \{i, j\}$ at two contexts with $V_{z_1}, V_{z_2} = \Theta(n)$, and consider the scheme in which i bids its capacity at z_1 and its lowest feasible target at z_2, and j the mirror image. Up to $O(\Lambda n s_n + \Lambda\sqrt{n \log n})$, member m's gain, with z_m its own context, is*

$$N_{z_m}(1 + \varphi_m(\mu^{z_m}) - \varphi_m(\theta_m)) - \tfrac{1}{2}\big[N_{z_1}(1 + \varphi_m(\mu^{z_1}) - \varphi_m(M_\star(z_1))) + N_{z_2}(1 + \varphi_m(\mu^{z_2}) - \varphi_m(M_\star(z_2)))\big],$$

where φ_m is the member's own cost function and θ_m is the highest remaining bid at z_m. The scheme is jointly profitable when this gain is at least cn for both members and some

$c > 0$. *For instance, if the two contexts have the same capabilities, intrinsic means and costs,* $V_{z_1} = V_{z_2}(1 + o(1))$, *and* $\varphi_m(M_\star) - \varphi_m(\theta_m)$ *is bounded away from zero, each member gains* $N_{z_1}(\varphi_m(M_\star) - \varphi_m(\theta_m)) - o(n)$. *It is not profitable in general: the owner of a context with much less traffic gives up half of a large tenure for all of a small one. Platform revenue falls by* $\sum_m N_{z_m}(M_\star(z_m) - \max\{\theta_m, b(z_m)\})$. *So on such instances* $S^\star$ *is not pair-proof even though the two-top condition holds at every context, and the local two-top condition does not protect against cross-context pairs.*

(iii) Across contexts, no market division is self-enforcing. *Suppose every context with* $V_z = \Omega(n)$ *has* $|\mathcal{T}(z)| \geq 2$. *Then no jointly profitable pair deviation in* Σ^{ctx} *is self-enforcing. Consequently* $S^\star$ *is pair-coalition-proof and every self-enforcing pair deviation leaves revenue at* $\sum_z V_z M_{(2)}(z) - o(n)$.

Proof. Parts (i) and (ii) are Proposition I.6 and a direct computation at the local primitives, using additive accounting to add the two contexts' contributions; for (i), the revenue rate at a unique-top context is $\theta = \max\{c_k, M_{(3)}(z), \mu_W^z \mathbf{1}\{W \text{ honest}\}\}$ with c_k the runner-up's lowest feasible bid, and $\theta = \max\{M_{(3)}(z), b(z)\}$ because every honest agent's intrinsic mean at z is at most θ while c_k and $\mu_W^z \mathbf{1}\{W \text{ honest}\}$ are at most $b(z)$. For the example in (ii), equal primitives give the two contexts the same per-pull utilities, and $N_{z_1} = N_{z_2}(1 + o(1))$ turns the displayed gain into $N_{z_1}\big[(1 + \varphi_m(\mu) - \varphi_m(\theta_m)) - (1 + \varphi_m(\mu) - \varphi_m(M_\star))\big] - o(n) = N_{z_1}(\varphi_m(M_\star) - \varphi_m(\theta_m)) - o(n)$.

For (iii), call z *price-conceded* under a pair deviation if some member of $\mathcal{T}(z)$ bids below $M_\star(z)$ there; such an agent is not the local winner on the good event, so the price at z is set below $M_\star(z)$ by its bid or by a lower one. Suppose the pair $\{1, 2\}$ deviates to c and the deviation is jointly profitable, each member gaining at least $\epsilon_0 n$ for all large n. Then some price-conceded context has $V_z \geq \epsilon_0 n$: the only sources of gain are saved Stage-1 effort, which is $O(QL_n) = o(n)$ in total, and a lower local price or a higher local win probability, and at a context whose two local top agents both bid $M_\star(z)$ the price is $M_{(2)}(z) - O(s_n)$ and no member can win more often than under $S^\star$; a context contributes at most $2V_z$ to any gain. A context can be conceded by a member that is not locally top, whose bid does not enter the price at all, since the two local top agents still bid capacity; such a concession is real, saving that member $L_n(\varphi_j(M_j^z) - \varphi_j(c_{j,z}))$, but it is invisible to the price, which is why the selection below is made among price-conceded contexts and not among conceded ones.

Choosing the context. Let C be the set of price-conceded contexts, for $z \in C$ let $L(z) := \sum\{V_{z'} : z' \in C,\ t_1(z') > t_1(z)\}$ be the total mass of the price-conceded contexts whose Stage 1 completes after that of z, and put $a := \kappa_\varphi/(4KQ) \in (0, 1)$. Start from $z_0 \in C$ with $V_{z_0} \geq \epsilon_0 n$. Given $z_r \in C$ with $V_{z_r} \geq a^r \epsilon_0 n$: if $L(z_r) \leq (\kappa_\varphi/(4K))V_{z_r}$, stop and put $z^* := z_r$; otherwise some $z' \in C$ with $t_1(z') > t_1(z_r)$ has $V_{z'} \geq L(z_r)/Q > (\kappa_\varphi/(4KQ))V_{z_r} = aV_{z_r} \geq a^{r+1}\epsilon_0 n$, and put $z_{r+1} := z'$. Each step moves to a strictly later context and $|C| \leq Q$, so the procedure stops after at most $Q - 1$ steps at a price-conceded z^* with

$$V_{z^*} \geq a^{Q-1}\epsilon_0 n \qquad \text{and} \qquad L(z^*) \leq \frac{\kappa_\varphi}{4K} V_{z^*}.$$

The defection. By definition some member $j \in \mathcal{T}(z^*)$ bids below $M_\star(z^*)$ at z^*. Let j deviate unilaterally by bidding $M_j^{z^*} = M_\star(z^*)$ at z^* and changing nothing else. On the local good event it is then among the largest bidders, so it wins at z^* with probability at least $1/K$, at a price at most $M_\star(z^*) = M_j^{z^*}$; by (E1) applied locally, and comparing with its utility at z^* under c through the cases of the scalar necessity proof, its gain at z^* is at least $(\kappa_\varphi/K)N_{z^*} - L_n - O(1) \geq (\kappa_\varphi/K)(V_{z^*} - KL_n) - L_n - O(1)$.

The loss elsewhere. Agent j's own bids elsewhere are unchanged, so by context locality the only way its payoff at another context z' can change is through the partner's reaction. No observable of the partner before $t_1(z^*)$ depends on j's Stage-1 targets at z^*, because the mechanism's only use of those targets is through β_{j,z^*}, which is formed at $t_1(z^*)$; so the partner can react only through its behaviour at contexts z' with $t_1(z') > t_1(z^*)$, and only through the part of that behaviour that comes after $t_1(z^*)$. At such a z' the member j plays a constant Stage-1 target followed by $S_j^\star$, so Lemma I.10 says that the partner's harshest reaction at z' is to bid capacity there. If z' is not price-conceded, both local top agents already bid $M_\star(z')$, the price is $M_{(2)}(z') - O(s_n)$ whatever the partner does, the partner cannot win if it is not locally top and already bids capacity if it is, and so the reaction changes j's payoff at z' by at most the bad-event terms $O(V_{z'} n^{-2})$. If z' is price-conceded with $t_1(z') > t_1(z^*)$, bound the change by $2V_{z'}$. The total loss elsewhere is therefore at most $2L(z^*) + O(n^{-1}) \leq (\kappa_\varphi/(2K))V_{z^*} + O(n^{-1})$.

Conclusion. The net gain of j is at least $(\kappa_\varphi/(2K))V_{z^*} - (\kappa_\varphi/K)KL_n - L_n - O(1) \geq (\kappa_\varphi/(2K))a^{Q-1}\epsilon_0 n - o(n)$, which is $\Omega(n)$ for fixed K and Q. Hence j has a unilateral gain of $\Omega(n)$ from the deviated profile, so the deviation is not self-enforcing. The revenue statement follows because a self-enforcing pair deviation price-concedes no context of linear mass, so every positive-frequency context has two bids at $M_\star(z)$ and its price is $M_{(2)}(z) - O(s_n)$, while for fixed Q the contexts of sublinear mass contribute $o(n)$. □

What (iii) adds to Proposition I.7 is worth isolating. If both members are restricted to constant bids, the deviated profile is itself a member of Σ^{ctx}, a market division violates (A_z) and (B_z) at some context, and part (ii) of that proposition already gives some agent an $\Omega(V_z)$ gain. The content of (iii) is that trigger strategies do not rescue the scheme: a partner whose bid at one context is a function of what it saw at another is not in Σ^{ctx}, and the two ingredients that kill it are the timing, since no observable depends on the deviator's bid at z^* before $t_1(z^*)$, and Lemma I.10, since after $t_1(z^*)$ the partner's harshest available action is the one it is already taking. The argument never uses the arrival order, although the order decides whether the signal exists at all: when the Stage-1 windows of two contexts overlap, as they do under i.i.d. arrivals with both contexts frequent, neither member can condition its bid at one context on the price at the other and the scheme is unenforceable for a second, independent reason. The hypothesis that every positive-frequency context has $|\mathcal{T}(z)| \geq 2$ is exactly the hypothesis of case (B_z), so the coalition result and the unilateral result hold under one and the same condition; where it fails, the concession at a unique-top context is self-enforcing, which is part (i), and the revenue loss there is already counted in Corollary I.8.

Remark I.12 (Why a deeper concession is more stable, and the second half of the reason). *The inversion of the usual cartel logic needs two halves, and the second is the load-bearing one. The first is that the cost of reversing a concession grows with its depth: buying back in at z means moving the whole Stage-1 bid up to $M_\star(z)$ and paying the elicitation effort again. The second is a second-price fact. In an ordinary cartel the defector's prize grows with the depth of the concession, because it undercuts a high cartel price and keeps most of it. Here it cannot: the low price at a conceded context is created by the conceder's own bid, being the second order statistic, and the act of contesting destroys it. The defector cannot both win the tenure and enjoy the price it was conceding, so its prize is, heuristically, $N_z(1+\varphi_j(\mu_j)-\varphi_j(M_\star(z)))/|\mathcal{T}(z)|$ up to $O(\Lambda N_z s_n)$, whatever it conceded. The accurate one-sentence version is therefore that the cost of reversing a concession grows with its depth while the prize does not, because the concession is the price and contesting it removes it.*

Remark I.13 (The honest benchmark survives collusion). *Part (iii) of Proposition H.2 is already stated for arbitrary joint behavioural play, and its proof contains no hidden independence across agents. Its four steps are a one-sided Azuma–Hoeffding bound on the honest agent's own local bid, which uses only honesty and the freshness of that agent's own reward seeds; the order-statistic inequality $\beta_{(2),z} \geq \beta_{h,z}$ when h is not the local winner; Lemma H.4, which holds for arbitrary predictable targets of whoever won and therefore for coordinated ones; and Lemma H.5 with the honest agent as comparator, which requires only that its conditional mean when selected is at least $b(z)$, again by honesty alone. The per-context split that precedes all four is additive accounting, which needs no independence at all. So $P(n;S) \geq \mathbb{E}\sum_t b(x_t) - o(n)$ holds under collusion by any subset of the agents, up to and including all K, and by Corollary I.8 it is attained at an equilibrium as soon as one positive-frequency context has a unique local top. The one hypothesis that matters is that arrivals must not react to realised rewards, which is a hypothesis about the world rather than about the agents.*

J Experimental details and additional results

Every table in this appendix is emitted directly from the retained summary files by `figures/make_tables.py`, so no number here is transcribed by hand. The supplementary artifact contains the code, the seed schedule, the summary files, the exact commands, and the environment specification.

Evidence categories. We keep four categories apart and never merge them. The exploitability results of Table 3 and Table 4 and the contextual results of Table 6 are *exact computations* on synthetic instances: no sampling enters them. The ladder of Table 5 and the public-UCB1 results of Tables 7 to 10 are *paired simulation* on synthetic instances. Tables 11 and 12 report *real outcomes*, that is, real labeled tasks answered by real algorithmic providers, with the effort choice scripted. No experiment involves human subjects, and none of them is evidence about human effort.

Statistical protocol. Seeds derive from

`SeedSequence([ 20260828, suite, phase, instance, n, rep ])`,

with the pilot, final, training, and evaluation phases disjoint by construction, so no pilot seed appears in a reported table and no training seed appears in a reported evaluation. Simulated comparisons are paired by replicate key and use nonparametric bootstrap 95% intervals on the paired differences. Once adaptive paths diverge, a shared seed no longer implies identical counterfactual noise, but the pairing by replicate key is preserved. Exact computations carry no interval because they involve no sampling; their reported interval is the gap between the certified lower and upper bounds.

Why the exploitability computation is exact. Every experiment reported here instantiates the linear case $\varphi_i(m) = m$ of the cost model of Section 2.1, for which $\Lambda = 1$ and the normalisation (1) holds with equality; no number below is recomputed for any other cost function. Fix the deviator's information state at the start of a round. Its one-round objective is affine in the target it selects, because the expected reward is the target and the effort cost is the target minus the intrinsic mean. An optimal target is therefore an endpoint of the feasible interval, so backward induction over two actions per state is exact for the bidding and primary stages: there is no target grid and no discretization error. The fallback phase is the only approximation, and it is bracketed rather than estimated, between a continuation value of zero and the EXP3 regret bound of Appendix F.3. The reported lower bound uses the zero continuation and the upper bound uses the EXP3 bound, so the true supremum over the behavioral class lies between them.

Decomposition and the non-top branch. The optimal deviation splits into adaptive underbidding during the bidding stage, whose value is of order L_n, and a primary-stage term that rides the monitoring boundary. Underbidding that adapts to whether the win is already secured or already hopeless is worth 676 utility at $n = 50{,}000$ in the unique-top instance. The fixed midpoint underbid of Proposition F.4 in Appendix F, the deviation the tightness argument uses, guarantees $L_n(M_i - V_i)/2 = 3092 \times 0.125 = 386.5$ there, so adapting the bid to the state of the race is worth roughly three quarters again as much. Fixed hand-picked deviations therefore understate finite-horizon exploitability, and by more than this comparison shows, since the certified bound in Table 3 covers Stage 2 as well. The theorem no longer has a non-top branch, so its constant applies to every focal agent at every horizon, which is what the last column of Table 3 prints. An earlier version of this section reported the two non-top instances by their certified lower bound only, because the old branch condition failed at every horizon we could run; that restriction is gone, and those instances are now covered by the same bound as the rest.

Detection delay. The anytime boundary predicts a median detection delay of $\log(2n^4)/(2(\tau - m)^2)$ once a deviator settles on a primary target m below the announced price. Over the 42 cells in which the deviator delivers nothing the observed median is 0.93 to 1.02 times that prediction. Over the 23 cells in which it shades by a fixed amount the ratio runs from 0.67 to 1.03, the low end at the smaller shading and the shorter horizons, where a first-order approximation to a boundary crossing is weakest. Quoting only the tightest subset would misrepresent the agreement, so both ranges are given.

Grid geometry. Two geometry findings motivate the exact tiling used in the paper. Under a grid whose last cell along each axis is clipped, the thin cells never accumulate enough visits to finish

Table 2: Scalar instances used in the main paper. Agents are numbered from one. The focal agent is the strategic deviator; every instance contains at least one honest agent. The capacity-gap grid varies M_2 only and is reported in Table 4.

instance	μ	M	honest	focal
unique top	$(0.1, 0.2, 0.35)$	$(0.9, 0.65, 0.4)$	3	1
tied top	$(0.1, 0.2, 0.35)$	$(0.8, 0.8, 0.45)$	3	1
near gap	$(0.1, 0.2, 0.45)$	$(0.8, 0.76, 0.5)$	3	1
honest top	$(0.75, 0.2, 0.15)$	$(0.75, 0.7, 0.45)$	1	2
low-gap non-top	$(0.1, 0.2, 0.45)$	$(0.8, 0.795, 0.5)$	3	2
$K = 2$ unique top	$(0.1, 0.35)$	$(0.9, 0.65)$	2	1
$K = 2$ tied top	$(0.1, 0.35)$	$(0.8, 0.8)$	2	1

Table 3: Exact per-round exploitability of MESP. LB and UB bracket the supremum unilateral gain over the behavioral class: the bidding and primary stages are computed exactly by backward induction and the fallback is bracketed between a zero continuation and the EXP3 bound. Stage columns give the two parts of the optimal deviation. The last column is the explicit constant of Theorem 3.2.

instance	n	LB	UB	stage 1	stage 2	Pr(win)	Thm. 3.2
near gap	2,000	0.0563	0.0797	0.0002	0.0562	0.891	0.4644
near gap	5,000	0.0449	0.0581	0.0002	0.0447	0.956	0.3203
near gap	8,000	0.0388	0.0488	0.0003	0.0385	0.978	0.2652
near gap	20,000	0.0280	0.0339	0.0005	0.0275	0.997	0.1840
near gap	50,000	0.0197	0.0231	0.0008	0.0189	1.000	0.1283
tied top	2,000	0.0347	0.0724	0.0032	0.0315	0.500	0.4644
tied top	5,000	0.0244	0.0487	0.0010	0.0234	0.500	0.3203
tied top	8,000	0.0203	0.0398	0.0007	0.0197	0.500	0.2652
tied top	20,000	0.0141	0.0267	0.0003	0.0138	0.500	0.1840
tied top	50,000	0.0096	0.0180	0.0001	0.0095	0.500	0.1283
unique top	2,000	0.0910	0.1198	0.0280	0.0630	1.000	0.4644
unique top	5,000	0.0702	0.0861	0.0234	0.0467	1.000	0.3203
unique top	8,000	0.0606	0.0723	0.0212	0.0394	1.000	0.2652
unique top	20,000	0.0447	0.0512	0.0171	0.0276	1.000	0.1840
unique top	50,000	0.0324	0.0360	0.0135	0.0189	1.000	0.1283

The primary-stage column equals $b_n(N_n)$ times the win probability to within 0.001 in 73 of the 96 distinct cells and 0.01 in 88; the exceptions are confined to $n = 200$ and $n = 500$.

exploring and supply the entire in-class gain at $n = 2{,}000{,}000$. Under a dyadic refinement, Kq_nJ_n exceeds the horizon at three of the five horizons tested, so the rule never commits at all and the guarantee is vacuous. Partitioning each axis into equal parts removes both. Horizons at which $Kq_nJ_n > n$ are omitted from Table 6 for the same reason.

Does the rule learn? The results above measure exploitability, which says nothing about whether the mechanism allocates well. Table 8 reports oracle regret per round for the contextual rules on four environment families at two profile dimensions, with every agent delivering its capacity, so these runs measure learning alone. Three things are worth stating plainly. Partitioning each axis into equal parts improves Static Grid-ETC substantially at $D = 2$, from 0.0355 to 0.0295 in the local-linear family and from 0.0506 to 0.0421 in the checkerboard family, which is the same clipped-cell effect that dominated the exploitability computation. The known-horizon confidence width, by contrast, costs Static Grid-UCB1 regret at $D = 2$, raising it from 0.0344 to 0.0421 in the local-linear family, because that width is the wider of the two and therefore explores more; exact monotonicity is bought with finite-horizon regret, and we do not claim otherwise. At $D = 6$ Static Grid-ETC is at or behind context-blind UCB1 in every family, and Static Grid-UCB1 is behind it only on the discontinuous checkerboard capacities; that is the dimensional cost of carrying the full profile dimension $D = Kd$. LinUCB receives the whole profile and assumes a global linear structure the grid rules do not, so it wins wherever that structure holds and is not a strategic mechanism.

Index and tie rule. Table 9 reports both tie rules, because the two give different answers and the difference is the point. In each cell one agent's fixed target is raised pointwise to its capacity on a shared reward tape and the other agents' pull counts are compared on the same sample point. Under the single fixed priority the proof assumes, the known-horizon index never violated monotonicity in 240,000 coupled paths, while the anytime index violated it 59 times, at $n = 8$, 50, and 200. Under a fresh tie coin drawn each round the known-horizon index violates as well, 36 times, because the coin can hand a pull to a different agent after the raise. The pathwise claim in this paper is therefore stated for one priority fixed before play, and no pathwise claim is made for the per-round coin. Table 10 separates symmetric from index-order initialization; they are indistinguishable at $n \geq 2{,}000$ and separate only at $n = 6$, where index-order initialization gives an interval excluding zero.

Table 4: Capacity-gap grid at $K = 3$. Exploitability grows with the gap while the theorem's constant does not depend on it, so the bound is loosest at a tie.

$M_1 - M_2$	n	LB	UB	Thm. 3.2
0.00	2,000	0.0347	0.0724	0.4644
0.00	8,000	0.0203	0.0398	0.2652
0.00	20,000	0.0141	0.0267	0.1840
0.01	2,000	0.0400	0.0707	0.4644
0.01	8,000	0.0275	0.0395	0.2652
0.01	20,000	0.0210	0.0272	0.1840
0.02	2,000	0.0464	0.0723	0.4644
0.02	8,000	0.0333	0.0428	0.2652
0.02	20,000	0.0254	0.0305	0.1840
0.05	2,000	0.0594	0.0835	0.4644
0.05	8,000	0.0399	0.0505	0.2652
0.05	20,000	0.0287	0.0348	0.1840
0.10	2,000	0.0673	0.0953	0.4644
0.10	8,000	0.0447	0.0562	0.2652
0.10	20,000	0.0326	0.0389	0.1840
0.20	2,000	0.0824	0.1117	0.4644
0.20	8,000	0.0550	0.0669	0.2652
0.20	20,000	0.0405	0.0471	0.1840

Table 5: Mechanism ladder at $n = 20{,}000$, 200 paired replicates per cell. The gain column maximises over deviation profiles inside each instance and then ranges over the three instances whose focal agent contests the bidding stage (unique top, tied top, near gap). It excludes the honest-top instance, in which the focal agent is the strategic runner-up and almost never wins, so no monitor is exercised and every variant shows the same bidding-stage term of 0.0123. The false-switch column is measured under the proposed profile $S^\star$. Every column is restricted to those same three instances. The last two report whether the platform stayed above $\mu_H^\star$ under sequential joint shading, and by how much per round; in the fourth instance every variant, MESP included, finishes 0.029 per round below $\mu_H^\star$, which is the mechanism's own bidding block and price correction and lies far inside the slack of Theorem 3.2.

variant	best tested gain	false switch	revenue gap	benchmark	headroom
MESP	0.0063 to 0.0123	0	+0.084	yes	+0.080 to +0.142
confidence exponent $1/n$	0.0063 to 0.0123	0	+0.045	yes	+0.080 to +0.142
exclusion	0.0063 to 0.0123	0	+0.084	yes	+0.054 to +0.078
UCB1 fallback	0.0063 to 0.0123	0	+0.084	yes	+0.088 to +0.149
UCB1 fallback, warm start	0.0063 to 0.0123	0	+0.084	yes	+0.031 to +0.081
EXP3 fallback, warm start	0.0063 to 0.0123	0	+0.084	yes	+0.061 to +0.117
no price correction	0.0060 to 0.0123	0	-0.001	yes	+0.080 to +0.142
monitor without slack	0.0066 to 0.0123	0.99 to 1.00	-0.173	yes	+0.081 to +0.143
round-robin fallback	0.0063 to 0.0123	0	+0.084	no	-0.156 to -0.070
unmonitored runner-up	0.0063 to 0.0123	0	+0.084	no	-0.282 to -0.183
windowed monitor	0.0589 to 0.1132	0	+0.084	yes	+0.076 to +0.138
no monitor	0.2680 to 0.4887	0	+0.084	no	-0.282 to -0.183

Reporting one range over all four instances would read 0.012 to 0.489 for the unmonitored mechanism and would hide the effect rather than show it. In that fourth instance the focal agent is the strategic runner-up and is never itself monitored, so eleven of the twelve variants show the same bidding-stage term of 0.0123; the slackless monitor shows 0.0000 there, because its false switches on the honest winner destroy the deviator's gain along with the platform's. Under the proposed profile a complying primary winner delivers exactly the announced price, so the monitored walk has zero drift and a test without slack convicts by ordinary noise; when every agent instead delivers its capacity the walk has positive drift and the slackless monitor's false-switch rate falls to 0.20 to 0.63 on the same three instances. Every other variant is exactly zero under both profiles.

Provider banks. Two banks of real labeled tasks are answered offline by real algorithmic providers whose compute budget is a controllable knob, and the recorded correctness of each answer is the Bernoulli outcome the mechanism observes. The tasks come from Fashion-MNIST (Zalando SE, MIT License) and from Covertype (Jock Blackard, UCI Machine Learning Repository, DOI 10.24432/C50K5N, CC BY 4.0, whose attribution requirement this sentence discharges); the providers are scikit-learn classifiers under the BSD 3-Clause licence, differing only in training budget. The effort choice is scripted, so the runs test the mechanism and not the providers' incentives. The banks are near-ties, and the regime in which effort strictly pays appears only at the supplementary horizon $n = 200{,}000$. Under detected sabotage the platform's real compute bill roughly triples, a cost the model's effort convention

Table 6: Exact per-round gain against Static Grid-ETC under a grid that tiles the cube exactly, computed cell by cell over the fixed-map class. Horizons at which $Kq_nJ_n > n$, where the rule never commits, are omitted. The last column is a count-dependent deviation outside the class, which the theorem does not cover and which grows with the horizon.

environment	deviator	n	cells	explore share	in class	outside class
current	top	8,000	64	0.480	0.0040	0.1745
current	top	32,000	216	0.648	0.0777	0.1218
current	top	128,000	343	0.410	0.0000	0.2087
current	top	512,000	729	0.342	0.0000	0.2355
current	top	2,000,000	1331	0.252	0.0000	0.2690
separator	top	8,000	64	0.480	0.0346	0.1490
separator	top	32,000	216	0.648	0.1029	0.1030
separator	top	128,000	343	0.410	0.0083	0.1762
separator	top	512,000	729	0.342	0.0000	0.2000
separator	top	2,000,000	1331	0.252	0.0000	0.2307
separator	mixed	8,000	64	0.480	0.0515	0.0889
separator	mixed	32,000	216	0.648	0.1140	0.0564
separator	mixed	128,000	343	0.410	0.0373	0.0882
separator	mixed	512,000	729	0.342	0.0279	0.0923
separator	mixed	2,000,000	1331	0.252	0.0199	0.0990

Table 7: Best fixed target against symmetric UCB1, over a grid of targets in $[\mu_i, M_i]$, for both confidence widths. The paper specifies the known-horizon rows; the anytime rows are shown because the two widths give different best targets and different gains, and because the paper's monotonicity claim distinguishes them. With a unique top agent the best target is strictly interior and the gain is linear in n under either width; with two tied top agents the best target is the capacity itself and the gain is exactly zero under either.

instance	index	n	best target	gain per round	95% interval
tied top	anytime	2,000	0.80	0.0000	[0.0000, 0.0000]
tied top	anytime	8,000	0.80	0.0000	[0.0000, 0.0000]
tied top	anytime	20,000	0.80	0.0000	[0.0000, 0.0000]
tied top	known horizon	2,000	0.80	0.0000	[0.0000, 0.0000]
tied top	known horizon	8,000	0.80	0.0000	[0.0000, 0.0000]
tied top	known horizon	20,000	0.80	0.0000	[0.0000, 0.0000]
unique top	anytime	2,000	0.74	0.0889	[0.0863, 0.0915]
unique top	anytime	8,000	0.72	0.1229	[0.1212, 0.1247]
unique top	anytime	20,000	0.72	0.1454	[0.1443, 0.1464]
unique top	known horizon	2,000	0.74	0.0726	[0.0705, 0.0746]
unique top	known horizon	8,000	0.74	0.1021	[0.1009, 0.1033]
unique top	known horizon	20,000	0.72	0.1271	[0.1259, 0.1282]

does not represent. The map from effort to accuracy is fitted on a measurement half of the task pool, disjoint from the half the mechanism plays on, and the realized accuracy on the play half differs from the fitted one. That drift is the sharpest evidence that these outcomes are real rather than Bernoulli by construction, and it has two visible consequences. At $n = 200{,}000$ the monitor eliminates a complying agent in 3 of 100 replicates of the proposed profile on the Fashion-MNIST bank, so the false-switch rate is zero at the horizons of Tables 11 and 12 but not at every horizon. Against partial shading the median detection delay matches the first-order formula when the formula is evaluated at the realized shortfall, at ratios 0.962 and 0.996 on the two banks, and fails when it is evaluated at the shortfall the calibration predicts, at 0.124 and 0.916. No Bernoulli simulator is used anywhere in this evaluation.

Searched cross-check. A covariance-matrix evolution strategy searched a phase-scaled threshold class and an interior-target class, training on training seeds, selecting on a validation block, and evaluating once on 2,000 held-out paired seeds. No searched policy in any completed cell exceeded the certified upper bound, which is the check that would have exposed an error in the backward induction or in the bound. Interior-target policies reach 0.25 to 0.78 of the certified lower bound and never approach the upper bound, which is the empirical counterpart of the affinity argument that makes the two-action recursion exact. The threshold class reaches 0.966 of the certified lower bound at $n = 2{,}000$ in the unique-top instance and 0.926 in the tied-top instance; the residual gap is expected, because the searched rule has a fixed shape while the exact optimum has a time-varying threshold. Two threshold cells at $n = 20{,}000$ did not finish within our compute window and are reported as outstanding rather than estimated. These searched values are

Table 8: Oracle regret per round at $n = 8{,}000$, 100 paired replicates. The first number in each cell uses the index and the grid the paper specifies, namely the known-horizon confidence width and a partition of each axis into equal parts; the number in parentheses uses the anytime width and the clipped grid that earlier versions of this code used, so that the effect of the change is visible rather than hidden. LinUCB receives the whole profile and exploits a global linear structure that the grid rules do not assume, so it is a reference point and not a strategic mechanism. The checkerboard capacities are discontinuous and lie outside the Lipschitz hypothesis. A dash means the method does not apply to that environment.

environment	D	Static Grid-UCB1	Static Grid-ETC	LinUCB	context-blind UCB1	exact-profile UCB1
local linear	2	0.0421 (0.0344)	0.0295 (0.0355)	0.0025 (0.0025)	0.0613 (0.0601)	–
local linear	6	0.0340 (0.0350)	0.0383 (0.0383)	0.0055 (0.0055)	0.0354 (0.0342)	–
smooth	2	0.0484 (0.0384)	0.0336 (0.0410)	0.0232 (0.0232)	0.0721 (0.0710)	–
smooth	6	0.0622 (0.0653)	0.0751 (0.0751)	0.0272 (0.0272)	0.0724 (0.0716)	–
checkerboard	2	0.0618 (0.0464)	0.0421 (0.0506)	0.0487 (0.0487)	0.0778 (0.0760)	–
checkerboard	6	0.0827 (0.0832)	0.0837 (0.0837)	0.0755 (0.0755)	0.0775 (0.0757)	–
finite profiles	2	0.0291 (0.0207)	0.0132 (0.0132)	0.0454 (0.0454)	0.0522 (0.0499)	0.0291 (0.0207)

These runs measure learning, not incentives. Every agent delivers its capacity, so nothing here is evidence about strategic behavior.

lower bounds for the class searched. They are not evidence of equilibrium.

What these experiments do not establish. They do not establish equilibrium in any class. They say nothing about instances, classes, or horizons not computed, and in particular nothing about uniformity as capacity gaps vanish. They do not establish coalition-proofness, since the joint profiles tested are specific and few. They make no claim about human behavior or about effort inferred from crowd-label data. They do not show that any constant in the mechanism is optimal. A positive finite-horizon gain does not contradict an $o(n)$ theorem, and a decaying finite-horizon gain does not prove one.

K Additional experiments

This appendix holds the figures that Section B.5 and Section B.4 quote, the details of the serving experiment of Section 5.2, and the per-round dynamics behind Figure 9. Unless stated, the runs use the ten RouterBench subsets at $n = 3 \times 10^6$ with twenty seeds; the substitution runs use one four-provider subset with ten seeds; MESP under the two effort-reduction schedules of Figure 8b uses the ten real-order subsets at both horizons under real and independent arrivals, with twenty and ten seeds. Every number is read from the committed aggregate files of the data package and of the serving experiment; the figure conventions are those of Section 5.

The dynamics runs. The instance is the RouterBench triple Mixtral-8x7B, GPT-4 and Yi-34B on one task type, scalarised so that a fixed shade is well defined; GPT-4 is the deviator and Mixtral is honest. Ten seeds with common random numbers at $n = 20{,}000$ and $n = 300{,}000$. The deviation is the same against every rule: bid the capability, then deliver 0.05 (or 0.10) below whatever the reference profile delivers in every later served round, which is the announced price while a price is in force and the capability otherwise. On the same instance the mechanism of Cao et al. [4] spends 0, 0, 7, 35 and 57% of the horizon in its priced phase at 2×10^4, 5×10^4, 10^5, 3×10^5 and 10^6 rounds, because its budget reserves about $39\, n^{2/3}$ rounds for exploration and blind trust. Under MESP the monitor convicts the 0.05 shade at pull 7,447 on average and the 0.10 shade at 1,946; the deviator's utility per round falls from 0.745 under honesty to

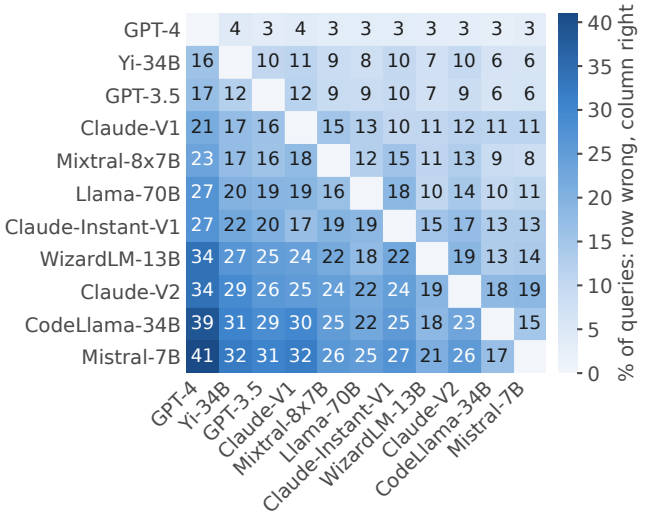


Figure 13: Pairwise complementarity on RouterBench: share of task families on which the row model is beaten by the column model.

0.569 and 0.360. Under Cao et al. [4] at $n = 20{,}000$ the deviator is never tested and its utility rises from 0.331 to 0.353 and 0.375. The UCB1 router's mild loss at $n = 20{,}000$ is incidental, a smaller gap to the runner-up makes it explore more, and at $n = 300{,}000$ the same deviation gains under both routers.

The serving experiment. Qwen3-4B (Hugging Face revision 1cfa9a72), bf16, thinking mode on, served by vLLM 0.32.0 in eager mode on one NVIDIA L40S or A100 per job. The customer-visible configuration is fixed: model identifier, the service tier `standard`, one system prompt asking for a final-answer line, temperature 0.6, top-p 0.95, top-k 20, a 7,936-token output cap. The backend control is vLLM's per-request thinking-token budget, which forces the token that ends thinking once the reasoning segment reaches the budget; reasoning and answer tokens are counted from the token

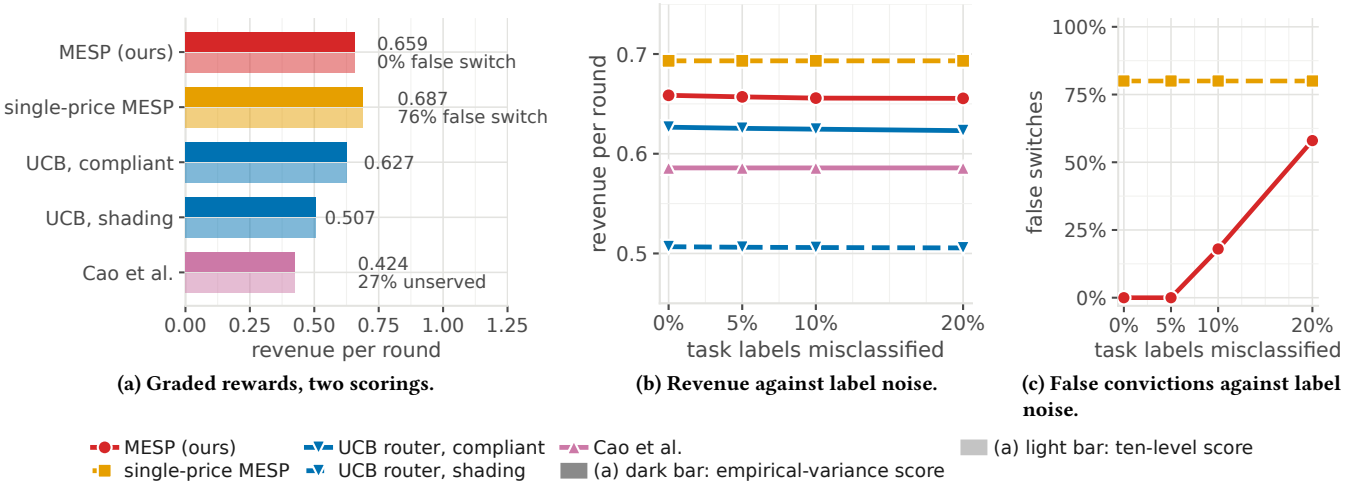
Table 9: Pathwise monotonicity: one agent's fixed target is raised pointwise to its capacity on a shared reward tape, and the other agents' pull counts are compared on the same sample point. Under the single priority fixed before play, which is what the proof assumes, the known-horizon index never violates monotonicity and the anytime index does; under a fresh tie coin each round both violate. This is why the paper specifies the known-horizon width and states its pathwise claim for a fixed priority.

index	tie rule	instance	n	paths	violating	rate
anytime	fresh coin each round	tied top	6	20000	0	0.00000
anytime	fresh coin each round	tied top	8	20000	52	0.00260
anytime	fresh coin each round	tied top	12	20000	4	0.00020
anytime	fresh coin each round	tied top	50	20000	8	0.00040
anytime	fresh coin each round	tied top	200	20000	7	0.00035
anytime	fresh coin each round	tied top	1,000	20000	0	0.00000
anytime	fresh coin each round	unique top	6	20000	0	0.00000
anytime	fresh coin each round	unique top	8	20000	20	0.00100
anytime	fresh coin each round	unique top	12	20000	5	0.00025
anytime	fresh coin each round	unique top	50	20000	7	0.00035
anytime	fresh coin each round	unique top	200	20000	0	0.00000
anytime	fresh coin each round	unique top	1,000	20000	0	0.00000
anytime	one fixed priority	tied top	6	20000	0	0.00000
anytime	one fixed priority	tied top	8	20000	43	0.00215
anytime	one fixed priority	tied top	12	20000	0	0.00000
anytime	one fixed priority	tied top	50	20000	1	0.00005
anytime	one fixed priority	tied top	200	20000	7	0.00035
anytime	one fixed priority	tied top	1,000	20000	0	0.00000
anytime	one fixed priority	unique top	6	20000	0	0.00000
anytime	one fixed priority	unique top	8	20000	8	0.00040
anytime	one fixed priority	unique top	12	20000	0	0.00000
anytime	one fixed priority	unique top	50	20000	0	0.00000
anytime	one fixed priority	unique top	200	20000	0	0.00000
anytime	one fixed priority	unique top	1,000	20000	0	0.00000
known horizon	fresh coin each round	tied top	6	20000	0	0.00000
known horizon	fresh coin each round	tied top	8	20000	19	0.00095
known horizon	fresh coin each round	tied top	12	20000	0	0.00000
known horizon	fresh coin each round	tied top	50	20000	2	0.00010
known horizon	fresh coin each round	tied top	200	20000	0	0.00000
known horizon	fresh coin each round	tied top	1,000	20000	0	0.00000
known horizon	fresh coin each round	unique top	6	20000	0	0.00000
known horizon	fresh coin each round	unique top	8	20000	15	0.00075
known horizon	fresh coin each round	unique top	12	20000	0	0.00000
known horizon	fresh coin each round	unique top	50	20000	0	0.00000
known horizon	fresh coin each round	unique top	200	20000	0	0.00000
known horizon	fresh coin each round	unique top	1,000	20000	0	0.00000
known horizon	one fixed priority	tied top	6	20000	0	0.00000
known horizon	one fixed priority	tied top	8	20000	0	0.00000
known horizon	one fixed priority	tied top	12	20000	0	0.00000
known horizon	one fixed priority	tied top	50	20000	0	0.00000
known horizon	one fixed priority	tied top	200	20000	0	0.00000
known horizon	one fixed priority	tied top	1,000	20000	0	0.00000
known horizon	one fixed priority	unique top	6	20000	0	0.00000
known horizon	one fixed priority	unique top	8	20000	0	0.00000
known horizon	one fixed priority	unique top	12	20000	0	0.00000
known horizon	one fixed priority	unique top	50	20000	0	0.00000
known horizon	one fixed priority	unique top	200	20000	0	0.00000
known horizon	one fixed priority	unique top	1,000	20000	0	0.00000

ids around the thinking delimiters. A seeded split draws 50 calibration and 150 disjoint evaluation questions per benchmark; subgroups were fixed before generation (GSM8K by reference-solution length, MATH-500 by its own level: 1–2, 3, 4–5). Experiment A/B: every evaluation question under every budget, three seeds, 2,700 generations, condition order shuffled per seed. Experiment C: one shuffled stream per seed replayed through five gateway policies, 4,500 generations; the task-adaptive rule uses only the benchmark label available before generation. Grading reads only the text after the thinking segment, prefers a boxed answer, and compares

Table 10: Fairness among top agents: the difference in pull share between two tied top agents, per round. The interval covers zero at every horizon of 2,000 or more under both indices and both initializations.

instance	index	initialization	n	share difference	95% interval
tied top	anytime	fixed	6	-0.00700	[-0.01395, -0.00005]
tied top	anytime	fixed	12	+0.00515	[-0.00036, +0.01065]
tied top	anytime	fixed	50	+0.00241	[-0.00275, +0.00757]
tied top	anytime	fixed	2,000	+0.01648	[-0.00410, +0.03706]
tied top	anytime	fixed	8,000	-0.00539	[-0.02369, +0.01292]
tied top	anytime	fixed	20,000	+0.00321	[-0.01401, +0.02042]
tied top	anytime	symmetric	6	-0.00571	[-0.01269, +0.00127]
tied top	anytime	symmetric	12	+0.00477	[-0.00074, +0.01028]
tied top	anytime	symmetric	50	+0.00220	[-0.00295, +0.00736]
tied top	anytime	symmetric	2,000	+0.01648	[-0.00410, +0.03706]
tied top	anytime	symmetric	8,000	-0.00539	[-0.02369, +0.01292]
tied top	anytime	symmetric	20,000	+0.00321	[-0.01401, +0.02042]
tied top	horizon	fixed	6	-0.00521	[-0.01058, +0.00016]
tied top	horizon	fixed	12	+0.00390	[-0.00022, +0.00801]
tied top	horizon	fixed	50	+0.00196	[-0.00148, +0.00539]
tied top	horizon	fixed	2,000	+0.01190	[-0.00185, +0.02566]
tied top	horizon	fixed	8,000	-0.00093	[-0.01390, +0.01203]
tied top	horizon	fixed	20,000	+0.00254	[-0.00984, +0.01492]
tied top	horizon	symmetric	6	-0.00404	[-0.00941, +0.00133]
tied top	horizon	symmetric	12	+0.00394	[-0.00018, +0.00806]
tied top	horizon	symmetric	50	+0.00197	[-0.00147, +0.00541]
tied top	horizon	symmetric	2,000	+0.01190	[-0.00185, +0.02566]
tied top	horizon	symmetric	8,000	-0.00093	[-0.01390, +0.01203]
tied top	horizon	symmetric	20,000	+0.00254	[-0.00984, +0.01492]



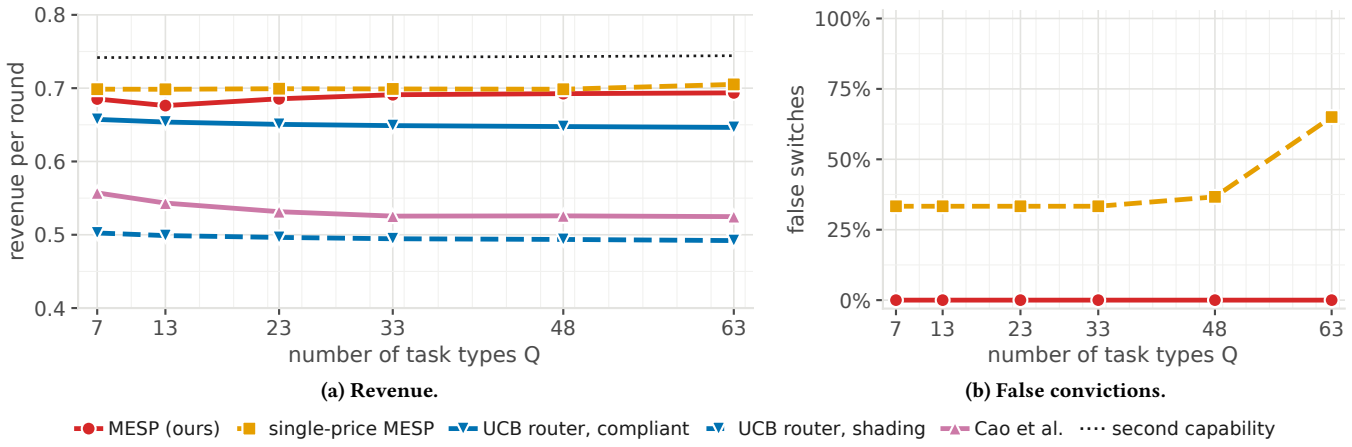
(a) Graded rewards, two scorings. **(b) Revenue against label noise.** **(c) False convictions against label noise.**

Figure 14: Robustness to graded rewards and label noise.

numerically for GSM8K and after a conservative LaTeX normalisation for MATH-500; malformed and truncated answers count as wrong. Intervals are question-level bootstrap percentiles with 2,000 resamples, paired within questions.

Table 11: Provider bank `covertype`: real labeled tasks answered by real algorithmic providers with a compute knob. Outcomes are real; the effort choice is scripted, so these runs are evidence about the mechanism and not about provider behavior. The last column divides the median detection delay by the first-order formula evaluated at the realized shortfall, and is reported only where the monitor fired. The price-in-band column is the fraction of replicates in which the announced price lay inside the band the proof requires.

variant	behavior	n	reward	price in band	false switch	delay / theory
exclusion	sabotage	2,000	0.8012	1.000	0.000	1.015
exclusion	sabotage	8,000	0.8055	1.000	0.000	1.017
exclusion	sabotage	20,000	0.7960	1.000	0.000	1.015
exclusion	profile	2,000	0.7677	1.000	0.000	n/a
exclusion	profile	8,000	0.7991	1.000	0.000	n/a
exclusion	profile	20,000	0.8083	1.000	0.000	n/a
no monitor	sabotage	2,000	0.5186	1.000	0.000	n/a
no monitor	sabotage	8,000	0.4050	1.000	0.000	n/a
no monitor	sabotage	20,000	0.2937	1.000	0.000	n/a
no monitor	profile	2,000	0.7677	1.000	0.000	n/a
no monitor	profile	8,000	0.7991	1.000	0.000	n/a
no monitor	profile	20,000	0.8083	1.000	0.000	n/a
UCB1 fallback	sabotage	2,000	0.8507	1.000	0.000	1.015
UCB1 fallback	sabotage	8,000	0.8714	1.000	0.000	1.017
UCB1 fallback	sabotage	20,000	0.8817	1.000	0.000	1.015
UCB1 fallback	profile	2,000	0.7677	1.000	0.000	n/a
UCB1 fallback	profile	8,000	0.7991	1.000	0.000	n/a
UCB1 fallback	profile	20,000	0.8083	1.000	0.000	n/a
MESP	sabotage	2,000	0.8508	1.000	0.000	1.015
MESP	sabotage	8,000	0.8729	1.000	0.000	1.017
MESP	sabotage	20,000	0.8833	1.000	0.000	1.015
MESP	profile	2,000	0.7677	1.000	0.000	n/a
MESP	profile	8,000	0.7991	1.000	0.000	n/a
MESP	profile	20,000	0.8083	1.000	0.000	n/a



(a) Revenue. **(b) False convictions.**

Figure 15: Scaling the number of task types, $Q = 7$ to 63.

Table 12: Provider bank `fashion_mnist`: real labeled tasks answered by real algorithmic providers with a compute knob. Outcomes are real; the effort choice is scripted, so these runs are evidence about the mechanism and not about provider behavior. The last column divides the median detection delay by the first-order formula evaluated at the realized shortfall, and is reported only where the monitor fired.

variant	behavior	n	reward	price in band	false switch	delay / theory
exclusion	sabotage	2,000	0.7636	1.000	0.000	1.013
exclusion	sabotage	8,000	0.7686	1.000	0.000	1.011
exclusion	sabotage	20,000	0.7590	1.000	0.000	1.014
exclusion	profile	2,000	0.7282	1.000	0.000	n/a
exclusion	profile	8,000	0.7609	1.000	0.000	n/a
exclusion	profile	20,000	0.7668	1.000	0.000	n/a
no monitor	sabotage	2,000	0.4750	1.000	0.000	n/a
no monitor	sabotage	8,000	0.3772	1.000	0.000	n/a
no monitor	sabotage	20,000	0.2765	1.000	0.000	n/a
no monitor	profile	2,000	0.7282	1.000	0.000	n/a
no monitor	profile	8,000	0.7609	1.000	0.000	n/a
no monitor	profile	20,000	0.7668	1.000	0.000	n/a
UCB1 fallback	sabotage	2,000	0.8222	1.000	0.000	1.013
UCB1 fallback	sabotage	8,000	0.8442	1.000	0.000	1.011
UCB1 fallback	sabotage	20,000	0.8535	1.000	0.000	1.014
UCB1 fallback	profile	2,000	0.7282	1.000	0.000	n/a
UCB1 fallback	profile	8,000	0.7609	1.000	0.000	n/a
UCB1 fallback	profile	20,000	0.7668	1.000	0.000	n/a
MESP	sabotage	2,000	0.8229	1.000	0.000	1.013
MESP	sabotage	8,000	0.8454	1.000	0.000	1.011
MESP	sabotage	20,000	0.8563	1.000	0.000	1.014
MESP	profile	2,000	0.7282	1.000	0.000	n/a
MESP	profile	8,000	0.7609	1.000	0.000	n/a
MESP	profile	20,000	0.7668	1.000	0.000	n/a

benchmark	budget	accuracy (95% CI)	reasoning tokens	total tokens	at cap	truncated
GSM8K	128	93.3 (89.8, 96.4)	128	578	100%	0.2%
GSM8K	512	94.9 (91.6, 97.8)	476	731	71%	0
GSM8K	2,048	95.6 (92.2, 98.4)	922	1,137	11%	0
MATH-500	128	81.3 (75.6, 86.4)	128	2,302	100%	9.6%
MATH-500	512	86.4 (81.6, 91.1)	512	1,948	99%	4.7%
MATH-500	2,048	86.4 (81.3, 91.1)	1,647	2,668	51%	5.1%

gateway policies, 300 questions, three seeds	
fixed low / medium / high	89.2 / 90.0 / 91.7% at 128 / 492 / 1,277 reasoning tokens
task-adaptive	91.7% at 880 reasoning tokens; vs fixed high 0.0 (−1.3, +1.3)
random, matched mix	90.1% at 711 reasoning tokens; adaptive minus random +1.6 (−0.1, +3.2)

Table 13: The serving experiment: accuracy and computation per budget, and the five gateway policies. Percentages in the "at cap" column are the share of requests whose reasoning segment reached the budget. A rerun of 150 requests from the saved configuration reproduced the per-budget accuracy and token means within sampling noise with 10% of completions byte-identical, as expected across GPU type and batch composition.

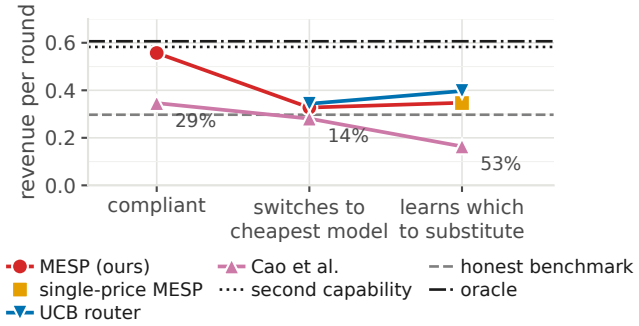


Figure 16: Substitution as shading: revenue when the top provider substitutes a cheaper model; numbers are the share of rounds the mechanism of Cao et al. [4] leaves unserved.

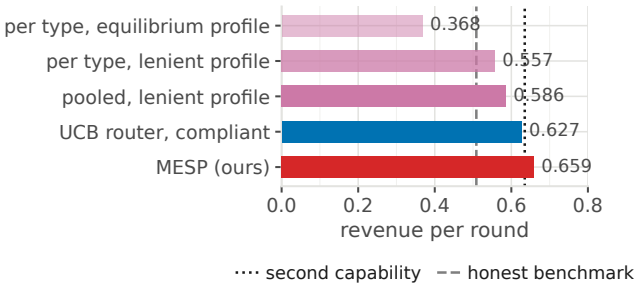
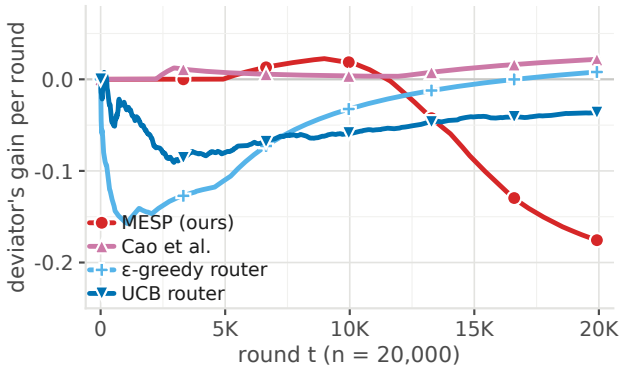
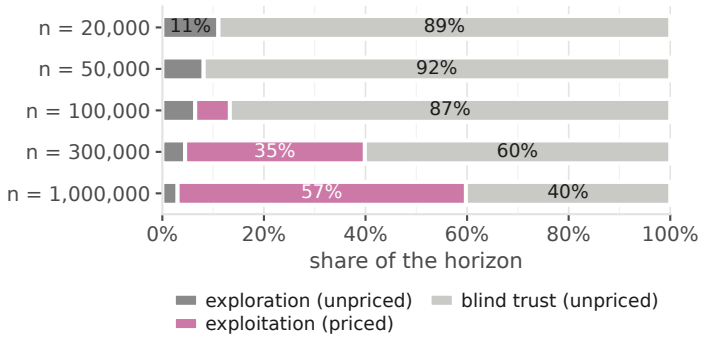


Figure 17: The mechanism of Cao et al. [4] run per task type, under its equilibrium and lenient profiles, beside its pooled form and MESP.

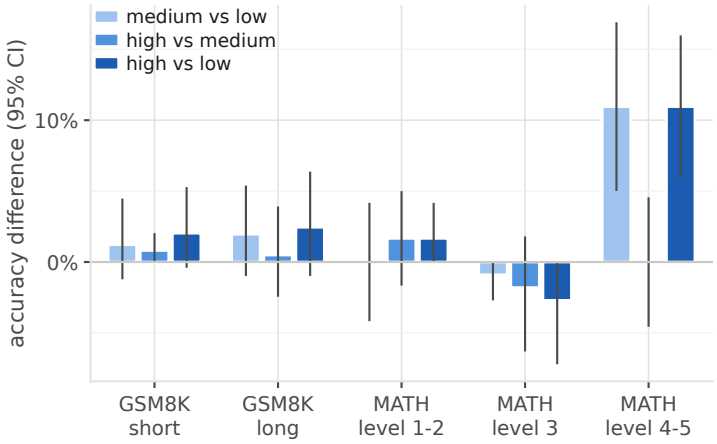


(a) Gain per round elapsed, $n = 20{,}000$.

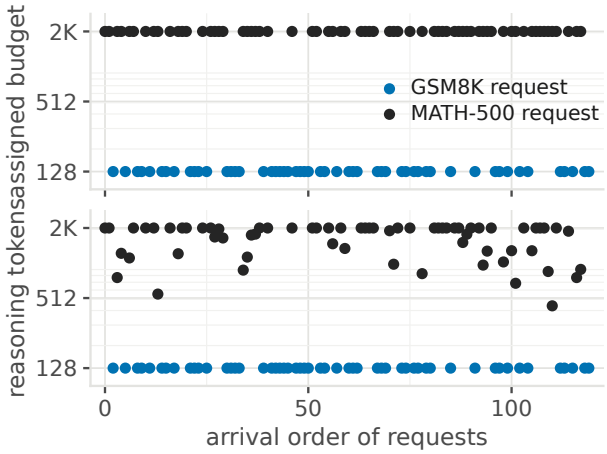


(b) Phase split of Cao et al. [4] against horizon.

Figure 18: Behind Figure 9.



(a) Paired accuracy gains between budgets.



(b) Budgets assigned to consecutive requests, and tokens generated.

Figure 19: The serving experiment, continued.